



Cambridge International Examinations
Cambridge International General Certificate of Secondary Education (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/21

May/June 2018

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

A copy of the Periodic Table is printed on page 16.

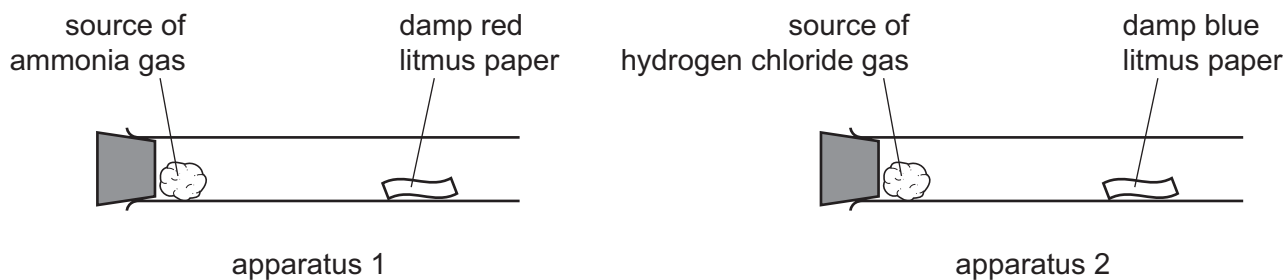
Electronic calculators may be used.

This document consists of **13** printed pages and **3** blank pages.



- 1 A student investigated the diffusion of ammonia gas, NH_3 , and hydrogen chloride gas, HCl .

Two sets of apparatus were set up as shown at room temperature and pressure.

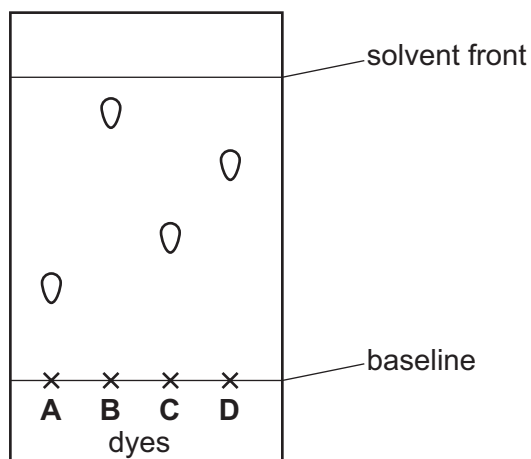


The damp red litmus paper in apparatus 1 changed colour after 30 seconds.

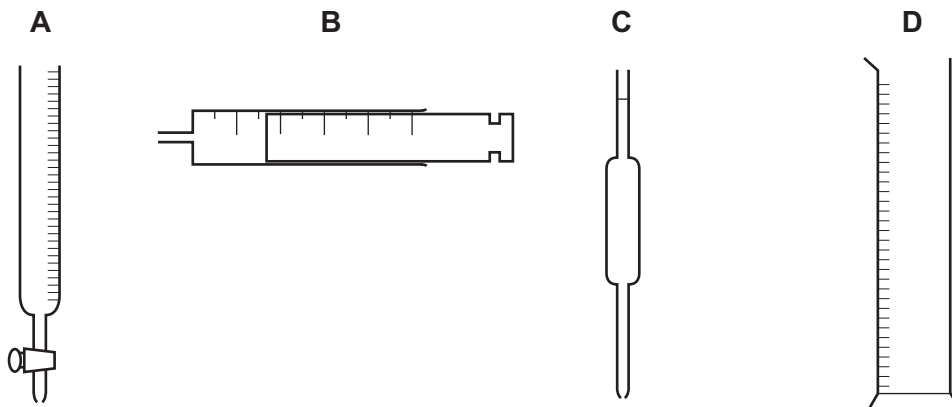
How long does it take for the damp blue litmus paper to change colour in apparatus 2?

- A** 64 seconds
 - B** 30 seconds
 - C** 21 seconds
 - D** The blue litmus paper would not change colour.
- 2 Chromatography is a technique used to separate coloured dyes.

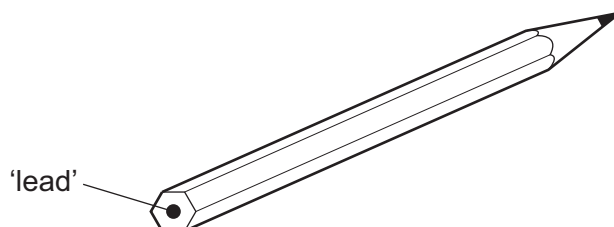
Which dye has an R_f value of 0.7?



- 3 Which piece of apparatus is used to measure exactly 26.3 cm^3 of a liquid?



- 4 The 'lead' in a pencil is made of a mixture of graphite and clay.



When the percentage of graphite is increased, the pencil slides across the paper more easily.

Which statement explains this observation?

- A Graphite has a high melting point.
- B Graphite is a form of carbon.
- C Graphite is a lubricant.
- D Graphite is a non-metal.

- 5 Chlorine exists as two common isotopes, ^{35}Cl and ^{37}Cl .

Information about these two isotopes is shown.

	number of protons	number of neutrons	number of electron shells
^{35}Cl	17	18	3
^{37}Cl	17	20	3

Which statement explains why the two isotopes are of the same element?

- A** Both have the same number of electron shells.
B Both have the same number of protons.
C Both have 7 outer shell electrons.
D ^{37}Cl has 2 more neutrons than ^{35}Cl .

- 6 Which substance is **not** a macromolecule?

- A** diamond
B graphite
C silicon(IV) oxide
D sulfur

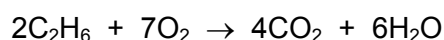
- 7 Copper is a metallic element.

Which statements about copper are correct?

- 1 Copper is malleable because layers of ions are in fixed positions and cannot move.
- 2 The structure of copper consists of negative ions in a lattice.
- 3 Copper conducts electricity because electrons can move through the metal.
- 4 Electrons hold copper ions together in a lattice by electrostatic attraction.

- A** 1 and 2 **B** 2, 3 and 4 **C** 2 and 3 only **D** 3 and 4 only

- 8 The equation for the combustion of ethane is shown.



Which volume of carbon dioxide, at room temperature and pressure, is formed when 0.5 moles of ethane burn?

- A** 48 dm³ **B** 24 dm³ **C** 12 dm³ **D** 6 dm³

- 9 A solution of ethanoic acid, CH_3COOH , has a concentration of 2 mol/dm^3 .

Which statement about this solution is correct?

- A 20 g of ethanoic acid is dissolved in 10 cm^3 of water.
- B 30 g of ethanoic acid is dissolved in 250 cm^3 of water.
- C 60 g of ethanoic acid is dissolved in 1 dm^3 of water.
- D 120 g of ethanoic acid is dissolved in 2 dm^3 of water.

- 10 Aqueous copper(II) sulfate is electrolysed using copper electrodes.

Which statement is correct?

- A A reduction reaction occurs at the positive electrode.
- B The blue colour of the solution becomes darker.
- C The concentration of copper ions in the solution decreases.
- D The mass of the negative electrode increases.

- 11 Dilute sulfuric acid is electrolysed using inert electrodes.

What are the ionic half-equations for the reactions that take place at each electrode?

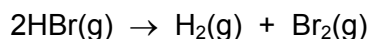
	positive electrode	negative electrode
A	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	$4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$
B	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	$4\text{OH}^- + 4\text{H}^+ \rightarrow 4\text{H}_2\text{O}$
C	$4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
D	$4\text{OH}^- + 4\text{H}^+ \rightarrow 4\text{H}_2\text{O}$	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$

- 12 Plant cells use energy from sunlight for photosynthesis.

Which row describes and explains the energy change that occurs?

	type of energy change	explanation
A	endothermic	less energy is released making bonds than is absorbed to break bonds
B	endothermic	more energy is released making bonds than is absorbed to break bonds
C	exothermic	less energy is released making bonds than is absorbed to break bonds
D	exothermic	more energy is released making bonds than is absorbed to break bonds

- 13** Hydrogen bromide decomposes to form hydrogen and bromine. The equation is shown.



The bond energies are shown in the table. The reaction is endothermic.

bond	bond energy in kJ/mol
Br–Br	+193
H–Br	+366
H–H	+436

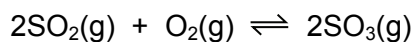
What is the energy change for the reaction?

- A** +263 kJ/mol **B** +103 kJ/mol **C** –103 kJ/mol **D** –263 kJ/mol
- 14** Which row describes the effects of increasing both concentration and temperature on the collisions between reacting particles?

	increasing concentration	increasing temperature
A	more collisions per second only	more collisions per second only
B	more collisions per second and more collisions with sufficient energy to react	more collisions per second only
C	more collisions per second only	more collisions per second and more collisions with sufficient energy to react
D	more collisions per second and more collisions with sufficient energy to react	more collisions per second and more collisions with sufficient energy to react

- 15** The formation of sulfur trioxide is a reversible reaction.

The equation is shown.

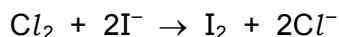


The forward reaction is exothermic.

Which conditions produce the highest equilibrium yield of sulfur trioxide?

	pressure	temperature
A	high	high
B	high	low
C	low	high
D	low	low

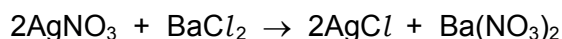
- 16 Chlorine displaces iodide ions from potassium iodide.



What is the oxidising agent?

- A chloride ions
 - B chlorine
 - C iodide ions
 - D iodine
- 17 Which statement about oxides is correct?
- A A solution of magnesium oxide has a pH less than pH 7.
 - B A solution of sulfur dioxide has a pH greater than pH 7.
 - C Magnesium oxide reacts with nitric acid to make a salt.
 - D Sulfur dioxide reacts with hydrochloric acid to make a salt.
- 18 Which solution has the lowest pH?
- A 0.1 mol/dm³ ammonia solution
 - B 0.1 mol/dm³ ethanoic acid
 - C 0.1 mol/dm³ lithium hydroxide
 - D 0.1 mol/dm³ nitric acid
- 19 A student mixes silver nitrate and barium chloride to form a white precipitate of silver chloride.

The equation is shown.



Which row describes the solubility of the salts?

	soluble	insoluble
A	silver nitrate	barium chloride, barium nitrate and silver chloride
B	silver nitrate and barium chloride	barium nitrate and silver chloride
C	silver nitrate, barium chloride and barium nitrate	silver chloride
D	silver nitrate, barium chloride and silver chloride	barium nitrate

20 Which methods are suitable for preparing **both** zinc sulfate and copper(II) sulfate?

- 1 reacting the metal oxide with warm dilute aqueous sulfuric acid
- 2 reacting the metal with dilute aqueous sulfuric acid
- 3 reacting the metal carbonate with dilute aqueous sulfuric acid

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

21 Which element is in the same period of the Periodic Table as silicon?

- A** germanium
B scandium
C sodium
D strontium

22 Which statement about the halogens is correct?

- A** A sample of bromine reacts with potassium chloride solution.
B A sample of bromine reacts with potassium iodide solution.
C A sample of chlorine has a higher density than a sample of bromine.
D A sample of chlorine is a darker colour than a sample of bromine.

23 Which row shows the catalytic activity of transition elements and their compounds?

	catalytic activity of transition elements	catalytic activity of compounds of transition elements
A	good	good
B	good	poor
C	poor	good
D	poor	poor

24 The following statements are made about the metals copper, iron, magnesium and zinc.

- 1 Their oxides are acidic.
- 2 They all conduct electricity in the solid state.
- 3 They all have high melting points.
- 4 They all react with dilute acids to form hydrogen.

Which statements are correct?

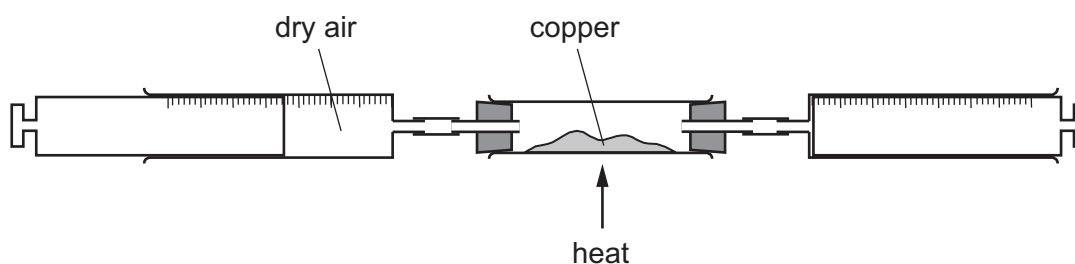
A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

- 25 Silver is a less reactive metal than cadmium.

Cadmium is a less reactive metal than barium.

Which statement is correct?

- A Barium does not react when heated with silver oxide.
 - B Cadmium displaces barium from a solution of barium chloride.
 - C Cadmium displaces silver from a solution of silver nitrate.
 - D Cadmium reacts when heated with barium oxide.
- 26 Aluminium metal is extracted from aluminium oxide using electrolysis.
- Which statement about the extraction process is **not** correct?
- A A large amount of electricity is required.
 - B Molten cryolite is used to dissolve the aluminium oxide.
 - C Oxygen gas is released which reacts to form carbon dioxide.
 - D The negative electrodes burn away and have to be replaced.
- 27 Which statement explains why aluminium is used in the manufacture of aircraft?
- A It conducts heat well.
 - B It has a low density.
 - C It is a good conductor of electricity.
 - D It is easy to recycle.
- 28 Dry air is passed over hot copper until all the oxygen has reacted.



The volume of gas at the end of the reaction is 120 cm^3 .

What is the starting volume of dry air?

- A 132 cm^3
- B 152 cm^3
- C 180 cm^3
- D 570 cm^3

29 A steel bicycle which had been left outdoors for several months was starting to rust.

What would **not** reduce the rate of corrosion?

- A** Remove the rust and paint the bicycle.
- B** Remove the rust and store the bicycle in a dry shed.
- C** Remove the rust and wipe the bicycle with a clean, damp cloth.
- D** Remove the rust and wipe the bicycle with an oily cloth.

30 Which statements about water are correct?

- 1 Household water contains dissolved salts.
- 2 Water for household use is filtered to remove soluble impurities.
- 3 Water is treated with chlorine to kill bacteria.
- 4 Water is used in industry for cooling.

- A** 1, 2, 3 and 4
- B** 1, 2 and 3 only
- C** 1, 3 and 4 only
- D** 2, 3 and 4 only

31 Ammonia is manufactured by reacting hydrogen with nitrogen in the Haber process.

Which row describes the sources of hydrogen and nitrogen and the conditions used in the manufacture of ammonia in the Haber process?

	source of hydrogen	source of nitrogen	temperature of reaction / °C	pressure of reaction / atm
A	air	natural gas	250	2
B	air	natural gas	250	200
C	natural gas	air	450	2
D	natural gas	air	450	200

32 Which statements about the carbon cycle are correct?

- 1 Carbon dioxide is added to the atmosphere by respiration.
- 2 Carbon dioxide is added to the atmosphere by combustion of coal.
- 3 Carbon dioxide is removed from the atmosphere by photosynthesis.

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

33 Which statement about sulfur and its compounds is **not** correct?

- A Sulfur dioxide is used as a food preservative.
- B Sulfur dioxide turns acidified aqueous potassium manganate(VII) from purple to colourless.
- C Sulfur forms a basic oxide.
- D Sulfur is used in the manufacture of sulfuric acid.

34 Which process is used to convert limestone (calcium carbonate) into lime?

- A electrolysis
- B fractional distillation
- C incomplete combustion
- D thermal decomposition

35 What is **not** the correct use of the fraction named?

	name of fraction	use
A	fuel oil	making waxes
B	gas oil	fuel in diesel engines
C	kerosene	jet fuel
D	naphtha	making chemicals

36 Which reaction is **not** a reaction which alkenes undergo?

- A bromination
- B hydration
- C hydrogenation
- D hydrolysis

37 Which substances can be obtained by cracking hydrocarbons?

- A ethanol and ethene
- B ethanol and hydrogen
- C ethene and hydrogen
- D ethene and poly(ethene)

38 Ethanol is produced by fermentation or from ethene.

What is a disadvantage of producing ethanol by fermentation?

- A Distillation is needed to purify the ethanol produced.
- B Fermentation uses glucose from plants.
- C Fermentation is catalysed by enzymes in yeast.
- D Fermentation occurs at a low temperature and pressure.

39 Which structural formula represents methyl propanoate?

- A $\text{CH}_3\text{CH}_2\text{COOCH}_3$
- B $\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3$
- C $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_3$
- D $\text{HCOOCH}_2\text{CH}_2\text{CH}_3$

40 Which row describes addition polymerisation and condensation polymerisation?

	addition polymerisation	condensation polymerisation
A	monomers have a C=C double bond and the polymer is the only product	monomers have a C=C double bond and the polymer is the only product
B	monomers have a C=C double bond and the polymer is the only product	the monomers react to form the polymer and a small molecule
C	the monomers react to form the polymer and a small molecule	monomers have a C=C double bond and the polymer is the only product
D	the monomers react to form the polymer and a small molecule	the monomers react to form the polymer and a small molecule

BLANK PAGE

BLANK PAGE

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

Group

lanthanoids

57	La	lanthanum	139	58	Ce	cerium	140	59	Pr	praseodymium	141	60	Nd	neodymium	144	61	Pm	promethium	—	62	Sm	samarium	150	63	Eu	euroium	152	64	Gd	gadolinium	157	65	Tb	terbium	159	66	Dy	dysprosium	163	67	Ho	holmium	165	68	Er	erbium	167	69	Tm	thulium	169	70	Yb	ytterbium	173	71	Lu	lutetium	175
----	----	-----------	-----	----	----	--------	-----	----	----	--------------	-----	----	----	-----------	-----	----	----	------------	---	----	----	----------	-----	----	----	---------	-----	----	----	------------	-----	----	----	---------	-----	----	----	------------	-----	----	----	---------	-----	----	----	--------	-----	----	----	---------	-----	----	----	-----------	-----	----	----	----------	-----

actinoids

89	Ac	actinium	—	90	Th	thorium	232	91	Pa	protactinium	231	92	U	uranium	238	93	Np	neptunium	—	94	Pu	plutonium	—	95	Am	americium	—	96	Cm	curium	—	97	Bk	berkelium	—	98	Cf	californium	—	99	Es	einsteinium	—	100	Fm	fermium	—	101	Md	mendelevium	—	102	No	nobelium	—	103	Lr	lawrencium	—
----	----	----------	---	----	----	---------	-----	----	----	--------------	-----	----	---	---------	-----	----	----	-----------	---	----	----	-----------	---	----	----	-----------	---	----	----	--------	---	----	----	-----------	---	----	----	-------------	---	----	----	-------------	---	-----	----	---------	---	-----	----	-------------	---	-----	----	----------	---	-----	----	------------	---

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education

CHEMISTRY

0971/21

Paper 2 Multiple Choice (Extended)

May/June 2018

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2018 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

IGCSE™ is a registered trademark.

This document consists of **3** printed pages.



Cambridge Assessment
International Education

Question	Answer	Marks
1	A	1
2	D	1
3	A	1
4	C	1
5	B	1
6	D	1
7	D	1
8	B	1
9	B	1
10	D	1
11	C	1
12	A	1
13	B	1
14	C	1
15	B	1
16	B	1
17	C	1
18	D	1
19	C	1
20	C	1
21	C	1
22	B	1
23	A	1
24	C	1
25	C	1
26	D	1
27	B	1
28	B	1

Question	Answer	Marks
29	C	1
30	C	1
31	D	1
32	A	1
33	C	1
34	D	1
35	A	1
36	D	1
37	C	1
38	A	1
39	A	1
40	B	1



Cambridge International Examinations
Cambridge International General Certificate of Secondary Education (9–1)

CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

October/November 2018

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

A copy of the Periodic Table is printed on page 16.

Electronic calculators may be used.

This document consists of **16** printed pages.



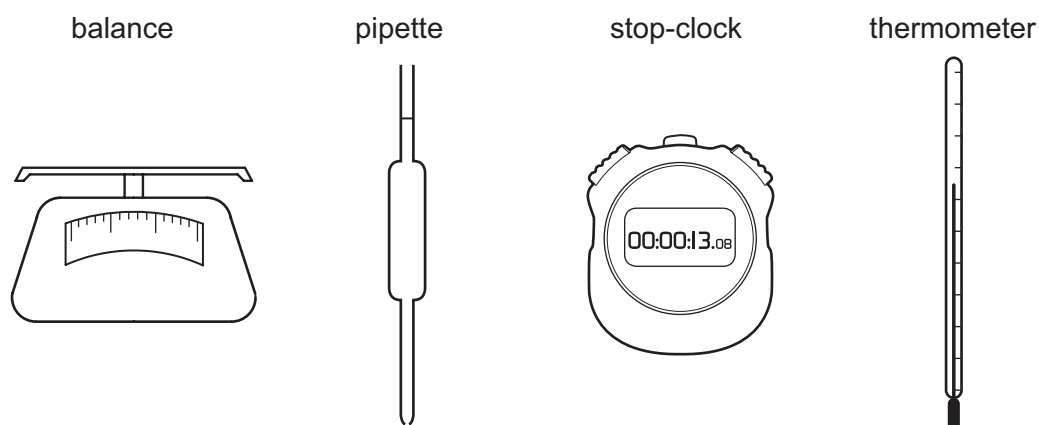
- 1 Oxygen and fluorine are gaseous elements next to each other in the Periodic Table.

Under the same conditions of temperature and pressure, oxygen diffuses1..... than fluorine because its2..... is less than that of fluorine.

Which words correctly complete gaps 1 and 2?

	1	2
A	faster	molecular mass
B	faster	reactivity
C	slower	molecular mass
D	slower	reactivity

- 2 The diagrams show four pieces of laboratory equipment.



Which equipment is essential to find out if dissolving a salt in water is an exothermic process?

	balance	pipette	stop-clock	thermometer
A	x	x	x	✓
B	✓	x	x	✓
C	x	✓	x	✓
D	✓	x	✓	x

- 3 How many neutrons are present in the atom ${}^{45}_{21}\text{X}$?

A 21 **B** 24 **C** 45 **D** 66

- 4 Two naturally occurring isotopes of oxygen are ^{16}O and ^{17}O .

Which statement is correct?

- A Both isotopes react with iron to form rust.
- B Neither isotope reacts with iron to form rust.
- C Only ^{16}O reacts with iron to form rust.
- D Only ^{17}O reacts with iron to form rust.

- 5 How many electrons are used to form covalent bonds in a molecule of methanol, CH_3OH ?

- A 5 B 6 C 8 D 10

- 6 Potassium bromide and methanol are both compounds.

Their melting points are different.

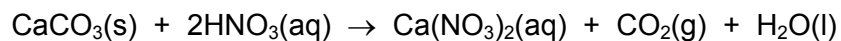
Which row is correct?

	substance with the higher melting point	reason why the melting points are different
A	methanol	the attractive forces between oppositely charged ions is greater than the attractive forces between molecules
B	methanol	the attractive forces between molecules is greater than the attractive forces between oppositely charged ions
C	potassium bromide	the attractive forces between oppositely charged ions is greater than the attractive forces between molecules
D	potassium bromide	the attractive forces between molecules is greater than the attractive forces between oppositely charged ions

- 7 Which gas sample contains the smallest number of molecules?

- A 4 g of helium
- B 16 g of oxygen
- C 28 g of carbon monoxide
- D 28 g of nitrogen

- 8 The equation for the reaction between calcium carbonate and dilute nitric acid is shown.



25 g of calcium carbonate is reacted with an excess of dilute nitric acid.

Which mass of calcium nitrate and which volume of carbon dioxide is produced at room temperature and pressure?

	mass of calcium nitrate / g	volume of carbon dioxide / dm ³
A	29	6
B	29	12
C	41	6
D	41	12

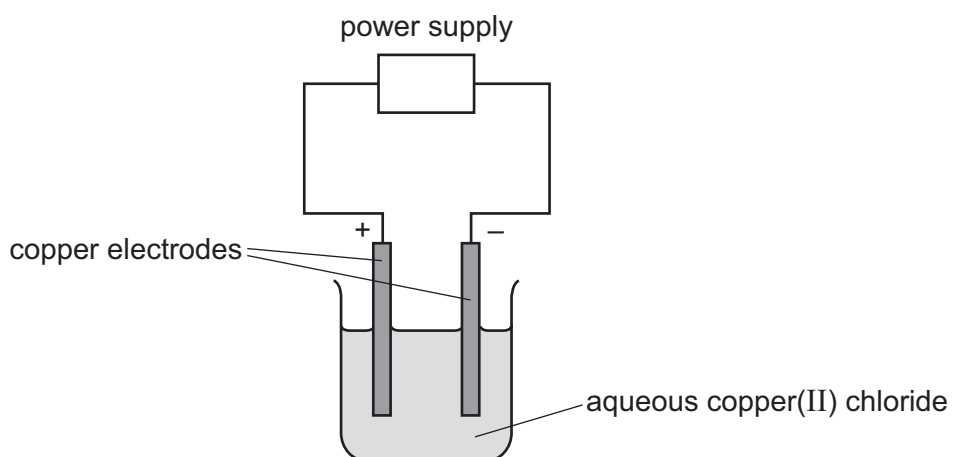
- 9 The formulae of some ions are shown.

positive ion	negative ion
Ti^{4+}	PO_4^{3-}
Al^{3+}	SO_4^{2-}
Mg^{2+}	NO_3^-
K^+	Cl^-

Which formula is **not** correct?

- A** $\text{Al}_3(\text{SO}_4)_2$ **B** K_3PO_4 **C** $\text{Mg}(\text{NO}_3)_2$ **D** TiCl_4

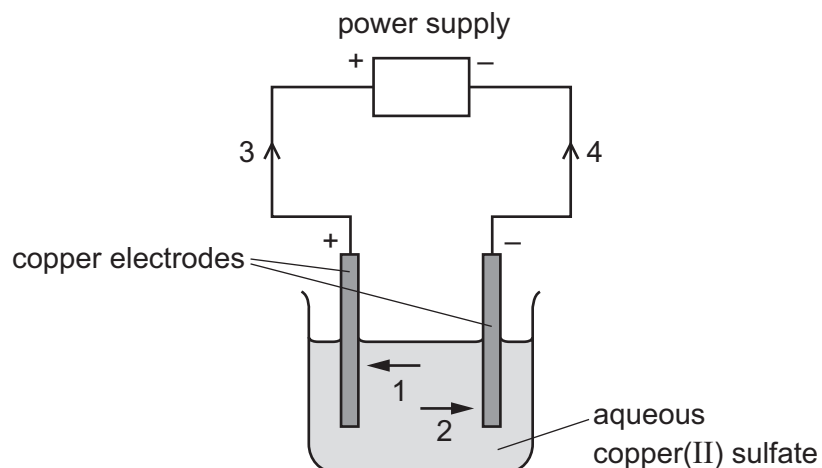
10 Concentrated aqueous copper(II) chloride is electrolysed using copper electrodes as shown.



What happens to the mass of each electrode during this process?

	positive electrode	negative electrode
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

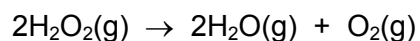
11 The diagram shows a circuit used to electrolyse aqueous copper(II) sulfate.



Which arrows indicate the movement of the copper ions in the electrolyte and of the electrons in the external circuit?

	copper ions	electrons
A	1	3
B	1	4
C	2	3
D	2	4

12 Hydrogen peroxide, H–O–O–H, decomposes to form water and oxygen.



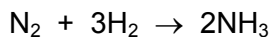
The bond energies are shown in the table. The reaction is exothermic.

bond	bond energy in kJ/mol
O–H	+460
O–O	+150
O=O	+496

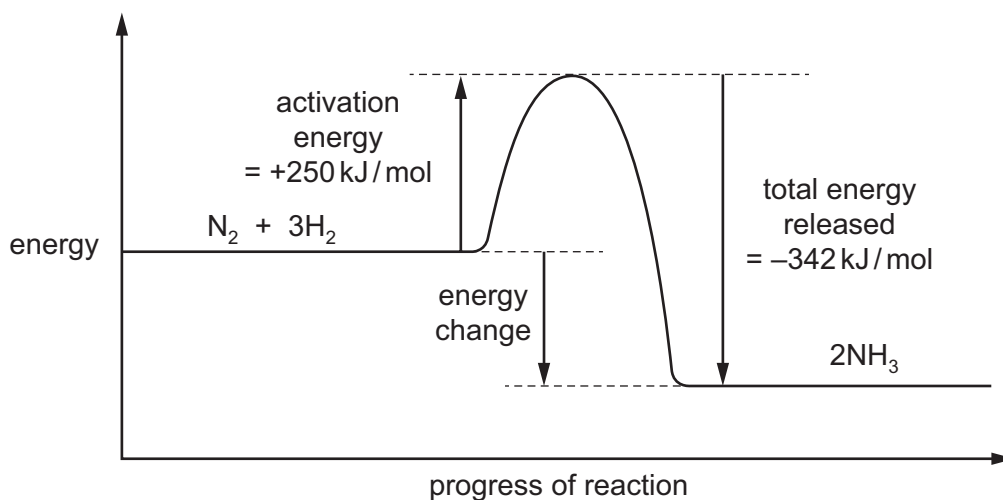
What is the energy change for the reaction?

A –346 kJ/mol **B** –196 kJ/mol **C** +196 kJ/mol **D** +346 kJ/mol

- 13 The equation for the formation of ammonia is shown.



The energy level diagram for the reaction is shown.



What is the energy change for the reaction?

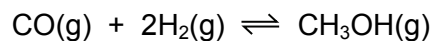
- A -592 kJ/mol
 - B -92 kJ/mol
 - C +92 kJ/mol
 - D +592 kJ/mol
- 14 The rate of reaction between magnesium ribbon and 2 mol/dm³ hydrochloric acid at 25 °C to produce hydrogen gas is measured.

In another experiment, either the concentration of the hydrochloric acid or the temperature is changed. All other conditions are kept the same.

Which conditions increase the rate of reaction?

- A 1 mol/dm³ hydrochloric acid at 25 °C
- B 2 mol/dm³ hydrochloric acid at 10 °C
- C 2 mol/dm³ hydrochloric acid at 20 °C
- D 3 mol/dm³ hydrochloric acid at 25 °C

15 Methanol is prepared by the reversible reaction shown.



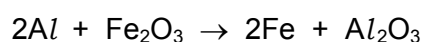
The forward reaction is exothermic.

Which conditions produce the highest equilibrium yield of methanol?

	temperature	pressure
A	high	high
B	high	low
C	low	high
D	low	low

16 The thermite reaction can be used to produce iron from iron(III) oxide.

The equation for the reaction is shown.



Which statements about this reaction are correct?

- 1 Aluminium is the oxidising agent.
- 2 Aluminium is less reactive than iron.
- 3 Electrons are transferred from aluminium to iron.
- 4 The iron in the iron(III) oxide is reduced.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

17 In which row are the oxides correctly identified?

	acidic	basic
A	magnesium oxide, calcium oxide	sulfur dioxide, carbon dioxide
B	magnesium oxide, sulfur dioxide	carbon dioxide, calcium oxide
C	sulfur dioxide, carbon dioxide	calcium oxide, magnesium oxide
D	sulfur dioxide, magnesium oxide	calcium oxide, carbon dioxide

- 18** When dilute sulfuric acid is added to solid X, a colourless solution is formed and a gas is produced.

What is X?

- A** copper(II) oxide
- B** sodium oxide
- C** copper(II) carbonate
- D** sodium carbonate

- 19** A few drops of methyl orange are added to a reaction mixture.

During the reaction, a gas is produced and the methyl orange turns from red to orange.

What are the reactants?

- A** aqueous sodium hydroxide and ammonium chloride
- B** aqueous sodium hydroxide and calcium carbonate
- C** dilute hydrochloric acid and magnesium
- D** dilute hydrochloric acid and aqueous sodium hydroxide

- 20** Some general rules for the solubility of salts in water are listed.

- Carbonates are insoluble (except ammonium carbonate, potassium carbonate and sodium carbonate).
- Chlorides are soluble (except lead(II) chloride and silver chloride).
- Nitrates are soluble.
- Sulfates are soluble (except barium sulfate, calcium sulfate and lead(II) sulfate).

Which substances produce an insoluble salt when aqueous solutions of them are mixed?

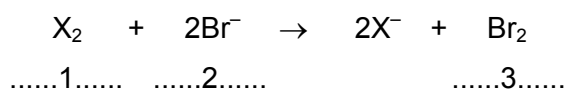
- A** barium chloride and magnesium nitrate
- B** calcium chloride and ammonium nitrate
- C** silver nitrate and zinc chloride
- D** sodium carbonate and potassium sulfate

21 Elements in Group I of the Periodic Table react with water.

Which row describes the products made in the reaction and the trend in reactivity of the elements?

	products	trend in reactivity
A	metal hydroxide and hydrogen	less reactive down the group
B	metal hydroxide and hydrogen	more reactive down the group
C	metal oxide and hydrogen	less reactive down the group
D	metal oxide and hydrogen	more reactive down the group

22 The equation shows the reaction between a halogen and aqueous bromide ions.



Which words complete gaps 1, 2 and 3?

	1	2	3
A	chlorine	brown	colourless
B	chlorine	colourless	brown
C	iodine	brown	colourless
D	iodine	colourless	brown

23 An inert gas R is used to fill weather balloons.

Which descriptions of R are correct?

	number of outer shell electrons in atoms of R	structure of gas R
A	2	diatomic molecules
B	2	single atoms
C	8	diatomic molecules
D	8	single atoms

24 Heating copper(II) carbonate produces copper(II) oxide and carbon dioxide.

Heating the copper(II) oxide formed with carbon produces copper.

Which processes are involved in this conversion of copper(II) carbonate to copper?

- A** sublimation followed by oxidation
- B** sublimation followed by reduction
- C** thermal decomposition followed by oxidation
- D** thermal decomposition followed by reduction

25 Four metals, W, X, Y and Z, are separately reacted with water and dilute hydrochloric acid.

The results are shown.

	metal			
	W	X	Y	Z
reaction with water	fizzes	no reaction	fizzes vigorously	no reaction
reaction with dilute hydrochloric acid	fizzes	no reaction	fizzes violently	fizzes

What is the order of reactivity of the four metals starting with the least reactive?

	least reactive $\longrightarrow$ most reactive			
A	X	W	Z	Y
B	X	Z	W	Y
C	Y	W	Z	X
D	Y	Z	W	X

26 Which statement about the uses of metals is **not** correct?

- A** Aluminium is used in aircraft because of its strength and good electrical conductivity.
- B** Copper is used in electrical wiring because of its good electrical conductivity.
- C** Stainless steel resists corrosion and is used to make cutlery.
- D** Transition elements are often used as catalysts.

27 Bauxite contains aluminium oxide.

Aluminium is extracted from aluminium oxide by electrolysis.

Why is cryolite added to the electrolytic cell used to extract aluminium?

- A** Cryolite prevents the carbon anodes being burned away.
- B** Cryolite removes impurities from the bauxite.
- C** Cryolite increases the rate at which aluminium ions are discharged.
- D** Molten cryolite dissolves the aluminium oxide.

28 Which statement about the Haber process is correct?

- A** The hydrogen used is obtained from the air.
- B** The nitrogen used is obtained from nitrates in the soil.
- C** Nitrogen reacts with hydrogen to make ammonia.
- D** The reaction takes place at room temperature and pressure.

29 Which statements about sulfur dioxide pollution are correct?

- 1 It increases the pH of rivers.
- 2 It damages limestone buildings.
- 3 It causes respiratory problems.

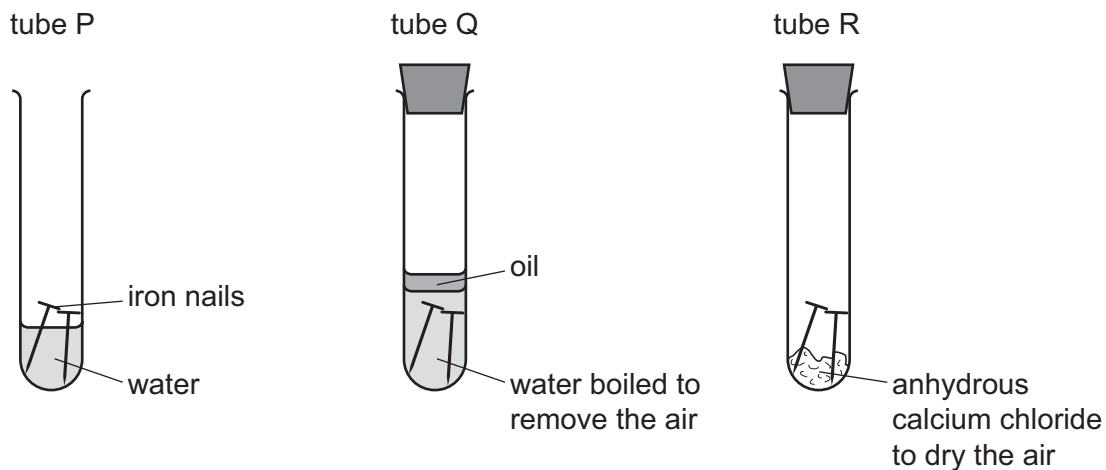
- A** 1 only **B** 2 only **C** 1 and 3 **D** 2 and 3

30 Argon is a noble gas used to fill light bulbs.

What is the approximate percentage of argon in air?

- A** 1% **B** 20% **C** 79% **D** 99%

31 The diagrams show experiments involving the rusting of iron.



A student predicted the following results.

- 1 In tube P, the iron nails rust.
- 2 In tube Q, the iron nails do not rust.
- 3 In tube R, the iron nails do not rust.

Which predictions are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

32 In the carbon cycle, which two processes add carbon dioxide to the atmosphere?

- A** combustion and carbonate formation
B combustion and photosynthesis
C combustion and respiration
D respiration and photosynthesis

33 Which statement about sulfur or one of its compounds is correct?

- A** Sulfur occurs naturally as the element sulfur.
B Sulfur dioxide is used to kill bacteria in drinking water.
C Sulfuric acid is a weak acid.
D Dilute sulfuric acid is a dehydrating agent.

34 What is **not** a use of lime?

- A It is used as a bleach in the manufacture of wood pulp.
- B It is used to desulfurise flue gases.
- C It is used to neutralise acidic industrial waste.
- D It is used to treat acidic soil.

35 Which equation representing a reaction of methane is correct?

- A $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$
- B $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_4\text{Cl}_2$
- C $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_2\text{Cl}_2 + \text{H}_2$
- D $2\text{CH}_4 + 2\text{Cl}_2 \rightarrow 2\text{CH}_3\text{Cl} + \text{Cl}_2 + \text{H}_2$

36 Which two compounds are molecules which both contain a double bond?

- A ethane and ethanoic acid
- B ethane and ethanol
- C ethene and ethanoic acid
- D ethene and ethanol

37 Ethanol can be formed by:

- 1 fermentation
- 2 reaction between steam and ethene.

Which of these processes use a catalyst?

	1	2
A	✓	✓
B	✓	x
C	x	✓
D	x	x

38 Ethanol is manufactured from ethene.

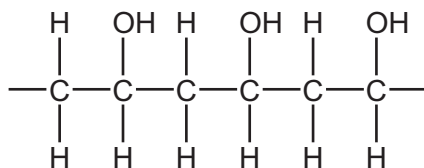
What is an advantage of this process?

- A It is a continuous process.
- B It has high labour costs.
- C It needs high temperature and pressure.
- D It uses non-renewable materials.

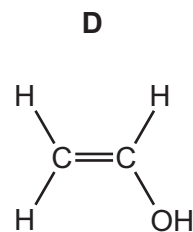
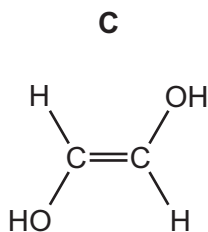
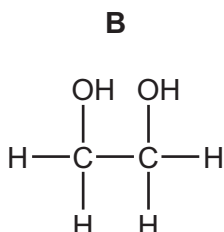
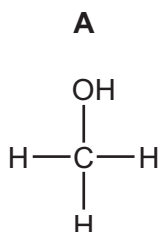
39 Which reaction can be used to make ethanoic acid?

- A oxidation of ethanol
- B oxidation of ethene
- C reduction of ethanol
- D reduction of ethene

40 The structure of an addition polymer is shown.



Which monomer is used to make this polymer?



Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.



CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

October/November 2018

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2018 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	A	1
3	B	1
4	A	1
5	D	1
6	C	1
7	B	1
8	C	1
9	A	1
10	B	1
11	C	1
12	B	1
13	B	1
14	D	1
15	C	1
16	D	1
17	C	1
18	D	1
19	C	1
20	C	1
21	B	1
22	B	1
23	B	1
24	D	1
25	B	1
26	A	1
27	D	1
28	C	1

Question	Answer	Marks
29	D	1
30	A	1
31	A	1
32	C	1
33	A	1
34	A	1
35	A	1
36	C	1
37	A	1
38	A	1
39	A	1
40	D	1



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/21

May/June 2019

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

A copy of the Periodic Table is printed on page 16.

Electronic calculators may be used.

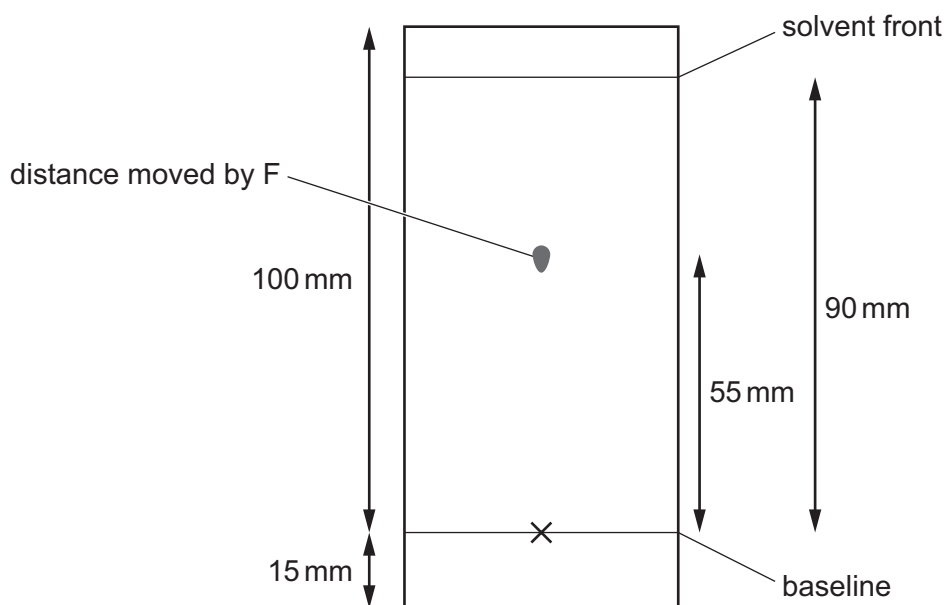
This document consists of **16** printed pages.

- 1 Which statement explains why ammonia gas, NH_3 , diffuses at a faster rate than hydrogen chloride gas, HCl ?
- A Ammonia expands to occupy all of the space available.
- B Ammonia has a smaller relative molecular mass than hydrogen chloride.
- C Ammonia is an alkali and hydrogen chloride is an acid.
- D Ammonia molecules diffuse in all directions at the same time.
- 2 2.00 g of powdered calcium carbonate is added to 50.0 cm^3 of hydrochloric acid.

Which apparatus is used to measure the calcium carbonate and the hydrochloric acid?

	calcium carbonate	hydrochloric acid
A	balance	burette
B	balance	thermometer
C	pipette	burette
D	pipette	thermometer

- 3 The measurements from a chromatography experiment using substance F are shown. The diagram is not drawn to scale.



What is the R_f value of F?

- A 0.55 B 0.61 C 0.90 D 1.64

4 Which statement about an atom of fluorine, $^{19}_{9}\text{F}$, is correct?

- A It contains more protons than neutrons.
- B It contains a total of 28 protons, neutrons and electrons.
- C Its isotopes contain different numbers of protons.
- D Its nucleus contains 9 neutrons.

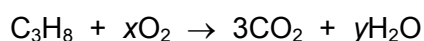
5 Which row describes the formation of single covalent bonds in methane?

A	atoms share a pair of electrons	both atoms gain a noble gas electronic structure
B	atoms share a pair of electrons	both atoms have the same number of electrons in their outer shell
C	electrons are transferred from one atom to another	both atoms gain a noble gas electronic structure
D	electrons are transferred from one atom to another	both atoms have the same number of electrons in their outer shell

6 Which statement describes the structure of an ionic compound?

- A It is a giant lattice of oppositely charged ions.
- B It is a giant lattice of positive ions in a 'sea' of electrons.
- C It is a giant molecule of oppositely charged ions.
- D It is a simple molecule of oppositely charged ions.

7 Propane burns in oxygen.



Which values of x and y balance the equation?

	x	y
A	5	4
B	7	4
C	10	8
D	13	8

- 8 A tablet contains 0.080 g of ascorbic acid ($M_r = 176$).

What is the concentration of ascorbic acid when one tablet is dissolved in 200 cm³ of water?

- A $9.1 \times 10^{-5} \text{ mol/dm}^3$
 B $4.5 \times 10^{-4} \text{ mol/dm}^3$
 C $9.1 \times 10^{-2} \text{ mol/dm}^3$
 D $2.3 \times 10^{-3} \text{ mol/dm}^3$

- 9 Which statement about the electrolysis of copper(II) sulfate solution using carbon electrodes is correct?

- A A colourless gas is produced at the anode.
 B A colourless gas is produced at the cathode.
 C The colour of the electrolyte remains the same.
 D The mass of both electrodes remains constant.

- 10 Aluminium metal is extracted from aluminium oxide by electrolysis.

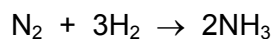
Which ionic half-equation describes a reaction that occurs at the named electrode?

	ionic half-equation	electrode
A	$2\text{O}^{2-} \rightarrow \text{O}_2 + 2\text{e}^-$	anode
B	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	anode
C	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	cathode
D	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	cathode

- 11 Which statement about the hydrogen fuel cell is **not** correct?

- A Chemical energy is converted into electrical energy.
 B Hydrogen is oxidised.
 C The reaction that takes place is endothermic.
 D Water is the only product.

- 12 Nitrogen reacts with hydrogen to produce ammonia.



The reaction is exothermic. The bond energies are shown in the table.

bond	bond energy in kJ/mol
$\text{N}\equiv\text{N}$	945
$\text{H}-\text{H}$	436
$\text{N}-\text{H}$	390

What is the energy change for this reaction?

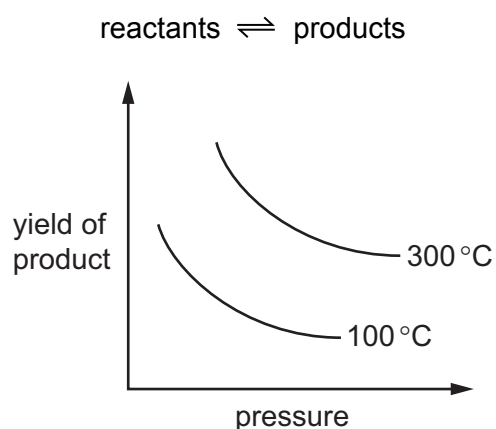
- A** -1473 kJ/mol
- B** -87 kJ/mol
- C** 87 kJ/mol
- D** 1473 kJ/mol
- 13 Which change in reaction conditions increases both the collision rate and the proportion of molecules with sufficient energy to react?
- A** addition of a catalyst
- B** increasing the concentration of a reactant
- C** increasing the surface area of a reactant
- D** increasing the temperature of the reaction
- 14 When blue-green crystals of nickel(II) sulfate are heated, water is produced and a yellow solid remains. When water is added to the yellow solid, the blue-green colour returns.

Which process describes these changes?

- A** combustion
- B** corrosion
- C** neutralisation
- D** reversible reaction

- 15 The graph shows how the yield of product in a reversible reaction changes as the temperature and pressure are changed.

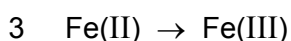
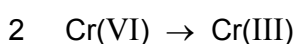
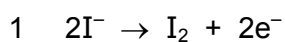
All reactants and products are gases.



Which row is correct for this reversible reaction?

	side of reaction with fewer moles	forward reaction
A	reactant	exothermic
B	reactant	endothermic
C	product	endothermic
D	product	exothermic

- 16 Which changes represent oxidation?



- A** 1 and 2 **B** 1 and 3 **C** 1 only **D** 2 only

- 17 Nitrogen(I) oxide, N_2O , nitrogen(II) oxide, NO , and carbon monoxide, CO , are all non-metal oxides.

They do not react with acids or bases.

Which statement is correct?

- A** They are acidic oxides.
B They are amphoteric oxides.
C They are basic oxides.
D They are neutral oxides.

18 The positions of elements W, X, Y and Z in the Periodic Table are shown.

A simplified periodic table with 18 columns and 4 rows. The first two columns are labeled 'W' and 'X' in the first row. The next six columns are empty. The seventh column is labeled 'Y' in the third row. The next six columns are empty. The last two columns are labeled 'Z' in the fourth row. A small square box is located above the seventh column.

Which elements form basic oxides?

- A** W, X and Y **B** W and X only **C** Y only **D** Z only

19 Ethanoic acid is a weak acid.

Hydrochloric acid is a strong acid.

Which statements are correct?

- 1 Ethanoic acid molecules are partially dissociated into ions.
- 2 1.0 mol/dm^3 ethanoic acid has a higher pH than 1.0 mol/dm^3 hydrochloric acid.
- 3 Ethanoic acid is always more dilute than hydrochloric acid.
- 4 Ethanoic acid is a proton acceptor.

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

20 The properties of an element are shown.

electrical conductivity	density	reaction with water
high	low	reacts violently with cold water

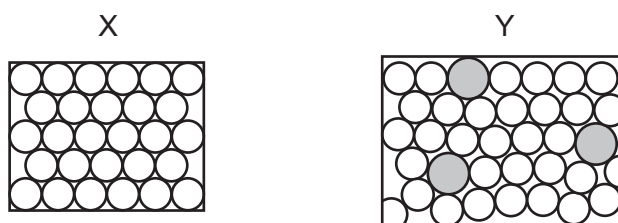
Which element has these properties?

A simplified periodic table grid is shown. The grid consists of 18 columns and 4 rows. The elements are marked as follows:

- A** is located in the 2nd column, 3rd row.
- B** is located in the 16th column, 3rd row.
- C** is located in the 10th column, 4th row.
- D** is located in the 1st column, 4th row.

There are also empty boxes at the top of the grid: one in the 10th column, 1st row, and one in the 18th column, 1st row.

- 21 Which statement about elements in Group I and Group VII of the Periodic Table is correct?
- A Bromine reacts with potassium chloride to produce chlorine.
 - B Iodine is a monoatomic non-metal.
 - C Lithium has a higher melting point than potassium.
 - D Sodium is more reactive with water than potassium.
- 22 Which statement about elements in Group VIII of the Periodic Table is correct?
- A They all have a full outer shell of electrons.
 - B They all react with Group I elements to form ionic compounds.
 - C They are all diatomic molecules.
 - D They are all liquids at room temperature.
- 23 The diagrams show the structure of two substances used to make electrical conductors.



Which statement correctly describes X and Y?

- A X is a pure metal and Y is a compound.
- B X is a pure metal and Y is an alloy.
- C X is a solid and Y is a liquid.
- D X is harder and stronger than Y.

24 Three metal compounds, P, Q and R, are heated using a Bunsen burner.

The results are shown.

P colourless gas produced, which relights a glowing splint

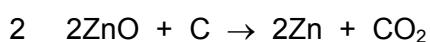
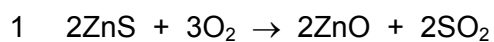
Q colourless gas produced, which turns limewater milky

R no reaction

Which row shows the identity of P, Q and R?

	P	Q	R
A	magnesium carbonate	potassium carbonate	potassium nitrate
B	magnesium carbonate	potassium nitrate	potassium carbonate
C	potassium nitrate	magnesium carbonate	potassium carbonate
D	potassium nitrate	potassium carbonate	magnesium carbonate

25 Zinc is extracted from its ore, zinc blende, using two chemical reactions.



Which substance is reduced in reactions 1 and 2?

	reaction 1	reaction 2
A	O ₂	C
B	O ₂	ZnO
C	ZnS	C
D	ZnS	ZnO

- 26** Four metals, zinc, M, copper and magnesium, are reacted with aqueous solutions of their nitrates.

The results are shown.

metal	magnesium nitrate	M nitrate	copper nitrate	zinc nitrate
magnesium		✓	✓	✓
zinc	X	✓	✓	
M	X		✓	X
copper	X	X		X

key

✓ = reacts

X = no reaction

What is the order of reactivity of these four metals starting with the most reactive?

- A** copper → zinc → M → magnesium
B copper → M → zinc → magnesium
C magnesium → M → zinc → copper
D magnesium → zinc → M → copper
- 27** Why is aluminium used to make containers for storing food?
- A** It conducts electricity.
B It has a high melting point.
C It is resistant to corrosion.
D It is strong.
- 28** Water can be treated by filtration then chlorination.

Which uses do **not** need water of this quality?

- 1 water for cooling in industry
- 2 water for washing clothes
- 3 water for drinking

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

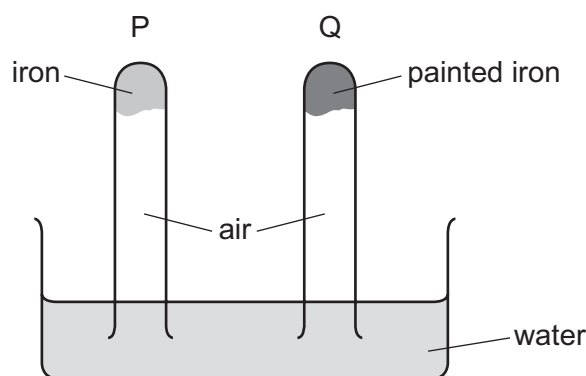
- 29** Oxides of nitrogen are formed in car engines and are a source of air pollution.

To decrease this pollution, catalytic converters are fitted to car exhausts.

What happens to the oxides of nitrogen in the catalytic converter?

- A** combustion
- B** cracking
- C** oxidation
- D** reduction

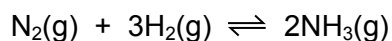
- 30** The diagram shows an experiment to investigate how paint affects the rusting of iron.



What happens to the water level in tubes P and Q?

	tube P	tube Q
A	falls	rises
B	no change	rises
C	rises	falls
D	rises	no change

- 31** Ammonia is manufactured by the Haber Process.

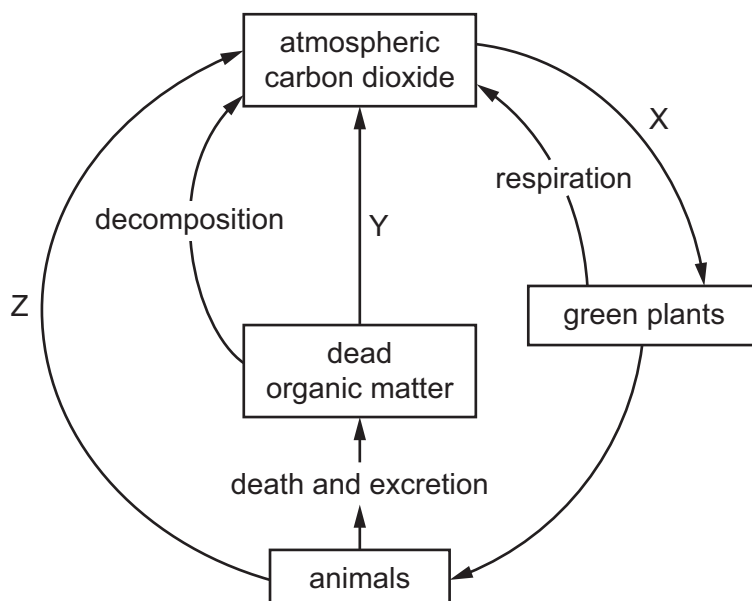


The forward reaction is exothermic.

Which conditions maximise the yield of ammonia?

	pressure	temperature
A	high	high
B	high	low
C	low	high
D	low	low

32 The carbon cycle is shown.



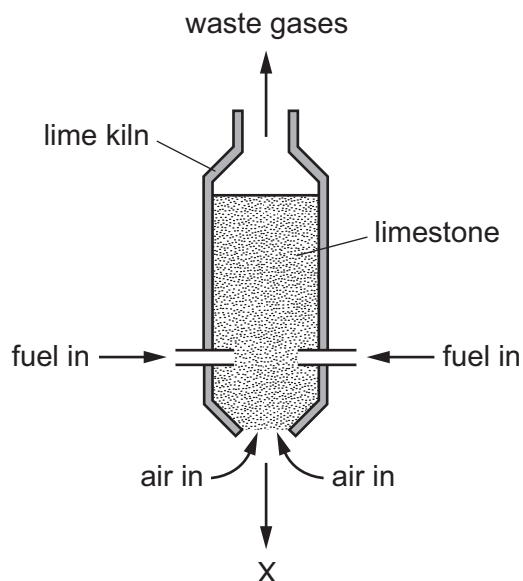
Which row describes processes X, Y and Z?

	X	Y	Z
A	respiration	combustion	photosynthesis
B	respiration	photosynthesis	combustion
C	photosynthesis	combustion	respiration
D	photosynthesis	respiration	combustion

33 Which row shows the conditions used in the Contact process?

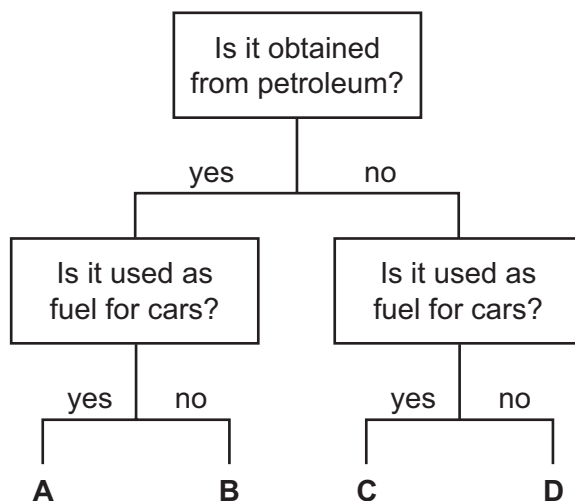
	temperature / °C	pressure / atm	catalyst
A	25	2	iron
B	25	200	iron
C	450	2	vanadium(V) oxide
D	450	200	vanadium(V) oxide

- 34 The diagram represents a lime kiln used to heat limestone to a very high temperature.



What leaves the kiln at X?

- A calcium carbonate
 - B calcium hydroxide
 - C calcium oxide
 - D calcium sulfate
- 35 Which fuel could be gasoline?



36 Why is ethanol a member of the homologous series of alcohols but propane is **not**?

- A Ethanol has two carbon atoms per molecule but propane has three.
- B Ethanol can be made from ethene but propane is obtained from petroleum.
- C Ethanol is a liquid but propane is a gas.
- D Ethanol contains the same functional group as other alcohols but propane does not.

37 Chlorine reacts with methane.

Which statements are correct?

- 1 The reaction takes place in the dark.
- 2 The reaction of chlorine with methane forms chloromethane.
- 3 Chloromethane reacts with chlorine to produce dichloromethane.
- 4 The reaction of chlorine with methane is an addition reaction.

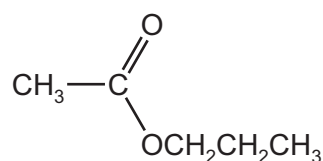
- A 1 and 2 B 1 and 3 C 2 and 3 D 3 and 4

38 Which statements about aqueous ethanoic acid are correct?

- 1 Ethanoic acid contains the functional group -COOH .
- 2 Ethanoic acid reacts with carbonates to produce hydrogen.
- 3 Ethanoic acid turns Universal Indicator paper blue.
- 4 Ethanoic acid has a pH lower than pH 7.

- A 1 and 2 B 1 and 3 C 1 and 4 D 2 and 4

39 The structure of an ester is shown.



What is the name of the ester?

- A ethyl propanoate
- B methyl propanoate
- C propyl ethanoate
- D propyl methanoate

40 The structure of a polymer is shown.



Which type of polymer is shown and by which process is it formed?

	type of polymer	formed by
A	carbohydrate	addition polymerisation
B	carbohydrate	condensation polymerisation
C	polyester	addition polymerisation
D	polyester	condensation polymerisation

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

The Periodic Table of Elements

Group																			
I	II	Key										III	IV	V	VI	VII	VIII		
		atomic number atomic symbol name relative atomic mass										1 H hydrogen 1							2 He helium 4
3 Li lithium 7	4 Be beryllium 9											5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19			
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40		
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —		
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —		114 Fl flerovium —	116 Lv livermorium —				

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education (9–1)

CHEMISTRY (9–1)

0971/21

Paper 2 Multiple Choice (Extended)

May/June 2019

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	A	1
3	B	1
4	B	1
5	A	1
6	A	1
7	A	1
8	D	1
9	A	1
10	D	1
11	C	1
12	B	1
13	D	1
14	D	1
15	B	1
16	B	1
17	D	1
18	A	1
19	A	1
20	D	1
21	C	1
22	A	1
23	B	1
24	C	1
25	B	1
26	D	1
27	C	1
28	B	1

Question	Answer	Marks
29	D	1
30	D	1
31	B	1
32	C	1
33	C	1
34	C	1
35	A	1
36	D	1
37	C	1
38	C	1
39	C	1
40	B	1



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education (9–1)

CHEMISTRY (9–1)

0971/22

Paper 2 Multiple Choice (Extended)

May/June 2019

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	C	1
3	C	1
4	B	1
5	A	1
6	A	1
7	C	1
8	A	1
9	B	1
10	C	1
11	B	1
12	B	1
13	D	1
14	D	1
15	D	1
16	B	1
17	C	1
18	A	1
19	D	1
20	C	1
21	C	1
22	D	1
23	B	1
24	C	1
25	B	1
26	D	1
27	C	1
28	A	1

Question	Answer	Marks
29	B	1
30	D	1
31	D	1
32	C	1
33	C	1
34	C	1
35	A	1
36	A	1
37	A	1
38	C	1
39	B	1
40	B	1



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/22

October/November 2019

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

A copy of the Periodic Table is printed on page 16.

Electronic calculators may be used.

This document consists of **15** printed pages and **1** blank page.

- 1 The rate of diffusion of a gas depends on its molecular mass and the temperature.

Which combination of molecular mass and temperature gives the slowest rate of diffusion?

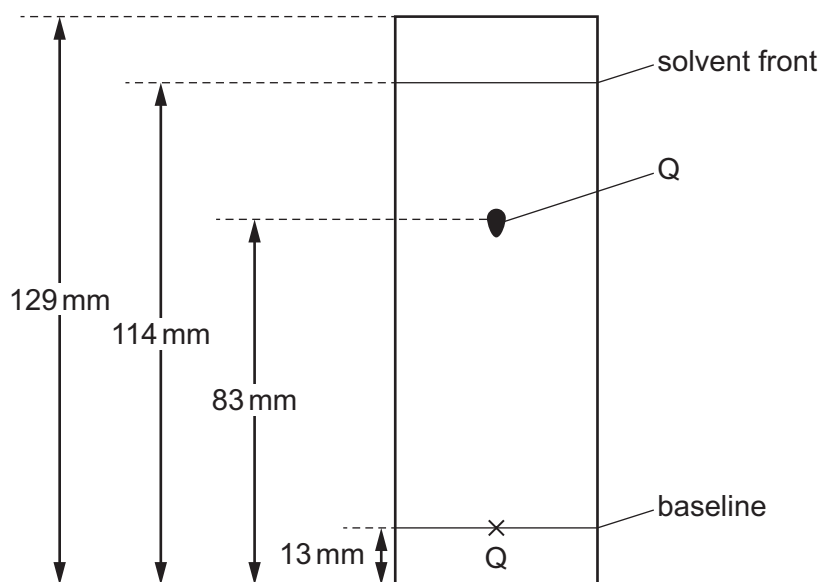
	molecular mass	temperature
A	high	high
B	high	low
C	low	high
D	low	low

- 2 A student is asked to measure the time taken for 0.4 g of magnesium carbonate to react completely with 25.0 cm³ of dilute hydrochloric acid.

Which pieces of apparatus does the student need?

- A** balance, stop-clock, pipette
B balance, stop-clock, thermometer
C balance, pipette, thermometer
D stop-clock, pipette, thermometer
- 3 Substance Q was investigated using chromatography.

The chromatogram is shown. The diagram is not drawn to scale.



What is the R_f value of Q?

- A** 0.60 **B** 0.64 **C** 0.69 **D** 0.72

- 4 Which statement about an ionic compound is **not** correct?
- A It conducts electricity when dissolved in water.
 - B It has a high melting point due to strong attractive forces between ions.
 - C It has a regular lattice of oppositely charged ions in a 'sea of electrons'.
 - D The ionic bonds are formed between metallic and non-metallic elements.

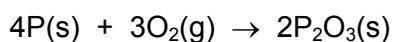
- 5 What is the total number of electrons in one molecule of ammonia, NH_3 ?
- A 6 B 8 C 10 D 11

- 6 Rubidium has two isotopes, $^{85}_{37}\text{Rb}$ and $^{87}_{37}\text{Rb}$.

Which statement explains why both isotopes have the same chemical properties?

- A They have the same number of protons.
 - B They have the same number of outer shell electrons.
 - C They have different numbers of neutrons.
 - D They have different mass numbers.
- 7 Which statement about the structure and properties of silicon(IV) oxide is **not** correct?
- A It has a giant structure similar to that of diamond.
 - B It has a high melting point due to the strong attractive force between molecules.
 - C There are strong covalent bonds between silicon and oxygen.
 - D There are no free electrons, so silicon(IV) oxide does not conduct electricity.
- 8 Which statement describes the structure of copper?
- A It has a lattice of negative ions in a 'sea of electrons'.
 - B It has a lattice of negative ions in a 'sea of protons'.
 - C It has a lattice of positive ions in a 'sea of electrons'.
 - D It has a lattice of positive ions in a 'sea of protons'.

- 9 Phosphorus reacts with oxygen to form phosphorus(III) oxide as shown.



Which mass of phosphorus(III) oxide is produced from 6.2 g of phosphorus?

- A 1.1 g B 5.5 g C 11.0 g D 22.0 g

- 10 Calcium carbonate is heated. Calcium oxide and carbon dioxide gas are formed.

The equation for the reaction is shown.



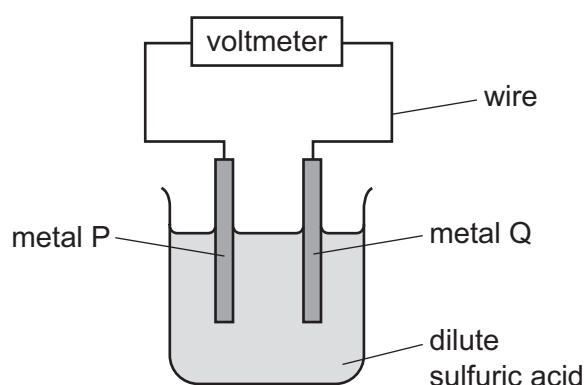
225 kg of calcium carbonate is heated until there is no further change in mass.

The yield of calcium oxide is 85 kg.

What is the percentage yield?

- A** 37.8% **B** 47.2% **C** 67.5% **D** 85.0%

- 11 The diagram shows a simple cell.



Which pair of metals produces the largest voltage?

	metal P	metal Q
A	magnesium	iron
B	magnesium	copper
C	zinc	iron
D	zinc	copper

- 12 What are the ionic half-equations for the electrode reactions during the electrolysis of concentrated aqueous sodium chloride?

	anode	cathode
A	$\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$	$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$
B	$2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
C	$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$	$\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$
D	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	$2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$

13 The temperature of the water in two beakers, X and Y, is measured as 21.5 °C.

5 g of sodium chloride is dissolved in the water in beaker X. The temperature changes to 18.0 °C.

5 g of calcium oxide is dissolved in the water in beaker Y. The temperature changes to 29.4 °C.

Which types of process are occurring in beakers X and Y?

	X	Y
A	endothermic	endothermic
B	endothermic	exothermic
C	exothermic	endothermic
D	exothermic	exothermic

14 Which gases are used to generate electricity in a fuel cell?

- A** carbon dioxide and oxygen
- B** hydrogen and methane
- C** hydrogen and oxygen
- D** methane and carbon dioxide

15 Which row identifies a chemical and a physical change?

	chemical change	physical change
A	boiling ethanol	burning ethanol
B	burning ethanol	evaporating ethanol
C	dissolving ethanol in water	burning ethanol
D	evaporating ethanol	dissolving ethanol in water

- 16** A sample of dilute nitric acid is added to lumps of limestone in a conical flask. The conical flask is placed on a balance and the loss in mass is measured.

A second sample of nitric acid of a different concentration is separately tested. All other conditions are kept the same.

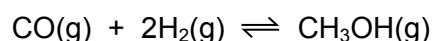
The loss in mass in 1 minute at each concentration of nitric acid is shown.

concentration in mol/dm ³	loss in mass in 1 minute/g
0.5	0.15
1.0	0.25

Which row describes and explains the results obtained using 1.0 mol/dm³ nitric acid compared with 0.5 mol/dm³ nitric acid?

	description	explanation
A	decrease in reaction rate	decrease in particle collision energy
B	decrease in reaction rate	increase in particle collision rate
C	increase in reaction rate	increase in particle collision rate
D	increase in reaction rate	increase in particle collision rate and collision energy

- 17** When carbon monoxide reacts with hydrogen, methanol is formed.



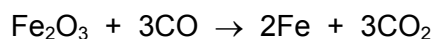
The forward reaction is exothermic.

Which statements are correct?

- There are more moles of gas on the left-hand side of the reaction.
- Increasing the temperature increases the amount of methanol at equilibrium.
- Increasing the pressure increases the amount of methanol at equilibrium.
- Increasing the initial amount of hydrogen decreases the amount of methanol at equilibrium.

A 1 and 2 only **B** 1 and 3 only **C** 2 and 4 only **D** 3 and 4 only

- 18** In the blast furnace, iron is formed when iron(III) oxide reacts with carbon monoxide in a redox reaction.



Which substance is the oxidising agent and which substance is the reducing agent?

	oxidising agent	reducing agent
A	CO	Fe_2O_3
B	CO_2	Fe
C	Fe	CO_2
D	Fe_2O_3	CO

- 19** Which oxide is classified as an amphoteric oxide?

- A** aluminium oxide
- B** calcium oxide
- C** copper(II) oxide
- D** nitrogen oxide

- 20** Which statement describes the properties of hydrochloric acid?

- A** Carbon dioxide is produced when limestone reacts with hydrochloric acid.
- B** Hydrogen is produced when sodium hydroxide reacts with hydrochloric acid.
- C** Methyl orange turns yellow in strong hydrochloric acid.
- D** Red litmus paper turns blue when dipped into hydrochloric acid.

21 A method used to make copper(II) sulfate crystals is shown.

- 1 Place dilute sulfuric acid in a beaker.
- 2 Warm the acid.
- 3 Add copper(II) oxide until it is in excess.
- 4 Filter the mixture.
- 5 Evaporate the filtrate until crystals start to form.
- 6 Leave the filtrate to cool.

What are the purposes of step 3 and step 4?

	step 3	step 4
A	to ensure all of the acid has reacted	to obtain solid copper(II) sulfate
B	to ensure all of the acid has reacted	to remove the excess of copper(II) oxide
C	to speed up the reaction	to obtain solid copper(II) sulfate
D	to speed up the reaction	to remove the excess of copper(II) oxide

22 Lead(II) sulfate is an insoluble salt.

Which reaction produces a mixture from which lead(II) sulfate is obtained by filtration?

- A** adding solid lead(II) carbonate to dilute sulfuric acid
- B** adding solid lead(II) hydroxide to dilute sulfuric acid
- C** adding metallic lead to dilute sulfuric acid
- D** adding aqueous lead(II) nitrate to dilute sulfuric acid

23 Helium is a noble gas.

Which statement about helium is correct?

- A** It has eight electrons in its outer shell.
- B** It is a diatomic gas.
- C** It is reactive.
- D** It is used for filling balloons.

24 Which pair of elements reacts together most violently?

- A** chlorine and lithium
- B** chlorine and potassium
- C** iodine and lithium
- D** iodine and potassium

25 Iron(II) ions can be oxidised to iron(III) ions by hydrogen peroxide.

Which statement explains why iron is a transition element?

- A** Iron is a transition element because it can be oxidised.
- B** Iron is a transition element because it has variable oxidation states.
- C** Iron is a transition element because it takes part in redox reactions.
- D** Iron is a transition element because it reacts with chlorine.

26 Some properties of substance X are listed.

- It conducts electricity when molten.
- It has a high melting point.
- It burns in oxygen and the oxide dissolves in water to give a solution with pH 11.

What is X?

- A** a covalent compound
- B** a macromolecule
- C** a metal
- D** an ionic compound

27 Which row describes the uses of aluminium, copper and mild steel?

	aluminium	copper	mild steel
A	aircraft bodies	electrical wiring	car bodies
B	car bodies	cooking utensils	electrical wiring
C	electrical wiring	aircraft bodies	food containers
D	food containers	aircraft bodies	cooking utensils

28 The properties of four metals are listed.

- Metal W does not react with dilute hydrochloric acid.
- Metal X reacts with dilute hydrochloric acid.
- Metal Y displaces metal X from an aqueous solution of its ions.
- Metal Z reacts with water and dilute hydrochloric acid.

What is the order of reactivity of the metals?

	most reactive → least reactive			
A	W	X	Y	Z
B	W	Y	X	Z
C	Z	X	Y	W
D	Z	Y	X	W

29 Which statement about the extraction of aluminium from aluminium oxide is correct?

- A** Aluminium is formed at the positive electrode during electrolysis.
- B** Pure aluminium oxide is dissolved in molten cryolite.
- C** Pure aluminium oxide is electrolysed using aluminium as the positive electrode.
- D** Pure aluminium oxide is heated with carbon to form carbon dioxide and aluminium.

30 River water contains soluble impurities, insoluble impurities and bacteria.

River water is made safe to drink by filtration and chlorination.

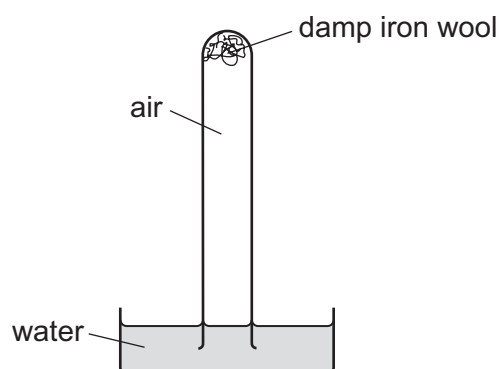
Which statement is correct?

- A** Filtration removes bacteria and insoluble impurities, and chlorination removes soluble impurities.
- B** Filtration removes insoluble impurities, and chlorination kills the bacteria.
- C** Filtration removes soluble and insoluble impurities, and chlorination kills the bacteria.
- D** Filtration removes soluble impurities and bacteria, and chlorination removes insoluble impurities.

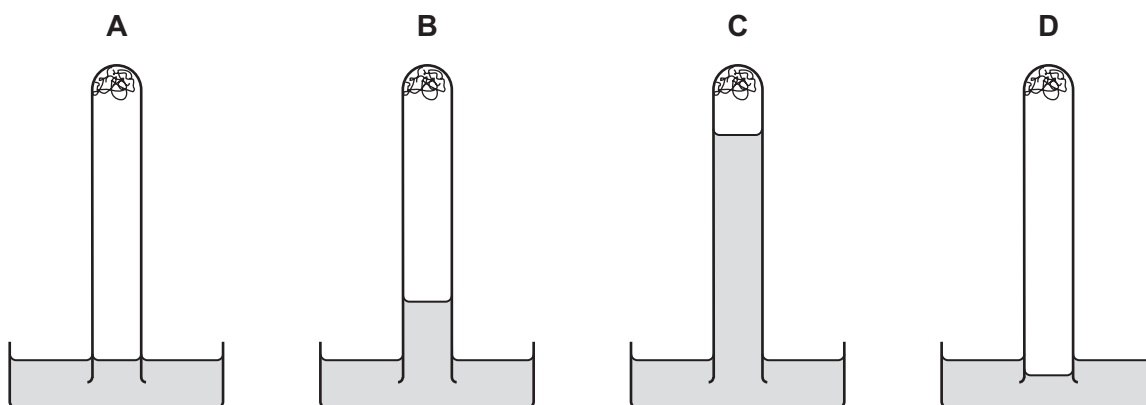
31 Which process is used to separate nitrogen and oxygen from air?

- A chromatography
- B evaporation
- C filtration
- D fractional distillation

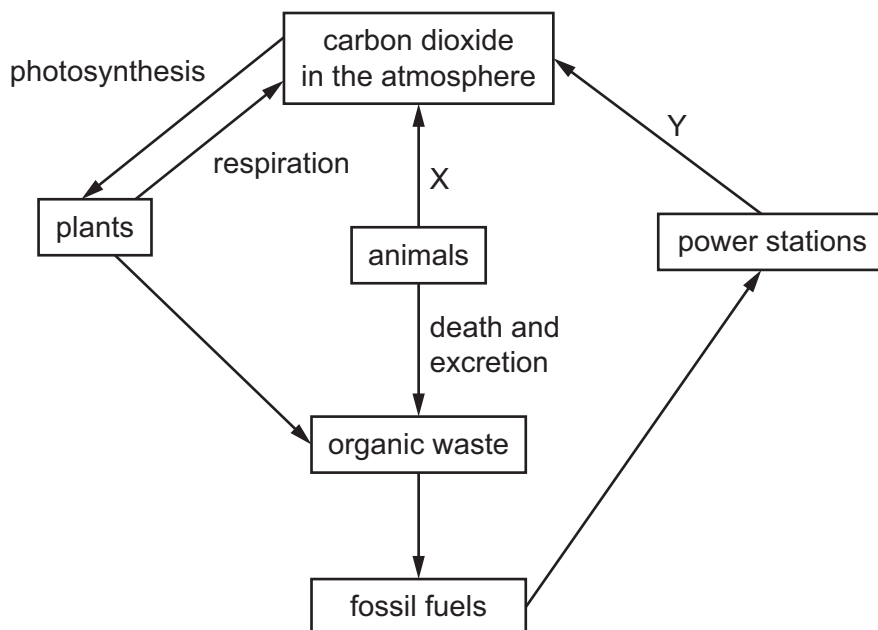
32 The apparatus shown is set up and left for a week.



Which diagram shows the level of the water at the end of the week?



33 The diagram represents an outline of the carbon cycle.



Which processes are X and Y?

	X	Y
A	combustion	respiration
B	decomposition	respiration
C	photosynthesis	combustion
D	respiration	combustion

34 Ammonium sulfate is used as a fertiliser.

It is made from ammonia and sulfuric acid.

Which words complete gaps 1, 2 and 3?

The1..... is made by the2..... process in which3..... is used as a catalyst.

	1	2	3
A	ammonia	Contact	iron
B	ammonia	Haber	vanadium(V) oxide
C	sulfuric acid	Contact	vanadium(V) oxide
D	sulfuric acid	Haber	iron

35 Which statement about limestone and lime is correct?

- A Limestone combines with water to produce slaked lime.
- B Lime is obtained from limestone by oxidation.
- C Lime is used in the desulfurisation of flue gases.
- D Lime is used in the treatment of alkaline soil.

36 Some fractions obtained from petroleum are listed.

	fraction	use	position collected in the fractionating column
1	gasoline	waxes and polishes	below refinery gas
2	bitumen	making roads	above kerosene
3	kerosene	jet fuel	below gasoline
4	refinery gas	heating and cooking	above gasoline

Which rows are correct?

- A 1, 3 and 4
- B 2, 3 and 4
- C 3 and 4 only
- D 4 only

37 Which products are obtained by the cracking of an alkane?

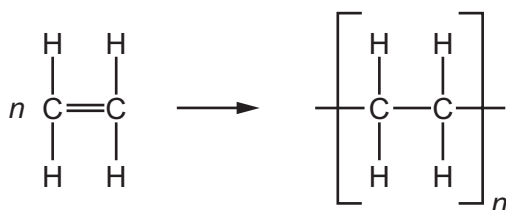
	alkene	hydrogen	water
A	✓	✓	✓
B	✓	✓	x
C	✓	x	✓
D	x	✓	✓

38 Ethanol is produced by fermentation or by the reaction of ethene with steam.

Which row is correct?

	by fermentation	from ethene
A	uses a temperature of 100 °C	uses a temperature of 350 °C
B	needs yeast as a catalyst	does not need a catalyst
C	very slow reaction	very fast reaction
D	high yield of ethanol	low yield of ethanol

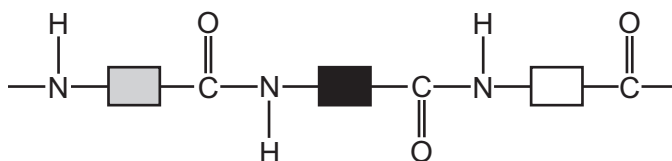
39 The diagram shows the structure of a monomer and of the polymer made from it.



What are the monomer and polymer?

	monomer	polymer
A	ethane	poly(ethane)
B	ethane	poly(ethene)
C	ethene	poly(ethane)
D	ethene	poly(ethene)

40 The structure of a naturally occurring polymer, X, is shown.



What is X?

- A** an amino acid
- B** a carbohydrate
- C** a protein
- D** a sugar

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

The Periodic Table of Elements

Group																																																																														
I	II											III	IV	V	VI	VII	VIII																																																													
3 Li lithium 7		4 Be beryllium 9		Key atomic number atomic symbol name relative atomic mass										1 H hydrogen 1							2 He helium 4																																																									
11 Na sodium 23	12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	55 Cs caesium 133	56 Ba barium 137	lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —	87 Fr francium —	88 Ra radium —	actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —	—

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

October/November 2019

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Mark
1	B	1
2	A	1
3	C	1
4	C	1
5	C	1
6	B	1
7	B	1
8	C	1
9	C	1
10	C	1
11	B	1
12	B	1
13	B	1
14	C	1
15	B	1
16	C	1
17	B	1
18	D	1
19	A	1
20	A	1
21	B	1
22	D	1
23	D	1
24	B	1
25	B	1
26	C	1
27	A	1
28	D	1

Question	Answer	Mark
29	B	1
30	B	1
31	D	1
32	B	1
33	D	1
34	C	1
35	C	1
36	C	1
37	B	1
38	C	1
39	D	1
40	C	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/21

May/June 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Blank pages are indicated.



- 1 A mixture of ice and water is left to stand and the ice melts.

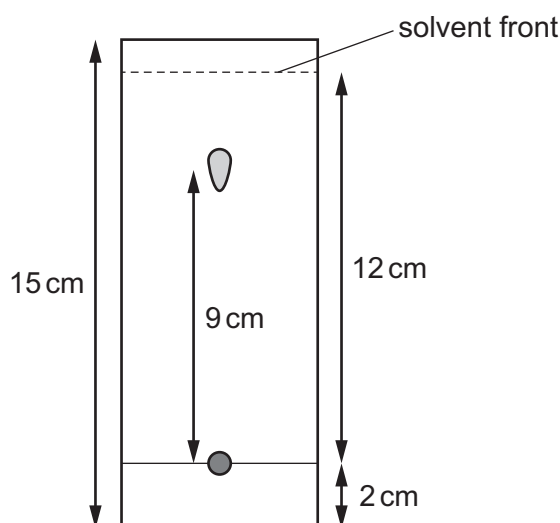
Which row describes what happens as the ice is melting?

	temperature of mixture	energy changes
A	increases	average kinetic energy of particles increases
B	increases	energy is used to overcome attractive forces
C	stays the same	average kinetic energy of particles increases
D	stays the same	energy is used to overcome attractive forces

- 2 Which piece of apparatus should be used to measure exactly 21.4 cm^3 of water?

- A** 25 cm^3 beaker
B 25 cm^3 pipette
C 50 cm^3 burette
D 50 cm^3 measuring cylinder

- 3 The chromatogram for an unknown dye is shown.



What is the R_f value of the dye?

- A** 0.60 **B** 0.64 **C** 0.75 **D** 0.82

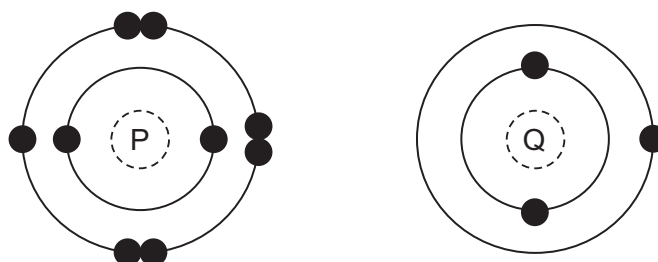
- 4 The atomic number and nucleon number of a potassium atom are shown.

	potassium atom
atomic number	19
nucleon number	39

How many protons, neutrons and electrons are in a potassium ion, K^+ ?

	protons	neutrons	electrons
A	19	20	18
B	19	20	20
C	20	19	18
D	20	19	19

- 5 The electronic structures of two atoms, P and Q, are shown.



P and Q combine together to form a compound.

What is the type of bonding in the compound and what is the formula of the compound?

	type of bonding	formula
A	ionic	PQ
B	ionic	PQ_2
C	covalent	PQ_2
D	covalent	PQ

- 6 Which row contains a description of metallic bonding and a property that is explained by reference to metallic bonding?

	description of metallic bonding	property explained by metallic bonding
A	a lattice of negative ions in a sea of electrons	a metal will react with an acid, producing hydrogen
B	a lattice of negative ions in a sea of electrons	a piece of a metal can be moulded into different shapes
C	a lattice of positive ions in a sea of electrons	a metal will react with an acid, producing hydrogen
D	a lattice of positive ions in a sea of electrons	a piece of a metal can be moulded into different shapes

- 7 Which statement explains why methane has a lower boiling point than water?

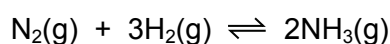
- A** Methane has weaker covalent bonds than water.
B Methane has weaker attractive forces than water.
C Methane molecules are heavier than water molecules.
D Methane molecules have more bonds than water molecules.

- 8 A solution of iron(III) sulfate reacts with aqueous sodium hydroxide to form a red-brown precipitate.

What is the balanced equation, including state symbols, for the reaction?

- A** $\text{FeSO}_4(\text{aq}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{Fe}(\text{OH})_2(\text{s}) + \text{Na}_2\text{SO}_4(\text{aq})$
B $\text{FeSO}_4(\text{l}) + 2\text{NaOH}(\text{l}) \rightarrow \text{Fe}(\text{OH})_2(\text{s}) + \text{Na}_2\text{SO}_4(\text{l})$
C $\text{Fe}_2(\text{SO}_4)_3(\text{aq}) + 6\text{NaOH}(\text{aq}) \rightarrow 2\text{Fe}(\text{OH})_3(\text{s}) + 3\text{Na}_2\text{SO}_4(\text{aq})$
D $\text{Fe}_2(\text{SO}_4)_3(\text{l}) + 6\text{NaOH}(\text{aq}) \rightarrow 2\text{Fe}(\text{OH})_3(\text{s}) + 3\text{Na}_2\text{SO}_4(\text{l})$

- 9 The Haber process is a reversible reaction.



The reaction has a 30% yield of ammonia.

Which volume of ammonia gas, NH_3 , measured at room temperature and pressure, is obtained by reacting 0.75 moles of hydrogen with excess nitrogen?

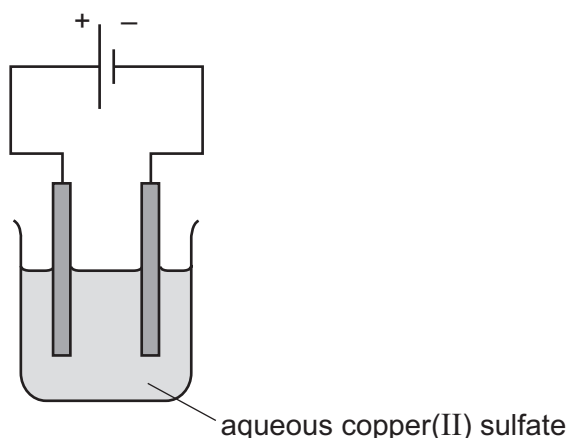
- A** 3600 cm³ **B** 5400 cm³ **C** 12 000 cm³ **D** 18 000 cm³

10 Dilute aqueous sodium chloride is electrolysed using platinum electrodes.

What is the half-equation for the reaction at the cathode?

- A $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
- B $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$
- C $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
- D $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$

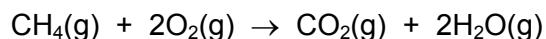
11 The electrolysis of aqueous copper(II) sulfate, using inert electrodes, is shown.



Which statement about a reaction at an electrode is correct?

- A Copper ions gain electrons at the negative electrode.
- B Copper ions gain electrons at the positive electrode.
- C Hydrogen ions gain electrons at the negative electrode.
- D Hydrogen ions gain electrons at the positive electrode.

12 The equation for the complete combustion of methane gas is shown.



Bond energies are shown.

bond	bond energy in kJ/mol
C–H	412
H–O	463
C=O	743
O=O	498

What is the overall energy change, in kJ/mol, for the above reaction?

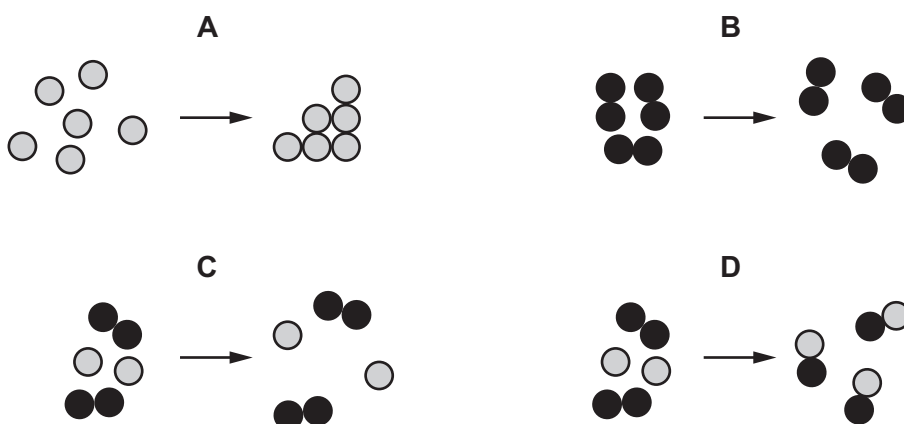
- A** –1192 **B** –694 **C** +694 **D** +1192

13 Which statements about hydrogen fuel cells are correct?

- 1 Water is formed as the only waste product.
- 2 Both water and carbon dioxide are formed as waste products.
- 3 The overall reaction is $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$.
- 4 The overall reaction is endothermic.

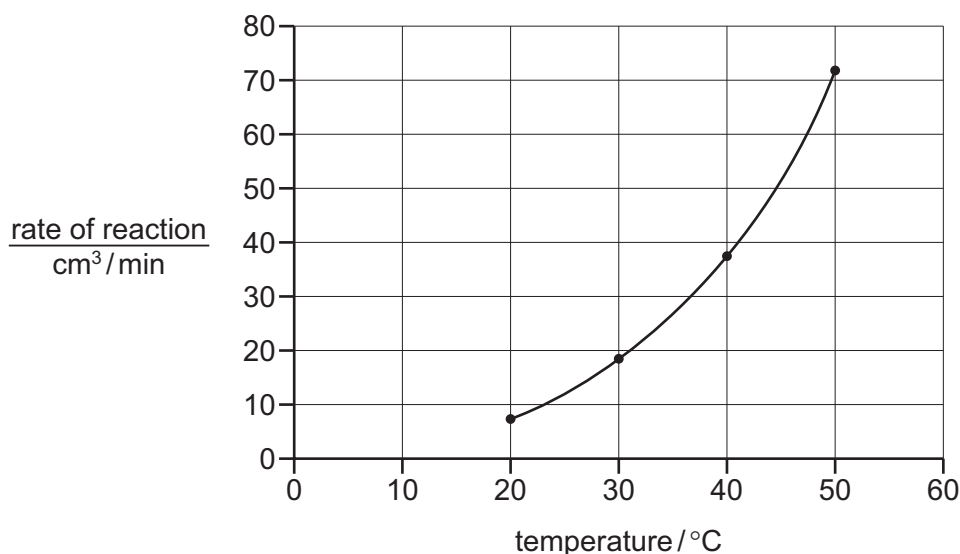
- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

14 Which diagram represents a chemical change?



- 15 The rate of reaction between calcium carbonate chips and hydrochloric acid is studied by collecting the volume of gas released in one minute at different temperatures.

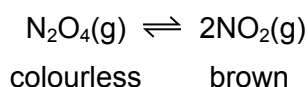
A graph of rate of reaction against temperature is shown.



Which statement fully explains why increasing the temperature has this effect on the rate?

- A The kinetic energy of the particles increases so the collisions are harder.
 - B The number of collisions between particles increases.
 - C The activation energy needed for the particles to react is reduced.
 - D There are more frequent collisions between particles with enough energy to react.
- 16 The equation shows the equilibrium between dinitrogen tetroxide, N_2O_4 , and nitrogen dioxide, NO_2 .

The colours of the reactant and product are also shown.

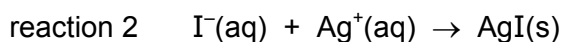
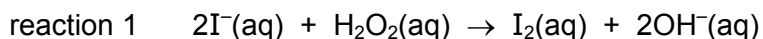


The forward reaction is endothermic.

Which statement is **not** correct?

- A At equilibrium the concentrations of the reactant and the product are constant.
- B At equilibrium the rate of the forward reaction is equal to the rate of the reverse reaction.
- C When the pressure is increased a darker brown colour is seen.
- D When the temperature is increased a darker brown colour is seen.

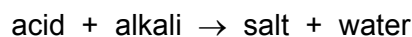
17 The equations for two reactions of iodide ions are shown.



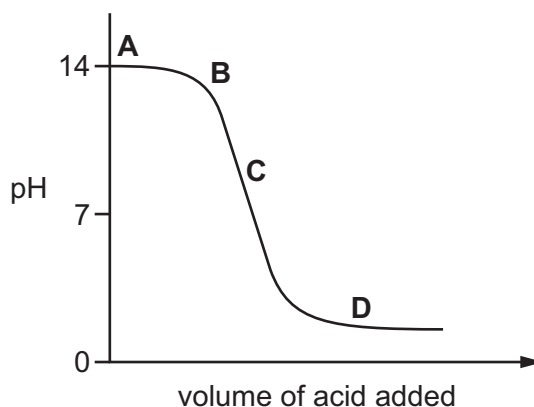
Which statement is correct?

- A Both reactions are redox reactions.
- B Neither reaction is a redox reaction.
- C Only reaction 1 is a redox reaction.
- D Only reaction 2 is a redox reaction.

18 The graph shows how the pH of a solution changes as an acid is added to an alkali.



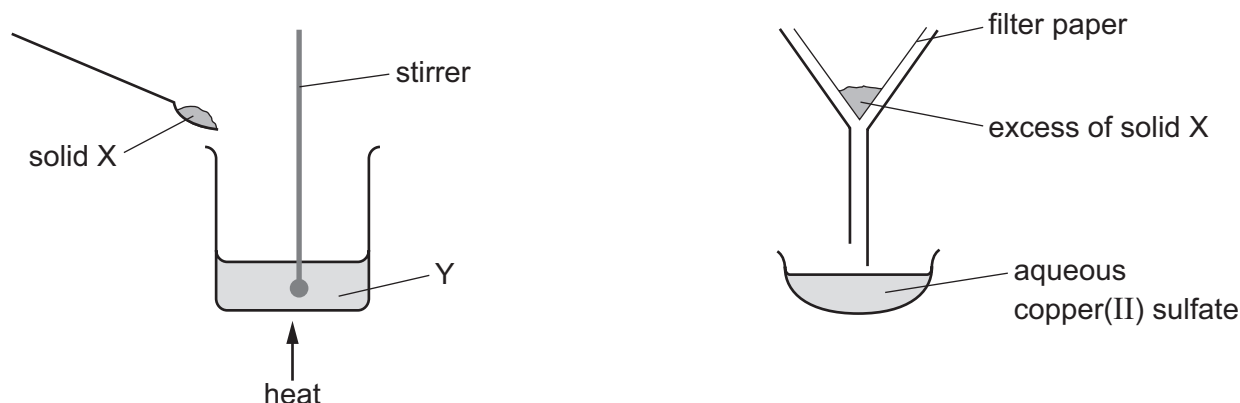
Which letter represents the area of the graph where both acid and salt are present?



19 Which statement describes a weak acid?

- A It is a proton acceptor and is fully ionised in aqueous solution.
- B It is a proton acceptor and is partially ionised in aqueous solution.
- C It is a proton donor and is fully ionised in aqueous solution.
- D It is a proton donor and is partially ionised in aqueous solution.

20 The apparatus shown is used to prepare aqueous copper(II) sulfate.



What are X and Y?

	X	Y
A	copper	aqueous iron(II) sulfate
B	copper(II) chloride	dilute sulfuric acid
C	copper(II) oxide	dilute sulfuric acid
D	sulfur	aqueous copper(II) chloride

21 Lead(II) sulfate is an insoluble salt.

Which method is suitable for obtaining solid lead(II) sulfate?

- A** Mix aqueous lead(II) nitrate and aqueous potassium sulfate, heat to evaporate all of the water, collect the solid and then wash and dry it.
- B** Mix aqueous lead(II) nitrate and aqueous potassium sulfate, filter, collect the filtrate, crystallise, then wash and dry the crystals.
- C** Mix aqueous lead(II) nitrate and dilute sulfuric acid, filter, then wash and dry the residue.
- D** Titrate aqueous lead(II) hydroxide with dilute sulfuric acid, crystallise, then wash and dry the crystals.

22 A Group I metal (lithium, sodium or potassium) is reacted with a Group VII element (chlorine, bromine or iodine).

Which compound is formed when the Group I metal of highest density reacts with the Group VII element of lowest density?

- A** lithium chloride
- B** potassium chloride
- C** potassium iodide
- D** lithium iodide

23 The properties of the element titanium, Ti, can be predicted from its position in the Periodic Table.

Which row identifies the properties of titanium?

	can be used as a catalyst	conducts electricity when solid	has low density	forms coloured compounds
A	✓	✓	✓	✗
B	✓	✓	✗	✓
C	✓	✗	✓	✓
D	✗	✓	✓	✓

24 A balloon is filled with helium. Helium is a noble gas and makes the balloon rise up in the air.

The density of air is 1.23 g/dm^3 .

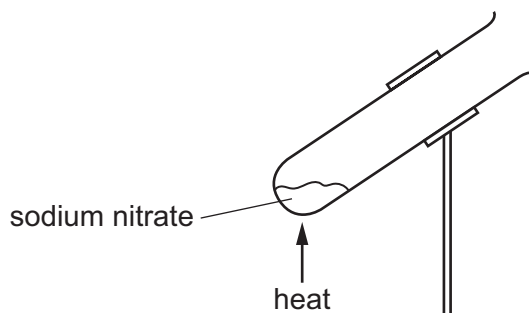
Which gas is helium?

	density in g/dm^3	reaction with oxygen
A	0.0899	burns rapidly
B	0.179	does not react with oxygen
C	1.78	does not react with oxygen
D	3.75	does not react with oxygen

25 Which property is shown by **all** metals?

- A** They are extracted from their ores by heating with carbon.
- B** They conduct electricity.
- C** They form acidic oxides.
- D** They react with hydrochloric acid to form hydrogen.

26 Sodium nitrate is a white crystalline solid that decomposes on heating.



Which row describes the decomposition products formed when sodium nitrate is heated strongly?

	solid products	gaseous products
A	sodium nitrite	NO_2 and O_2
B	sodium nitrite	O_2 only
C	sodium oxide	NO_2 and O_2
D	sodium oxide	O_2 only

27 Molten iron from the blast furnace contains impurities.

The process of turning the impure iron into steel involves blowing oxygen into the molten iron and adding calcium oxide.

What are the reasons for blowing in oxygen and adding calcium oxide?

	blowing in oxygen	adding calcium oxide
A	carbon is removed by reacting with oxygen	reacts with acidic impurities making slag
B	carbon is removed by reacting with oxygen	reacts with slag and so removes it
C	iron reacts with the oxygen	reacts with acidic impurities making slag
D	iron reacts with the oxygen	reacts with slag and so removes it

28 Element Y reacts with copper(II) oxide to form copper.

Element Y will not react with zinc oxide. Copper has no reaction with zinc oxide.

What is the order of reactivity of these three elements, most reactive first?

- A** $\text{Cu} \rightarrow \text{Y} \rightarrow \text{Zn}$
- B** $\text{Cu} \rightarrow \text{Zn} \rightarrow \text{Y}$
- C** $\text{Zn} \rightarrow \text{Cu} \rightarrow \text{Y}$
- D** $\text{Zn} \rightarrow \text{Y} \rightarrow \text{Cu}$

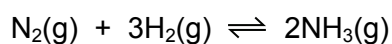
29 Which statement shows that a liquid is pure water?

- A It boils at 100 °C.
- B It has a pH value of 7.
- C It turns blue cobalt(II) chloride pink.
- D It turns white copper(II) sulfate blue.

30 Which process removes carbon dioxide from the atmosphere?

- A combustion
- B decomposition
- C photosynthesis
- D respiration

31 Ammonia is manufactured by the Haber process.



What are the conditions used in the Haber process?

	temperature / °C	pressure / atm
A	400	100
B	400	300
C	20	300
D	20	100

32 Coating iron helps to prevent rusting.

Which coating will continue to protect the iron even when the coating is damaged?

- A copper
- B paint
- C plastic
- D zinc

33 A student suggests three uses of calcium carbonate (limestone).

- 1 manufacture of cement
- 2 manufacture of iron
- 3 treating alkaline soils

Which suggestions are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

34 The Contact process is used to manufacture concentrated sulfuric acid and consists of four steps.

Which step involves a catalyst?

- A** production of sulfur dioxide gas
B production of sulfur trioxide gas
C production of oleum
D production of concentrated sulfuric acid

35 Which row about the production of ethanol by fermentation is correct?

	raw materials	energy requirement	rate of reaction
A	non-renewable	high	slow
B	renewable	low	slow
C	non-renewable	low	fast
D	renewable	high	fast

36 Which statement about homologous series is correct?

- A** Members of a homologous series have the same structural formula.
B Members of a homologous series all have similar chemical properties.
C Members of a homologous series all have similar physical properties.
D Members of all homologous series are hydrocarbons.

- 37** Increasing the number of atoms in one molecule of a hydrocarbon increases the amount of energy released when it burns.

What is the correct order?

	less energy released	—————→	more energy released
A	ethene	ethane	methane
B	ethene	methane	ethane
C	methane	ethane	ethene
D	methane	ethene	ethane

- 38** Some properties of an organic compound J are listed.

- It is a liquid at room temperature.
- It is soluble in water.
- A solution of J reacts with calcium carbonate to form carbon dioxide.
- A solution of J has a pH of 3.

In which homologous series does J belong?

- A** alkane
B alkene
C alcohol
D carboxylic acid

- 39** Ethane, C_2H_6 , reacts with chlorine in a substitution reaction.

What are the products of this reaction?

- A** chloroethane and hydrogen
B chloroethane and hydrogen chloride
C chloroethene and hydrogen
D chloroethene and hydrogen chloride

40 Which polymers or types of polymer are synthetic?

- 1 carbohydrates
- 2 nylon
- 3 proteins
- 4 *Terylene*

A 1 and 3

B 1 and 4

C 2 and 3

D 2 and 4

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										1 H hydrogen 1							
	2 He helium 4																		
	11 Na sodium 23											12 Mg magnesium 24							
	19 K potassium 39											20 Ca calcium 40							
37 Rb rubidium 85	38 Sr strontium 88	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —						

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™ (9–1)

CHEMISTRY (9–1)

0971/21

Paper 2 Multiple Choice (Extended)

May/June 2020

MARK SCHEME

Maximum Mark: 40

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	C	1
3	C	1
4	A	1
5	A	1
6	D	1
7	B	1
8	C	1
9	A	1
10	A	1
11	A	1
12	B	1
13	A	1
14	D	1
15	D	1
16	C	1
17	C	1
18	D	1
19	D	1
20	C	1
21	C	1
22	B	1
23	B	1
24	B	1
25	B	1
26	B	1
27	A	1
28	D	1

Question	Answer	Marks
29	A	1
30	C	1
31	B	1
32	D	1
33	A	1
34	B	1
35	B	1
36	B	1
37	D	1
38	D	1
39	B	1
40	D	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

May/June 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Blank pages are indicated.

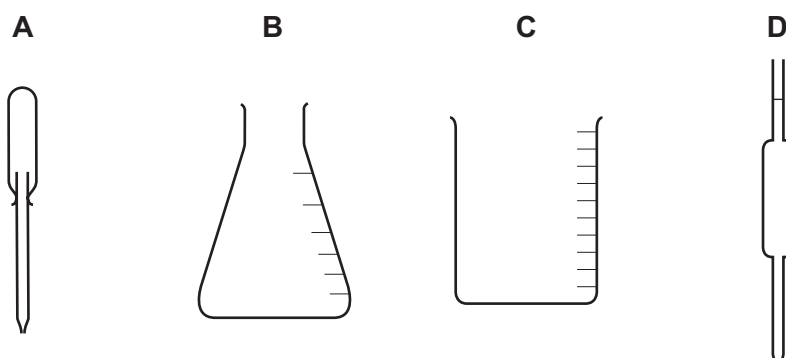


- 1 A mixture of ice and water is left to stand and the ice melts.

Which row describes what happens as the ice is melting?

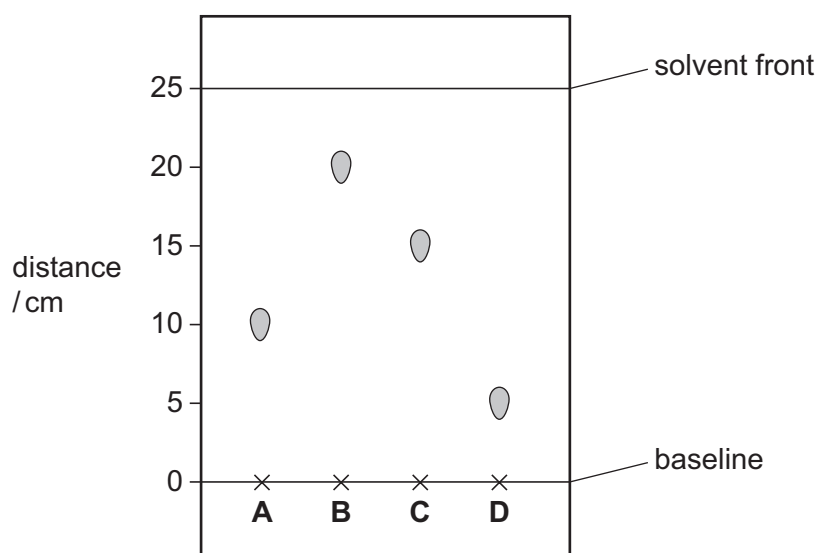
	temperature of mixture	energy changes
A	increases	average kinetic energy of particles increases
B	increases	energy is used to overcome attractive forces
C	stays the same	average kinetic energy of particles increases
D	stays the same	energy is used to overcome attractive forces

- 2 Which piece of apparatus is used to measure 25.0 cm³ of aqueous sodium hydroxide?

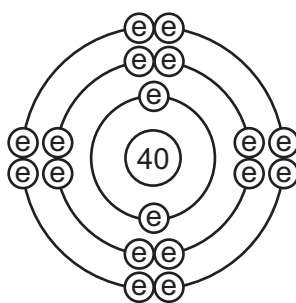


- 3 Paper chromatography is used to determine the R_f values for four different food colourings.

Which food colouring has an R_f value of 0.6?



- 4 The diagram shows the electronic structure of a particle with a nucleon number (mass number) of 40.



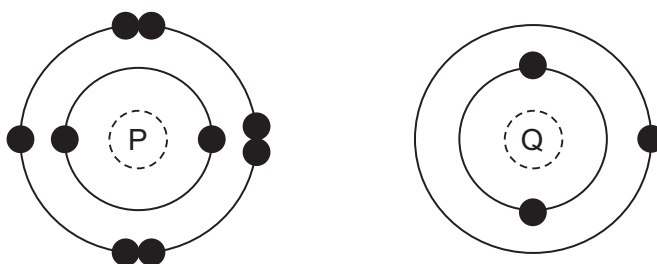
The table shows the suggestions that three students, 1, 2 and 3, made to identify the particle.

	student		
	1	2	3
particle	Ar	Cl	Ca ²⁺

Which students are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 5 The electronic structures of two atoms, P and Q, are shown.



P and Q combine together to form a compound.

What is the type of bonding in the compound and what is the formula of the compound?

	type of bonding	formula
A	ionic	PQ
B	ionic	PQ ₂
C	covalent	PQ ₂
D	covalent	PQ

- 6 Which statement about the structure of a metal explains why metals are malleable?
- A** The electrons can move freely throughout the lattice.
- B** The layers of metal ions can slide over each other.
- C** The metal ions are positively charged.
- D** There is a strong force of attraction between the metal ions and the electrons.
- 7 The bonding, structure and melting point of sodium chloride and sulfur dichloride are shown.

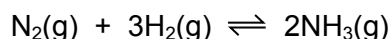
compound	bonding	structure	melting point / °C
sodium chloride	ionic	giant lattice	801
sulfur dichloride	covalent	simple molecular	–121

Why does sulfur dichloride have a lower melting point than sodium chloride?

- A** The covalent bonds in sulfur dichloride are weaker than the attractive forces between molecules in sodium chloride.
- B** The covalent bonds in sulfur dichloride are weaker than the ionic bonds in sodium chloride.
- C** The attractive forces between molecules in sulfur dichloride are weaker than the attractive forces between molecules in sodium chloride.
- D** The attractive forces between molecules in sulfur dichloride are weaker than the ionic bonds in sodium chloride.
- 8 Lead(II) nitrate, $\text{Pb}(\text{NO}_3)_2$, reacts with potassium iodide, KI , to form a yellow precipitate, PbI_2 , and a soluble salt, KNO_3 .

What is the equation for the reaction?

- A** $\text{Pb}(\text{NO}_3)_2 + \text{KI} \rightarrow \text{PbI}_2 + \text{KNO}_3$
- B** $\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 + \text{KNO}_3$
- C** $2\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 + 2\text{KNO}_3$
- D** $\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 + 2\text{KNO}_3$
- 9 The Haber process is a reversible reaction.



The reaction has a 30% yield of ammonia.

Which volume of ammonia gas, NH_3 , measured at room temperature and pressure, is obtained by reacting 0.75 moles of hydrogen with excess nitrogen?

- A** 3600 cm^3 **B** 5400 cm^3 **C** $12\,000 \text{ cm}^3$ **D** $18\,000 \text{ cm}^3$

10 Electrolytes can be broken down by electrolysis.

Which rows are correct for each electrolyte?

	electrolyte	reaction at cathode	product at anode
1	dilute aqueous sodium chloride	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	oxygen
2	concentrated hydrochloric acid	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	chlorine
3	molten aluminium oxide	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	aluminium
4	concentrated aqueous sodium bromide	$\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$	bromine

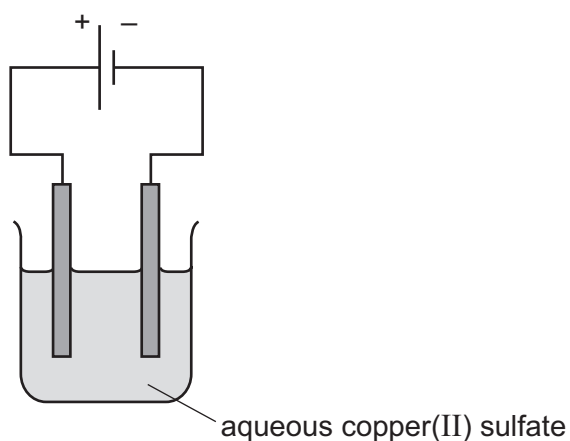
A 1 and 2

B 1 and 4

C 2 and 3

D 3 and 4

11 The electrolysis of aqueous copper(II) sulfate, using inert electrodes, is shown.



Which statement about a reaction at an electrode is correct?

A Copper ions gain electrons at the negative electrode.

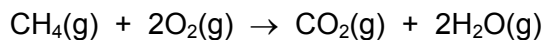
B Copper ions gain electrons at the positive electrode.

C Hydrogen ions gain electrons at the negative electrode.

D Hydrogen ions gain electrons at the positive electrode.

12 Methane burns in excess oxygen.

The equation is shown.



Bond energies are shown.

bond	bond energy / kJ mol ⁻¹
C=O	805
C–H	410
O=O	496
O–H	460

What is the energy change for the reaction?

- A** $(4 \times 410 + 2 \times 496) - (2 \times 805 + 4 \times 460)$
B $(2 \times 805 + 2 \times 460) - (410 + 2 \times 496)$
C $(410 + 2 \times 496) - (805 + 2 \times 460)$
D $(410 + 496) - (805 + 460)$

13 Which statements about hydrogen fuel cells are correct?

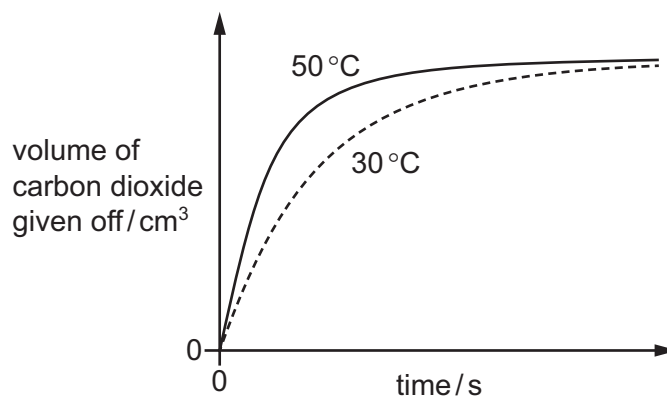
- 1 Water is formed as the only waste product.
- 2 Both water and carbon dioxide are formed as waste products.
- 3 The overall reaction is $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$.
- 4 The overall reaction is endothermic.

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

14 Which list contains **only** chemical changes?

- A** melting, evaporating, dissolving
B rusting, freezing, subliming
C neutralisation, polymerisation, combustion
D boiling, condensing, distillation

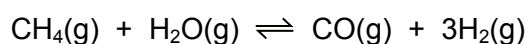
- 15** The results of adding excess marble chips (calcium carbonate) to hydrochloric acid at 50 °C and at 30 °C are shown. Only the temperature is changed.



Which row describes the reacting particles at 30 °C compared to those at 50 °C?

	collision rate	collision energy
A	higher	higher
B	higher	lower
C	lower	higher
D	lower	lower

- 16** Methane reacts with steam and an equilibrium is reached.



The forward reaction is endothermic.

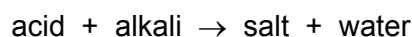
Which row shows how the amount of hydrogen at equilibrium changes when the pressure or temperature is changed as indicated?

	change in temperature	change in pressure	amount of hydrogen
A	decrease	no change	increase
B	increase	no change	decrease
C	no change	increase	decrease
D	no change	decrease	decrease

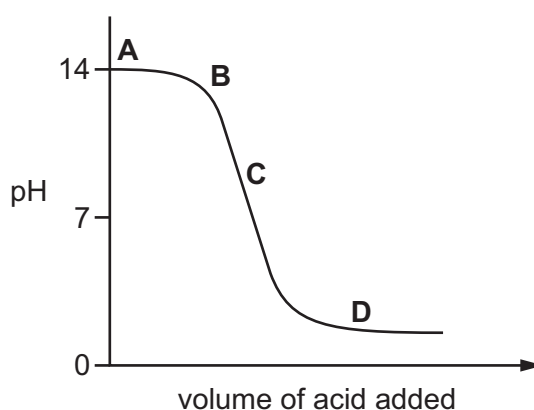
- 17 When aqueous iron(III) chloride is added to aqueous potassium iodide a chemical reaction occurs and iodine is formed.

Which statement is correct?

- A Iodide ions are oxidised, they gain electrons in this reaction.
 - B Iodide ions are oxidised, they lose electrons in this reaction.
 - C Iron(III) chloride is oxidised in this reaction.
 - D Neither iodide ions nor iron(III) chloride is oxidised in this reaction.
- 18 The graph shows how the pH of a solution changes as an acid is added to an alkali.

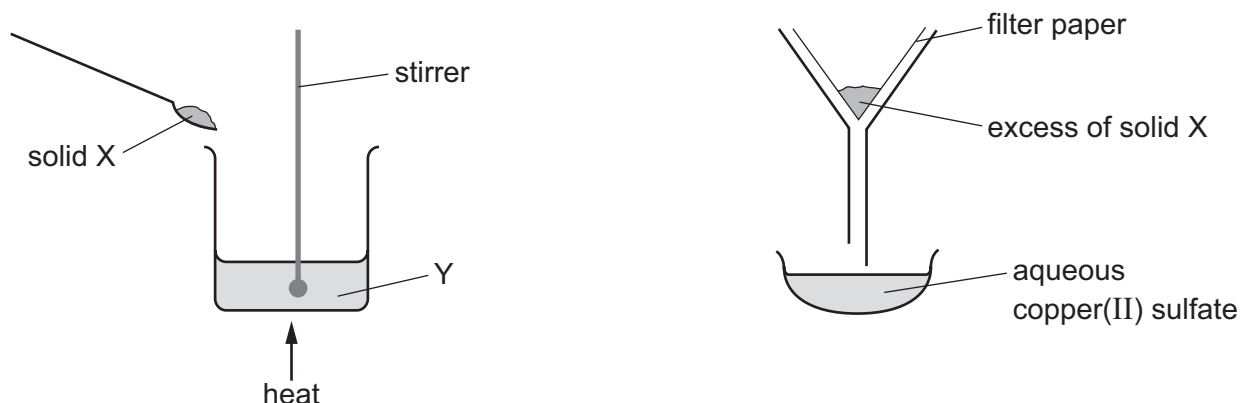


Which letter represents the area of the graph where both acid and salt are present?



- 19 Which statement describes a weak acid?
- A It is a proton acceptor and is fully ionised in aqueous solution.
 - B It is a proton acceptor and is partially ionised in aqueous solution.
 - C It is a proton donor and is fully ionised in aqueous solution.
 - D It is a proton donor and is partially ionised in aqueous solution.

20 The apparatus shown is used to prepare aqueous copper(II) sulfate.



What are X and Y?

	X	Y
A	copper	aqueous iron(II) sulfate
B	copper(II) chloride	dilute sulfuric acid
C	copper(II) oxide	dilute sulfuric acid
D	sulfur	aqueous copper(II) chloride

21 Which process is **not** used in the preparation of an insoluble salt?

- A** filtration
- B** washing
- C** crystallisation
- D** drying

22 Which statement about Group I and Group VII elements is correct?

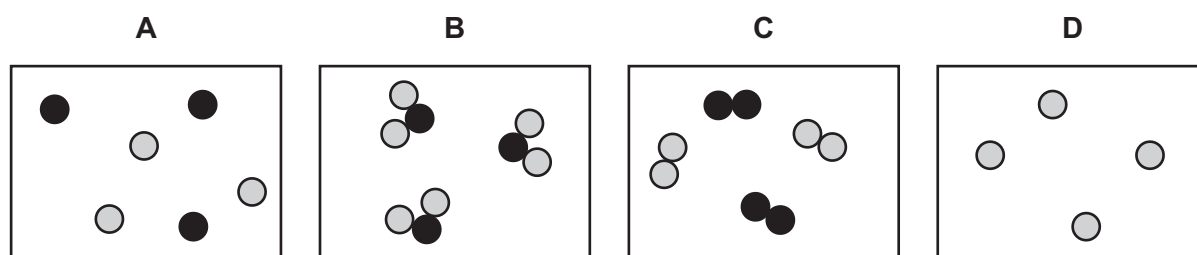
- A** Group VII elements are monoatomic non-metals.
- B** Lithium is more reactive with water than caesium.
- C** The melting points of Group I metals increase down the group.
- D** Potassium bromide reacts with chlorine to produce an orange solution.

23 The properties of the element titanium, Ti, can be predicted from its position in the Periodic Table.

Which row identifies the properties of titanium?

	can be used as a catalyst	conducts electricity when solid	has low density	forms coloured compounds
A	✓	✓	✓	✗
B	✓	✓	✗	✓
C	✓	✗	✓	✓
D	✗	✓	✓	✓

24 Which diagram shows a mixture of noble gases?



25 Which property is shown by **all** metals?

- A** They are extracted from their ores by heating with carbon.
- B** They conduct electricity.
- C** They form acidic oxides.
- D** They react with hydrochloric acid to form hydrogen.

26 Many metal carbonates decompose when they are heated.

Which row describes what happens when potassium carbonate, calcium carbonate and copper(II) carbonate are heated using a Bunsen burner?

	decomposes easily	decomposes with difficulty	does not decompose at Bunsen temperatures
A	calcium carbonate	copper(II) carbonate	potassium carbonate
B	copper(II) carbonate	calcium carbonate	potassium carbonate
C	copper(II) carbonate	potassium carbonate	calcium carbonate
D	potassium carbonate	calcium carbonate	copper(II) carbonate

27 Molten iron from the blast furnace contains impurities.

The process of turning the impure iron into steel involves blowing oxygen into the molten iron and adding calcium oxide.

What are the reasons for blowing in oxygen and adding calcium oxide?

	blowing in oxygen	adding calcium oxide
A	carbon is removed by reacting with oxygen	reacts with acidic impurities making slag
B	carbon is removed by reacting with oxygen	reacts with slag and so removes it
C	iron reacts with the oxygen	reacts with acidic impurities making slag
D	iron reacts with the oxygen	reacts with slag and so removes it

28 Four iron nails are added to four different metal sulfate solutions.

In which solution does a displacement reaction occur?

- A** copper(II) sulfate
- B** magnesium sulfate
- C** sodium sulfate
- D** zinc sulfate

29 Which statement about pure water is **not** correct?

- A** It condenses at 100 °C.
- B** It freezes at 0 °C.
- C** It turns cobalt(II) chloride paper blue.
- D** It turns anhydrous copper(II) sulfate blue.

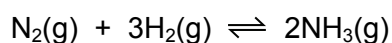
30 Three processes in the carbon cycle are shown.

- 1 Methane reacts with oxygen producing carbon dioxide and water.
- 2 Carbon dioxide and water are absorbed and used by plants to make oxygen.
- 3 Oxygen is used by living things to release energy.

Which processes have taken place?

	1	2	3
A	combustion	photosynthesis	respiration
B	combustion	respiration	photosynthesis
C	photosynthesis	combustion	respiration
D	respiration	photosynthesis	combustion

31 In the Haber process, nitrogen and hydrogen are reacted to make ammonia.



The forward reaction is exothermic.

Which conditions produce the maximum yield of ammonia?

	pressure	temperature
A	high	high
B	high	low
C	low	high
D	low	low

32 Which process, used to prevent iron from rusting, involves sacrificial protection?

- A** alloying
- B** electroplating
- C** galvanising
- D** painting

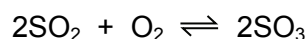
33 A student suggests three uses of calcium carbonate (limestone).

- 1 manufacture of cement
- 2 manufacture of iron
- 3 treating alkaline soils

Which suggestions are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

34 One of the reactions used in the manufacture of sulfuric acid is shown.



Which catalyst is used to increase the rate of this reaction?

- A** iron
B manganese(IV) oxide
C vanadium(V) oxide
D nickel

35 Ethanol is made on an industrial scale by the fermentation of sugars or by the reaction of ethene with steam in the presence of a suitable catalyst.

What is a **disadvantage** of making ethanol from ethene rather than by fermentation?

- A** A continuous production process is used.
B A non-renewable raw material is used.
C The product is very pure.
D The rate of reaction is very high.

36 Which statement about compounds in the same homologous series is correct?

- A** They have the same chemical properties because they have the same number of carbon atoms.
B They have the same physical properties because they have the same number of carbon atoms.
C They have different chemical properties because they have different numbers of carbon atoms.
D They have different physical properties because they have different numbers of carbon atoms.

- 37 Increasing the number of atoms in one molecule of a hydrocarbon increases the amount of energy released when it burns.

What is the correct order?

	less energy released	→	more energy released
A	ethene	ethane	methane
B	ethene	methane	ethane
C	methane	ethane	ethene
D	methane	ethene	ethane

- 38 An organic compound, P, reacts with zinc to produce a gas, Q.

What are P and Q?

	P	Q
A	ethanoic acid	carbon dioxide
B	ethanoic acid	hydrogen
C	ethanol	carbon dioxide
D	ethanol	hydrogen

- 39 Alkanes undergo substitution reactions in the presence of UV light.

Which equation represents a substitution reaction of ethane?

- A** $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_4 + 2\text{HCl}$
B $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{HCl}$
C $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_4\text{Cl}_2 + \text{H}_2$
D $\text{C}_2\text{H}_6 + \text{HCl} \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{H}_2$

- 40 Which substances are natural polymers?

- 1 proteins
- 2 carbohydrates
- 3 nylon
- 4 poly(ethene)

- A** 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 3 and 4

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

The Periodic Table of Elements

Group																		
I	II	Key										III	IV	V	VI	VII	VIII	
		atomic number atomic symbol name relative atomic mass										1 H hydrogen 1						
3 Li lithium 7	4 Be beryllium 9																	
11 Na sodium 23	12 Mg magnesium 24											5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —		114 Fl flerovium —		116 Lv livermorium —	—	

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™ (9–1)

CHEMISTRY (9–1)

0971/22

Paper 2 Multiple Choice (Extended)

May/June 2020

MARK SCHEME

Maximum Mark: 40

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	D	1
3	C	1
4	B	1
5	A	1
6	B	1
7	D	1
8	D	1
9	A	1
10	A	1
11	A	1
12	A	1
13	A	1
14	C	1
15	D	1
16	C	1
17	B	1
18	D	1
19	D	1
20	C	1
21	C	1
22	D	1
23	B	1
24	A	1
25	B	1
26	B	1
27	A	1
28	A	1

Question	Answer	Marks
29	C	1
30	A	1
31	B	1
32	C	1
33	A	1
34	C	1
35	B	1
36	D	1
37	D	1
38	B	1
39	B	1
40	A	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

October/November 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Blank pages are indicated.



1 Which gas has the slowest rate of diffusion?

A H_2

B NH_3

C CH_4

D CO_2

2 A chromatography experiment is carried out to analyse the pigments present in four different types of leaf. The student carrying out the experiment forgot to complete his table of results, which is shown.

plant leaf	number of pigments identified	colour of identified pigments	distance travelled by the solvent front (cm)	distance travelled from the origin by each pigment (cm)	R_f value
maple	F	green / yellow	3.7	green: 3.0 yellow: 3.1	green: 0.81 yellow: 0.83
laurel	2	green / yellow	G	green: 2.5 yellow: 2.5	green: 0.78 yellow: 0.78
lime	3	green / yellow / orange	3.5	green: 2.9 yellow: 3.0 orange: 2.7	green: 0.83 yellow: 0.86 yellow: 0.77
ash	3	green / yellow / orange	3.5	green: 2.8 yellow: 3.0 orange: 2.7	green: 0.80 yellow: H orange: 0.77

Which row identifies the values of **F**, **G** and **H**?

	F	G	H
A	2	3.2	0.80
B	3	3.5	0.83
C	2	3.2	0.86
D	3	3.5	0.78

3 Which statement about isotopes is correct?

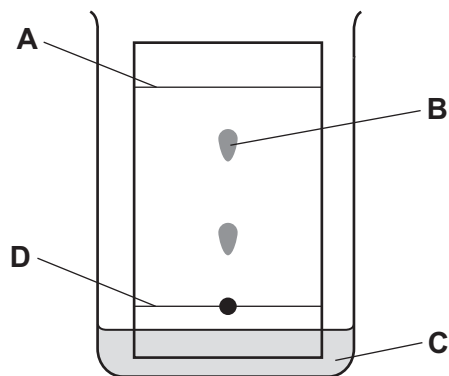
A They have different proton numbers.

B They have different chemical properties.

C They have the same nucleon number.

D They have the same number of electrons in their outer shell.

- 4 In the chromatography experiment shown, which label represents the solvent front?

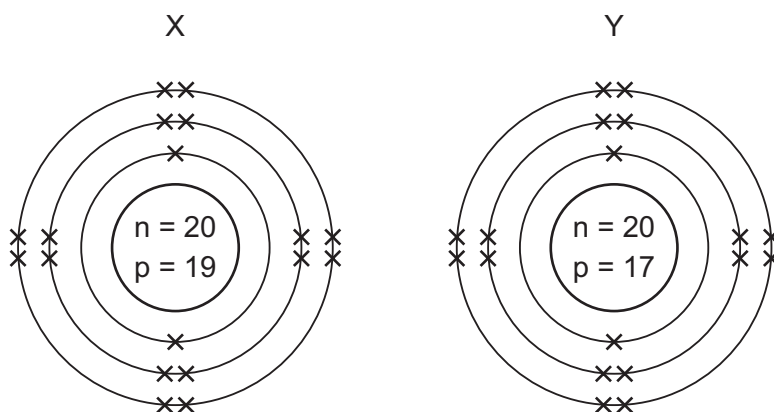


- 5 Different methods of separation rely on substances having different properties.

Which property does distillation make use of?

- A boiling point
- B colour
- C particle size
- D solubility in different solvents

- 6 The arrangements of the electrons in two ions formed from elements X and Y are shown.



Which equation represents the reaction between elements X and Y?

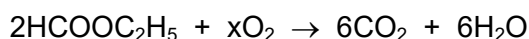
- A $X_2 + 2Y \rightarrow 2X^+ + 2Y^-$
- B $X_2 + 2Y \rightarrow 2X^- + 2Y^+$
- C $2X + Y_2 \rightarrow 2X^+ + 2Y^-$
- D $2X + Y_2 \rightarrow 2X^- + 2Y^+$

- 7 Which row identifies compounds that contain single covalent bonds only, double covalent bonds only or both single and double covalent bonds?

	single covalent bonds only	double covalent bonds only	both single and double covalent bonds
A	C ₂ H ₄	CH ₃ OH	CO ₂
B	CH ₃ OH	C ₂ H ₄	CO ₂
C	CH ₃ OH	CO ₂	C ₂ H ₄
D	CO ₂	C ₂ H ₄	CH ₃ OH

- 8 Ethyl methanoate, HCOOC₂H₅, burns in excess oxygen to produce carbon dioxide and water.

The equation is shown.



What is the value of x?

- A** 2 **B** 7 **C** 9 **D** 18

- 9 Rubidium is in Group I of the Periodic Table and bromine is in Group VII.

Rubidium reacts with bromine to form an ionic compound.

Which row shows the electron change taking place for rubidium and the correct formula of the rubidium ion?

	electron change	formula of ion formed
A	electron gained	Rb ⁺
B	electron gained	Rb ⁻
C	electron lost	Rb ⁺
D	electron lost	Rb ⁻

- 10 Which statement explains why graphite is used as a lubricant?

- A** All bonds between the atoms are weak.
B It conducts electricity.
C It has a low melting point.
D Layers in the structure can slide over each other.

- 11 The relative atomic mass of chlorine is 35.5.

When calculating relative atomic mass, which particle is the mass of a chlorine atom compared to?

- A a neutron
- B a proton
- C an atom of carbon-12
- D an atom of hydrogen-1

- 12 Universal indicator solution is added to a neutral solution of concentrated aqueous sodium chloride.

The solution, which contains H^+ (hydrogen), Na^+ (sodium), Cl^- (chloride) and OH^- (hydroxide) ions, is electrolysed.

The product at the cathode is hydrogen gas and the product at the anode is chlorine gas.

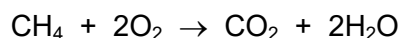
What happens to the colour of the indicator **in the solution** during electrolysis?

- A The colour changes from blue to green.
- B The colour changes from blue to red.
- C The colour changes from green to blue.
- D The colour changes from green to red.

- 13 What is the empirical formula of an oxide of iron, formed by reacting 2.24 g of iron with 0.96 g of oxygen?

- A FeO B Fe_2O C Fe_2O_3 D Fe_3O_4

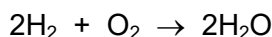
- 14 The combustion of methane is exothermic.



Which statement about this reaction is correct?

- A The energy needed to break the bonds in methane and oxygen is greater than the energy released in making new bonds in carbon dioxide and water.
- B The energy needed to break the bonds in methane and oxygen is less than the energy released in making new bonds in carbon dioxide and water.
- C The energy released in breaking bonds in methane and oxygen is greater than the energy needed to make new bonds in carbon dioxide and water.
- D The energy released in breaking bonds in methane and oxygen is less than the energy needed to make new bonds in carbon dioxide and water.

- 15 Hydrogen reacts with oxygen in a fuel cell.



The reaction is exothermic.

286 kJ of energy is released for every mole of water formed.

Which volume of hydrogen gas, measured at room temperature and pressure, would react with oxygen with the release of 7000 J of energy?

- A 587 cm³ B 1175 cm³ C 587 dm³ D 1175 dm³

- 16 Which substance does **not** require oxygen in order to produce energy?

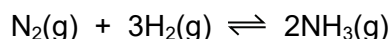
- A coal
B hydrogen
C natural gas
D ²³⁵U

- 17 Nitrogen, N₂, and hydrogen, H₂, can be converted into ammonia, NH₃, using a catalyst.

What is the purpose of the catalyst?

- A to increase the amount of ammonia produced
B to increase the rate of reaction
C to reduce the amount of reactants needed
D to reduce the rate of reaction

- 18 Ammonia is produced by the Haber process. The equation is shown.

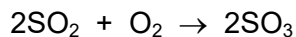


The forward reaction is exothermic.

Which statement is correct?

- A Increasing pressure decreases the yield of ammonia, but speeds up the reaction.
B Increasing temperature decreases the yield of ammonia, but speeds up the reaction.
C Increasing the concentration of hydrogen and nitrogen results in a lower yield of ammonia.
D Increasing the temperature increases the yield of ammonia and speeds up the reaction.

- 19 During the manufacture of sulfuric acid, sulfur dioxide is converted to sulfur trioxide.



Which type of reaction is this?

- A displacement
- B neutralisation
- C oxidation
- D thermal decomposition

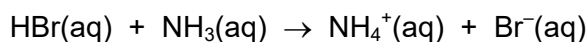
- 20 The equation for a redox reaction is shown.



Which element is reduced?

- A chlorine
- B iron
- C oxygen
- D sulfur

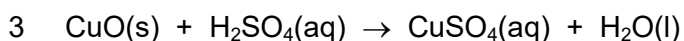
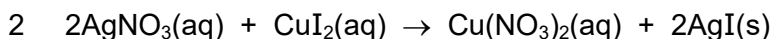
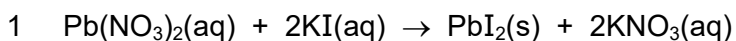
- 21 The equation shows a reaction between aqueous hydrogen bromide and aqueous ammonia.



Which statement describes the role of aqueous hydrogen bromide?

- A It is a catalyst.
- B It is a reducing agent.
- C It is a proton acceptor.
- D It is a proton donor.

- 22 The equations for three reactions are shown.



Which reactions are suitable for making a salt by precipitation?

- A 1 and 2 only B 1 and 3 only C 2 and 3 only D 1, 2 and 3

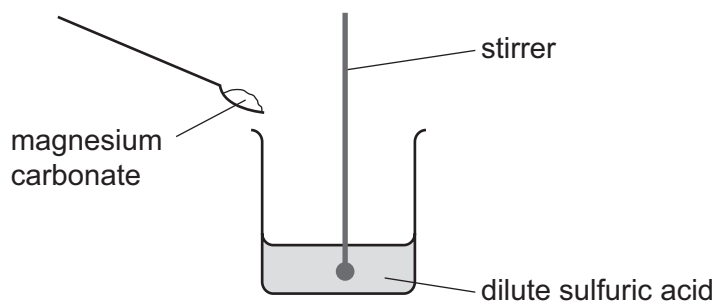
23 Zinc oxide is an amphoteric oxide.

Which row describes the reactions of zinc oxide?

	reaction with alkalis	reaction with acids
A	✓	✓
B	✓	x
C	x	✓
D	x	x

24 A student carries out an experiment to prepare pure magnesium sulfate crystals.

The diagram shows the first stage of the preparation.



He adds magnesium carbonate until no more reacts.

Which process should he use for the next stage?

- A** crystallisation
- B** evaporation
- C** filtration
- D** neutralisation

25 Which row about elements in the Periodic Table is correct?

	statement 1	statement 2
A	two elements in the same group have similar chemical properties	metals are on the left of the table
B	two elements in the same group have similar chemical properties	metals are on the right of the table
C	two elements in the same period have similar chemical properties	metals are on the left of the table
D	two elements in the same period have similar chemical properties	metals are on the right of the table

26 A new element oxfordium, Ox, was discovered with the following properties.

solubility	electrical conduction	formula of element	bonding in a molecule of Ox ₂
insoluble in water	doesn't conduct	Ox ₂	Ox≡Ox

In which group of the Periodic Table should the new element be placed?

- A** Group III
- B** Group V
- C** Group VII
- D** Group VIII

27 A flammable gas needs to be removed from a tank at an industrial plant.

For safety reasons, an inert gas is used.

Which gas is suitable?

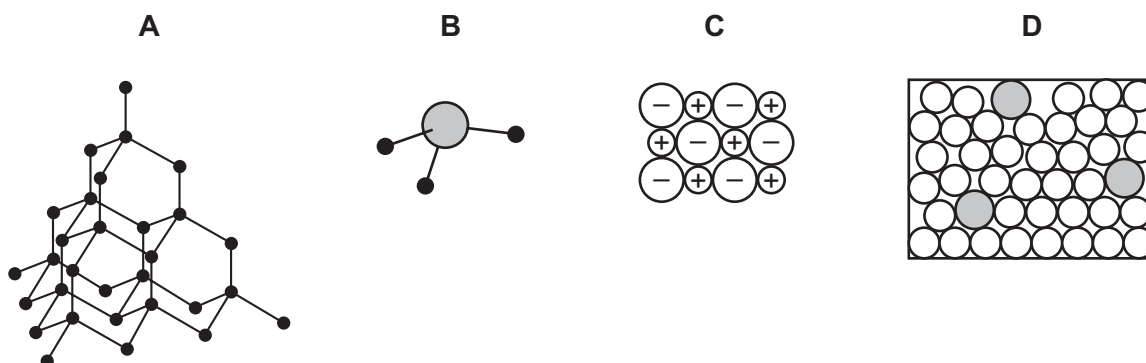
- A** argon
- B** hydrogen
- C** methane
- D** oxygen

28 Transition elements can have variable oxidation states.

Which pair of compounds shows a transition element in two different oxidation states?

- A** Cr₂O₃ and Cr₂(SO₄)₃
- B** Cu₂O and CuCO₃
- C** ZnS and ZnSO₄
- D** NiO and Ni(NO₃)₂

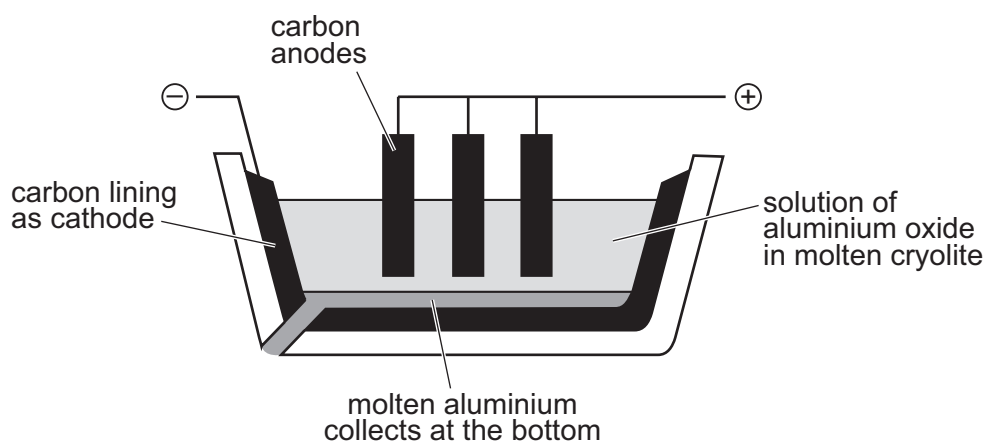
- 29 Which diagram best represents the structure of a substance that is a good conductor of electricity at 25 °C?



- 30 Why is aluminium metal unreactive with air?

- A** It is covered with a layer of oxide.
B It is low in the reactivity series.
C It is produced by electrolysis of its oxide.
D It melts at a high temperature.

- 31 The apparatus used for the extraction of aluminium oxide by electrolysis is shown.



Which equation represents a reaction taking place at the anode?

- A** $\text{O} + 2\text{e}^- \rightarrow \text{O}^{2-}$
B $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$
C $\text{Al}^{3+} \rightarrow \text{Al} + 3\text{e}^-$
D $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$

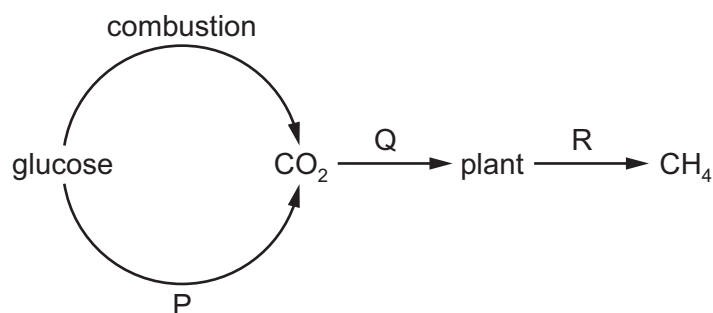
32 The results of tests on solid S and its aqueous solution are shown.

tests on solid S	tests on aqueous solution of S	
effect of heat	effect of aqueous sodium hydroxide	effect of aqueous ammonia
brown gas given off, together with a gas which relights a glowing splint	white ppt., soluble in excess, giving a colourless solution	white ppt., soluble in excess, giving a colourless solution

What is S?

- A aluminium nitrate
- B aluminium sulfate
- C zinc sulfate
- D zinc nitrate

33 Part of the carbon cycle is shown.



What are processes P, Q and R?

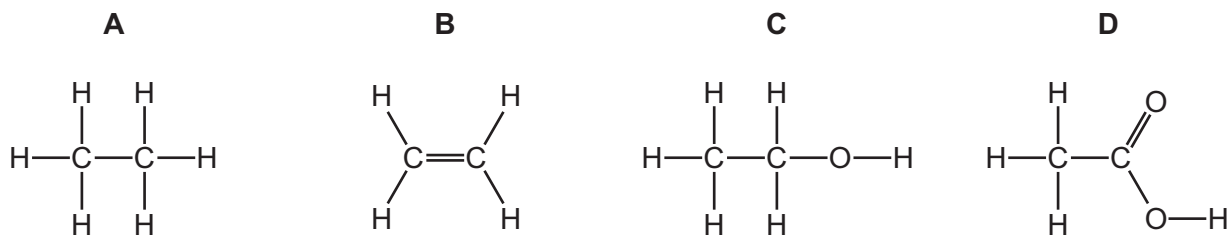
	P	Q	R
A	decomposition	respiration	photosynthesis
B	respiration	photosynthesis	decomposition
C	respiration	decomposition	photosynthesis
D	photosynthesis	respiration	decomposition

34 The element sulfur is found in a number of different minerals.

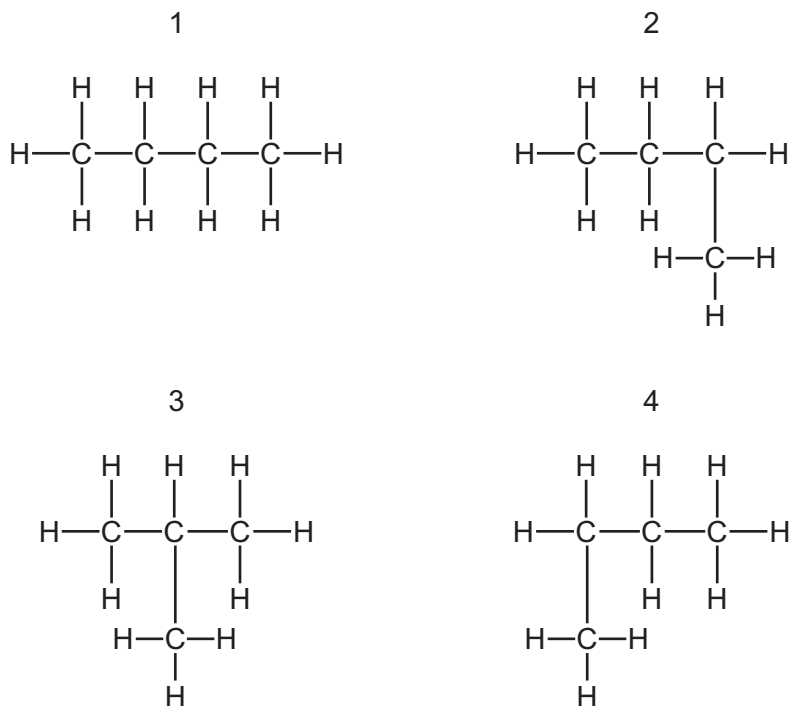
Which mineral contains the greatest percentage by mass of sulfur?

- A barite, BaSO_4
- B galena, PbS
- C gypsum, CaSO_4
- D pyrite, FeS_2

35 Which structure represents a molecule of ethanol?



36 Which structures are structural isomers of each other?

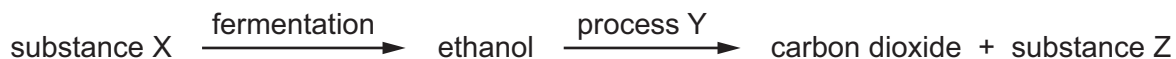


- A** 1, 2, 3 and 4
B 1, 2 and 4 only
C 1 and 3 only
D 2 and 4 only

37 Which molecule is **not** produced by an addition reaction of ethene?

- A** CH_3CH_3 **B** $\text{CH}_2\text{BrCH}_2\text{Br}$ **C** $\text{CH}_3\text{CH}_2\text{OH}$ **D** $\text{CH}_3\text{CH}_2\text{CH}_3$

- 38 The flow chart shows the preparation of ethanol and some important chemistry of ethanol.



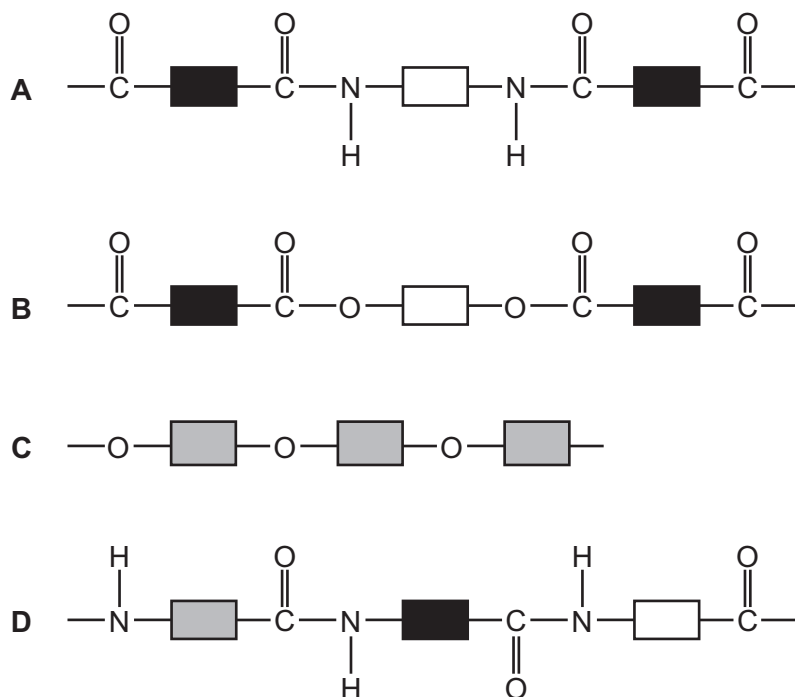
What are X, Y and Z?

	X	Y	Z
A	yeast	combustion	oxygen
B	glucose	combustion	steam
C	glucose	polymerisation	water
D	yeast	fermentation	glucose

- 39 Which statement about nylon and *Terylene* is correct?

- A** Nylon and *Terylene* are made from monomers with C=C bonds.
B Nylon and *Terylene* contain the same linkage.
C Nylon is a polyester.
D *Terylene* is made from two different monomers.

- 40 Which diagram represents the structure of a protein?



BLANK PAGE

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

The Periodic Table of Elements

Group																		
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII	
		Key atomic number atomic symbol name relative atomic mass																
3 Li lithium 7	4 Be beryllium 9																	
11 Na sodium 23	12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40											
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —	116 Lv livermorium —	—	—	

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™ (9–1)

CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

October/November 2020

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	C	1
3	D	1
4	A	1
5	A	1
6	C	1
7	C	1
8	B	1
9	C	1
10	D	1
11	C	1
12	C	1
13	C	1
14	B	1
15	A	1
16	D	1
17	B	1
18	B	1
19	C	1
20	A	1
21	D	1
22	A	1
23	A	1
24	C	1
25	A	1
26	B	1
27	A	1
28	B	1

Question	Answer	Marks
29	D	1
30	A	1
31	B	1
32	D	1
33	B	1
34	D	1
35	C	1
36	C	1
37	D	1
38	B	1
39	D	1
40	D	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/21

May/June 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

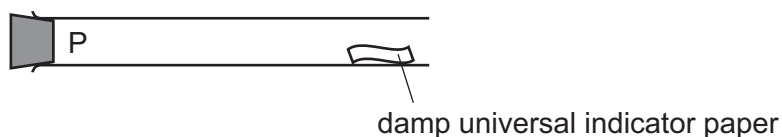
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.

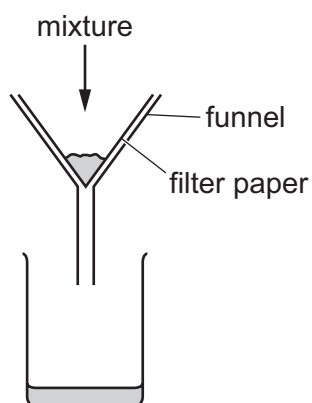


- 1 A gas is released at point P in the apparatus shown.



Which gas turns the damp universal indicator paper red most quickly?

- A ammonia, NH_3
 - B chlorine, Cl_2
 - C hydrogen chloride, HCl
 - D sulfur dioxide, SO_2
- 2 A mixture is separated using the apparatus shown.



What is the mixture?

- A aqueous copper(II) sulfate and aqueous sodium chloride
 - B aqueous copper(II) sulfate and copper
 - C copper and sulfur
 - D ethanol and ethanoic acid
- 3 Which statement about paper chromatography is correct?
- A A solvent is needed to dissolve the paper.
 - B Paper chromatography separates mixtures of solvents.
 - C The solvent should cover the baseline.
 - D The baseline should be drawn in pencil.

- 4 Element X has 7 protons.

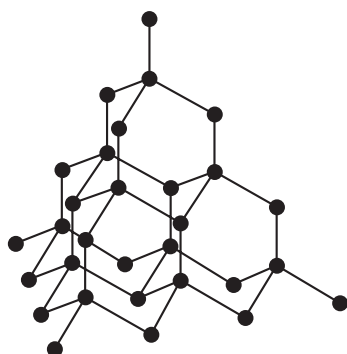
Element Y has 8 more protons than X.

Which statement about element Y is correct?

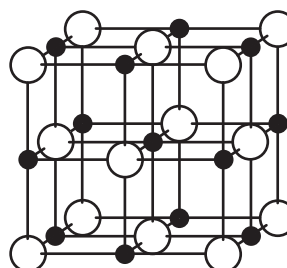
- A** Y has more electron shells than X.
- B** Y has more electrons in its outer shell than X.
- C** Y is in a different group of the Periodic Table from X.
- D** Y is in the same period of the Periodic Table as X.
- 5 A covalent molecule Q contains only six shared electrons.

What is Q?

- A** ammonia, NH_3
- B** chlorine, Cl_2
- C** methane, CH_4
- D** water, H_2O
- 6 The arrangement of particles in each of two solids, S and T, are shown.



S



T

What are S and T?

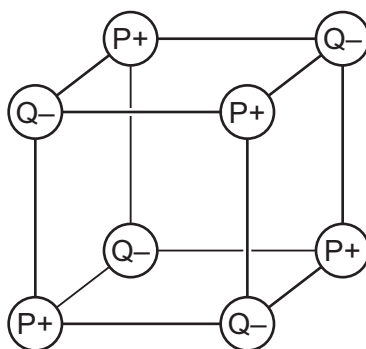
	S	T
A	diamond	silicon(IV) oxide
B	diamond	sodium chloride
C	graphite	silicon(IV) oxide
D	graphite	sodium chloride

7 Which statement about metals is correct?

- A Metals conduct electricity when molten because negative ions are free to move.
- B Metals conduct electricity when solid because positive ions are free to move.
- C Metals are malleable because the bonds between the atoms are weak.
- D Metals are malleable because the layers of ions can slide over each other.

8 Two elements, P and Q, are in the same period of the Periodic Table.

P and Q react together to form an ionic compound. Part of the lattice of this compound is shown.



Which statement is correct?

- A An ion of P has more electrons than an ion of Q.
- B Element P is non-metallic.
- C P is to the left of Q in the Periodic Table.
- D The formula of the compound is P_4Q_4 .

9 2.56 g of a metal oxide, MO_2 , is reduced to 1.92 g of the metal, M.

What is the relative atomic mass of M?

- A 48 B 96 C 128 D 192

10 In separate experiments, electricity was passed through concentrated aqueous sodium chloride and molten lead(II) bromide.

What would happen in **both** experiments?

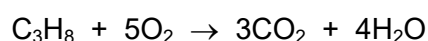
- A A halogen would be formed at the anode.
- B A metal would be formed at the cathode.
- C Hydrogen would be formed at the anode.
- D Hydrogen would be formed at the cathode.

11 What is the ionic half-equation for the reaction that occurs at the cathode when molten lead(II) bromide is electrolysed?

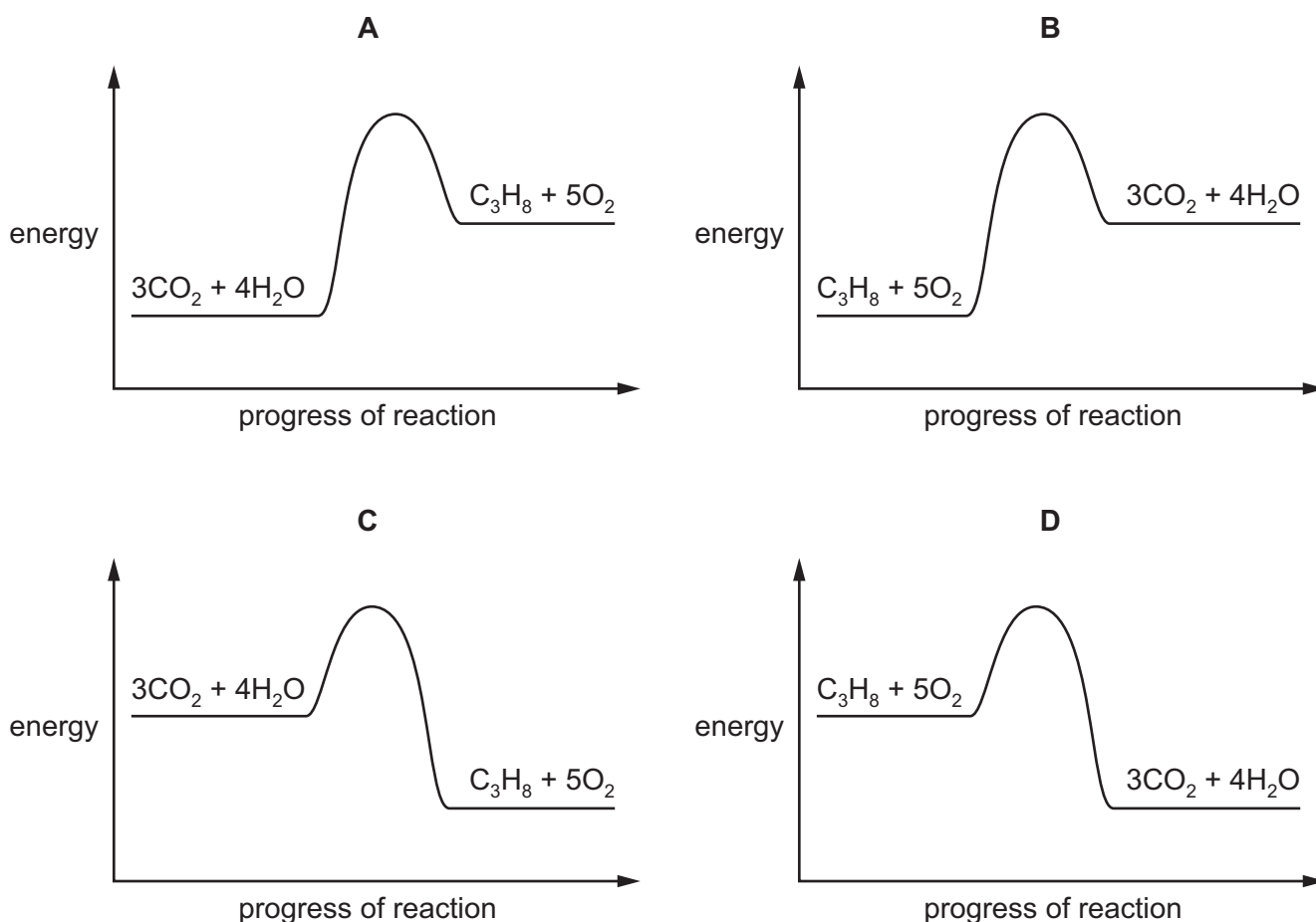
- A $\text{Pb}^{2+} + 2\text{e}^{-} \rightarrow \text{Pb}$
 B $2\text{Br}^{-} \rightarrow \text{Br}_2 + 2\text{e}^{-}$
 C $\text{Br}_2 + 2\text{e}^{-} \rightarrow 2\text{Br}^{-}$
 D $\text{Pb} \rightarrow \text{Pb}^{2+} + 2\text{e}^{-}$

12 The complete combustion of propane is exothermic.

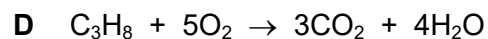
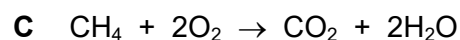
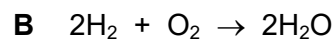
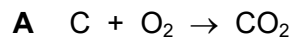
The equation for this reaction is shown.



Which energy level diagram represents the complete combustion of propane?



13 Which equation represents a reaction that takes place in a fuel cell?



14 When sulfur is heated it undergoes a1..... change as it melts.

Further heating causes the sulfur to undergo a2..... change and form sulfur dioxide.

Which words complete gaps 1 and 2?

	1	2
A	chemical	chemical
B	chemical	physical
C	physical	chemical
D	physical	physical

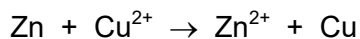
15 Four statements about the effect of increasing temperature on a reaction are shown.

- 1 The activation energy becomes lower.
- 2 The particles move faster.
- 3 There are more collisions between reacting particles per second.
- 4 There are more collisions which have energy greater than the activation energy.

Which statements are correct?

- A** 1, 2 and 3 **B** 1, 3 and 4 **C** 2, 3 and 4 **D** 2 and 3 only

16 An example of a redox reaction is shown.



Which statement about the reaction is correct?

- A Zn is the oxidising agent and it oxidises Cu^{2+} .
- B Zn is the oxidising agent and it reduces Cu^{2+} .
- C Zn is the reducing agent and it oxidises Cu^{2+} .
- D Zn is the reducing agent and it reduces Cu^{2+} .

17 Which statement about a reaction in equilibrium is correct?

- A Both the forward and the backward reactions are proceeding at the same rate.
- B Neither the forward nor the backward reaction is proceeding.
- C The amount of product present is no longer affected by changes in temperature or pressure.
- D The amount of product present is only affected by a change in pressure.

18 Element X forms an oxide, XO, that neutralises sulfuric acid.

Which row describes X and XO?

	element X	nature of oxide, XO
A	metal	acidic
B	metal	basic
C	non-metal	acidic
D	non-metal	basic

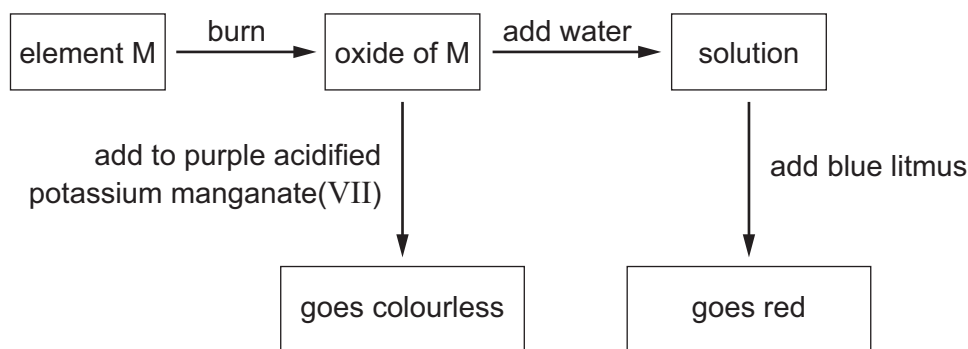
19 Copper(II) sulfate is prepared by adding excess copper(II) oxide to warm dilute sulfuric acid.

Which purification methods are used to obtain pure solid copper(II) sulfate from the reaction mixture?

- 1 crystallisation
- 2 filtration
- 3 chromatography
- 4 distillation

- A 1 and 4
- B 1 and 2
- C 2 and 3
- D 3 and 4

20 Some reactions of element M are shown.



What is element M?

- A carbon
- B iron
- C magnesium
- D sulfur

21 In which equation is the underlined reactant acting as a base?

- A $\text{CH}_3\text{COO}^- + \underline{\text{H}_3\text{O}^+} \rightarrow \text{CH}_3\text{COOH} + \text{H}_2\text{O}$
- B $\underline{\text{NH}_4^+} + \text{OH}^- \rightarrow \text{NH}_3 + \text{H}_2\text{O}$
- C $\text{CO}_2 + 2\underline{\text{H}_2\text{O}} \rightarrow \text{H}_3\text{O}^+ + \text{HCO}_3^-$
- D $\underline{\text{H}^+} + \text{OH}^- \rightarrow \text{H}_2\text{O}$

22 Why is helium used to fill balloons?

- A Helium is monoatomic.
- B Helium is in Group VIII of the Periodic Table.
- C Helium has a full outer electron shell.
- D Helium is less dense than air.

23 Which elements in the table are transition elements?

element	property
E	forms E^{3+} ions only
F	forms F^+ and F^{2+} ions
G	forms only white salts
H	used in catalytic converters

- A E and G
- B E and H
- C F and G
- D F and H

24 Element R forms a covalent compound R_2Si with silicon.

Which row describes R?

	metallic or non-metallic character	group number in the Periodic Table
A	metallic	II
B	metallic	VI
C	non-metallic	II
D	non-metallic	VI

25 Some properties of metal J are listed.

- J does not react with cold water.
- J reacts with dilute hydrochloric acid.
- No reaction occurs when the oxide of J is heated with carbon.

What is J?

- A** copper
- B** iron
- C** magnesium
- D** sodium

26 Some metal nitrates and carbonates decompose when heated strongly.

Metal Q has a nitrate that decomposes to give a salt and a colourless gas only.

The carbonate of metal Q does not decompose when heated with a Bunsen burner.

What is metal Q?

- A** calcium
- B** copper
- C** sodium
- D** zinc

27 Which substances are used in the extraction of aluminium?

- A bauxite and cryolite
- B bauxite and hematite
- C cryolite and zinc blende
- D hematite and zinc blende

28 Different types of steel alloys are manufactured by changing the percentage of carbon in the alloy.

The properties of four steel alloys are shown.

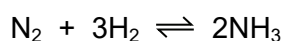
alloy mixture	percentage of carbon in the alloy	strength of the alloy	hardness of the alloy
1	0.00 to 0.20	high	low
2	0.21 to 0.30	high	medium
3	0.31 to 0.40	medium	high
4	0.41 to 1.50	low	high

What are the properties of the steel alloy containing 0.23% of carbon?

	strength	hardness
A	high	low
B	low	high
C	high	medium
D	medium	high

29 Ammonia is made by reacting nitrogen with hydrogen in the Haber process.

The equation for the process is shown.



Which changes in reaction conditions would produce a greater yield of ammonia?

- 1 adding more iron catalyst
- 2 increasing the reaction pressure
- 3 increasing the particle size of the iron catalyst

- A 1 only
- B 2 only
- C 1 and 2
- D 2 and 3

- 30 Which process removes carbon dioxide from the atmosphere?
- A combustion of fossil fuels
 - B fermentation
 - C photosynthesis
 - D respiration
- 31 Which catalyst is used in the Contact process?
- A calcium oxide
 - B iron
 - C manganese(II) oxide
 - D vanadium(V) oxide
- 32 A white solid Z reacts with dilute hydrochloric acid to produce a gas.
The same gas is produced when compound Z is heated strongly.
What is Z?
- A calcium
 - B calcium carbonate
 - C calcium hydroxide
 - D calcium oxide
- 33 What is the structure of butanoic acid?
- A $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
 - B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$
 - C $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$
 - D $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{CH}_3$

34 Compound Z contains carbon, hydrogen and oxygen.

Molecules of compound Z have four hydrogen atoms and two carbon atoms.

Compound Z can be made by oxidation of an alcohol.

What is compound Z?

- A** ethene
- B** ethanol
- C** ethanoic acid
- D** methyl methanoate

35 Which statement about homologous series and isomerism is correct?

- A** Butane and butene are structural isomers.
- B** Compounds in the same homologous series have the same general formula.
- C** Compounds in the same homologous series have the same molecular formula.
- D** Structural isomers have different molecular formulae.

36 Which statement about alkanes is correct?

- A** They burn in oxygen.
- B** They contain carbon, hydrogen and oxygen atoms.
- C** They contain double bonds.
- D** They contain ionic bonds.

37 What is an advantage of manufacturing ethanol by fermentation?

- A** The process is very fast.
- B** The ethanol requires no separation.
- C** The raw materials used are renewable.
- D** There are no other products formed.

38 P, Q, R and S are four organic compounds.

P is an unsaturated hydrocarbon.

Q burns but otherwise is unreactive.

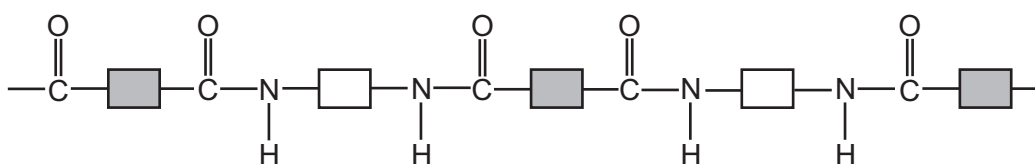
R contains a C–C single bond and a C=C double bond.

S undergoes addition polymerisation.

Which compounds are alkenes?

A P and R only **B** P, R and S **C** P, Q and S **D** Q, R and S

39 The structure of a synthetic polymer is shown.



The structure shows that it is a1..... . It is formed by2..... polymerisation.

Which words complete gaps 1 and 2?

	1	2
A	polyamide	addition
B	polyamide	condensation
C	polyester	addition
D	polyester	condensation

40 Which substance is a natural polymer?

- A** ethene
- B** *Terylene*
- C** nylon
- D** protein

BLANK PAGE

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										1 H hydrogen 1							2 He helium 4
11 Na sodium 23	12 Mg magnesium 24											5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20		
												13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40		
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —		114 Fl flerovium —		116 Lv livermorium —				

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™ (9–1)

CHEMISTRY (9–1)

0971/21

Paper 2 Multiple Choice (Extended)

May/June 2021

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	B	1
3	D	1
4	A	1
5	A	1
6	B	1
7	D	1
8	C	1
9	B	1
10	A	1
11	A	1
12	D	1
13	B	1
14	C	1
15	C	1
16	D	1
17	A	1
18	B	1
19	B	1
20	D	1
21	C	1
22	D	1
23	D	1
24	D	1
25	C	1
26	C	1
27	A	1
28	C	1

Question	Answer	Marks
29	B	1
30	C	1
31	D	1
32	B	1
33	B	1
34	C	1
35	B	1
36	A	1
37	C	1
38	B	1
39	B	1
40	D	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/22

May/June 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

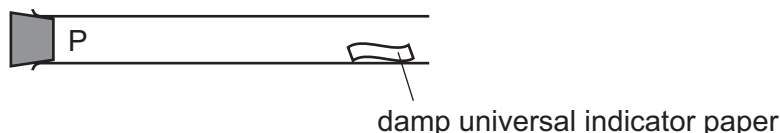
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 A gas is released at point P in the apparatus shown.



Which gas turns the damp universal indicator paper red most quickly?

- A ammonia, NH_3
 - B chlorine, Cl_2
 - C hydrogen chloride, HCl
 - D sulfur dioxide, SO_2
- 2 A mixture of colourless compounds is separated using chromatography.
- Which type of reagent is used to detect these compounds after separation?
- A a dehydrating agent
 - B a locating agent
 - C an oxidising agent
 - D a reducing agent
- 3 Which statement about paper chromatography is correct?
- A A solvent is needed to dissolve the paper.
 - B Paper chromatography separates mixtures of solvents.
 - C The solvent should cover the baseline.
 - D The baseline should be drawn in pencil.
- 4 Element X has 7 protons.

Element Y has 8 more protons than X.

Which statement about element Y is correct?

- A Y has more electron shells than X.
- B Y has more electrons in its outer shell than X.
- C Y is in a different group of the Periodic Table from X.
- D Y is in the same period of the Periodic Table as X.

- 5 A covalent molecule Q contains only six shared electrons.

What is Q?

- A** ammonia, NH_3
B chlorine, Cl_2
C methane, CH_4
D water, H_2O

- 6 Information about four substances E, F, G and H is shown.

	melting point / °C	electrical conductivity
E	1710	does not conduct when solid
F	3500	conducts when solid
G	120	does not conduct
H	801	conducts when molten

E, F, G and H are graphite, poly(ethene), sodium chloride and silicon(IV) oxide but not in that order.

What are E, F, G and H?

	E	F	G	H
A	graphite	poly(ethene)	silicon(IV) oxide	sodium chloride
B	sodium chloride	graphite	poly(ethene)	silicon(IV) oxide
C	poly(ethene)	sodium chloride	graphite	silicon(IV) oxide
D	silicon(IV) oxide	graphite	poly(ethene)	sodium chloride

- 7 Chemical compounds formed from a Group I element and a Group VII element contain ionic bonds.

How are the ionic bonds formed?

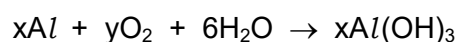
- A** Electrons are transferred from Group VII atoms to Group I atoms.
B Electrons are shared between Group I atoms and Group VII atoms.
C Electrons are lost by Group I atoms and Group VII atoms.
D Electrons are transferred from Group I atoms to Group VII atoms.

- 8 Some information about particles P, Q, R and S is shown.

	nucleon number	number of neutrons	number of electrons
P	12	6	6
Q	24	12	10
R	16	8	10
S	14	8	6

Which two particles are isotopes of the same element?

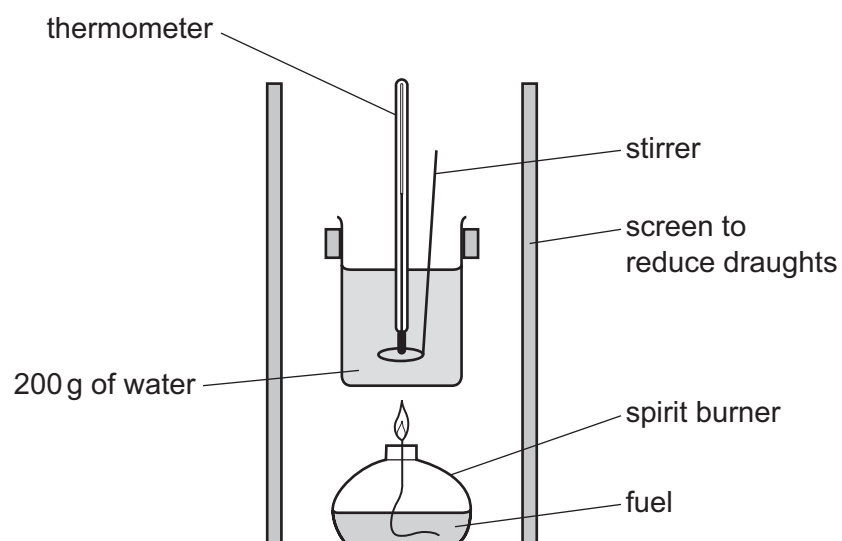
- A** P and Q **B** P and S **C** Q and R **D** R and S
- 9 Chlorine gas will react with iron metal.
- Exactly 21.3 g of chlorine reacts with 11.2 g of iron.
- How many iron atoms react with 30 molecules of chlorine?
- A** 10 **B** 15 **C** 20 **D** 30
- 10 In separate experiments, electricity was passed through concentrated aqueous sodium chloride and molten lead(II) bromide.
- What would happen in **both** experiments?
- A** A halogen would be formed at the anode.
B A metal would be formed at the cathode.
C Hydrogen would be formed at the anode.
D Hydrogen would be formed at the cathode.
- 11 A reaction involving aluminium is shown.



Which values of x and y balance the equation?

	x	y
A	2	3
B	3	2
C	3	4
D	4	3

- 12 Four different fuels are used to heat a beaker of water, for the same amount of time, using the apparatus shown.



The initial temperature of the water and the temperature after heating by the fuel are recorded.

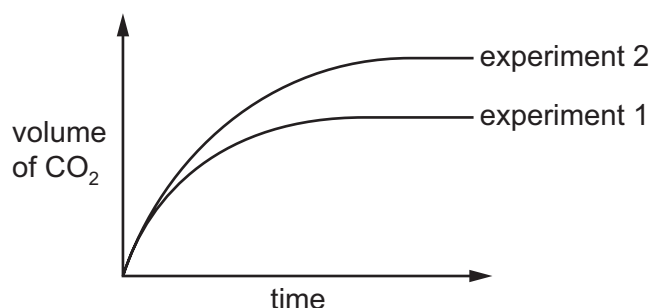
Which fuel releases the most heat energy?

	initial temperature / °C	temperature after heating / °C
A	17	46
B	24	52
C	26	61
D	30	62

- 13** An excess of calcium carbonate reacts with dilute hydrochloric acid. The volume of carbon dioxide produced is measured at regular time intervals. The results are shown as experiment 1.

The experiment is repeated with only **one** change to the reaction conditions.

The results are shown as experiment 2.



Which change is made in experiment 2?

- A** The concentration of the acid is increased.
 - B** The volume of acid is increased.
 - C** The mass of calcium carbonate is increased.
 - D** The calcium carbonate is powdered.
- 14** When sulfur is heated it undergoes a1..... change as it melts.
- Further heating causes the sulfur to undergo a2..... change and form sulfur dioxide.

Which words complete gaps 1 and 2?

	1	2
A	chemical	chemical
B	chemical	physical
C	physical	chemical
D	physical	physical

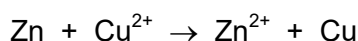
15 Four statements about the effect of increasing temperature on a reaction are shown.

- 1 The activation energy becomes lower.
- 2 The particles move faster.
- 3 There are more collisions between reacting particles per second.
- 4 There are more collisions which have energy greater than the activation energy.

Which statements are correct?

- A** 1, 2 and 3 **B** 1, 3 and 4 **C** 2, 3 and 4 **D** 2 and 3 only

16 An example of a redox reaction is shown.



Which statement about the reaction is correct?

- A** Zn is the oxidising agent and it oxidises Cu^{2+} .
B Zn is the oxidising agent and it reduces Cu^{2+} .
C Zn is the reducing agent and it oxidises Cu^{2+} .
D Zn is the reducing agent and it reduces Cu^{2+} .

17 When bismuth(III) chloride, BiCl_3 , reacts with water, a white precipitate of bismuth(III) oxychloride, BiOCl , is formed. The equation for the reaction is shown.



The reaction is in equilibrium.

Which changes cause the white precipitate to dissolve?

- 1 adding acid
- 2 adding water
- 3 adding sodium chloride solution

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

18 Element X forms an oxide, XO, that neutralises sulfuric acid.

Which row describes X and XO?

	element X	nature of oxide, XO
A	metal	acidic
B	metal	basic
C	non-metal	acidic
D	non-metal	basic

19 Information about the solubility of salts is shown.

salt	solubility
chlorides	soluble (except for lead(II) chloride and silver chloride)
nitrates	soluble
sulfates	soluble (except for barium sulfate and lead(II) sulfate)

Aqueous solutions of which two compounds would produce a precipitate when added together?

- A** $\text{Ba}(\text{NO}_3)_2$ and CaCl_2
- B** CuSO_4 and $\text{Zn}(\text{NO}_3)_2$
- C** KCl and Na_2SO_4
- D** $\text{Pb}(\text{NO}_3)_2$ and MgSO_4

- 20 The equation shows the reaction between hydrogen and oxygen.



The bond energies are shown.

	bond energy in kJ/mol
H-H	436
O=O	495
O-H	463

Which row shows the energy change and the type of reaction?

	energy change in kJ/mol	type of reaction
A	441	exothermic
B	441	endothermic
C	485	exothermic
D	485	endothermic

- 21 Burning fossil fuels releases sulfur dioxide which leads to acid rain.

Which ion in the rain water causes it to be acidic?

- A** H^+ **B** OH^- **C** O^{2-} **D** SO_4^{2-}

- 22 Which statement about the trends shown by the elements of Period 3 in the Periodic Table is **not** correct?

- A** The elements become less metallic across the period.
B The group number increases across the period.
C The number of electron shells increases across the period.
D The number of outer electrons increases across the period.

[illegible]

- 1 E has a higher density than F.
- 2 E has a higher melting point than F.
- 3 G has a darker colour than H.
- 4 G has a lower melting point than H.

- 24** When aqueous iodine is added to a solution of vanadium ions, V^{2+} , the V^{2+} ions each lose one electron.

A Transition elements have variable oxidation states.

B Transition elements form a stable $1+$ ion.

C Transition elements are oxidising agents.

D Transition elements can act as catalysts.

No immediate reaction is observed.

- A** Aluminium does not neutralise acids.
- B** Aluminium is a non-metal so does not react with acids.
- C** Aluminium is below hydrogen in the reactivity series.
- D** Aluminium is covered in an unreactive oxide layer.

- 26** Some metal nitrates and carbonates decompose when heated strongly.

Metal Q has a nitrate that decomposes to give a salt and a colourless gas only.

The carbonate of metal Q does not decompose when heated with a Bunsen burner.

What is metal Q?

- A** calcium
- B** copper
- C** sodium
- D** zinc

- 27** Aluminium is extracted from its ore by electrolysis.

Which equation represents the reaction that occurs at the anode during the electrolysis?

- A** $Al^{3+} + 3e^{-} \rightarrow Al$
- B** $Al^{3+} \rightarrow Al + 3e^{-}$
- C** $2O^{2-} \rightarrow O_2 + 4e^{-}$
- D** $2O^{2-} + 2e^{-} \rightarrow O_2$

- 28** Mild steel consists mostly of iron. Mild steel can be prevented from rusting by a process called galvanising.

Copper is not a very strong metal, however if it is mixed with a suitable metal a strong alloy called brass is produced.

Which statement is correct?

- A** Copper corrodes very quickly when wet and brass does not.
- B** Copper is mixed with zinc to produce brass.
- C** Galvanising mild steel changes it from a pure metal into an alloy.
- D** When a steel object is galvanised this means it is coated with a thin layer of tin.

- 29** Water is used for the irrigation of crops and for drinking water.

For which uses must water be chlorinated?

	irrigation	drinking
A	✓	✓
B	✓	x
C	x	✓
D	x	x

30 Which natural resource **cannot** provide a raw material for the manufacture of ammonia?

- A air
- B limestone
- C petroleum
- D water

31 Ammonia is made in the Haber process.

Which conditions are used in the Haber process?

	temperature / °C	pressure / atmospheres	catalyst used
A	450	200	iron
B	450	5	vanadium(V) oxide
C	200	450	iron
D	200	5	vanadium(V) oxide

32 Which process in the carbon cycle is responsible for removing carbon dioxide from the atmosphere?

- A combustion
- B decomposition
- C photosynthesis
- D respiration

33 The equations represent two reactions, P and Q, of lime (calcium oxide).

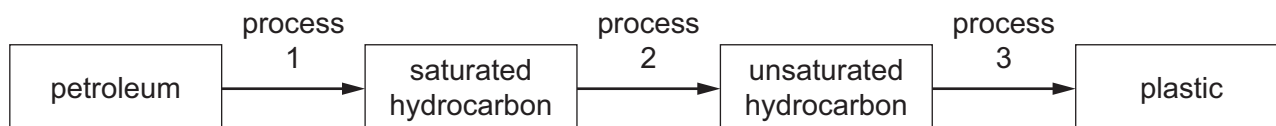


In which processes do the reactions occur?

	P	Q
A	extraction of iron	extraction of iron
B	extraction of iron	flue gas desulfurisation
C	flue gas desulfurisation	extraction of iron
D	flue gas desulfurisation	flue gas desulfurisation

- 34 Which statement about ethanol is **not** correct?
- A Ethanol can be made by fermentation.
 - B Ethanol is oxidised to make ethanoic acid.
 - C Ethanol reacts with oxygen exothermically, making it a good fuel.
 - D Ethanol reacts with propanoic acid to make propyl ethanoate.
- 35 Which pair of formulae represents two alkanes?
- A CH_4 and C_8H_{18}
 - B C_2H_6 and C_5H_8
 - C C_3H_6 and C_5H_{12}
 - D C_{10}H_8 and C_4H_8
- 36 Which statement about alkanes is correct?
- A They burn in oxygen.
 - B They contain carbon, hydrogen and oxygen atoms.
 - C They contain double bonds.
 - D They contain ionic bonds.
- 37 Which statements about ethanoic acid are correct?
- 1 It is a strong acid.
 - 2 It reacts with ethanol to form an ester.
 - 3 It has the formula CH_3COOH .
- A 1 and 2 only B 1 and 3 only C 2 and 3 only D 1, 2 and 3

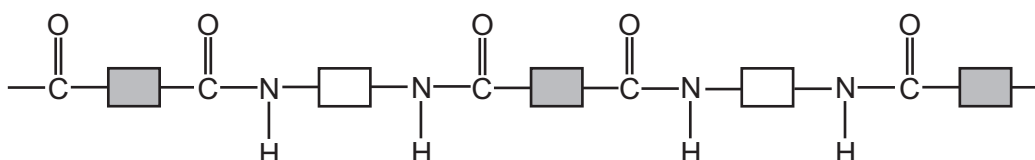
38 The flow chart shows how petroleum may be turned into a plastic.



What are processes 1, 2 and 3?

	process 1	process 2	process 3
A	cracking	fractional distillation	polymerisation
B	cracking	polymerisation	fractional distillation
C	fractional distillation	cracking	polymerisation
D	fractional distillation	polymerisation	cracking

39 The structure of a synthetic polymer is shown.



The structure shows that it is a1..... . It is formed by2..... polymerisation.

Which words complete gaps 1 and 2?

	1	2
A	polyamide	addition
B	polyamide	condensation
C	polyester	addition
D	polyester	condensation

40 Which substance is a natural polymer?

- A** ethene
- B** Terylene
- C** nylon
- D** protein

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										1 H hydrogen 1							2 He helium 4
11 Na sodium 23	12 Mg magnesium 24											5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20		
												13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40		
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —		114 Fl flerovium —		116 Lv livermorium —				

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™ (9–1)

CHEMISTRY (9–1)

0971/22

Paper 2 Multiple Choice (Extended)

May/June 2021

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	B	1
3	D	1
4	A	1
5	A	1
6	D	1
7	D	1
8	B	1
9	C	1
10	A	1
11	D	1
12	C	1
13	B	1
14	C	1
15	C	1
16	D	1
17	B	1
18	B	1
19	D	1
20	C	1
21	A	1
22	C	1
23	D	1
24	A	1
25	D	1
26	C	1
27	C	1
28	B	1

Question	Answer	Marks
29	C	1
30	B	1
31	A	1
32	C	1
33	B	1
34	D	1
35	A	1
36	A	1
37	C	1
38	C	1
39	B	1
40	D	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

October/November 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

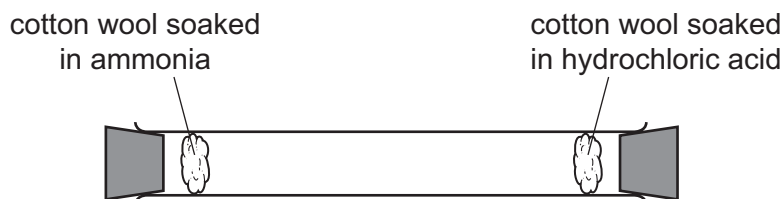
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

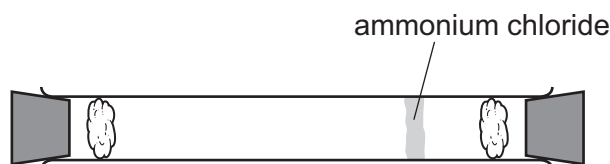
This document has **16** pages. Any blank pages are indicated.



- 1 An experiment is set up as shown.



After several minutes, a white ring of ammonium chloride appears as shown.



Which statement explains the observation after several minutes?

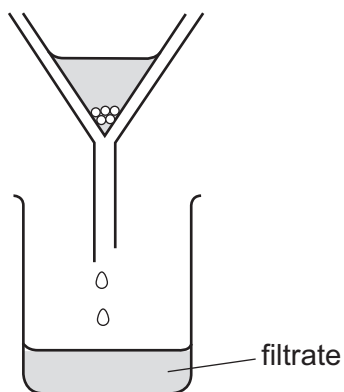
- A Ammonia gas diffuses faster than hydrogen chloride gas because its molecules have a lower molecular mass.
 - B Ammonia gas diffuses faster than hydrogen chloride gas because its molecules have a higher molecular mass.
 - C Ammonia gas diffuses slower than hydrogen chloride gas because its molecules have a lower molecular mass.
 - D Ammonia gas diffuses slower than hydrogen chloride gas because its molecules have a higher molecular mass.
- 2 A student put exactly 25.00 cm^3 of dilute hydrochloric acid into a conical flask.

The student added 2.5 g of solid sodium carbonate and measured the change in temperature of the mixture.

Which apparatus does the student need to use?

- A balance, measuring cylinder, thermometer
- B balance, pipette, stopwatch
- C balance, pipette, thermometer
- D burette, pipette, thermometer

- 3 A student separates sugar from pieces of broken glass by dissolving the sugar in water and filtering off the broken glass.



What is the filtrate?

- A broken glass only
 - B broken glass and sugar solution
 - C pure water
 - D sugar solution
- 4 How many protons, neutrons and electrons are there in one atom of the isotope ${}_{13}^{27}\text{Al}$?

	protons	neutrons	electrons
A	13	13	13
B	13	14	13
C	14	13	13
D	14	14	13

- 5 Which description of brass is correct?

- A alloy
- B compound
- C element
- D non-metal

6 Some properties of diamond are shown.

- 1 It is very hard.
- 2 Every atom forms four bonds.
- 3 It does not conduct electricity.

Which properties are also shown by silicon(IV) oxide?

- A** 1 only **B** 1 and 2 **C** 1 and 3 **D** 2 and 3

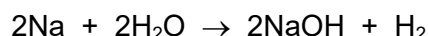
7 Which statement describes the attractive forces between molecules?

- A** They are strong covalent bonds which hold molecules together.
B They are strong ionic bonds which hold molecules together.
C They are weak forces formed between covalently-bonded molecules.
D They are weak forces which hold ions together in a lattice.

8 Which substance is described as a macromolecule?

- A** ammonia
B graphite
C iron
D sodium chloride

9 The equation for the reaction of sodium with water is shown.



What is the volume of hydrogen gas, measured at r.t.p., produced when 18.4 g of sodium reacts with excess water?

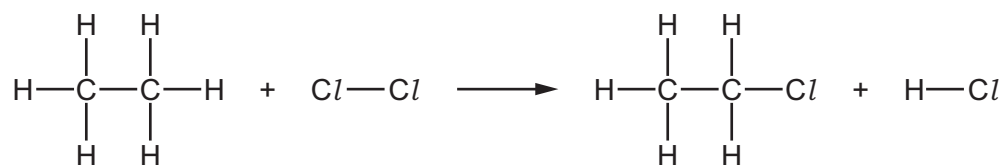
- A** 9.6 dm³ **B** 15.0 dm³ **C** 19.2 dm³ **D** 30.0 dm³

10 Iron can be electroplated with zinc to make it resistant to corrosion.

Which row about electroplating iron with zinc is correct?

	positive electrode (anode)	negative electrode (cathode)	electrolyte
A	iron	zinc	iron nitrate
B	iron	zinc	zinc nitrate
C	zinc	iron	iron nitrate
D	zinc	iron	zinc nitrate

- 11 Chlorine reacts with ethane to produce chloroethane and hydrogen chloride.



The reaction is exothermic.

The bond energies are shown in the table.

bond	bond energy in kJ/mol
C-Cl	+340
C-C	+350
C-H	+410
Cl-Cl	+240
H-Cl	+430

What is the energy change for the reaction?

- A -1420 kJ/mol
 B -120 kJ/mol
 C +120 kJ/mol
 D +1420 kJ/mol
- 12 Chlorine gas is bubbled into aqueous potassium iodide.

What is the ionic equation for the reaction that takes place?

- A $\text{Cl} + \text{I}^- \rightarrow \text{Cl}^- + \text{I}$
 B $\text{Cl}_2 + 2\text{I}^- \rightarrow \text{Cl}_2^- + \text{I}_2$
 C $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2$
 D $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + 2\text{I}$

- 13 Concentrated aqueous sodium chloride is electrolysed.

Which equation represents the reaction at the cathode?

- A $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$
 B $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$
 C $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
 D $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$

14 Which statements about hydrogen are correct?

- 1 When hydrogen is burned, heat energy is released.
- 2 When hydrogen is used in a fuel cell, electrical energy is generated.
- 3 When hydrogen is used as a fuel, water is the only product.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 only **D** 3 only

15 Solid X is heated strongly.

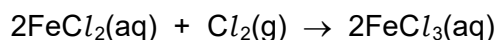
The colour of the solid changes from blue to white.

What is solid X?

- A** anhydrous cobalt(II) chloride
B calcium carbonate
C hydrated copper(II) sulfate
D lead(II) bromide

16 Iron(II) chloride solution reacts with chlorine gas.

The equation is shown.



Which statements about this reaction are correct?

- 1 Fe^{2+} ions are reduced to Fe^{3+} ions.
- 2 Chlorine acts as a reducing agent.
- 3 Fe^{2+} ions each lose an electron.
- 4 Cl_2 molecules are reduced to Cl^- ions.

A 1 and 2 **B** 2 and 3 **C** 2 and 4 **D** 3 and 4

17 Which statements about acids and bases are correct?

- 1 An acid reacts with a metal to give off hydrogen.
- 2 A base reacts with an ammonium salt to give off ammonia.
- 3 An acid reacts with a carbonate to give off carbon dioxide.
- 4 Alkaline solutions are orange in methyl orange.

A 1, 2 and 3 **B** 1, 2 and 4 **C** 1, 3 and 4 **D** 2, 3 and 4

18 Oxide 1 is a solid that reacts with dilute hydrochloric acid.

Oxide 2 is a gas that reacts with sodium hydroxide solution.

What are the formulae of the oxides?

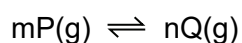
	oxide 1	oxide 2
A	CaO	MgO
B	MgO	NO ₂
C	NO ₂	SO ₂
D	SO ₂	CaO

19 Which reaction is a photochemical reaction?

- A** addition of bromine to propene
- B** esterification of ethanol and ethanoic acid
- C** oxidation of ethanol
- D** substitution of methane with chlorine

20 The equation shown represents a reaction at equilibrium.

m and n represent the balancing numbers for the reactant and product respectively.



A high temperature increases the concentration of Q.

A high pressure increases the concentration of Q.

Which statement about the reaction is correct?

- A** The forward reaction is exothermic and m is greater than n.
- B** The forward reaction is exothermic and m is less than n.
- C** The forward reaction is endothermic and m is greater than n.
- D** The forward reaction is endothermic and m is less than n.

21 A period of the Periodic Table is shown.

group	I	II	III	IV	V	VI	VII	VIII
element	R	S	T	V	W	X	Y	Z

The letters are not their chemical symbols.

Which statement is correct?

- A** Element R does not conduct electricity.
- B** Elements R and Y react together to form an ionic compound.
- C** Element Z exists as a diatomic molecule.
- D** Element Z reacts with element T.

22 All metal nitrates are soluble in water.

All metal chlorides are soluble except silver and lead.

All metal carbonates are insoluble except sodium and potassium.

Which aqueous solutions produce a precipitate when mixed together?

- 1 silver nitrate + sodium carbonate
- 2 silver nitrate + sodium chloride
- 3 barium nitrate + potassium chloride

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

23 Which row describes properties of transition elements?

	property 1	property 2	property 3
A	coloured compounds	high density	variable oxidation states
B	high density	high melting point	one oxidation state
C	high melting point	coloured compounds	one oxidation state
D	low melting point	high density	variable oxidation states

- 24** The noble gases are in Group VIII of the Periodic Table.

Which statement explains why noble gases are unreactive?

- A** They all have eight electrons in their outer shells.
- B** They all have full outer shells.
- C** They are all gases.
- D** They are all monoatomic.

- 25** Which statement is correct for **all** metals?

- A** They conduct electricity when molten.
- B** They gain electrons when they form ions.
- C** They have a low density.
- D** They have a low melting point.

- 26** Carbon dioxide is produced during the extraction of aluminium from bauxite.

Which statement describes how this carbon dioxide is made?

- A** Carbon monoxide reduces aluminium oxide forming carbon dioxide and aluminium.
- B** Carbon is burned in the blast furnace to release heat energy.
- C** Oxygen made in the process reacts with the carbon electrode.
- D** The ore of aluminium undergoes thermal decomposition.

- 27** Aluminium objects do not need protection from corrosion.

Iron objects must be protected from corrosion.

Which statement explains why aluminium resists corrosion?

- A** Aluminium does not form ions easily.
- B** Aluminium does not react with water or air.
- C** Aluminium has a protective oxide layer.
- D** Aluminium is below iron in the reactivity series.

28 Which statements explain why zinc is used to protect iron from rusting?

- 1 Zinc is more reactive than iron.
- 2 Zinc is less reactive than iron.
- 3 Zinc can form alloys with iron.
- 4 Zinc acts as a sacrificial metal.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

29 Which conditions are used in the Haber process?

	temperature / °C	pressure / atmospheres
A	100	10
B	450	10
C	450	200
D	1000	500

30 Which process does **not** produce a greenhouse gas?

- A** acid rain on limestone buildings
- B** combustion of wood
- C** digestion in cows
- D** zinc reacting with sulfuric acid

31 Which reaction involving sulfur dioxide is correct?

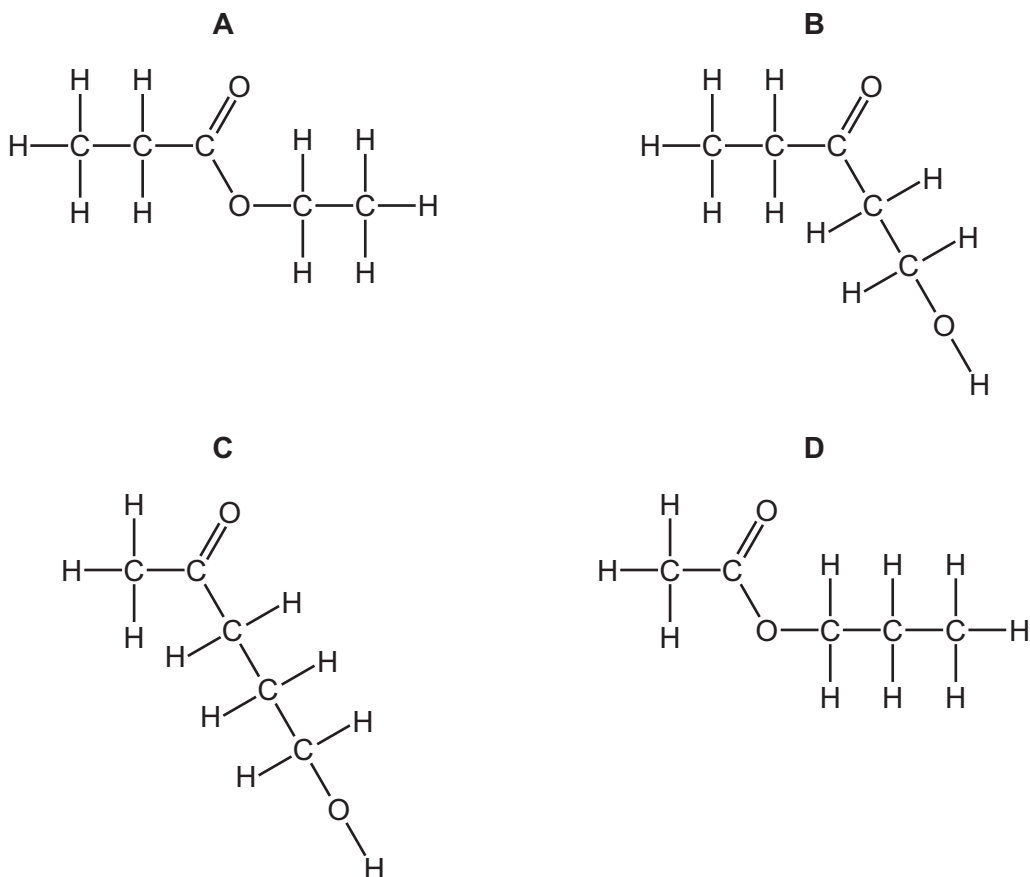
- A** It is produced during the extraction of zinc from zinc blende.
- B** It reacts with concentrated sulfuric acid to form oleum.
- C** It reacts with sulfur to form sulfur trioxide.
- D** It turns an acidified solution of potassium manganate(VII) purple.

32 Lime (calcium oxide) is used to treat waste water from a factory.

Which substance is removed by the lime?

- A** ammonia
- B** sodium chloride
- C** sodium hydroxide
- D** sulfuric acid

33 What is the structure of the ester formed from ethanoic acid and propanol?

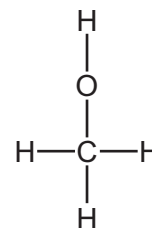
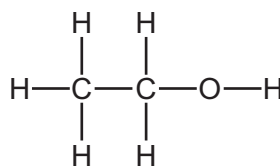
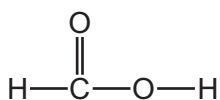
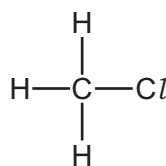


34 Fuel X produces carbon dioxide and water when it is burned in air. So does fuel Y.

What could X and Y be?

	X	Y
A	C	H ₂
B	C	C ₈ H ₁₈
C	CH ₄	H ₂
D	CH ₄	C ₈ H ₁₈

35 The structures of four organic molecules are shown.



How many different homologous series are represented by these molecules?

- A** 1 **B** 2 **C** 3 **D** 4

36 Which statement about ethene is correct?

- A** It has the chemical formula C_2H_6 .
B It burns in excess oxygen producing carbon dioxide and water.
C It reacts with Br_2 to produce an orange solution.
D It reacts with oxygen to form ethanol.

37 Ethanol is manufactured by fermentation of sugars or by catalytic hydration of ethene.

Which row states an advantage of each method?

	fermentation	hydration
A	produces purer ethanol	is a batch process
B	produces purer ethanol	is a continuous process
C	uses a renewable resource	is a batch process
D	uses a renewable resource	is a continuous process

38 Which statements about unsaturated hydrocarbons are correct?

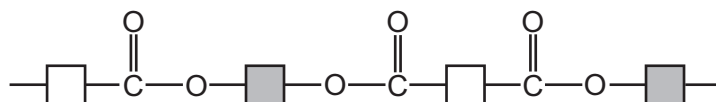
- 1 They contain both single and double bonds.
- 2 They turn aqueous bromine from colourless to brown.
- 3 They can be manufactured by cracking.

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

39 Which polymers have the same linkage between monomer units?

- A carbohydrate and polyamide
- B carbohydrate and polyester
- C protein and polyamide
- D protein and polyester

40 The diagram shows the partial structure of *Terylene*.



From which pair of compounds is it made?

- A $\text{HO}-\text{C}(=\text{O})-\square-\text{C}(=\text{O})-\text{OH}$ + $\text{HO}-\square-\text{OH}$
- B $\text{HO}-\square-\text{C}(=\text{O})-\text{OH}$ + $\text{HO}-\square-\text{C}(=\text{O})-\text{OH}$
- C $\text{HO}-\square-\text{OH}$ + $\text{HO}-\text{C}(=\text{O})-\square-\text{C}(=\text{O})-\text{OH}$
- D $\text{HO}-\text{C}(=\text{O})-\square-\text{C}(=\text{O})-\text{OH}$ + $\text{HO}-\text{C}(=\text{O})-\square-\text{C}(=\text{O})-\text{OH}$

BLANK PAGE

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

The Periodic Table of Elements

Group																	
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
	11 Na sodium 23											12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —	119 Uue unbinilium —	120 Uuh ununilium —

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™ (9–1)

CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

October/November 2021

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	C	1
3	D	1
4	B	1
5	A	1
6	C	1
7	C	1
8	B	1
9	A	1
10	D	1
11	B	1
12	C	1
13	C	1
14	A	1
15	C	1
16	D	1
17	A	1
18	B	1
19	D	1
20	C	1
21	B	1
22	A	1
23	A	1
24	B	1
25	A	1
26	C	1
27	C	1
28	B	1

Question	Answer	Marks
29	C	1
30	D	1
31	A	1
32	D	1
33	D	1
34	D	1
35	C	1
36	B	1
37	D	1
38	B	1
39	C	1
40	A	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/21

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Which two gases will diffuse at the same rate, at the same temperature?
- A** carbon monoxide and carbon dioxide
B carbon monoxide and nitrogen
C chlorine and fluorine
D nitrogen and oxygen
- 2 A student measures the time taken for 2.0 g of magnesium to dissolve in 50 cm³ of dilute sulfuric acid.

Which apparatus is essential to complete the experiment?

- 1 stop-clock
 2 measuring cylinder
 3 thermometer
 4 balance

- A** 1, 2 and 4 **B** 1 and 2 only **C** 1 and 4 only **D** 2, 3 and 4
- 3 The numbers of protons and neutrons and the electronic structures of four particles, W, X, Y and Z, are shown.

	number of protons	number of neutrons	electronic structure
W	8	8	2,8
X	8	10	2,6
Y	8	8	2,6
Z	10	8	2,8

Which particles have the same chemical properties?

- A** W and Y **B** W and Z **C** X and Y **D** X and Z
- 4 Which substance should be pure for the intended use?
- A** a drug for curing disease
B limestone for iron extraction
C petroleum for fractional distillation
D water for washing a car

- 5 Metals and ionic compounds have similarities and differences in their structure and properties.

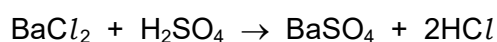
Which row about metals and ionic compounds is correct?

	similarity	difference
A	both contain positive ions	only ionic compounds contain anions
B	both contain positive ions	ionic compounds conduct using a 'sea of electrons'
C	both are malleable	only ionic compounds contain anions
D	both are malleable	ionic compounds conduct using a 'sea of electrons'

- 6 Which diagram represents the outer-shell electron arrangement in a nitrogen molecule?



- 7 The equation for the reaction between barium chloride and dilute sulfuric acid is shown.



Which row shows the state symbols for this equation?

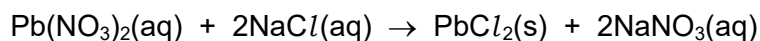
	BaCl_2	H_2SO_4	BaSO_4	2HCl
A	(aq)	(aq)	(s)	(aq)
B	(aq)	(l)	(s)	(aq)
C	(l)	(aq)	(s)	(l)
D	(aq)	(l)	(aq)	(l)

- 8 The relative atomic mass, A_r , of an element is determined by comparing the mass of one atom of the element with the mass of one atom of element Q.

What is Q?

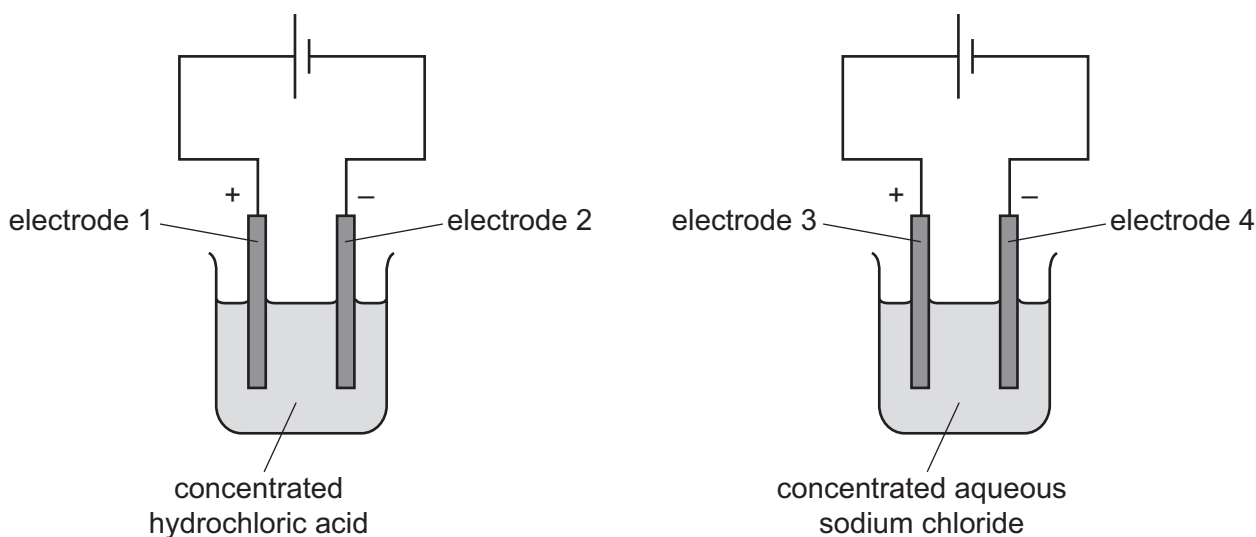
- A** carbon
- B** chlorine
- C** hydrogen
- D** oxygen

- 9 The equation for the reaction between aqueous lead(II) nitrate and aqueous sodium chloride is shown.



If 100 cm^3 of aqueous lead(II) nitrate of concentration 0.1 mol/dm^3 is reacted with an excess of aqueous sodium chloride, which mass of lead(II) chloride is obtained?

- A 1.16 g B 2.42 g C 2.78 g D 3.31 g
- 10 The diagram shows the electrolysis of concentrated hydrochloric acid and concentrated aqueous sodium chloride using carbon electrodes.



At which electrodes is hydrogen produced?

- A electrode 1 only
 B electrodes 1 and 3
 C electrode 2 only
 D electrodes 2 and 4
- 11 Aqueous copper(II) sulfate is electrolysed using copper electrodes.

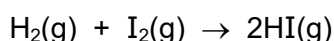
What is the ionic half-equation for the reaction at the cathode?

- A $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$
 B $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$
 C $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
 D $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$

12 Which row identifies a chemical change and a physical change?

	chemical change	physical change
A	boiling ethanol	burning ethanol
B	burning ethanol	evaporating ethanol
C	dissolving ethanol in water	burning ethanol
D	evaporating ethanol	dissolving ethanol in water

13 The equation for the reaction between gaseous hydrogen and gaseous iodine to form gaseous hydrogen iodide is shown.



The reaction is exothermic.

Which statement explains why the reaction is exothermic?

- A** Energy is released when H–H and I–I bonds are broken.
- B** The bond energies of the reactants are larger than the bond energies of the products.
- C** The products are at a higher energy level than the reactants.
- D** More energy is released when two HI bonds are formed than is used when the H–H and I–I bonds are broken.

14 Acidified aqueous silver nitrate is added to a test-tube containing aqueous chloride ions.

The test-tube is then left in direct sunlight.

Which row describes the observations and explains what happens to the reaction mixture?

	observation on adding aqueous silver nitrate	observation after leaving in sunlight	explanation
A	yellow precipitate	precipitate dissolves	silver chloride forms
B	yellow precipitate	precipitate turns grey	silver ions are reduced
C	white precipitate	precipitate dissolves	silver chloride forms
D	white precipitate	precipitate turns grey	silver ions are reduced

15 Water is added to anhydrous copper(II) sulfate.

What happens during the reaction?

- A** The copper(II) sulfate turns blue and the solution formed gets colder.
- B** The copper(II) sulfate turns blue and the solution formed gets hotter.
- C** The copper(II) sulfate turns white and the solution formed gets colder.
- D** The copper(II) sulfate turns white and the solution formed gets hotter.

16 Aqueous iron(III) chloride, FeCl_3 , reacts with aqueous potassium iodide, KI.



Which statements are correct?

- 1 In the balanced equation, v , w , x and y have the same value.
- 2 Potassium iodide is an oxidising agent.
- 3 A dark brown solution is produced in the reaction.

- A** 1 and 2 **B** 1 and 3 **C** 2 only **D** 2 and 3

17 Which statement about acids is correct?

- A** A strong acid has a higher pH than a weak acid of the same concentration.
- B** A strong acid is a proton acceptor.
- C** A weak acid is a proton donor.
- D** A weak acid is fully ionised in aqueous solution.

- 18** The oxides of two elements, X and Y, are separately dissolved in water and the pH of each solution tested.

oxide tested	pH of solution
X	1
Y	13

Which information about X and Y is correct?

	oxide is acidic	oxide is basic	metal	non-metal
A	X	Y	X	Y
B	X	Y	Y	X
C	Y	X	X	Y
D	Y	X	Y	X

- 19** An acid is neutralised by adding an excess of an insoluble solid base.

A soluble salt is formed.

How is the pure salt obtained from the reaction mixture?

- A** crystallisation → evaporation → filtration
B evaporation → crystallisation → filtration
C filtration → crystallisation → evaporation
D filtration → evaporation → crystallisation

- 20** The electronic structure of element Z is 2,8,1.

Which statements about Z are correct?

- 1 It is a metal.
 2 It has two outer-shell electrons.
 3 It is in Period 3.

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 only

21 Elements in Group IV of the Periodic Table are shown.

carbon
silicon
germanium
tin
lead

What does **not** occur in Group IV as it is descended?

- A The proton number of the elements increases.
- B The elements become more metallic.
- C The elements have more electrons in their outer shell.
- D The elements have more electron shells.

22 Element M forms both M^+ and M^{2+} ions.

In which part of the Periodic Table is M placed?

- A Group I
- B Group II
- C Group III
- D transition elements

23 In the extraction of aluminium by electrolysis, cryolite is added to the bauxite ore.

Which row describes the role of cryolite and gives the ionic half-equation at the cathode?

	role of cryolite	ionic half-equation at the cathode
A	catalyst	$Al^{3+} + 3e^- \rightarrow Al$
B	catalyst	$Al^{3+} + 3e^- \rightarrow 3Al$
C	lowers melting point of electrolyte	$Al^{3+} + 3e^- \rightarrow Al$
D	lowers melting point of electrolyte	$Al^{3+} + 3e^- \rightarrow 3Al$

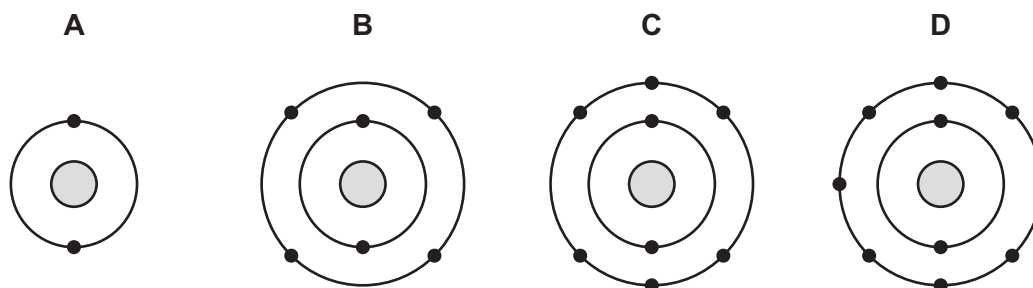
24 Mild steel is galvanised to prevent corrosion of the iron.

Which statements about galvanising are correct?

- 1 Galvanising prevents corrosion because the zinc forms an alloy.
- 2 If the coating is damaged, water and oxygen do not corrode the iron.
- 3 Zinc is a sacrificial metal and corrodes in preference to iron.

A 1 and 2 **B** 1 and 3 **C** 2 only **D** 2 and 3

25 Which diagram represents the arrangement of the outer-shell electrons of a noble gas?



26 Which statements about the general properties of metals are correct?

- 1 They are good conductors of heat and electricity.
- 2 They have low melting points.
- 3 They react with dilute acids to form a salt and water.
- 4 They react with oxygen to form basic oxides.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

27 Reactions of three metals and their oxides are shown.

metal	add dilute hydrochloric acid to metal	heat metal oxide with carbon	key ✓ = reacts ✗ = does not react
1	✓	✓	
2	✓	✗	
3	✗	✓	

What is the order of reactivity of these metals, from most reactive to least reactive?

A 1 → 2 → 3 **B** 1 → 3 → 2 **C** 2 → 1 → 3 **D** 2 → 3 → 1

28 Three metal compounds, J, K and L, are heated using a Bunsen burner.

The results are shown.

J colourless gas produced, which relights a glowing splint

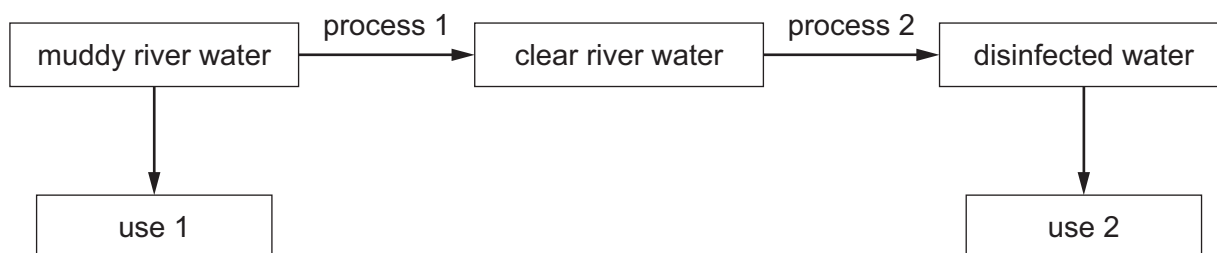
K colourless gas produced, which turns limewater milky

L no reaction

Which row identifies J, K and L?

	J	K	L
A	magnesium carbonate	potassium carbonate	potassium nitrate
B	magnesium carbonate	potassium nitrate	potassium carbonate
C	potassium nitrate	magnesium carbonate	potassium carbonate
D	potassium nitrate	potassium carbonate	magnesium carbonate

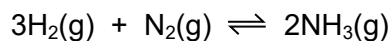
29 The diagram shows the uses and treatment processes of muddy river water.



Which row identifies uses 1 and 2 and processes 1 and 2?

	use 1	use 2	process 1	process 2
A	drinking	watering crops	chlorination	filtration
B	drinking	watering crops	filtration	chlorination
C	watering crops	drinking	chlorination	filtration
D	watering crops	drinking	filtration	chlorination

- 30 The equation for the manufacture of ammonia in the Haber process is shown.



The forward reaction is exothermic.

Which row describes the effect of the stated change on the reaction rate and the yield of ammonia?

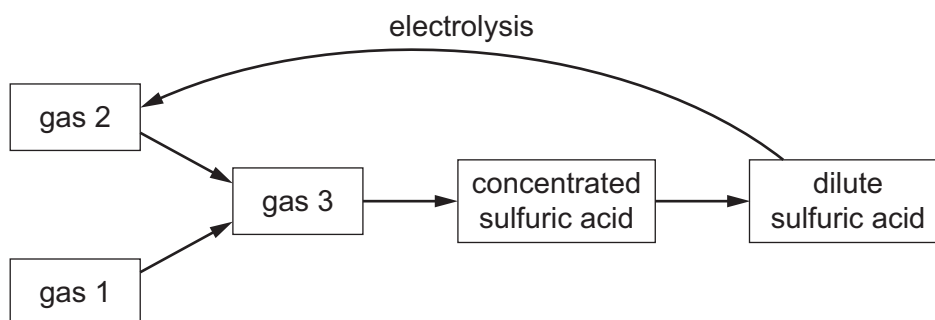
	change	effect on reaction rate	effect on yield of ammonia
A	decrease pressure	increases	decreases
B	decrease temperature	decreases	increases
C	increase pressure	increases	decreases
D	increase temperature	increases	increases

- 31 Fertilisers are used to provide three of the elements needed for plant growth.

Which two compounds would give a fertiliser containing all three of these elements?

- A** $\text{Ca}(\text{NO}_3)_2$ and $(\text{NH}_4)_2\text{SO}_4$
B $\text{Ca}(\text{NO}_3)_2$ and $(\text{NH}_4)_3\text{PO}_4$
C KNO_3 and $(\text{NH}_4)_2\text{SO}_4$
D KNO_3 and $(\text{NH}_4)_3\text{PO}_4$

- 32 The flow chart shows part of the process for the manufacture of sulfuric acid and its electrolysis.



What are gases 1, 2 and 3?

	gas 1	gas 2	gas 3
A	sulfur dioxide	hydrogen	sulfur trioxide
B	sulfur dioxide	oxygen	sulfur trioxide
C	sulfur trioxide	hydrogen	sulfur dioxide
D	sulfur trioxide	oxygen	sulfur dioxide

33 Which statements about sulfur dioxide are correct?

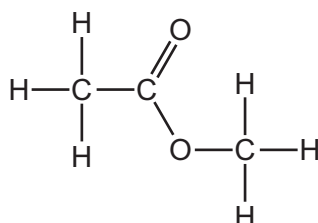
- 1 Sulfur dioxide decolourises acidified potassium manganate(VII).
- 2 Sulfur dioxide forms when acids react with carbonates.
- 3 Sulfur dioxide is used as a bleach.
- 4 Sulfur dioxide is used to treat acidic soil.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

34 What are the products when limestone (calcium carbonate) is heated strongly?

- A** calcium hydroxide and carbon dioxide
B calcium hydroxide and carbon monoxide
C calcium oxide and carbon dioxide
D calcium oxide and carbon monoxide

35 The structure of ester W is shown.



Which row gives the names of ester W and the carboxylic acid and alcohol from which it is made?

	name of ester W	carboxylic acid	alcohol
A	ethyl methanoate	ethanoic acid	methanol
B	ethyl methanoate	methanoic acid	ethanol
C	methyl ethanoate	ethanoic acid	methanol
D	methyl ethanoate	methanoic acid	ethanol

36 Ethanol is made industrially by the fermentation of glucose or by the catalytic addition of steam to ethene.

Which statement describes an advantage of fermentation compared to catalytic addition?

- A** Ethanol is the only product of fermentation.
B Fermentation uses a batch process but catalytic addition is continuous.
C Fermentation uses a higher temperature than catalytic addition.
D Fermentation uses a renewable resource.

37 Some properties of colourless liquid L are listed.

- It boils at 65 °C.
- When added to water, two layers form which do not mix.
- It does not react with sodium carbonate.
- It has no effect on bromine water.

What is L?

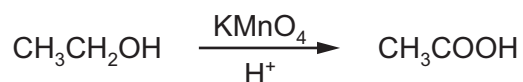
- A** ethanol
B hexane
C hexene
D ethanoic acid

38 A molecule of compound P contains two carbon atoms and four hydrogen atoms.

Which row represents P?

	name of compound	M_r	reacts with aqueous bromine
A	ethane	30	$\times$
B	ethene	16	✓
C	ethene	28	✓
D	ethene	28	$\times$

39 The reaction of ethanol with acidified potassium manganate(VII) is shown.



Which type of reaction is taking place?

- A** addition
B condensation
C hydrolysis
D oxidation

40 Which polymer is a synthetic polyamide?

- A nylon
- B poly(ethene)
- C protein
- D *Terylene*

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

The Periodic Table of Elements

Group																	
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	2 He helium 4
	11 Na sodium 23											12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —	127 Ts tennessine —	118 Og oganesson —	119 Nh nihonium —	120 Ds darmstadtium —

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™ (9–1)

CHEMISTRY (9–1)

0971/21

Paper 2 Multiple Choice (Core)

May/June 2022

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	A	1
3	C	1
4	A	1
5	A	1
6	D	1
7	A	1
8	A	1
9	C	1
10	D	1
11	B	1
12	B	1
13	D	1
14	D	1
15	B	1
16	B	1
17	C	1
18	B	1
19	D	1
20	C	1
21	C	1
22	D	1
23	C	1
24	D	1
25	A	1
26	B	1
27	C	1
28	C	1

Question	Answer	Marks
29	D	1
30	B	1
31	D	1
32	B	1
33	A	1
34	C	1
35	C	1
36	D	1
37	B	1
38	C	1
39	D	1
40	A	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/22

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages.



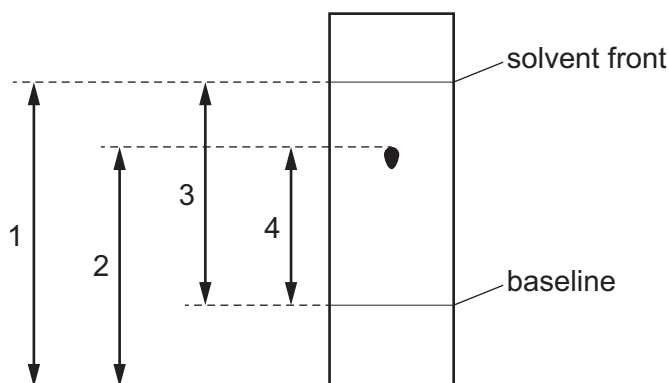
- 1 Which two gases will diffuse at the same rate, at the same temperature?
- A** carbon monoxide and carbon dioxide
B carbon monoxide and nitrogen
C chlorine and fluorine
D nitrogen and oxygen
- 2 A student measures the time taken for 2.0 g of magnesium to dissolve in 50 cm³ of dilute sulfuric acid.

Which apparatus is essential to complete the experiment?

- 1 stop-clock
2 measuring cylinder
3 thermometer
4 balance

A 1, 2 and 4 **B** 1 and 2 only **C** 1 and 4 only **D** 2, 3 and 4

- 3 A chromatogram of a single substance T is shown.



Which measurements are used to find the R_f value of T?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

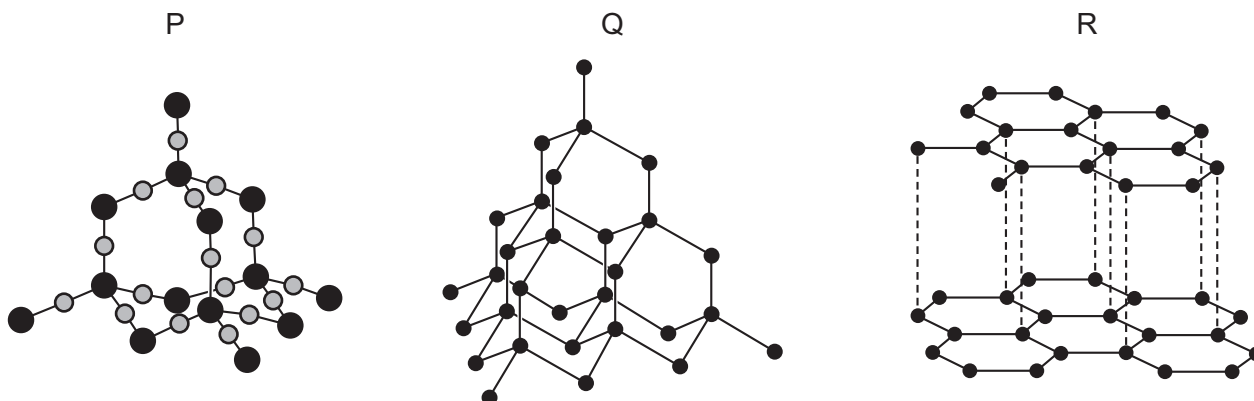
- 4 X and Y are two different elements.

X and Y have the same number of nucleons.

Which statement about X and Y is correct?

- A** They have the same physical properties.
B Their atoms have the same number of electrons.
C They are in different groups of the Periodic Table.
D They have different relative masses.

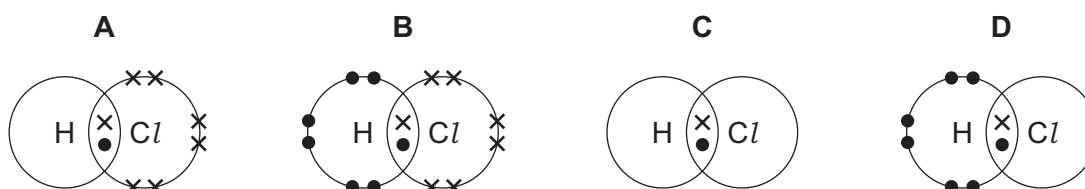
- 5 The diagrams show the structures of three macromolecules P, Q and R.



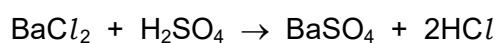
What are P, Q and R?

	P	Q	R
A	diamond	silicon(IV) oxide	graphite
B	graphite	diamond	silicon(IV) oxide
C	silicon(IV) oxide	diamond	graphite
D	silicon(IV) oxide	graphite	diamond

- 6 Which dot-and-cross diagram shows the arrangement of outer shell electrons in a molecule of hydrogen chloride?



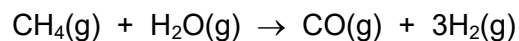
- 7 The equation for the reaction between barium chloride and dilute sulfuric acid is shown.



Which row shows the state symbols for this equation?

	BaCl_2	H_2SO_4	BaSO_4	2HCl
A	(aq)	(aq)	(s)	(aq)
B	(aq)	(l)	(s)	(aq)
C	(l)	(aq)	(s)	(l)
D	(aq)	(l)	(aq)	(l)

- 8 Methane and steam react in the presence of a catalyst.



0.5 mol of methane reacts completely with 0.5 mol of steam.

What is the volume of carbon monoxide and hydrogen produced, measured at room temperature and pressure?

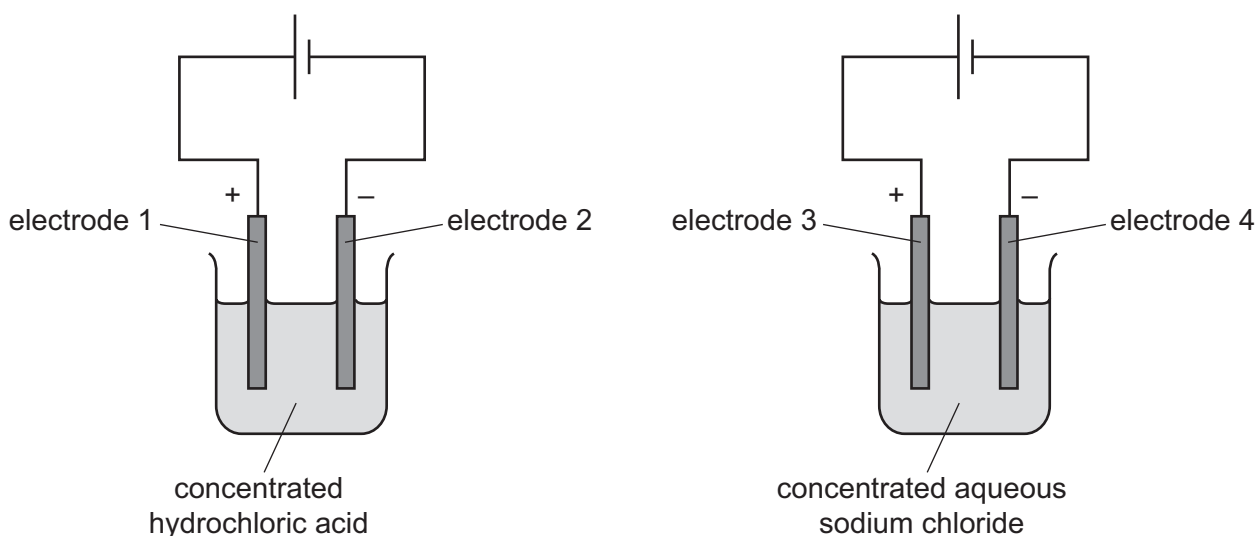
	volume of CO / dm ³	volume of H ₂ / dm ³
A	0.5	1.5
B	1.0	3.0
C	12.0	12.0
D	12.0	36.0

- 9 A compound of element X has the formula X₂O and a relative formula mass of 144.

What is element X?

- A** copper, Cu
- B** gadolinium, Gd
- C** sulfur, S
- D** tellurium, Te

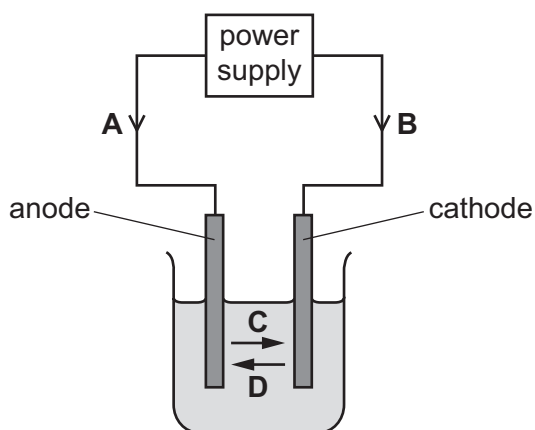
- 10 The diagram shows the electrolysis of concentrated hydrochloric acid and concentrated aqueous sodium chloride using carbon electrodes.



At which electrodes is hydrogen produced?

- A electrode 1 only
 - B electrodes 1 and 3
 - C electrode 2 only
 - D electrodes 2 and 4
- 11 The diagram shows the electrolysis of aqueous copper(II) sulfate using inert electrodes.

Which arrow shows the movement of electrons in the circuit?



12 Which row identifies a chemical change and a physical change?

	chemical change	physical change
A	boiling ethanol	burning ethanol
B	burning ethanol	evaporating ethanol
C	dissolving ethanol in water	burning ethanol
D	evaporating ethanol	dissolving ethanol in water

13 Which statements explain why increasing the concentration of a reactant increases the rate of reaction?

- 1 It increases the collision rate of particles.
- 2 It lowers the activation energy.
- 3 A greater proportion of the colliding molecules have the required activation energy.
- 4 There are more particles per unit volume.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

14 When the colourless gas N_2O_4 is heated, it forms the brown gas NO_2 .

When the reaction mixture is cooled, the brown colour fades and turns back to colourless.

Which type of reaction is described by these observations?

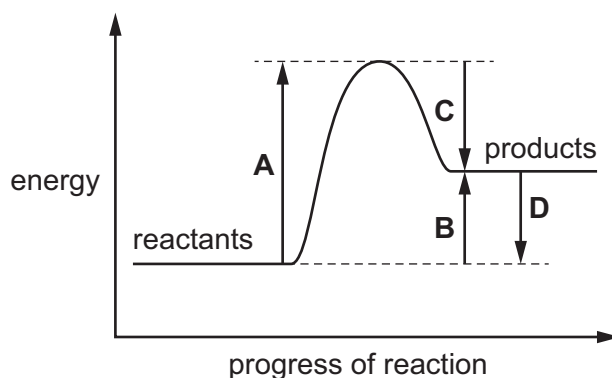
- A** decomposition
- B** displacement
- C** reduction
- D** reversible

15 Water is added to anhydrous copper(II) sulfate.

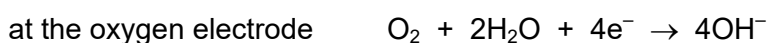
What happens during the reaction?

- A** The copper(II) sulfate turns blue and the solution formed gets colder.
- B** The copper(II) sulfate turns blue and the solution formed gets hotter.
- C** The copper(II) sulfate turns white and the solution formed gets colder.
- D** The copper(II) sulfate turns white and the solution formed gets hotter.

- 16 Which arrow on the energy level diagram shows the overall energy change for an endothermic reaction?



- 17 When a hydrogen–oxygen fuel cell is in operation, a different reaction happens at each electrode.



The electrons that are lost at the hydrogen electrode travel through the external circuit to the oxygen electrode, where they are gained by the oxygen and water.

A hydrogen–oxygen fuel cell is operated for a period of time and four moles of oxygen molecules are consumed.

Which mass of hydrogen is consumed?

- A** 2.0g **B** 4.0g **C** 8.0g **D** 16.0g

- 18 The oxides of two elements, X and Y, are separately dissolved in water and the pH of each solution tested.

oxide tested	pH of solution
X	1
Y	13

Which information about X and Y is correct?

	oxide is acidic	oxide is basic	metal	non-metal
A	X	Y	X	Y
B	X	Y	Y	X
C	Y	X	X	Y
D	Y	X	Y	X

- 19 An acid is neutralised by adding an excess of an insoluble solid base.

A soluble salt is formed.

How is the pure salt obtained from the reaction mixture?

- A crystallisation → evaporation → filtration
- B evaporation → crystallisation → filtration
- C filtration → crystallisation → evaporation
- D filtration → evaporation → crystallisation

- 20 Substance J takes part in a redox reaction.

In the reaction, J gains electrons.

Which statement is correct?

- A J is the oxidising agent and it is oxidised in the reaction.
- B J is the oxidising agent and it is reduced in the reaction.
- C J is the reducing agent and it is oxidised in the reaction.
- D J is the reducing agent and it is reduced in the reaction.

- 21 Elements in Group IV of the Periodic Table are shown.

carbon
silicon
germanium
tin
lead

What does **not** occur in Group IV as it is descended?

- A The proton number of the elements increases.
- B The elements become more metallic.
- C The elements have more electrons in their outer shell.
- D The elements have more electron shells.

22 Which statement about acids is correct?

- A Acids are proton acceptors.
- B Acids transfer electrons to bases in aqueous solution.
- C Hydrochloric acid reacts with ammonium hydroxide to produce ammonia.
- D Ethanoic acid partially ionises in aqueous solution.

23 Which elements have both a high melting point and variable oxidation states?

- A alkali metals
- B transition elements
- C halogens
- D noble gases

24 Lithium, sodium and potassium are elements in Group I of the Periodic Table.

Chlorine, bromine and iodine are elements in Group VII of the Periodic Table.

Which row identifies the **least** dense of these elements in each group?

	Group I	Group VII
A	lithium	chlorine
B	lithium	iodine
C	potassium	chlorine
D	potassium	iodine

25 The reactions of metals P, Q, R and S are shown.

metal	reaction with water	reaction with hydrochloric acid	reduction of the metal oxide with carbon
P	no reaction	no reaction	reduced
Q	slow	vigorous	no reaction
R	vigorous	vigorous	no reaction
S	very slow	vigorous	reduced

What is the order of reactivity of the metals?

	least reactive → most reactive			
A	P	S	Q	R
B	P	Q	S	R
C	R	S	Q	P
D	R	Q	S	P

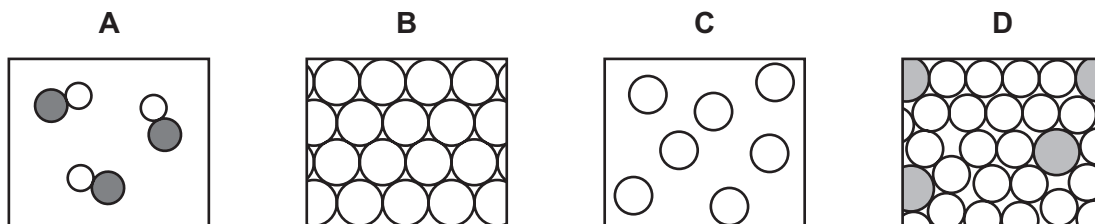
26 The number of protons and the number of neutrons in the atoms of elements X, Y and Z are shown.

	number of protons	number of neutrons
X	6	6
Y	7	6
Z	8	10

Which statement about the elements is correct?

- A** X and Y are isotopes of the same element.
- B** Z forms an ion with a +2 charge.
- C** X and Z react together to form an ionic compound.
- D** X, Y and Z are non-metals.

27 Which diagram represents the arrangement of atoms in an alloy?



28 Three metal compounds, J, K and L, are heated using a Bunsen burner.

The results are shown.

J colourless gas produced, which relights a glowing splint

K colourless gas produced, which turns limewater milky

L no reaction

Which row identifies J, K and L?

	J	K	L
A	magnesium carbonate	potassium carbonate	potassium nitrate
B	magnesium carbonate	potassium nitrate	potassium carbonate
C	potassium nitrate	magnesium carbonate	potassium carbonate
D	potassium nitrate	potassium carbonate	magnesium carbonate

29 Processes involved in the extraction of zinc are listed.

1 Heat zinc oxide with carbon.

2 Condense zinc vapour.

3 Vaporise the zinc.

4 Roast zinc ore in air.

In which order are the processes carried out?

A 1 → 2 → 3 → 4

B 4 → 3 → 1 → 2

C 4 → 1 → 3 → 2

D 1 → 4 → 3 → 2

30 Which process uses sacrificial protection to prevent steel from rusting?

- A** galvanising
- B** oiling
- C** copper plating
- D** painting

31 Fertilisers are used to provide three of the elements needed for plant growth.

Which two compounds would give a fertiliser containing all three of these elements?

- A** $\text{Ca}(\text{NO}_3)_2$ and $(\text{NH}_4)_2\text{SO}_4$
- B** $\text{Ca}(\text{NO}_3)_2$ and $(\text{NH}_4)_3\text{PO}_4$
- C** KNO_3 and $(\text{NH}_4)_2\text{SO}_4$
- D** KNO_3 and $(\text{NH}_4)_3\text{PO}_4$

32 Which processes produce carbon dioxide?

- 1 respiration
- 2 photosynthesis
- 3 fermentation
- 4 combustion of hydrogen

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

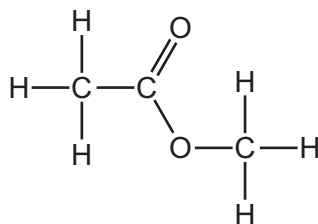
33 Which reaction in the Contact process requires the use of a catalyst?

- A** $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$
- B** $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$
- C** $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$
- D** $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2\text{H}_2\text{SO}_4$

34 What are the products when limestone (calcium carbonate) is heated strongly?

- A** calcium hydroxide and carbon dioxide
- B** calcium hydroxide and carbon monoxide
- C** calcium oxide and carbon dioxide
- D** calcium oxide and carbon monoxide

35 The structure of ester W is shown.



Which row gives the names of ester W and the carboxylic acid and alcohol from which it is made?

	name of ester W	carboxylic acid	alcohol
A	ethyl methanoate	ethanoic acid	methanol
B	ethyl methanoate	methanoic acid	ethanol
C	methyl ethanoate	ethanoic acid	methanol
D	methyl ethanoate	methanoic acid	ethanol

36 Ethene reacts with substance X to form ethanol.

What is X?

- A** ethanoic acid
- B** glucose
- C** hydrogen
- D** steam

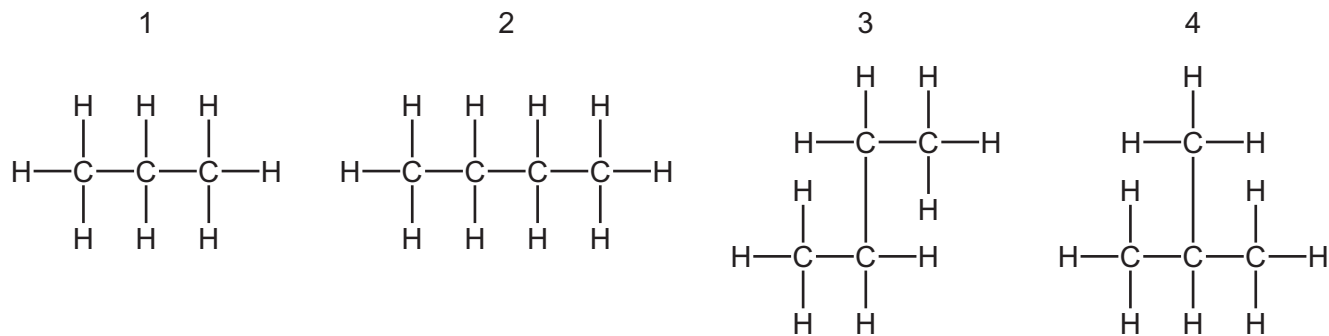
37 Alkenes can be produced by cracking large hydrocarbon molecules to form smaller hydrocarbon molecules.

Which equations represent possible reactions when tetradecane, $C_{14}H_{30}$, is cracked?

- 1 $C_{14}H_{30} \rightarrow C_2H_6 + C_3H_6 + C_4H_8 + C_5H_{10}$
- 2 $C_{14}H_{30} \rightarrow H_2 + C_2H_4 + C_3H_6 + C_4H_8 + C_5H_{10}$
- 3 $C_{14}H_{30} \rightarrow C_2H_6 + 4C_3H_6$
- 4 $C_{14}H_{30} \rightarrow C_2H_6 + C_3H_8 + C_9H_{18}$

- A** 1 only
- B** 1 and 4
- C** 1, 2 and 3
- D** 3 and 4

38 The structures of some hydrocarbons are shown.

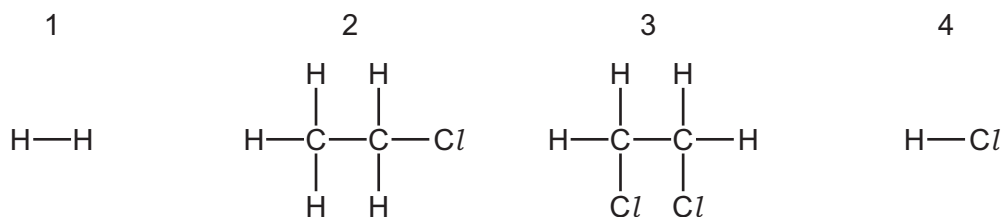


Which statement about the hydrocarbons is correct?

- A** 1 and 2 have a different general formula.
- B** 1 and 4 are in different homologous series.
- C** 2 and 3 are structural isomers.
- D** 3 and 4 have the same empirical formula.

39 Ethane reacts with chlorine in the presence of ultraviolet light.

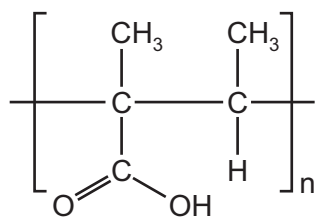
Which substances are produced in the reaction?



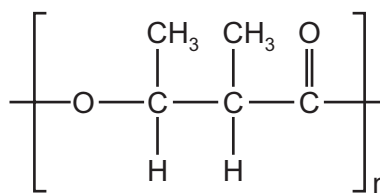
- A** 1, 2 and 3 **B** 1 and 3 only **C** 2, 3 and 4 **D** 2 and 4 only

40 Which polymer structure has the same linkages as *Terylene*?

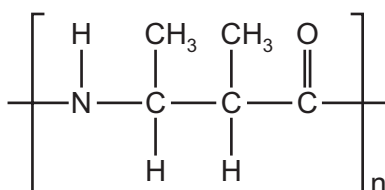
A



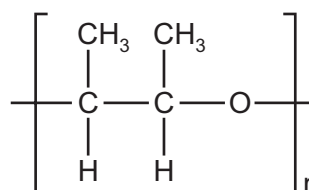
B



C



D



Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

The Periodic Table of Elements

Group																	
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
	11 Na sodium 23																
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65						
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —		114 Fl flerovium —		116 Lv livermorium —		

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™ (9–1)

CHEMISTRY (9–1)

0971/22

Paper 2 Multiple Choice (Core)

May/June 2022

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	A	1
3	D	1
4	C	1
5	C	1
6	A	1
7	A	1
8	D	1
9	A	1
10	D	1
11	B	1
12	B	1
13	B	1
14	D	1
15	B	1
16	B	1
17	D	1
18	B	1
19	D	1
20	B	1
21	C	1
22	D	1
23	B	1
24	A	1
25	A	1
26	D	1
27	D	1
28	C	1

Question	Answer	Marks
29	C	1
30	A	1
31	D	1
32	A	1
33	B	1
34	C	1
35	C	1
36	D	1
37	C	1
38	D	1
39	C	1
40	B	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

October/November 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

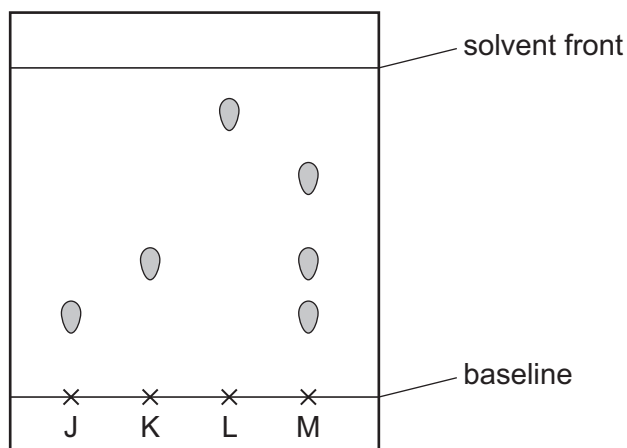
This document has **16** pages.



- 1 The rate of diffusion of three gases, ammonia, carbon dioxide and methane, is measured.

What is the order of the rate of diffusion of the gases from slowest to fastest?

- A $\text{CO}_2 \rightarrow \text{NH}_3 \rightarrow \text{CH}_4$
B $\text{CO}_2 \rightarrow \text{CH}_4 \rightarrow \text{NH}_3$
C $\text{CH}_4 \rightarrow \text{NH}_3 \rightarrow \text{CO}_2$
D $\text{NH}_3 \rightarrow \text{CH}_4 \rightarrow \text{CO}_2$
- 2 Which description of Brownian motion is correct?
- A random movement of particles due to bombardment by larger particles
B random movement of particles due to bombardment by smaller particles
C random movement of particles from a high concentration to a low concentration
D random movement of particles from a low concentration to a high concentration
- 3 The chromatogram obtained using four substances, J, K, L and M, is shown.



Which statement about M is correct?

- A It is a mixture of J and K only.
B It is a pure substance.
C It is a mixture of J, K and L.
D It is a mixture of J, K and an unknown substance.

4 Which statements about isotopes of the same element are correct?

- 1 They are atoms which have the same chemical properties because they have the same number of electrons in their outer shell.
- 2 They are atoms which have the same number of electrons and neutrons but different numbers of protons.
- 3 They are atoms which have the same number of electrons and protons but different numbers of neutrons.

A 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

5 Which statement about solid magnesium oxide is correct?

- A** It is a giant structure made up of magnesium and oxygen atoms bonded covalently.
- B** It is an electrical conductor with mobile magnesium ions and oxygen ions.
- C** Magnesium loses electrons and these electrons move freely through a lattice.
- D** Oxygen ions and magnesium ions are attracted to each other in a giant lattice.

6 Which molecule contains only three shared pairs of electrons?

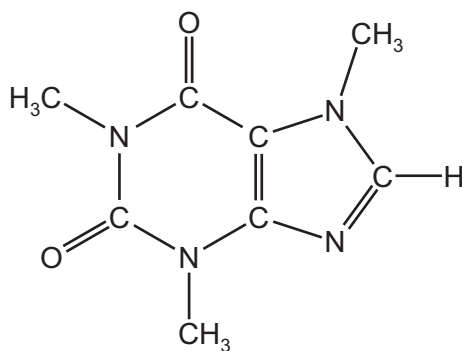
A CH₃OH **B** Cl₂ **C** H₂O **D** N₂

7 Which particles are present in the structure of metals?

- 1 positive ions
- 2 negative ions
- 3 shared pairs of electrons
- 4 mobile electrons

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

- 8 Caffeine is a stimulant found in coffee.

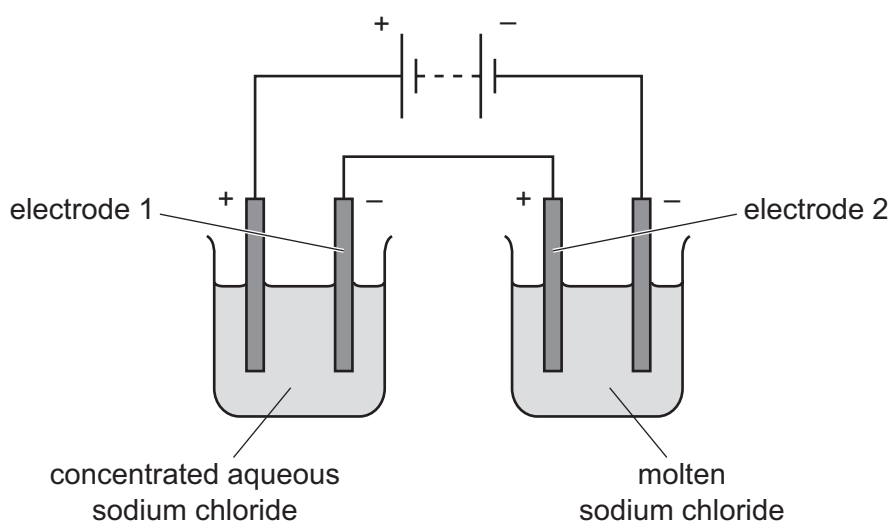


caffeine

Which formula represents caffeine?

- A** $C_7H_{10}N_4O_2$ **B** $C_8H_{10}N_3O_2$ **C** $C_8H_{10}N_4O_2$ **D** $C_8H_{11}N_4O_2$
- 9 Which sample does **not** contain a number of atoms equal to the Avogadro constant?
- A** 14 g of nitrogen, N_2
B 6 g of water, H_2O
C 4 g of helium, He
D 28 g of carbon monoxide, CO

- 10 The electrolysis of concentrated aqueous sodium chloride and molten sodium chloride is shown.



What are the products at electrodes 1 and 2?

	electrode 1	electrode 2
A	chlorine	chlorine
B	hydrogen	chlorine
C	hydrogen	sodium
D	sodium	sodium

- 11 When an acid is added to an alkali, the temperature of the reaction mixture rises.

Which words describe this reaction?

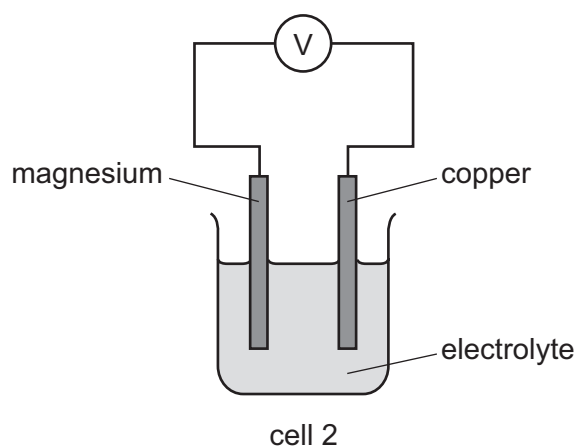
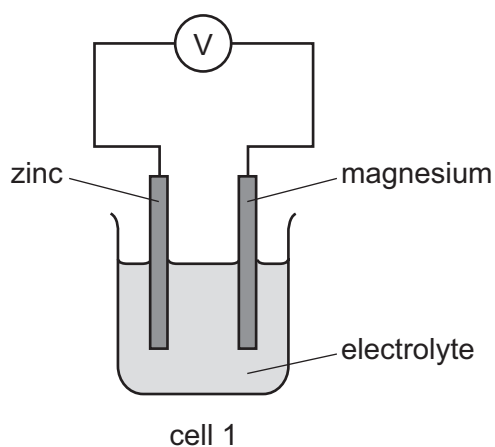
- A** decomposition and endothermic
- B** decomposition and exothermic
- C** neutralisation and endothermic
- D** neutralisation and exothermic

12 Some properties of four fuels are shown.

Which fuel is a gas at room temperature and makes two products when it burns in a plentiful supply of air?

	fuel	formula	melting point / °C	boiling point / °C
A	hydrogen	H ₂	-259	-253
B	methane	CH ₄	-182	-164
C	octane	C ₈ H ₁₈	-57	126
D	wax	C ₃₁ H ₆₄	60	400

13 The electrical energy, or voltage, of two simple cells is measured.



statement 1 The voltage of cell 1 is greater than cell 2.

statement 2 Zinc is more reactive than copper.

statement 3 Magnesium is oxidised in both cells.

statement 4 Magnesium atoms lose electrons to form magnesium ions.

Which option is correct?

- A** All the statements are correct.
- B** Only statements 1 and 3 are correct.
- C** Statement 2 is correct and explains statement 1.
- D** Statement 4 is correct and explains statement 3.

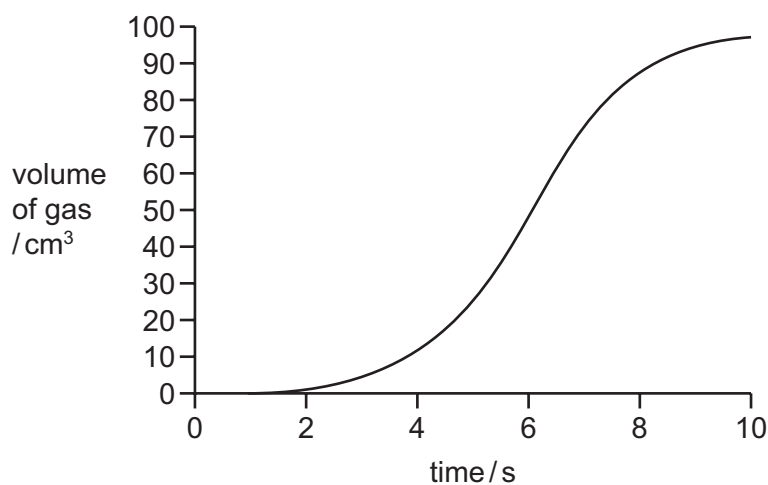
- 14 Dilute aqueous sodium chloride is electrolysed using carbon electrodes.

What is the product at the anode?

- A carbon dioxide
- B hydrogen
- C oxygen
- D sodium

- 15 The volume of gas given off in a chemical reaction is measured over time.

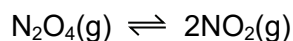
The results are shown.



At which time is the rate of reaction greatest?

- A 0 s
- B 4 s
- C 6 s
- D 10 s

- 16 Dinitrogen tetroxide, N_2O_4 , is converted into nitrogen dioxide, NO_2 , in a reversible reaction.



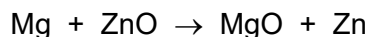
The forward reaction is endothermic.

Which conditions give the highest equilibrium yield of nitrogen dioxide?

	pressure / atmospheres	temperature
A	2	high
B	2	low
C	50	high
D	50	low

17 When magnesium is heated with zinc oxide a reaction occurs.

The equation is shown.



Which substance is oxidised?

- A** magnesium
- B** magnesium oxide
- C** zinc
- D** zinc oxide

18 X and Y are oxides of two different elements.

- X reacts with water to produce aqueous solution Z.
- Z turns universal indicator paper blue.
- An aqueous solution of Y reacts with sodium carbonate to produce carbon dioxide gas.

Which statement is correct?

- A** X and Y are both the oxides of metals.
- B** X and Y are both the oxides of non-metals.
- C** X is the oxide of a metal and Y is the oxide of a non-metal.
- D** X is the oxide of a non-metal and Y is the oxide of a metal.

19 Ethanoic acid reacts with water to produce an acidic solution.

Which row describes the roles of ethanoic acid and water in this reaction?

	ethanoic acid	water
A	accepts a proton	donates a proton
B	accepts an electron	donates an electron
C	donates a proton	accepts a proton
D	donates an electron	accepts an electron

20 Copper(II) sulfate is a soluble salt.

Calcium sulfate is an insoluble salt.

Which row shows suitable reactants for preparing a pure sample of the named salt?

	salt	reactants
A	calcium sulfate	calcium carbonate + dilute sulfuric acid
B	calcium sulfate	aqueous calcium chloride and aqueous sodium sulfate
C	copper(II) sulfate	copper + dilute sulfuric acid
D	copper(II) sulfate	aqueous copper(II) chloride and aqueous sodium sulfate

21 Strontium displaces magnesium from molten magnesium chloride.

Bromine displaces iodine from aqueous potassium iodide.

Which row describes the change in reactivity down both Group II and Group VII of the Periodic Table?

	reactivity down the group	
	Group II	Group VII
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

22 Elements J and K are in the same period in the Periodic Table.

J reacts with acids to produce a salt and hydrogen.

K reacts with sodium to form an ionic compound.

Which statement about J and K is correct?

- A** An atom of J has more electrons than an atom of K.
- B** J and K are both metals.
- C** J and K are both non-metals.
- D** J is to the left of K in the Periodic Table.

26 Silver is below copper in the reactivity series.

Which row describes the reactions of silver?

	reaction with steam	reaction with dilute hydrochloric acid
A	no reaction	no reaction
B	no reaction	reacts to produce hydrogen gas
C	reacts to produce hydrogen gas	no reaction
D	reacts to produce hydrogen gas	reacts to produce hydrogen gas

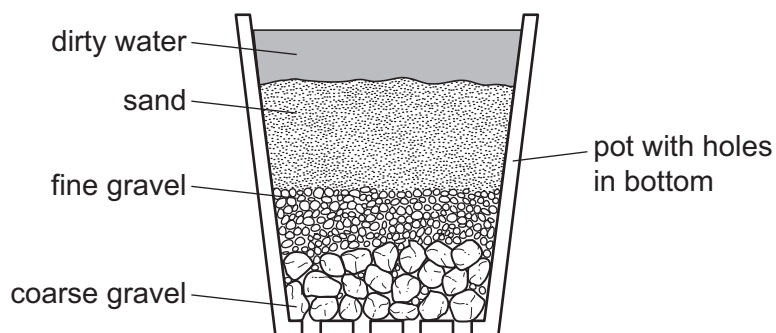
27 Iron is galvanised by coating it in zinc.

Brass is made by mixing copper with zinc.

Which row gives the reasons for each of these uses of zinc?

	reason for galvanising iron	reason for making brass
A	prevents corrosion	produces a softer metal
B	prevents corrosion	produces a harder metal
C	produces a harder metal	produces a softer metal
D	produces a harder metal	produces a harder metal

28 The diagram shows a stage in the purification of dirty water.



Which process does this apparatus show?

- A** chlorination
- B** condensation
- C** distillation
- D** filtration

29 Which substance in polluted air damages stonework and kills trees?

- A carbon dioxide
- B carbon monoxide
- C lead compounds
- D sulfur dioxide

30 Ammonium nitrate, NH_4NO_3 , is a fertiliser and is added to fields to help crops grow.

Slaked lime, $\text{Ca}(\text{OH})_2$, is an alkali and is added to fields to reduce the acidity of the soil.

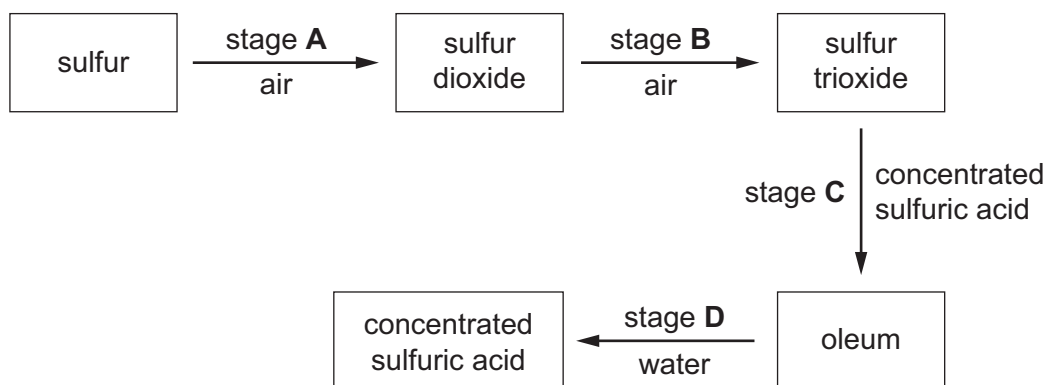
Ammonium nitrate and slaked lime should not be added to a field at the same time because they react with each other to form a gas, Z.

What is Z?

- A ammonia
- B hydrogen
- C nitrogen
- D oxygen

31 The scheme shows four stages in the conversion of sulfur to sulfuric acid.

In which stage is a catalyst used?



32 Which element has an oxide that is used as a food preservative?

- A helium
- B hydrogen
- C iron
- D sulfur

33 Which substance gives off carbon dioxide on heating?

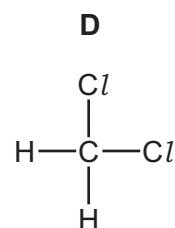
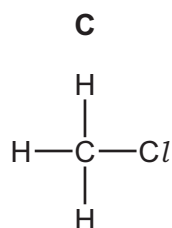
- A lime
- B limestone
- C limewater
- D slaked lime

34 Which compound has the most $\text{—CH}_2\text{—}$ groups in one molecule?

- A butane
- B butanoic acid
- C butan-1-ol
- D but-1-ene

35 Methane reacts with chlorine in the presence of ultraviolet light.

Which substance is **not** produced in this reaction?



36 Ethene reacts with both hydrogen and steam.

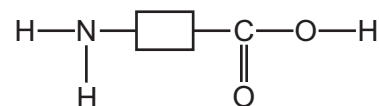
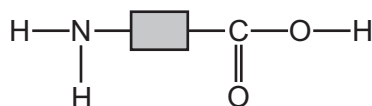
Which row about these reactions is correct?

	reactant with ethene	type of reaction	catalyst used
A	hydrogen	substitution	phosphoric acid
B	hydrogen	addition	nickel
C	steam	substitution	phosphoric acid
D	steam	addition	nickel

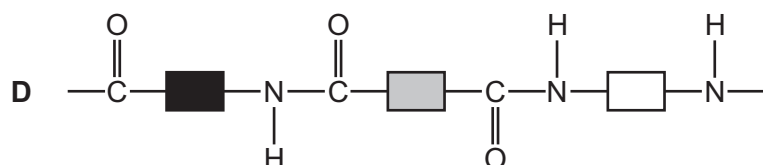
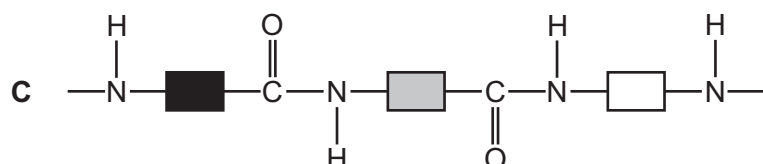
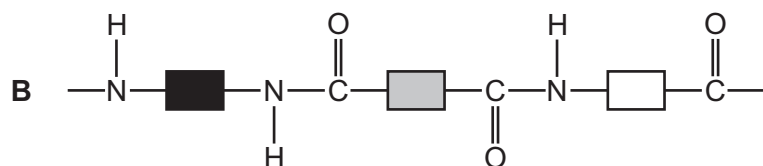
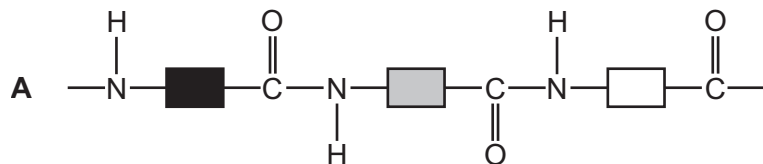
37 Which type of reaction occurs when ethanol is converted to ethanoic acid?

- A combustion
- B decomposition
- C neutralisation
- D oxidation

38 Hydrolysis of polymer P produces the three compounds shown.



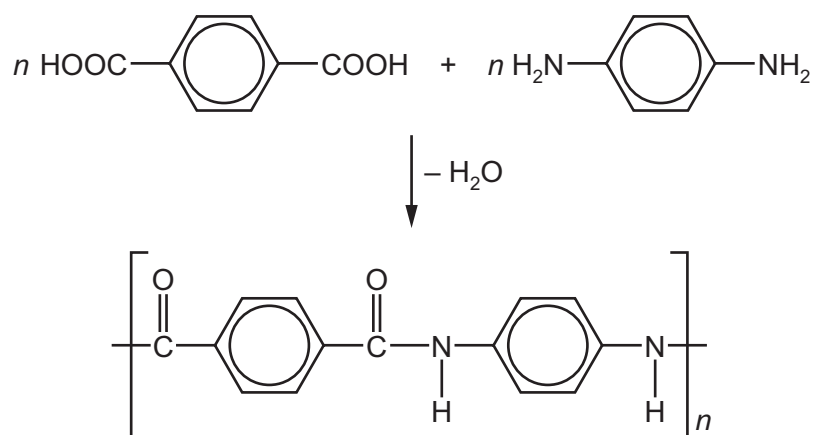
What is the structure of polymer P?



39 Which statement about unsaturated hydrocarbons is correct?

- A** $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3$ is an unsaturated hydrocarbon.
- B** Ethene has more hydrogen atoms per molecule than ethane.
- C** Unsaturated hydrocarbons have double bonds between carbon and hydrogen atoms.
- D** Unsaturated hydrocarbons turn aqueous bromine from colourless to brown.

40 The equation shows the formation of a polymer called *Kevlar*.



Which row describes *Kevlar*?

	how the polymer is formed	type of polymer
A	addition polymerisation	polyamide
B	addition polymerisation	polyester
C	condensation polymerisation	polyamide
D	condensation polymerisation	polyester

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

The Periodic Table of Elements

Group																		
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII	
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20	
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —	119 Nh nihonium —	120 Lr lawrencium —

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™ (9–1)

CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

October/November 2022

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	B	1
3	D	1
4	B	1
5	D	1
6	D	1
7	B	1
8	C	1
9	D	1
10	B	1
11	D	1
12	B	1
13	D	1
14	C	1
15	C	1
16	A	1
17	A	1
18	C	1
19	C	1
20	B	1
21	C	1
22	D	1
23	B	1
24	D	1
25	C	1
26	A	1
27	B	1
28	D	1

Question	Answer	Marks
29	D	1
30	A	1
31	B	1
32	D	1
33	B	1
34	C	1
35	A	1
36	B	1
37	D	1
38	A	1
39	A	1
40	C	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/21

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 The diagram shows the result of dropping a purple crystal into water.



Which processes take place in this experiment?

	chemical reaction	diffusing	dissolving
A	✓	✓	✗
B	✓	✗	✗
C	✗	✗	✓
D	✗	✓	✓

- 2 Which row about elements, mixtures and compounds is correct?

	metallic element	non-metallic element	mixture	compound
A	copper	methane	brass	sulfur
B	brass	sulfur	copper	methane
C	copper	sulfur	brass	methane
D	brass	methane	copper	sulfur

- 3 The atomic structures of four particles, W, X, Y and Z, are shown.

	electrons	neutrons	protons
W	2	2	2
X	2	2	3
Y	2	3	2
Z	3	2	3

Which particles are isotopes of the same element?

- A** W and X **B** W and Y **C** X and Y **D** X and Z

- 4 Which statement explains why isotopes of the same element have the same chemical properties?
- A** They have the same number of outer shell electrons.
- B** They have the same number of neutrons.
- C** They have different numbers of protons.
- D** They have different mass numbers.

- 5 Nitrogen forms a nitride ion with the formula N^{3-} .

Which particle does **not** have the same electronic configuration as the nitride ion?

- A** Al^{3+} **B** Cl^- **C** Na^+ **D** O^{2-}

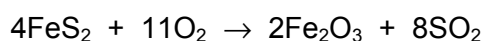
- 6 Which row describes the formation of single covalent bonds in methane?

A	atoms share a pair of electrons	both atoms gain a noble gas electronic structure
B	atoms share a pair of electrons	both atoms have the same number of electrons in their outer shell
C	electrons are transferred from one atom to another	both atoms gain a noble gas electronic structure
D	electrons are transferred from one atom to another	both atoms have the same number of electrons in their outer shell

- 7 Which formula is an empirical formula?

- A** $\text{C}_2\text{H}_4\text{O}$
- B** $\text{C}_4\text{H}_8\text{O}_2$
- C** $\text{C}_3\text{H}_7\text{COOH}$
- D** $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$

- 8 Heating iron sulfide, FeS_2 , in air produces sulfur dioxide.



What is the maximum mass of sulfur dioxide produced from 120 kg of iron sulfide?

- A** 64 kg **B** 128 kg **C** 240 kg **D** 512 kg

9 Which substance produces hydrogen and bromine when electrolysed?

- A concentrated aqueous copper(II) bromide
- B concentrated aqueous sodium bromide
- C dilute aqueous potassium bromide
- D molten lead(II) bromide

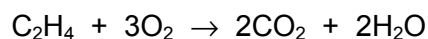
10 Which statements about hydrogen fuel cells are correct?

- 1 Water is formed as the only waste product.
- 2 Both water and carbon dioxide are formed as waste products.
- 3 The overall reaction is $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$.
- 4 The overall reaction is endothermic.

- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

11 Ethene gas, C_2H_4 , is completely burned in excess oxygen to form carbon dioxide and water.

The equation for this exothermic reaction is shown.



The table shows the bond energies involved in the reaction.

bond	bond energy in kJ/mol
C=C	614
C-H	413
O=O	495
C=O	799
O-H	467

What is the total energy change in this reaction?

- A -954 kJ/mol
- B -1010 kJ/mol
- C -1313 kJ/mol
- D -1369 kJ/mol

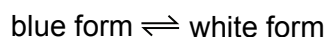
- 12 Which row describes the effect on the activation energy and the frequency of particle collisions when the temperature of a chemical reaction is increased?

	activation energy	frequency of collisions
A	increases	increases
B	no change	increases
C	increases	no change
D	no change	no change

- 13 Solid copper(II) sulfate exists in two different forms, anhydrous and hydrated.

One of these forms is blue and the other is white.

The change between these two forms is reversible.



What is the blue form and how is the change from the blue form to the white form brought about?

	blue form	change to white form
A	anhydrous	add water
B	anhydrous	heat
C	hydrated	add water
D	hydrated	heat

- 14 Sodium ions, Na^+ , and oxygen ions, O^{2-} , combine with chromium ions to form a salt.

The salt sodium dichromate has the formula $\text{Na}_2\text{Cr}_2\text{O}_7$.

What is the oxidation state of chromium in this salt?

- A** +2 **B** +3 **C** +6 **D** +12

- 15 The concentration of hydrogen ions in 100 cm^3 of 0.1 mol/dm^3 hydrochloric acid is higher than the concentration of hydrogen ions in 100 cm^3 of 0.1 mol/dm^3 ethanoic acid.

Which statement explains the difference in hydrogen ion concentration?

- A** Ethanoic acid is an organic acid.
B Ethanoic acid has a lower pH than hydrochloric acid.
C Ethanoic acid is partially dissociated.
D Ethanoic acid is a strong acid.

16 Which oxide is classified as an amphoteric oxide?

- A** aluminium oxide
- B** calcium oxide
- C** copper(II) oxide
- D** nitrogen oxide

17 Which method produces the salt copper(II) carbonate?

- A** Add copper(II) oxide to water, then add excess aqueous sodium carbonate. Filter off the precipitate.
- B** Add copper(II) oxide to dilute sulfuric acid, then add excess aqueous sodium carbonate. Filter off the precipitate.
- C** Add copper to dilute hydrochloric acid, then add aqueous sodium carbonate. Filter off the precipitate.
- D** Add copper(II) oxide to excess aqueous sodium carbonate. Filter off the precipitate.

18 Which statements about the trends across a period of the Periodic Table are correct?

- 1 Aluminium is more metallic than sodium.
- 2 Beryllium is more metallic than carbon.
- 3 Boron is more metallic than lithium.
- 4 Magnesium is more metallic than silicon.

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

19 Some information about elements in Group II of the Periodic Table is shown.

element	time taken to make 10 cm ³ of hydrogen gas when 1 g of metal is added to cold water	density in g/cm ³	melting point/°C
beryllium	no reaction	1.85	1280
magnesium	>300 seconds	1.74	650
calcium	60 seconds	1.54	850
strontium	30 seconds	2.62	768
barium	10 seconds	3.51	714

Which row shows the correct trends in reactivity, density and melting point of the elements going down Group II of the Periodic Table?

	reactivity	density	melting point
A	decreases down group	increases down group	decreases down group
B	decreases down group	decreases down group	no clear trend
C	increases down group	no clear trend	increases down group
D	increases down group	no clear trend	no clear trend

20 A new element oxfordium, Ox, was discovered with the following properties.

solubility	electrical conduction	formula of element	bonding in a molecule of Ox ₂
insoluble in water	does not conduct	Ox ₂	Ox≡Ox

In which group of the Periodic Table should the new element be placed?

- A** Group III
- B** Group V
- C** Group VII
- D** Group VIII

21 Which row describes a similarity and a difference between chlorine and bromine?

	similarity	difference
A	both are gases at room temperature and pressure	chlorine and bromine have different colours
B	both exist as diatomic molecules	chlorine is more dense than bromine
C	both have atoms with seven outer-shell electrons	only bromine will react with aqueous sodium chloride
D	both react with aqueous potassium iodide	chlorine is more reactive than bromine

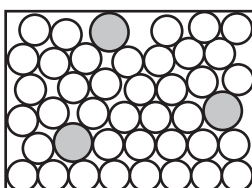
22 Which statement describes transition elements?

- A** They have high densities and high melting points.
- B** They have high densities and low melting points.
- C** They have low densities and high melting points.
- D** They have low densities and low melting points.

23 Which gas is made when powdered zinc is added to dilute hydrochloric acid?

- A** carbon dioxide
- B** chlorine
- C** hydrogen
- D** oxygen

24 The diagram represents the structure of a solid.



Which solids does the diagram represent?

	brass	graphite	sodium chloride
A	✓	✓	x
B	✓	x	x
C	x	✓	✓
D	x	x	✓

25 Steel is an alloy of iron.

Which statement explains why steel is stronger than iron?

- A** Steel contains carbon which is a very hard substance.
- B** The carbon atoms in steel bond together very strongly.
- C** The carbon atoms in steel make the iron atoms bond together very strongly.
- D** The carbon atoms prevent layers of iron atoms from sliding over each other.

26 Three students, X, Y and Z, are told that solid P reacts with dilute acids and also conducts electricity.

The table shows the students' suggestions about the identity of P.

X	Y	Z
copper	iron	graphite

Which students are correct?

- A** X, Y and Z **B** X only **C** Y only **D** Z only

27 Which statement explains why aluminium appears to be unreactive?

- A** It is coated in an oxide layer.
- B** It has a low density.
- C** It is low in the reactivity series.
- D** It is solid at room temperature.

28 During the electrolysis of aluminium oxide, the mass of the carbon anode changes.

Which row describes the change and gives a reason for this change?

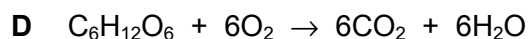
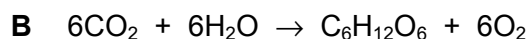
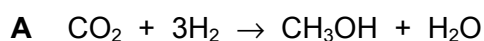
	mass change of the anode	reason
A	decreases	carbon reacts to form carbon dioxide
B	decreases	carbon dissolves in molten cryolite
C	increases	electrodes become coated with cryolite
D	increases	electrodes become coated with aluminium

29 Several processes are used to treat domestic water.

Which row identifies a reason for the given process?

	process	reason
A	chlorination	removes impurities
B	filtration	removes insoluble solids
C	sedimentation	removes soluble solids
D	use of carbon	kills bacteria

30 What is the equation for photosynthesis?



31 Which statement describes how the C–H bonds in methane gas in the atmosphere contribute to global warming?

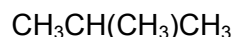
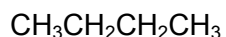
A They absorb thermal energy from the Sun and emit some of this energy into space.

B They absorb thermal energy from the Sun and emit all of this energy towards the Earth.

C They absorb thermal energy from the Earth and emit all of this energy towards the Earth.

D They absorb thermal energy from the Earth and emit some of this energy towards the Earth.

32 The structural formulae of two hydrocarbons are shown.



Which statement about the hydrocarbons is correct?

A They are both alkenes.

B They decolourise aqueous bromine.

C They are structural isomers.

D They undergo addition reactions.

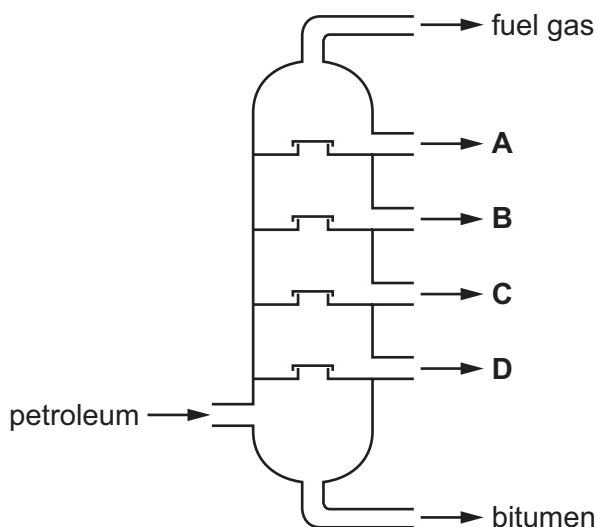
- 33 The structural formula of compound Q is given.



What is compound Q?

- A** butyl butanoate
B butyl propanoate
C propyl butanoate
D propyl propanoate
- 34 The fractional distillation of petroleum is shown.

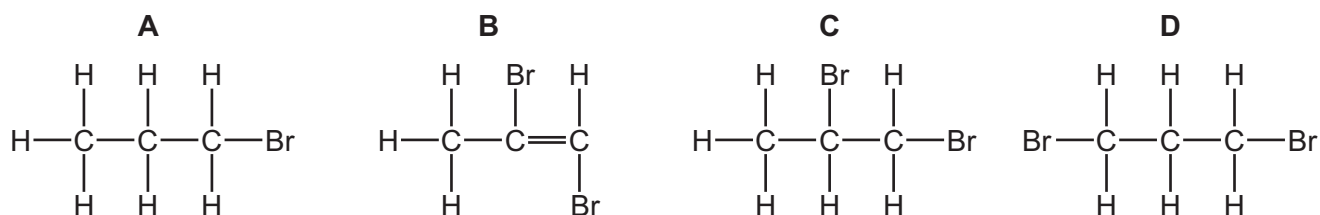
Which fraction contains hydrocarbons with the longest chain length?



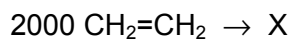
- 35 Which equation represents the cracking of an alkane?

- A** $3\text{C}_2\text{H}_4 \rightarrow \text{C}_6\text{H}_{12}$
B $\text{C}_6\text{H}_{12} + \text{H}_2 \rightarrow \text{C}_6\text{H}_{14}$
C $\text{C}_6\text{H}_{14} \rightarrow 6\text{C} + 7\text{H}_2$
D $\text{C}_6\text{H}_{14} \rightarrow \text{C}_2\text{H}_4 + \text{C}_4\text{H}_{10}$

- 36 What is the structure of the product of the reaction of propene with bromine?



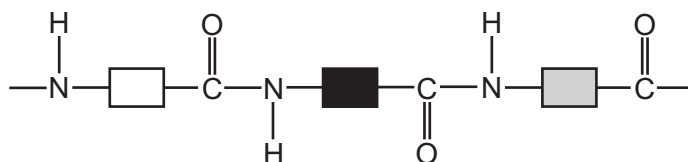
37 In reaction R, 2000 molecules of $\text{CH}_2=\text{CH}_2$ react to form a single molecule X only.



Which terms describe reaction R, $\text{CH}_2=\text{CH}_2$ and X?

	reaction R	$\text{CH}_2=\text{CH}_2$	X
A	addition	monomer	polymer
B	addition	polymer	monomer
C	substitution	monomer	polymer
D	substitution	polymer	monomer

38 Part of the structure of a polymer is shown.



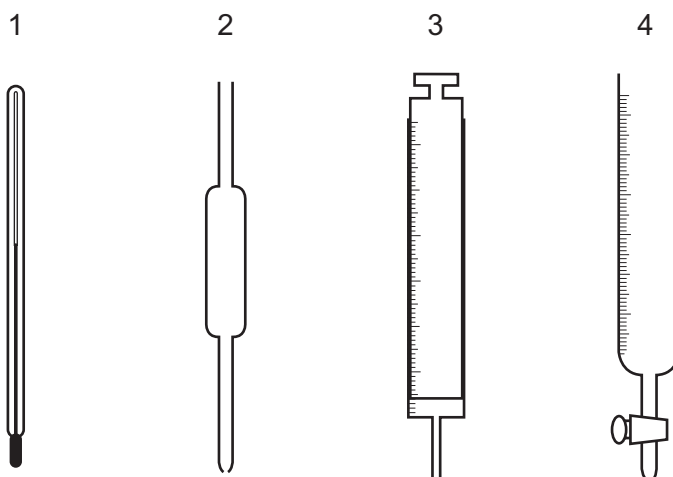
Which statements about the polymer are correct?

- 1 The polymer is nylon.
- 2 The polymer is formed by condensation polymerisation.
- 3 There are ester linkages between the monomers.

A 1 and 2 **B** 2 and 3 **C** 2 only **D** 3 only

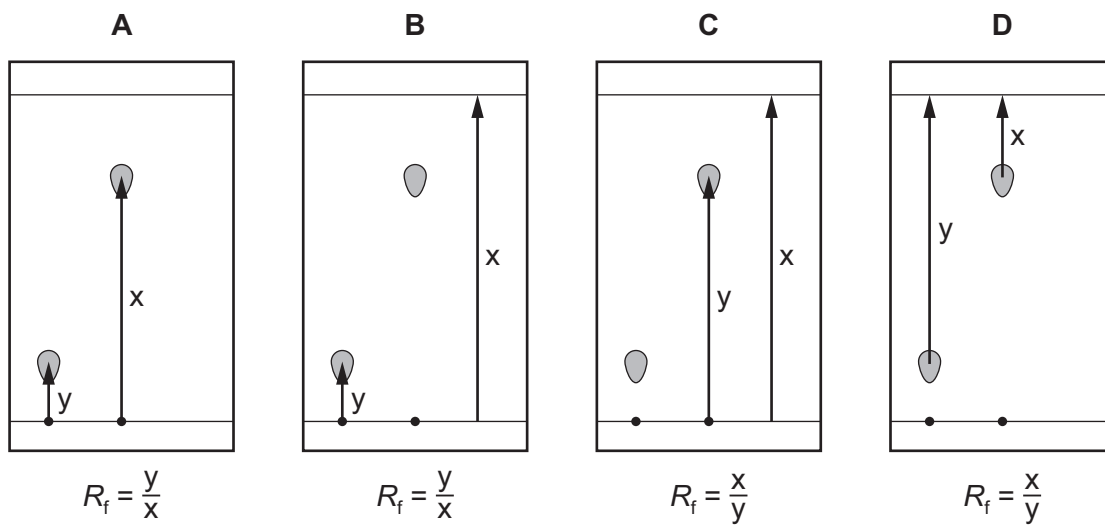
39 The concentration of acids and alkalis can be determined by titration.

Which pieces of equipment are needed to perform a titration?



A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 2 and 4

40 Which chromatogram shows how the R_f value of a substance is calculated?



BLANK PAGE

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

The Periodic Table of Elements

Group																				
I	II											III	IV	V	VI	VII	VIII			
		1 H hydrogen 1																		
		Key atomic number atomic symbol name relative atomic mass																		
3 Li lithium 7	4 Be beryllium 9													5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19		
11 Na sodium 23	12 Mg magnesium 24													13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84			
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131			
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —			
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —			

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™ (9–1)

CHEMISTRY (9–1)

0971/21

Paper 2 Multiple Choice (Extended)

May/June 2023

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

Question	Answer	Marks
1	D	1
2	C	1
3	B	1
4	A	1
5	B	1
6	A	1
7	A	1
8	B	1
9	B	1
10	A	1
11	C	1
12	B	1
13	D	1
14	C	1
15	C	1
16	A	1
17	B	1
18	C	1
19	D	1
20	B	1
21	D	1
22	A	1
23	C	1
24	B	1
25	D	1
26	C	1
27	A	1
28	A	1

Question	Answer	Marks
29	B	1
30	B	1
31	D	1
32	C	1
33	C	1
34	D	1
35	D	1
36	C	1
37	A	1
38	C	1
39	D	1
40	B	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/22

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages.



1 Four physical changes of ethanol are listed.

- 1 condensation
- 2 evaporation
- 3 freezing
- 4 boiling

In which changes do the particles move further apart?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

2 An atom of element X contains:

- 5 protons
- 6 neutrons
- 5 electrons.

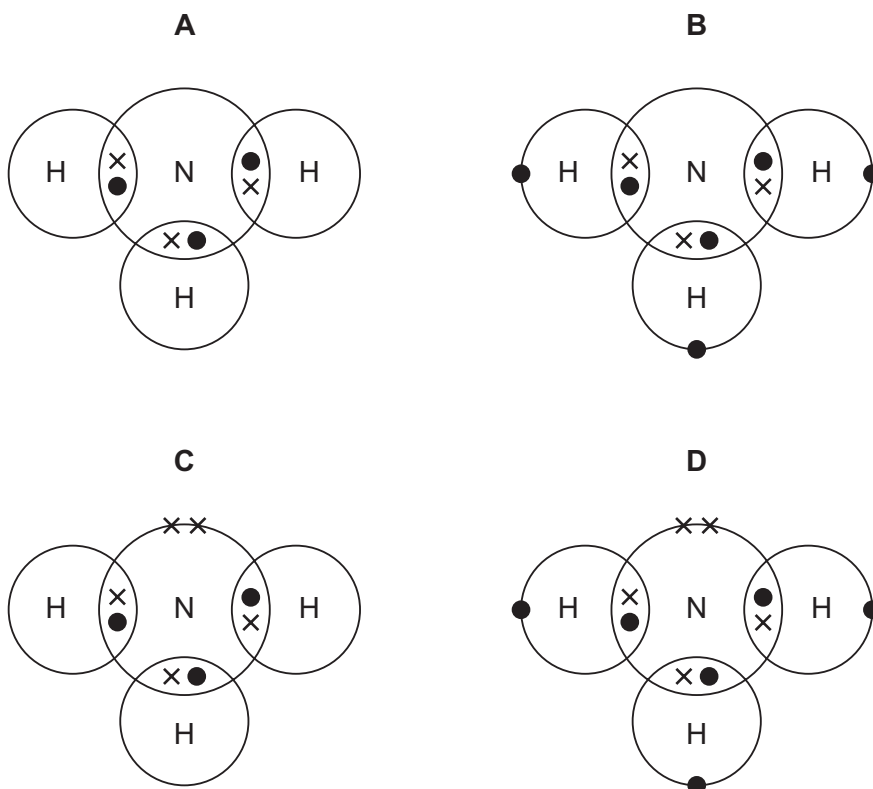
Which statements about element X are correct?

- 1 X has an atomic number of 6.
- 2 X has a nucleon number of 11.
- 3 X is in Group II of the Periodic Table.
- 4 X is in the second period of the Periodic Table.

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

- 3 Ammonia, NH_3 , is a covalent molecule.

Which diagram shows the outer-shell electron arrangement in a molecule of ammonia?



- 4 Which structure does silicon(IV) oxide most closely resemble?

- A** carbon dioxide
B diamond
C graphite
D sodium chloride

- 5 Substance P conducts electricity when solid.

Which particles move in solid P so that it can conduct electricity?

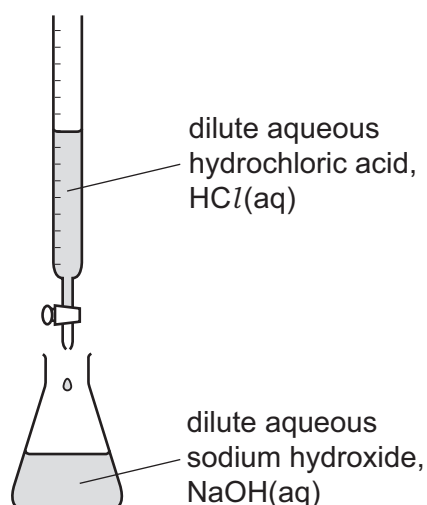
- 1 anions
 2 cations
 3 electrons

- A** 1 and 2 **B** 1 only **C** 2 and 3 **D** 3 only

- 6 Which equation represents a chemical change?
- A $\text{BaCl}_2(\text{s}) \rightarrow \text{BaCl}_2(\text{l})$
 - B $\text{Ca}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{CaSO}_4(\text{s})$
 - C $\text{KCl}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{K}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
 - D $\text{Na}^+(\text{aq}) + \text{NO}_3^-(\text{aq}) \rightarrow \text{NaNO}_3(\text{aq})$
- 7 Which sample contains the largest number of molecules?
- A 16 g of methane, $\text{CH}_4(\text{g})$
 - B 16 g of oxygen, $\text{O}_2(\text{g})$
 - C 16 g of phosphorus, $\text{P}_4(\text{s})$
 - D 16 dm^3 of methane at r.t.p., $\text{CH}_4(\text{g})$

- 8 The concentration of a sample of dilute aqueous sodium hydroxide is found by titration.

The apparatus used is shown.



Which information is needed to calculate the concentration of the dilute aqueous sodium hydroxide in mol/dm^3 ?

	concentration of HCl	volume of HCl used	molar mass of HCl	volume of NaOH used	molar mass of NaOH
A	✓	✓	✓	✓	✓
B	✓	✓	x	✓	x
C	x	✓	✓	✓	x
D	✓	x	x	x	✓

key

✓ = needed

x = not needed

- 9 In experiment 1, aqueous copper(II) sulfate is electrolysed using graphite electrodes.

In experiment 2, aqueous copper(II) sulfate is electrolysed using copper electrodes.

Which statement identifies a half-equation for a reaction at one of the electrodes?

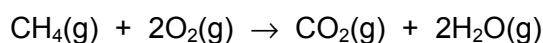
- A** In experiment 1, the half-equation for the anode reaction is $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$.
- B** In experiment 1, the half-equation for the cathode reaction is $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$.
- C** In experiment 2, the half-equation for the anode reaction is $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$.
- D** In experiment 2, the half-equation for the cathode reaction is $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$.

10 Which substance is **not** produced during the electrolysis of concentrated aqueous sodium chloride?

- A chlorine
- B hydrogen
- C sodium
- D sodium hydroxide

11 Methane burns in excess oxygen.

The equation is shown.



Bond energies are shown.

bond	bond energy in kJ/mol
C=O	805
C–H	410
O=O	496
O–H	460

What is the energy change for the reaction?

- A $(4 \times 410 + 2 \times 496) - (2 \times 805 + 4 \times 460)$
- B $(2 \times 805 + 4 \times 460) - (4 \times 410 + 2 \times 496)$
- C $(410 + 2 \times 496) - (805 + 2 \times 460)$
- D $(410 + 496) - (805 + 460)$

12 Which change increases the rate of reaction by decreasing the activation energy, E_a ?

- A addition of a catalyst
- B decrease in size of solid reactants
- C increase in concentration of solutions
- D increase in temperature

- 13** In the Contact process, sulfur dioxide is reacted with oxygen to form sulfur trioxide.

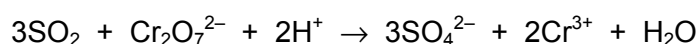
Which conditions are used in this reaction?

	temperature / °C	pressure / kPa	catalyst
A	300	200	iron
B	300	20 000	vanadium(V) oxide
C	450	200	vanadium(V) oxide
D	450	20 000	iron

- 14** Which reaction is reversible?

- A** an iron nail rusting when left in moist air
- B** limestone reacting with an acid to form carbon dioxide gas
- C** magnesium burning in air to produce a white ash
- D** white anhydrous copper(II) sulfate turning blue when water is added

- 15** The equation for the reaction of sulfur dioxide with acidified potassium dichromate(VI) is shown.



What is oxidised and what is the oxidising agent?

	oxidised	oxidising agent
A	SO_2	$\text{Cr}_2\text{O}_7^{2-}$
B	SO_2	H^+
C	$\text{Cr}_2\text{O}_7^{2-}$	H^+
D	$\text{Cr}_2\text{O}_7^{2-}$	$\text{Cr}_2\text{O}_7^{2-}$

- 16** What is the definition of a strong acid?

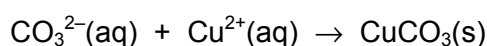
- A** a proton acceptor that is completely dissociated in aqueous solution
- B** a proton acceptor that is partially dissociated in aqueous solution
- C** a proton donor that is completely dissociated in aqueous solution
- D** a proton donor that is partially dissociated in aqueous solution

17 Which statement about amphoteric oxides is correct?

- A They are made by combining an acidic oxide with a basic oxide.
- B They react with water to give a solution of pH 7.
- C They react with both acids and bases.
- D They do not react with acids or bases.

18 Copper(II) carbonate is formed when aqueous sodium carbonate is added to aqueous copper(II) nitrate.

The ionic equation for the reaction is shown.



How is pure copper(II) carbonate obtained from the reaction mixture?

- A evaporate → filter → dry
- B evaporate → wash → crystallise
- C filter → evaporate → crystallise
- D filter → wash → dry

19 Q and R are elements in the same period of the Periodic Table.

Q has 7 electrons in its outer shell and R has 2 electrons in its outer shell.

Which statement about Q and R is correct?

- A Q is a metal and R is a non-metal.
- B Q and R have different numbers of electron shells.
- C R is found to the right of Q in the Periodic Table.
- D The proton number of R is less than the proton number of Q.

20 Lead(II) sulfate is an insoluble salt.

Which reaction produces a mixture from which lead(II) sulfate is obtained by filtration?

- A adding solid lead(II) carbonate to dilute sulfuric acid
- B adding solid lead(II) hydroxide to dilute sulfuric acid
- C adding metallic lead to dilute sulfuric acid
- D adding aqueous lead(II) nitrate to dilute sulfuric acid

21 Which statement about alkali metals is correct?

- A Lithium is more dense than sodium.
- B Sodium is more reactive than potassium.
- C Sodium has a higher melting point than potassium.
- D They are in Group II of the Periodic Table.

22 Which row describes the properties of a transition element?

	melting point	density	forms coloured compounds
A	high	low	no
B	high	high	yes
C	low	low	no
D	low	low	yes

23 Which row identifies the properties of zinc?

	thermal conductivity	reacts with dilute acid
A	good	yes
B	good	no
C	poor	yes
D	poor	no

24 Uses of metals depend on their properties.

Which property is necessary for the use given?

	use of the metal	property of the metal
A	car bodies	ductile
B	cutlery	conducts heat
C	food containers	resists corrosion
D	overhead electrical cables	high density

25 Which compounds **both** contribute to acid rain?

- A** carbon monoxide and carbon dioxide
- B** carbon monoxide and oxides of nitrogen
- C** oxides of nitrogen and sulfur dioxide
- D** sulfur dioxide and carbon dioxide

26 P, Q, R and S are metals.

P reacts with dilute hydrochloric acid, forming hydrogen.

Q reacts violently with water.

R reacts with water to give hydrogen.

S is formed by heating its oxide with carbon.

Which row identifies the metals?

	P	Q	R	S
A	copper	sodium	potassium	iron
B	zinc	magnesium	calcium	iron
C	zinc	sodium	calcium	magnesium
D	iron	potassium	sodium	zinc

27 Which compound is formed when iron rusts?

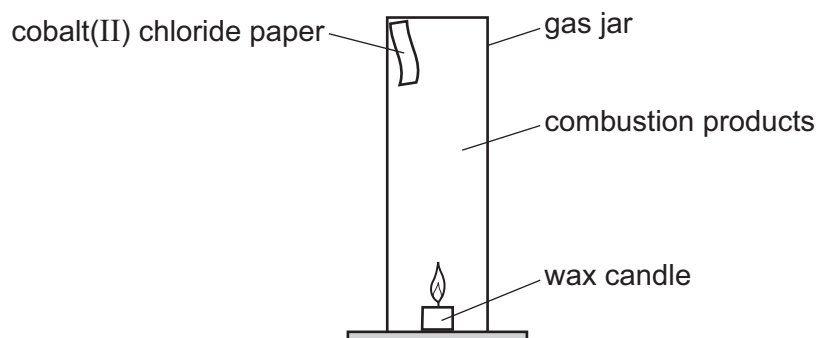
- A** anhydrous iron(II) oxide
- B** anhydrous iron(III) oxide
- C** hydrated iron(III) hydroxide
- D** hydrated iron(III) oxide

28 Why is cryolite used in the extraction of aluminium by electrolysis?

- A** It dissolves the aluminium oxide.
- B** It protects the anodes from corrosion.
- C** It changes bauxite to aluminium oxide.
- D** It decreases the melting point of the aluminium.

- 29 A wax candle is made from a mixture of hydrocarbons.

The candle is lit and placed in a gas jar along with a strip of cobalt(II) chloride test paper as shown.



After a short time, the oxygen in the jar is used up and the candle flame goes out.

Which substance does the cobalt(II) chloride paper identify?

- A carbon dioxide
 - B carbon monoxide
 - C sulfur dioxide
 - D water
- 30 The hydrocarbon C_4H_8 has two structural isomers, but-1-ene and but-2-ene.
- Which statement is correct?
- A But-2-ene has the structural formula $CH_3CH=CHCH_3$ and the same general formula as butane.
 - B But-2-ene has the structural formula $CH_3CH=CHCH_3$ and the same empirical formula as ethene.
 - C But-1-ene has the structural formula $CH_3CH_2CH=CH_2$ and the same general formula as butane.
 - D But-1-ene has the structural formula $CH_3CHCH_2=CH$ and the same empirical formula as ethene.
- 31 Which compound rapidly decolourises aqueous bromine?
- A propane
 - B propanoic acid
 - C propanol
 - D propene

32 What are the products of the addition reactions of ethene with bromine and hydrogen?

	bromine	hydrogen
A	$\text{CH}_2\text{BrCH}_2\text{Br}$	CH_3CH_3
B	$\text{CH}_2\text{BrCH}_2\text{Br}$	CH_2CH_2
C	$\text{CH}_3\text{CH}_2\text{Br}$	CH_3CH_3
D	$\text{CH}_3\text{CH}_2\text{Br}$	CH_2CH_2

33 Ethanol is manufactured by fermentation and the catalytic addition of steam to ethene.

Which row describes an advantage of both methods?

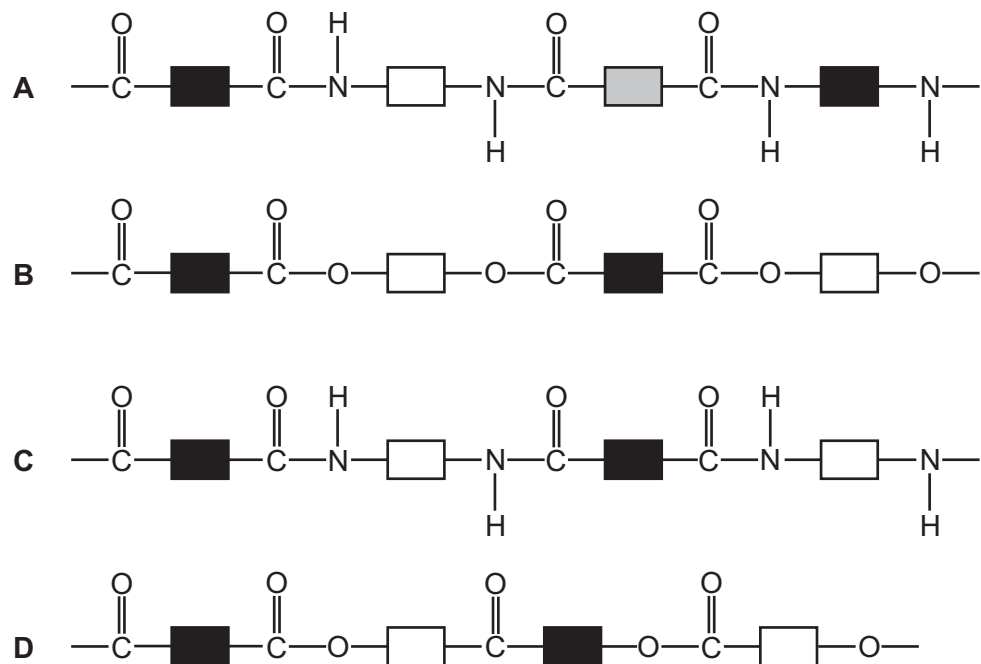
	from sugar by fermentation	from ethene and steam
A	ethanol needs to be purified	the process is continuous
B	it is a batch process	ethene comes from petroleum
C	the process is slow	the process is rapid
D	renewable resources are used	the ethanol produced is pure

34 Methanoic acid and propan-1-ol react to form an ester.

What is the structural formula of the ester?

- A** $\text{HCOOCH}_2\text{CH}_2\text{CH}_3$
- B** $\text{CH}_3\text{CH}_2\text{COOCH}_3$
- C** $\text{CH}_3\text{COOCH}_2\text{CH}_3$
- D** $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$

35 What is the correct structure of PET?



36 Alkanes undergo substitution reactions in the presence of UV light.

Which equation represents a substitution reaction of ethane?

- A $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_4 + 2\text{HCl}$
- B $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{HCl}$
- C $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_4\text{Cl}_2 + \text{H}_2$
- D $\text{C}_2\text{H}_6 + \text{HCl} \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{H}_2$

37 Methane reacts with chlorine in substitution reactions.

How many different products, containing a single carbon atom, can be made during the reactions?

- A 2 B 3 C 4 D 5

38 Rock salt is a mixture of salt and sand.

The method used to separate the sand from the salt is listed.

step 1 Crush the rock salt, add to warm water and stir.

step 2 Pour the mixture through a filter paper held in a funnel.

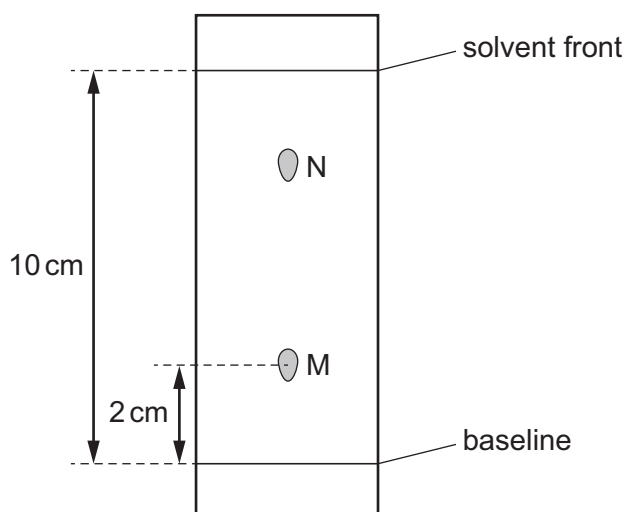
step 3 Evaporate the water to crystallise the salt.

Which statement about the method is correct?

- A The filtrate in step 2 is pure water.
- B The residue in step 2 is pure crystals of salt.
- C The solute is salt.
- D The solvent is a mixture of salt and water.

39 Two compounds, M and N, are dissolved in water and separated by chromatography.

The results are shown.



What is the R_f value of M and which compound is most soluble in water?

	R_f value of M	most soluble compound
A	0.2	M
B	0.2	N
C	5.0	M
D	5.0	N

40 When acid is added to salt X, a gas is produced which turns limewater milky.

When sodium hydroxide is added to salt X, a gas is produced which turns litmus paper blue.

What is X?

A CaCO_3

B $(\text{NH}_4)_2\text{CO}_3$

C NH_4NO_3

D ZnCO_3

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
		1 H hydrogen 1																	
		Key atomic number atomic symbol name relative atomic mass																	
3 Li lithium 7	4 Be beryllium 9													5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24													13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —		

lanthanoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

actinoids

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™ (9–1)

CHEMISTRY (9–1)

0971/22

Paper 2 Multiple Choice (Extended)

May/June 2023

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

Question	Answer	Marks
1	C	1
2	D	1
3	C	1
4	B	1
5	D	1
6	B	1
7	A	1
8	B	1
9	A	1
10	C	1
11	A	1
12	A	1
13	C	1
14	D	1
15	A	1
16	C	1
17	C	1
18	D	1
19	D	1
20	D	1
21	C	1
22	B	1
23	A	1
24	C	1
25	C	1
26	D	1
27	D	1
28	A	1

Question	Answer	Marks
29	D	1
30	B	1
31	D	1
32	A	1
33	D	1
34	A	1
35	B	1
36	B	1
37	C	1
38	C	1
39	B	1
40	B	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

October/November 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

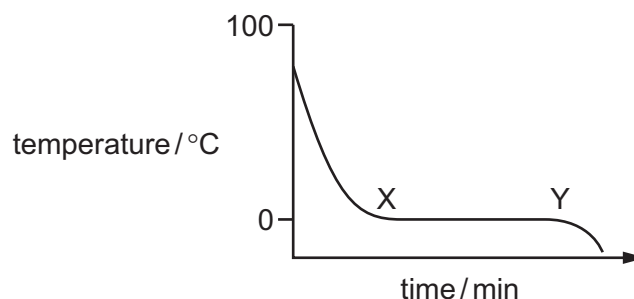
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages.

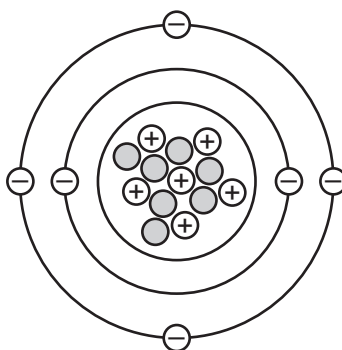


- 1 Part of a cooling curve for water is shown.



What is occurring between points X and Y?

- A** Steam is condensing into water.
 - B** The temperature of the water is decreasing.
 - C** Ice is melting.
 - D** Particles are losing heat to the surroundings.
- 2 Which statements about clean, dry air are correct?
- 1 It is a mixture of elements only.
 - 2 It is a mixture of elements and compounds.
 - 3 It contains only non-metals.
- A** 1 and 3 **B** 1 only **C** 2 and 3 **D** 2 only
- 3 A representation of an atom is shown.



What is the nucleon number of this atom?

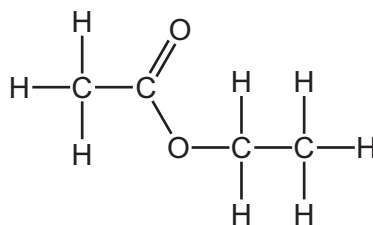
- A** 6 **B** 7 **C** 12 **D** 13

- 4 The percentage abundances of three isotopes in a sample of neon are shown.

isotope	percentage abundance / %
$^{20}_{10}\text{Ne}$	90.48
$^{21}_{10}\text{Ne}$	0.27
$^{22}_{10}\text{Ne}$	9.25

What is the relative atomic mass, A_r , of this sample of neon?

- A** 10.19 **B** 20.19 **C** 21.00 **D** 30.19
- 5 Potassium reacts with iodine to form potassium iodide.
- Which statement about potassium iodide is correct?
- A** Each potassium atom shares a pair of electrons with an iodine atom.
- B** In potassium iodide, the particles of potassium have more protons than electrons.
- C** Potassium iodide has a high melting point because it is a covalent compound.
- D** Potassium iodide has a low melting point because it is an ionic compound.
- 6 Which substance has the lowest melting point?
- A** graphite
- B** methanol
- C** silicon(IV) oxide
- D** sodium chloride
- 7 The diagram shows the structure of a molecule of ethyl ethanoate.



What is the molecular formula of a molecule of ethyl ethanoate?

- A** CHO **B** $\text{C}_4\text{H}_8\text{O}_2$ **C** $\text{C}_4(\text{H}_2)_2(\text{O}_2)$ **D** $\text{C}_2\text{H}_4\text{O}$

- 8 A hydrocarbon contains 85.7% of carbon by mass.

What is the empirical formula of the hydrocarbon?

- A** CH₂ **B** CH₄ **C** C₂H₅ **D** C₃H₆

- 9 The formula of a compound containing element X is Na₂X₂O₃.

The relative formula mass of the compound is 158.

What is the relative atomic mass of X?

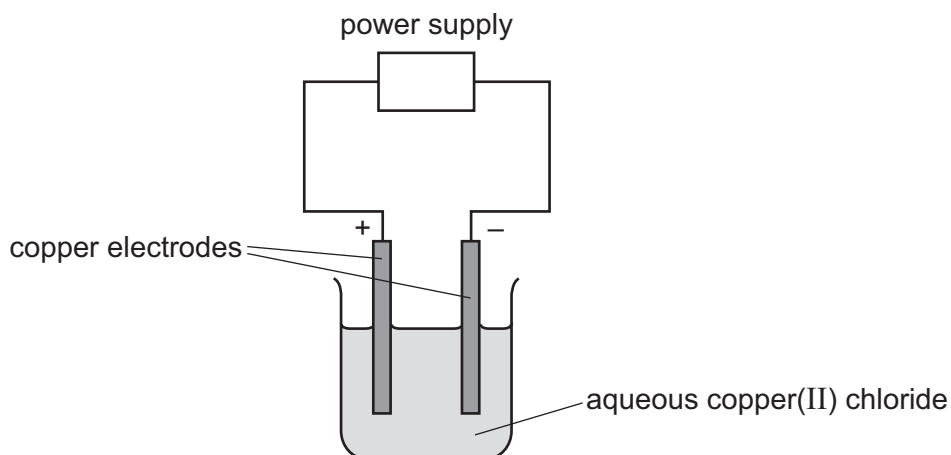
- A** 32 **B** 59.5 **C** 64 **D** 119

- 10 Dilute aqueous potassium chloride is electrolysed using platinum electrodes.

Which row identifies the product at each electrode?

	anode	cathode
A	chlorine	hydrogen
B	chlorine	potassium
C	oxygen	hydrogen
D	oxygen	potassium

- 11 Concentrated aqueous copper(II) chloride is electrolysed using copper electrodes, as shown.



What happens to the mass of each electrode during this process?

	positive electrode	negative electrode
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 12 The initial and final temperatures of four different reactions are measured.

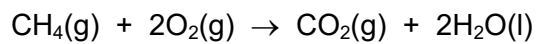
Which reaction is the **least** exothermic?

	initial temperature / °C	final temperature / °C
A	19	25
B	21	18
C	22	17
D	22	26

- 13 Which equation represents an endothermic reaction?

- A** $\text{Cl}_2(\text{g}) \rightarrow 2\text{Cl}(\text{g})$
B $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$
C $\text{H}(\text{g}) + \text{H}(\text{g}) \rightarrow \text{H}_2(\text{g})$
D $2\text{K}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{KOH}(\text{aq}) + \text{H}_2(\text{g})$

14 Methane burns in oxygen to form carbon dioxide and water.



The bond energies are shown.

bond	bond energy in kJ/mol
C–H	410
C–O	360
C=O	805
O–H	460
O–O	146
O=O	496

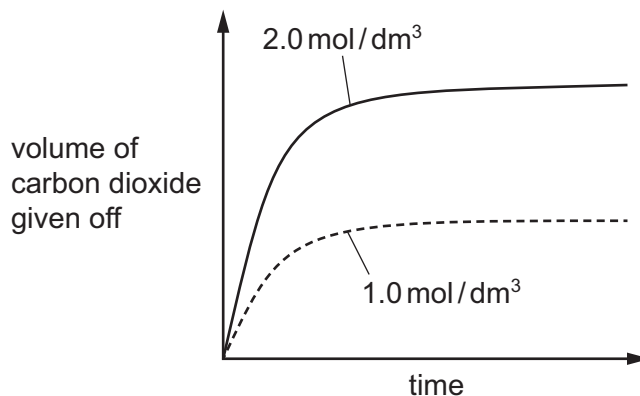
What is the energy change for this reaction?

A –818 kJ/mol **B** –102 kJ/mol **C** +102 kJ/mol **D** +818 kJ/mol

- 15** Hydrochloric acid is added to excess calcium carbonate in two separate experiments.

Two different concentrations of hydrochloric acid are used but the temperature is the same in both experiments.

The graph of the results shows the volume of carbon dioxide gas given off over time.

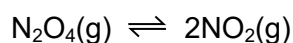


Which row is correct?

	particles in 2.0 mol/dm ³ compared to 1.0 mol/dm ³	
	collision rate	collision energy
A	higher	no change
B	higher	higher
C	lower	no change
D	lower	higher

- 16** The decomposition of dinitrogen tetroxide, N₂O₄, into nitrogen dioxide, NO₂, is a reversible reaction.

The equation for the reaction is shown.

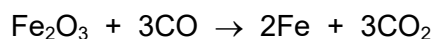


The forward reaction is endothermic.

Which row shows the effect on the position of equilibrium and the rate of the reverse reaction when the temperature is increased?

	position of equilibrium	rate of the reverse reaction
A	shifts to the left	decreases
B	shifts to the left	increases
C	shifts to the right	decreases
D	shifts to the right	increases

- 17 In a blast furnace, iron(III) oxide is converted to iron and carbon monoxide is converted to carbon dioxide.



What happens to each of these reactants?

- A Both iron(III) oxide and carbon monoxide are oxidised.
- B Both iron(III) oxide and carbon monoxide are reduced.
- C Iron(III) oxide is oxidised and carbon monoxide is reduced.
- D Iron(III) oxide is reduced and carbon monoxide is oxidised.

- 18 Which row describes what happens to Fe^{2+} ions when they are oxidised?

	electron movement	oxidation number of iron
A	they gain electrons	decreases
B	they gain electrons	increases
C	they lose electrons	decreases
D	they lose electrons	increases

- 19 In which reaction does an acid react with a base?

- A Dilute sulfuric acid is added to a piece of magnesium ribbon producing hydrogen.
- B Dilute sulfuric acid is added to aqueous barium chloride producing a white precipitate of barium sulfate.
- C Aqueous sodium hydroxide is added to aqueous copper(II) sulfate producing a blue precipitate of copper(II) hydroxide.
- D Aqueous sodium hydroxide is added to solid ammonium sulfate producing gaseous ammonia.

- 20 Which element forms an oxide that reacts with an aqueous solution of a base?

- A argon
- B sulfur
- C magnesium
- D copper

21 Which method is used to produce insoluble salts?

- A addition of excess insoluble base to an acid
- B addition of excess metal to an acid
- C precipitation using two aqueous solutions
- D titration using an acid and an alkali

22 The noble gases are in Group VIII of the Periodic Table.

Some properties of the first four noble gases are shown.

noble gas	boiling point in °C	density in g/dm ³
helium	−267	0.179
neon	−246	0.900
argon	−186	1.782
krypton	−152	3.708

Which row identifies the trends in boiling point and in density as Group VIII is descended?

	boiling point	density
A	decreasing	increasing
B	increasing	increasing
C	decreasing	decreasing
D	increasing	decreasing

23 Some properties of element R are shown.

melting point in °C	98
boiling point in °C	883
reaction with cold water	gives off H ₂ gas
reaction when heated with oxygen	burns to give a white solid

In which part of the Periodic Table is R found?

- A Group I
- B Group VII
- C Group VIII
- D transition elements

24 Which pair of compounds shows that transition elements have variable oxidation states?

- A** Cr_2O_3 and CrBr_3
- B** CuSO_4 and CuCl_2
- C** Fe_2O_3 and FeCl_2
- D** NiO and NiCl_2

25 The list gives the order of some metals and hydrogen in the reactivity series.

Metal X is also included.

most reactive	K
	Mg
	Zn
	H
	X
least reactive	Cu

Which row shows the properties of metal X?

	reacts with dilute acids	oxide reduced by carbon
A	no	no
B	no	yes
C	yes	no
D	yes	yes

26 When zinc is added to an aqueous solution containing magnesium ions, there is no reaction.

Which species has the greatest tendency to lose electrons?

- A** Mg
- B** Mg^{2+}
- C** Zn
- D** Zn^{2+}

27 Which gas in the air is needed for iron to rust?

- A** argon
- B** carbon dioxide
- C** nitrogen
- D** oxygen

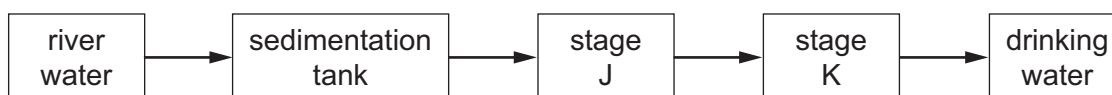
28 Which coating prevents iron from rusting even when the coating is damaged?

- A grease
- B paint
- C plastic
- D zinc

29 Why is limestone added to the blast furnace?

- A It neutralises the molten slag produced.
- B It reacts with impurities to form slag.
- C It releases carbon dioxide which reduces the iron(III) oxide.
- D It removes acidic gases such as carbon dioxide.

30 The flow chart shows stages in the treatment of river water to produce drinking water.



What occurs at stages J and K?

	J	K
A	distillation	chlorination
B	distillation	filtration
C	filtration	chlorination
D	filtration	distillation

- 31** Carbon dioxide acts as a greenhouse gas by interacting with a particular type of energy that radiates from the Earth's surface into the atmosphere.

Which type of energy is involved and what happens when this energy interacts with carbon dioxide molecules?

	type of energy involved	what happens
A	thermal	carbon dioxide molecules increase the Earth's energy loss to space
B	thermal	carbon dioxide molecules absorb the energy
C	light	carbon dioxide molecules increase the Earth's energy loss to space
D	light	carbon dioxide molecules absorb the energy

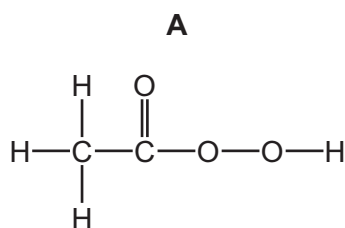
- 32** Oxides of nitrogen, such as NO and NO₂, are formed in the petrol engines of cars.

They are removed from the exhaust gases by reactions in the car's catalytic converter.

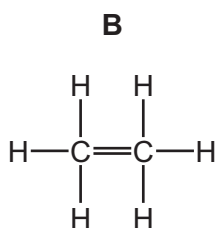
Which row describes how oxides of nitrogen are formed in a petrol engine and a reaction that happens in the catalytic converter?

	how oxides of nitrogen are formed	a reaction that happens in the catalytic converter
A	by the reaction between nitrogen and oxygen from the air	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$
B	by the reaction between nitrogen and oxygen from the air	$2\text{NO} + 2\text{H}_2 \rightarrow \text{N}_2 + 2\text{H}_2\text{O}$
C	by the reaction between nitrogen compounds in petrol and oxygen from the air	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$
D	by the reaction between nitrogen compounds in petrol and oxygen from the air	$2\text{NO} + 2\text{H}_2 \rightarrow \text{N}_2 + 2\text{H}_2\text{O}$

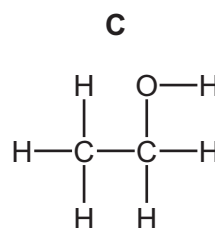
- 33** Which diagram shows the displayed formula for the named organic compound?



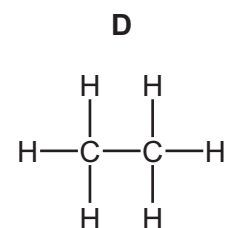
ethanoic acid



ethene



ethanol



methane

34 What is the total number of covalent bonds in a molecule of butane, C_4H_{10} ?

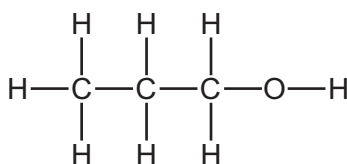
- A 3 B 10 C 13 D 14

35 Propane reacts with chlorine in a substitution reaction.

Which reaction condition is required for the reaction to occur?

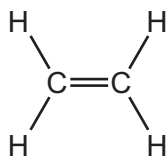
- A acid catalyst
B iron catalyst
C temperature of 400°C
D ultraviolet light

36 The structure of an organic compound is shown.

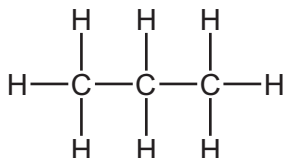


Which structure represents a molecule that reacts with steam to produce this product?

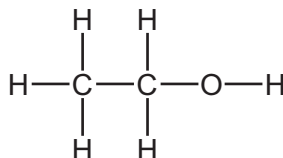
A



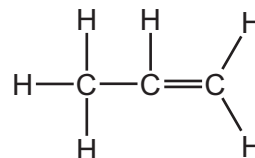
B



C



D

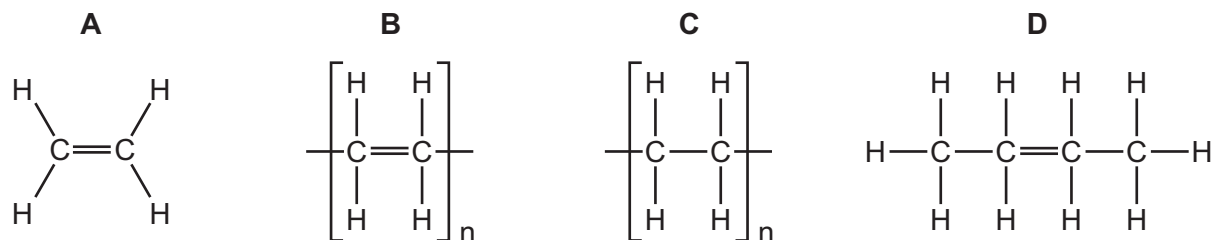


37 Which term describes nylon?

- A addition polymer
B natural polymer
C polyamide
D polyester

38 Ethene can be polymerised.

Which diagram represents the structure of the product formed?



39 An acid–base titration is described.

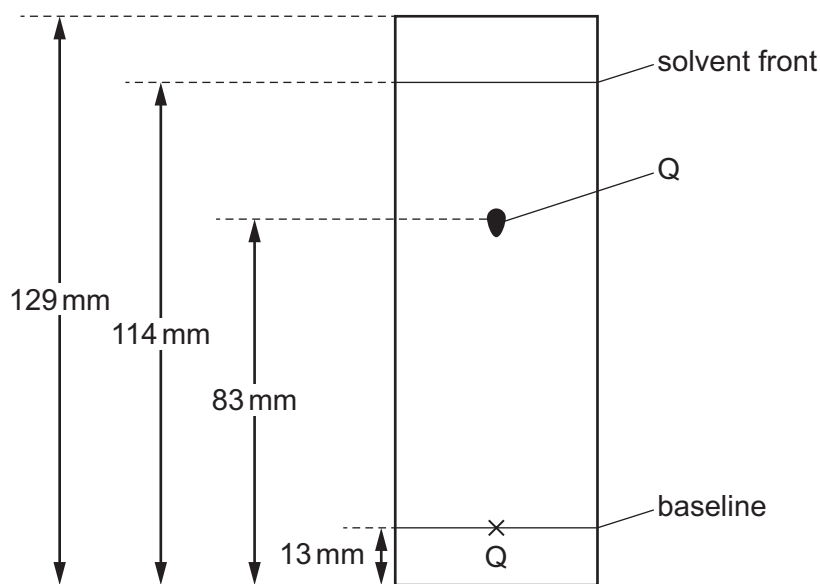
- 25.0 cm³ of dilute aqueous alkali is put into a conical flask.
- Indicator is added to the flask.
- Dilute acid is added to the aqueous alkali until the indicator changes colour.
- The volume of acid used is then recorded.

Which use of apparatus is correct?

- A** The 25.0 cm³ of aqueous alkali is measured using a volumetric pipette.
- B** The 25.0 cm³ of aqueous alkali is measured using the lines on the conical flask.
- C** The volume of acid is measured using a measuring cylinder.
- D** The volume of acid is measured using a volumetric pipette.

40 Substance Q is investigated using chromatography.

The chromatogram is shown. The diagram is not drawn to scale.



What is the R_f value of Q?

- A** 0.60 **B** 0.64 **C** 0.69 **D** 0.72

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

The Periodic Table of Elements

Group																				
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII			
		Key atomic number atomic symbol name relative atomic mass																		
3 Li lithium 7	4 Be beryllium 9													5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19		
11 Na sodium 23	12 Mg magnesium 24													13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5		
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84			
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131			
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —			
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —			



Cambridge IGCSE™ (9–1)

CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

October/November 2023

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

Question	Answer	Marks
1	D	1
2	C	1
3	D	1
4	B	1
5	B	1
6	B	1
7	B	1
8	A	1
9	A	1
10	C	1
11	B	1
12	D	1
13	A	1
14	A	1
15	A	1
16	D	1
17	D	1
18	D	1
19	D	1
20	B	1
21	C	1
22	B	1
23	A	1
24	C	1
25	B	1
26	A	1
27	D	1
28	D	1

Question	Answer	Marks
29	B	1
30	C	1
31	B	1
32	A	1
33	C	1
34	C	1
35	D	1
36	D	1
37	C	1
38	C	1
39	A	1
40	C	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/21

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 A gas is heated. The pressure is kept constant.

Which statement describes the behaviour of the particles in the gas?

- A The particles move faster and become closer together.
- B The particles move faster and become further apart.
- C The particles move more slowly and become closer together.
- D The particles move more slowly and become further apart.

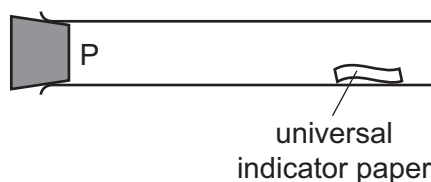
- 2 A mixture of ice and water is left to stand and the ice melts.

Which row describes what happens as the ice is melting?

	temperature of mixture	energy change
A	increases	average kinetic energy of particles decreases
B	increases	energy is used to overcome attractive forces
C	stays the same	average kinetic energy of particles decreases
D	stays the same	energy is used to overcome attractive forces

- 3 Hydrogen chloride gas [M_r : HCl, 36.5] is released at P in the apparatus shown.

The universal indicator paper turns red after 38 s.



The experiment is repeated using sulfur dioxide gas [M_r : SO₂, 64].

What is the result for sulfur dioxide gas?

	universal indicator paper turns	time for universal indicator paper to change colour / s
A	blue	26
B	blue	51
C	red	26
D	red	51

4 Four statements about atoms are listed.

- 1 The centre of an atom is positively charged.
- 2 Protons and electrons are located in the nucleus.
- 3 Protons and electrons have the same mass.
- 4 Most of the mass of an atom is in the nucleus.

Which statements are correct?

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

5 The electronic configurations of two elements are given.

element L: 2,8,8,1

element M: 2,8,4

Which row identifies the group number and the period number for element L and element M?

	element L		element M	
	group number	period number	group number	period number
A	I	4	IV	3
B	I	4	III	4
C	IV	1	III	4
D	IV	1	IV	3

6 Which statement explains why isotopes of the same element have the same chemical properties?

- A** They have different numbers of protons in their nucleus.
- B** They have different numbers of neutrons in their nucleus.
- C** They have the same electronic configuration.
- D** They have the same number of electrons as protons.

7 Which statements about potassium chloride are correct?

- 1 It conducts electricity when solid because its ions are free to move.
- 2 It has a high melting point because it has strong intermolecular forces.
- 3 Its structure is a giant lattice of alternating positive and negative ions.
- 4 It is soluble in water.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

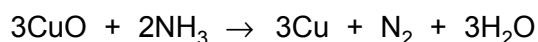
8 How many electrons are shared in **one** molecule of nitrogen and in **one** molecule of ethene?

	nitrogen	ethene
A	2	12
B	2	8
C	6	12
D	6	8

9 What is the total number of electrons in **one** molecule of ammonia, NH_3 ?

A 6 **B** 8 **C** 10 **D** 11

10 When heated, copper(II) oxide, CuO , reacts with ammonia, NH_3 .



8.5 g of ammonia reacts with an excess of copper(II) oxide to produce 26.4 g of copper.

What is the percentage yield of copper in this reaction?

A 27.5% **B** 32.2% **C** 55.0% **D** 82.5%

11 What is the empirical formula of ethanoic acid?

A CHO **B** CH_2O **C** $\text{C}_2\text{H}_2\text{O}$ **D** $\text{C}_2\text{H}_4\text{O}_2$

- 12** Magnesium chloride, MgCl_2 , contains magnesium ions and chloride ions.

How many chloride ions are present in **two** moles of magnesium chloride?

- A** 6.02×10^{23}
B 1.204×10^{24}
C 2.408×10^{24}
D 3.612×10^{24}

- 13** A metal object is electroplated with copper.

One electrode is the metal object and the other electrode is copper. The electrolyte is aqueous copper(II) sulfate.

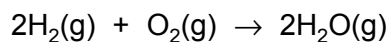
Which row shows the ionic half-equation for the reaction at the anode and the observation of the electrolyte?

	anode	electrolyte
A	$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	blue colour fades
B	$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	blue colour does not change
C	$\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$	blue colour fades
D	$\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$	blue colour does not change

- 14** Which statement about electrolysis is correct?

- A** Chemical energy is converted to electrical energy.
B Electrons flow through the electrolyte.
C Ionic compounds are broken down.
D Metals are formed at the positive electrode.

- 15 The reaction between hydrogen and oxygen releases 486 kJ/mol of energy.

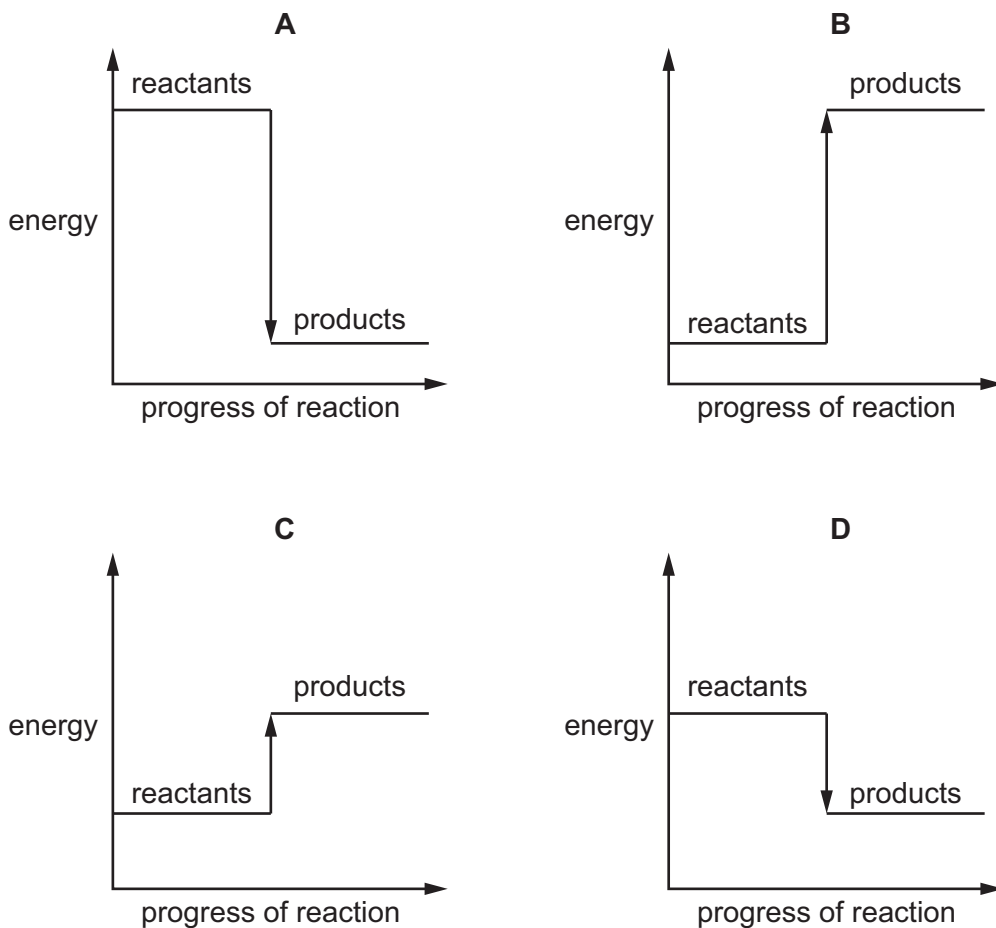


The bond energy of H–H is 436 kJ/mol and that of H–O is 464 kJ/mol.

What is the bond energy of O=O?

- A 430 kJ/mol
B 458 kJ/mol
C 498 kJ/mol
D 984 kJ/mol
- 16 Which reaction pathway diagram shows the reaction that will give out the most energy?

The scale on the y-axis is the same in each diagram.



- 17 When calcium carbonate is heated strongly, a gas is given off.

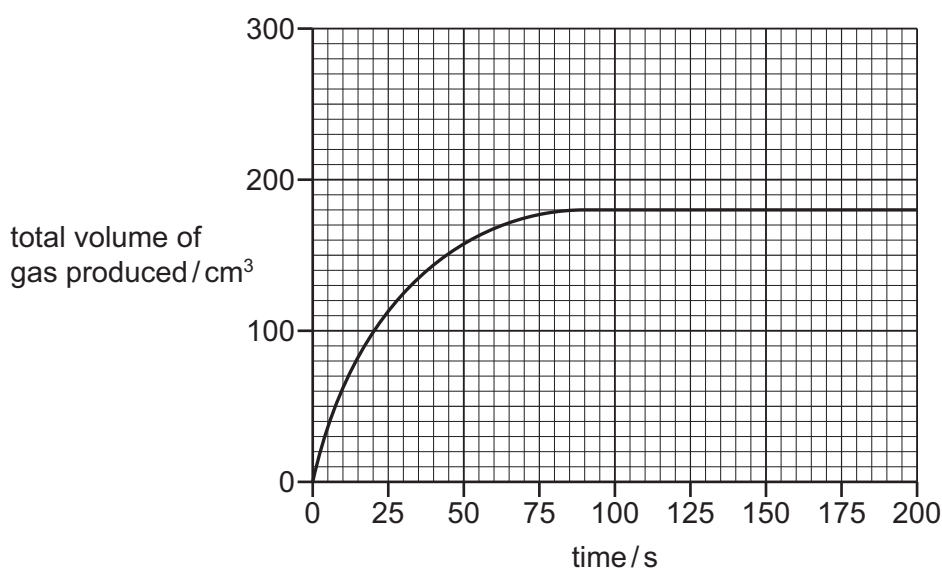
Which word describes this change?

- A chemical
- B exothermic
- C physical
- D reduction

- 18 Powdered magnesium carbonate is added to excess dilute hydrochloric acid.

The total volume of gas produced is measured over time.

A graph of the results is shown.



The experiment is repeated but the concentration of the hydrochloric acid is doubled.

All other conditions are kept the same.

Which statements about the second experiment are correct?

- 1 The final volume of gas is 360 cm^3 .
- 2 The reaction finishes before 90 seconds.
- 3 The activation energy of the reaction is lower.

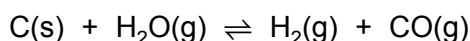
- A 1 and 2
- B 1 and 3
- C 2 and 3
- D 2 only

19 Which statements explain why increasing the temperature changes the rate of a chemical reaction?

- 1 It increases the activation energy.
- 2 It increases the frequency of collisions between the reacting particles.
- 3 It increases the kinetic energy of the reacting particles.
- 4 It increases the number of particles per unit volume.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

20 Hydrogen is made by reacting carbon with steam. The equation for the reaction is shown.



The forward reaction is endothermic.

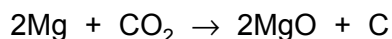
Which row describes changes in the pressure and the temperature that will **both** shift the position of equilibrium to the right?

	pressure	temperature
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

21 Which row shows the conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process?

	pressure / atm	temperature / °C	catalyst
A	250	200	vanadium(V) oxide
B	2	450	vanadium(V) oxide
C	250	200	iron
D	2	450	iron

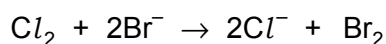
- 22 The equation for the reaction of magnesium with carbon dioxide is shown.



Which statement about this reaction is correct?

- A Magnesium is oxidised and carbon dioxide is reduced.
 - B Magnesium is reduced and carbon dioxide is oxidised.
 - C Magnesium and carbon dioxide are both oxidised.
 - D Magnesium and carbon dioxide are neither oxidised nor reduced.
- 23 Chlorine displaces bromine from aqueous potassium bromide.

The ionic equation for the reaction is shown.



Which statement about this reaction is correct?

- A Bromide ions act as an oxidising agent.
 - B Bromide ions are oxidised as electrons are lost.
 - C Chlorine acts as a reducing agent.
 - D Chlorine is reduced as electrons are lost.
- 24 Which gas is produced when ammonium chloride is warmed with aqueous sodium hydroxide?
- A ammonia
 - B chlorine
 - C hydrogen
 - D nitrogen
- 25 Which equation represents a solution of ethanoic acid in water?
- A $\text{HCOOH}(\text{aq}) \rightleftharpoons \text{HCOO}^-(\text{aq}) + \text{H}^+(\text{aq})$
 - B $\text{HCOOH}(\text{aq}) \rightarrow \text{HCOO}^-(\text{aq}) + \text{H}^+(\text{aq})$
 - C $\text{CH}_3\text{COOH}(\text{aq}) \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}^+(\text{aq})$
 - D $\text{CH}_3\text{COOH}(\text{aq}) \rightarrow \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}^+(\text{aq})$

26 Four statements about the reactions of oxides with dilute hydrochloric acid and with aqueous sodium hydroxide are listed.

- 1 Aluminium oxide reacts with both dilute hydrochloric acid and aqueous sodium hydroxide.
- 2 Calcium oxide reacts with both dilute hydrochloric acid and aqueous sodium hydroxide.
- 3 Copper(II) oxide reacts with dilute hydrochloric acid but **not** with aqueous sodium hydroxide.
- 4 Sulfur dioxide does **not** react with either dilute hydrochloric acid or aqueous sodium hydroxide.

Which statements are correct?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

27 Which statement about elements in Period 3 of the Periodic Table is correct?

- A** Aluminium is a non-metal in Group III.
B Argon is in Group VIII and has eight electrons in its outer electron shell.
C Magnesium is in Group II and has three electrons in its outer electron shell.
D Sulfur is a metal in Group VI.

28 Which row describes the structure of Group VII elements and the trend in their reactivity down the group?

	structure	reactivity down Group VII
A	diatomic	increases
B	diatomic	decreases
C	monatomic	increases
D	monatomic	decreases

29 Some information about four elements, P, Q, R and S, is shown.

	melting point in °C	density in g/cm ³	colour of chloride
P	1247	7.43	pink
Q	1410	2.33	white
R	1910	6.11	purple
S	115	2.07	red

Which elements are transition elements?

- A** P and R **B** P and S **C** Q and R **D** R and S

30 Propanoic acid is a carboxylic acid. It has similar chemical properties to ethanoic acid.

Which statements are correct?

- 1 Aqueous propanoic acid is a weaker acid than dilute hydrochloric acid.
- 2 Propanoic acid partially ionises in aqueous solution.
- 3 Propanoic acid reacts with ethanol to form propyl ethanoate.

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

31 Iron rusts in the presence of oxygen and water.

Which statements about the rusting of iron are correct?

- 1 Anhydrous iron(II) oxide is produced when iron rusts.
- 2 Iron rusts more quickly when attached to a piece of zinc.
- 3 Coating the iron with plastic prevents the iron from rusting.
- 4 Iron loses electrons when it rusts.

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

- 32** An iron nail is added to aqueous copper(II) sulfate and a different iron nail is added to aqueous magnesium sulfate.

The results are shown.

experiment	result
iron nail in aqueous copper(II) sulfate	nail is coated with a brown solid
iron nail in aqueous magnesium sulfate	no reaction

Which statement is correct?

- A** Copper atoms are oxidised more easily than magnesium atoms.
 - B** Copper atoms are reduced more easily than iron ions.
 - C** Iron atoms are oxidised more easily than copper atoms.
 - D** Iron atoms are reduced more easily than copper ions.
- 33** Which pollutant leads to the deoxygenation of water in ponds and lakes?
- A** fertilisers containing nitrates and phosphates
 - B** toxic metal compounds
 - C** combustion products of fossil fuels
 - D** acid rain
- 34** Which statement identifies a sample of water as pure?
- A** It melts at room temperature.
 - B** It turns anhydrous copper(II) sulfate blue.
 - C** It turns hydrated cobalt(II) chloride from blue to pink.
 - D** It boils at 100 °C.

35 Oxides of nitrogen are produced by car engines.

In a catalytic converter oxides of nitrogen are removed by reacting them with compound X.

Which row describes the type of reaction oxides of nitrogen undergo and identifies compound X?

	type of reaction	compound X
A	oxidation	carbon dioxide
B	oxidation	carbon monoxide
C	reduction	carbon dioxide
D	reduction	carbon monoxide

36 What is a disadvantage of producing ethanol using the catalytic addition of steam to ethene?

- A** the energy cost is low
- B** the process is continuous
- C** the process uses a non-renewable raw material
- D** the ethanol is pure

37 Which statement about the polymer PET is correct?

- A** It can be broken down into its monomers and re-polymerised.
- B** It is an addition polymer.
- C** It is a polyamide.
- D** It is made from amino acid monomers.

38 The formulae of five compounds are listed.

- 1 C_4H_{10}
- 2 $\text{C}_2\text{H}_5\text{OH}$
- 3 $\text{C}_4\text{H}_9\text{OH}$
- 4 $\text{C}_4\text{H}_9\text{COOH}$
- 5 $\text{C}_5\text{H}_{11}\text{OH}$

Which compounds are in the same homologous series?

- A** 1, 3 and 4
B 2, 3 and 5
C 3 and 4 only
D 3 and 5 only

39 Propane reacts with chlorine.

Which statements about this reaction are correct?

- 1 Ultraviolet light is used to provide the activation energy.
- 2 Propane undergoes an addition reaction.
- 3 One of the products is $\text{CH}_3\text{CH}_2\text{Cl}$.
- 4 One of the products is HCl .

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

40 Which statement about chromatography is correct?

- A** It is **not** possible for two different substances to have the same R_f value.
B It is only possible to use chromatography on substances which have a colour.
C It is possible to use chromatography on colourless substances using a locating agent.
D The R_f value of a substance = $\frac{\text{the distance travelled by the solvent}}{\text{the distance travelled by the substance}}$

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

The Periodic Table of Elements

Group																		
I	II											III	IV	V	VI	VII	VIII	
1 H hydrogen 1 Key atomic number atomic symbol name relative atomic mass																		
3 Li lithium 7	4 Be beryllium 9											5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20	2 He helium 4
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —	
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —	

lanthanoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
------------------------------	---------------------------	---------------------------------	------------------------------	-----------------------------	-----------------------------	-----------------------------	-------------------------------	----------------------------	-------------------------------	----------------------------	---------------------------	----------------------------	------------------------------	-----------------------------

actinoids

89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —
---------------------------	----------------------------	---------------------------------	---------------------------	----------------------------	----------------------------	----------------------------	-------------------------	----------------------------	------------------------------	------------------------------	---------------------------	-------------------------------	----------------------------	------------------------------

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p).

Cambridge IGCSE™ (9–1)

CHEMISTRY (9–1)**0971/21**

Paper 2 Multiple Choice (Extended)

May/June 2024**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	D	1
3	D	1
4	B	1
5	A	1
6	C	1
7	D	1
8	C	1
9	C	1
10	C	1
11	B	1
12	C	1
13	D	1
14	C	1
15	C	1
16	A	1
17	A	1
18	D	1
19	C	1
20	B	1
21	B	1
22	A	1
23	B	1
24	A	1
25	C	1
26	B	1
27	B	1
28	B	1

Question	Answer	Marks
29	A	1
30	B	1
31	D	1
32	C	1
33	A	1
34	D	1
35	D	1
36	C	1
37	A	1
38	B	1
39	B	1
40	C	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/22

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



1 Which gas has the slowest rate of diffusion?

A H_2

B NH_3

C CH_4

D CO_2

2 Which statements about the position of the elements in the Periodic Table are correct?

- 1 Elements in the same group have similar chemical properties.
- 2 Elements in the same period have similar chemical properties.
- 3 Elements in the same group have the same number of electron shells.
- 4 Elements in the same group have the same number of outer shell electrons.

A 1 and 3

B 1 and 4

C 2 and 3

D 2 and 4

3 Which statements about isotopes are correct?

- 1 Isotopes are atoms of different elements with the same number of protons.
- 2 Isotopes of the same element have the same chemical properties.
- 3 Isotopes are atoms with the same relative atomic mass.
- 4 Isotopes of the same element have the same electronic configuration.

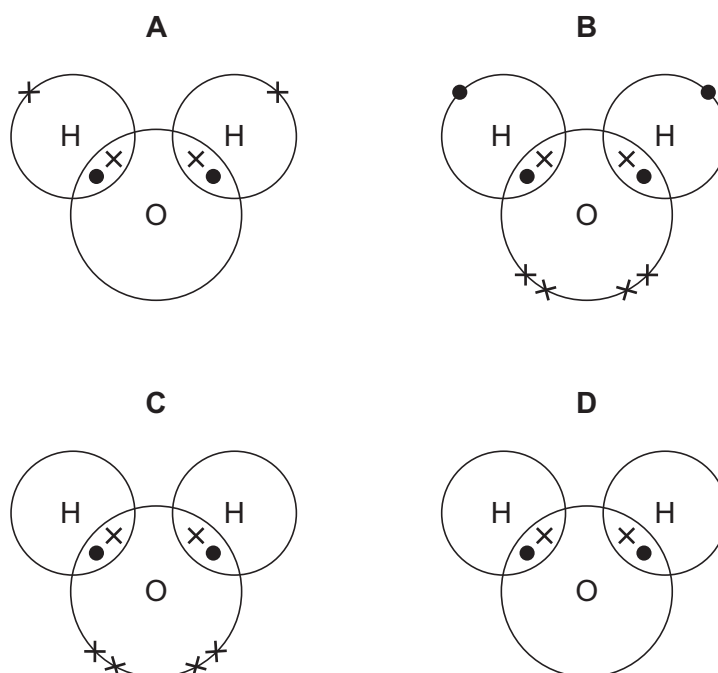
A 1 and 2

B 1 and 3

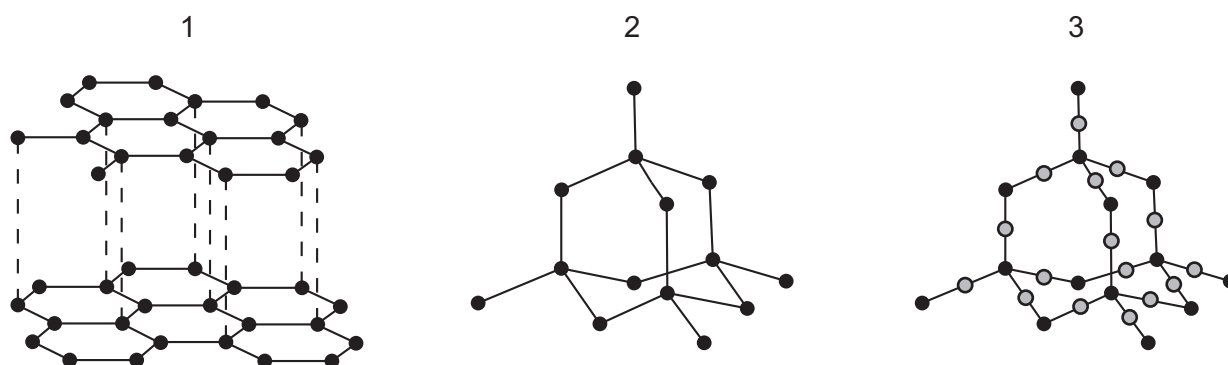
C 2 and 4

D 3 and 4

4 Which diagram shows the arrangement of the outer shell electrons in a molecule of water?



5 The structures of three substances are shown.



Which substances are hard and have a high melting point?

- A** 1, 2 and 3 **B** 1 only **C** 2 only **D** 2 and 3 only

6 Information about four substances, W, X, Y and Z, is shown.

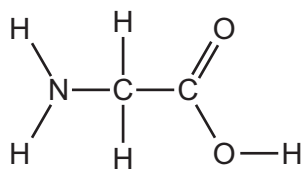
	melting point/ °C	electrical conductivity
W	1710	does not conduct when solid
X	3500	conducts when solid
Y	120	does not conduct
Z	801	conducts when molten

W, X, Y and Z are graphite, poly(ethene), sodium chloride and silicon(IV) oxide but not in that order.

What are W, X, Y and Z?

	W	X	Y	Z
A	graphite	poly(ethene)	silicon(IV) oxide	sodium chloride
B	sodium chloride	graphite	poly(ethene)	silicon(IV) oxide
C	poly(ethene)	sodium chloride	graphite	silicon(IV) oxide
D	silicon(IV) oxide	graphite	poly(ethene)	sodium chloride

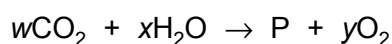
- 7 The structure of glycine is shown.



Which row is correct?

	formula of glycine	number of different elements in glycine
A	$\text{CH}_5\text{O}_2\text{N}$	10
B	$\text{C}_2\text{H}_5\text{O}_2\text{N}$	4
C	$\text{C}_2\text{H}_5\text{O}_2\text{N}$	10
D	$\text{H}_2\text{NCHCOOH}$	4

- 8 The incomplete equation for photosynthesis is shown.



Compound P is a product of the reaction.

Which row describes the values of w , x and y and gives the empirical formula of compound P?

	values of w , x and y	empirical formula of compound P
A	w , x and y are all the same	CH_2O
B	w , x and y are all the same	$\text{C}_6\text{H}_{12}\text{O}_6$
C	w and x are the same and both are greater than y	CH_2O
D	w and x are the same and both are greater than y	$\text{C}_6\text{H}_{12}\text{O}_6$

- 9 The concentration and volume of an aqueous alkali are known.

Which additional information is required to calculate the number of moles of acid needed to neutralise the aqueous alkali?

- A** the concentration of the acid
- B** the equation for the acid–alkali reaction
- C** the formula of the acid
- D** the volume of the acid required for neutralisation

10 Which statement about electrolysis is correct?

- A Electrons move through the electrolyte from the cathode to the anode.
- B Electrons move in the external circuit towards the cathode.
- C Negative ions move in the external circuit towards the anode.
- D Positive ions move through the electrolyte towards the anode.

11 Aqueous copper(II) sulfate is electrolysed using copper electrodes.

What is the half-equation for the reaction at the cathode?

- A $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
- B $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$
- C $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$
- D $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$

12 Three statements about activation energy, E_a , are listed.

- 1 Colliding particles must have at least E_a before they can react.
- 2 E_a for exothermic reactions is always greater than for endothermic reactions.
- 3 E_a is always endothermic.

Which statements are correct?

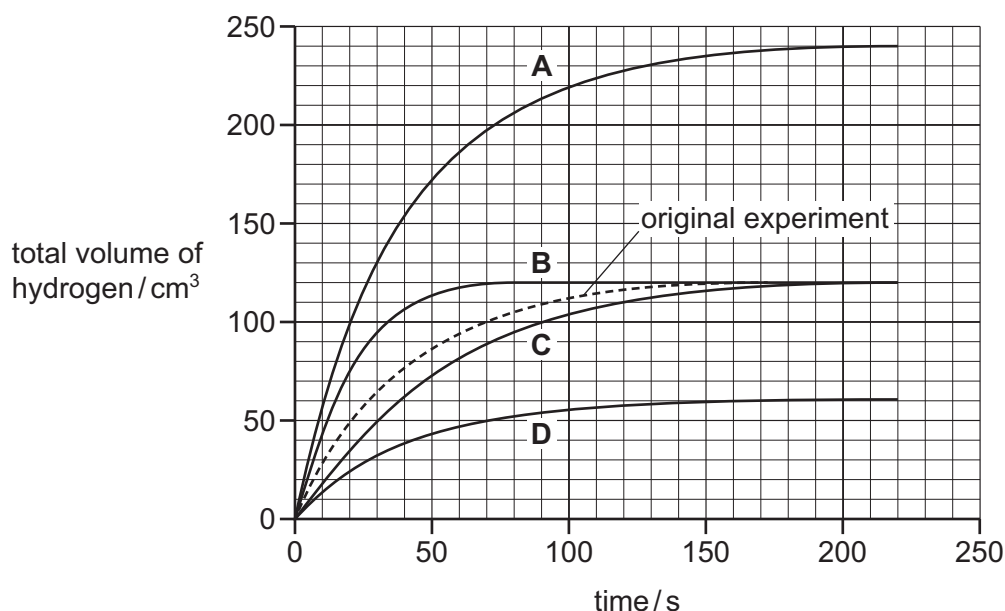
- A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only

- 13** A student adds excess magnesium ribbon to 10 cm^3 of 0.5 mol/dm^3 sulfuric acid.

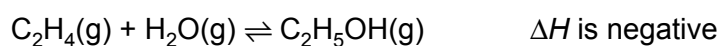
The hydrogen gas produced is collected and its total volume is measured every 10 seconds.

The experiment is repeated with 5 cm^3 of 0.5 mol/dm^3 sulfuric acid added to 5 cm^3 of water using the same mass of magnesium ribbon.

Which line on the graph shows the results of the second experiment?



- 14** The equation represents the reversible reaction between ethene and steam.



Which row describes the conditions that produce the greatest yield of ethanol?

	pressure	temperature
A	low	low
B	low	high
C	high	low
D	high	high

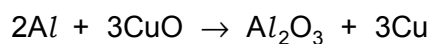
- 15 Which row identifies the pressure and the catalyst used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process?

	pressure / atm	catalyst
A	2	iron
B	2	vanadium(V) oxide
C	200	iron
D	200	vanadium(V) oxide

- 16 Which equation represents a redox reaction?

- A** $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$
B $\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$
C $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$
D $\text{MgCO}_3 + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$

- 17 The equation for the reaction between aluminium and copper(II) oxide is shown.



Which statements about this equation are correct?

- 1 The oxidation number of the aluminium reactant is +2.
- 2 The oxidation number of the aluminium in the product is +6.
- 3 The oxidation number of the copper in the reactant is +2.
- 4 The oxidation number of the copper product is 0.

- A** 1, 2 and 3 **B** 1 and 2 only **C** 3 and 4 **D** 4 only

- 18 What is the colour of thymolphthalein in dilute hydrochloric acid?

- A** blue
B red
C yellow
D colourless

- 19** Two acids, P and Q, with the same concentration and volume are reacted separately with the same mass of magnesium ribbon.

The reactions produce the same total volume of hydrogen gas but acid Q reacts much more slowly than acid P.

Which explanation for the difference between P and Q is correct?

- A** Acid P has a higher pH than acid Q.
 - B** Acid P has a lower concentration of hydrogen ions.
 - C** Acid Q is partially dissociated and acid P is fully dissociated.
 - D** Acid Q is a proton acceptor.
- 20** Which process is **not** used in the preparation of an insoluble salt?
- A** filtration
 - B** washing
 - C** crystallisation
 - D** drying

- 21** Lithium and potassium are metals in Group I of the Periodic Table.

Lithium has a melting point of 181 °C and a density of 0.53 g/cm³.

Which row describes the melting point and density of potassium?

	melting point in °C	density in g/cm ³
A	less than 181	less than 0.53
B	less than 181	greater than 0.53
C	greater than 181	less than 0.53
D	greater than 181	greater than 0.53

- 22** Which statements about transition elements are correct?

- 1 They have a low density.
- 2 They form ions with variable oxidation numbers.
- 3 They have a high melting point.
- 4 They form only colourless compounds.

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

23 Which statements describe the elements in Group VIII of the Periodic Table?

- 1 Their atoms have full outer electron shells.
- 2 They are unreactive metals.
- 3 They are monatomic gases.
- 4 They are diatomic gases.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

24 Four metals, Q, R, S and T, are each added to separate samples of water, steam and dilute hydrochloric acid.

The results are shown.

	observation with water	observation with steam	observation with dilute hydrochloric acid
Q	slow reaction	fast reaction	fast reaction
R	no reaction	no reaction	no reaction
S	no reaction	very slow reaction	slow reaction
T	fast reaction	explodes	explodes

Which statements are correct?

- 1 R is the least reactive metal.
- 2 T could be potassium.
- 3 S is more reactive than Q and R.
- 4 Metals generally react faster with steam than they react with water.

A 1, 2 and 4 **B** 1 and 2 only **C** 2 and 3 **D** 3 and 4

25 Metal X acts as a sacrificial metal to prevent iron from corroding.

Metal X does **not** act as a sacrificial metal to prevent aluminium from corroding.

What is X?

- A** copper
B magnesium
C silver
D zinc

26 Which equation represents the reduction of a compound found in hematite in the blast furnace?

- A $2\text{Al}_2\text{O}_3 \rightarrow 4\text{Al} + 3\text{O}_2$
- B $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$
- C $2\text{FeO} + \text{CO} \rightarrow \text{Fe}_2\text{O}_3 + \text{C}$
- D $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$

27 Which statements about the treatment of domestic water supplies are correct?

- 1 Filtration is used to remove insoluble substances from the water.
- 2 Sedimentation is used to remove soluble substances from the water.
- 3 Carbon is used to remove tastes and odours from the water.
- 4 Chlorine is used to lower the pH of the water.

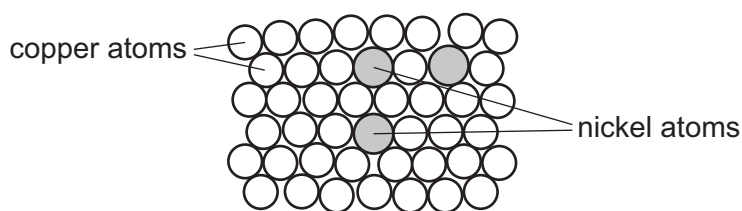
- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

28 Which statement about water is correct?

- A It turns anhydrous copper(II) sulfate from pink to blue.
- B It turns anhydrous copper(II) sulfate from white to blue.
- C It turns cobalt(II) chloride paper from blue to white.
- D It turns cobalt(II) chloride paper from pink to blue.

29 Cupronickel is used to make coins.

The arrangement of atoms in cupronickel is shown.



Which kind of substance is cupronickel?

- A an alloy
- B an isotope
- C a compound
- D a transition element

30 Which physical properties are typical of **all** metals?

- 1 good heat conductivity
- 2 low density
- 3 malleability

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 3 only

31 Which oxide is used to neutralise acidic gases in flue gas desulfurisation?

- A** calcium oxide
B carbon dioxide
C nitrogen oxide
D sulfur dioxide

32 Some information about three gases, P, Q and R, is listed.

- Gas P forms when magnesium reacts with dilute hydrochloric acid.
- Gas Q makes up 78% of the air.
- Gas R forms when gas P reacts with gas Q.

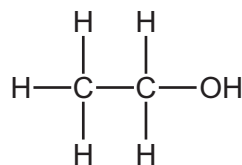
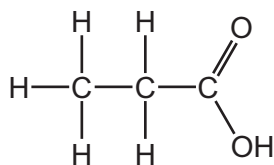
What is gas R?

- A** ammonia
B methane
C nitrogen dioxide
D water vapour

33 Which formula represents an alkene?

A CH₄ **B** C₂H₄ **C** C₂H₆ **D** C₂H₅OH

34 The structures of two compounds used to make an ester are shown.



What is the name of the ester?

- A ethyl propanoate
 - B propyl ethanoate
 - C ethyl ethanoate
 - D propyl propanoate
- 35 Which statement about a homologous series is correct?
- A All members have the same general formula.
 - B All members have the same molecular formula.
 - C All members have similar physical properties.
 - D Members show a trend in their chemical properties.
- 36 Which statements about aqueous ethanoic acid are correct?
- 1 It contains the functional group -COOH .
 - 2 It reacts with carbonates to produce hydrogen.
 - 3 It turns universal indicator paper blue.
 - 4 It has a pH lower than pH 7.
- A 1 and 2 B 1 and 3 C 1 and 4 D 2 and 4

37 Alkenes react with steam in an addition reaction.

Some alkenes produce only one alcohol product.

Some alkenes produce two different alcohols which are structural isomers.

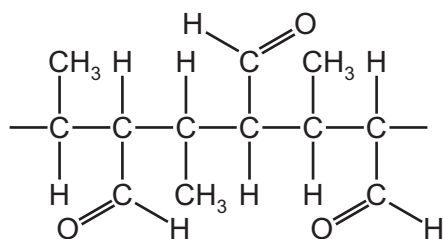
Which row gives the number of alcohol isomers formed when ethene reacts with steam and when propene reacts with steam?

	number of alcohol isomers formed	
	ethene + steam	propene + steam
A	1	1
B	1	2
C	2	1
D	2	2

38 What is an advantage of manufacturing ethanol by fermentation rather than by the addition of steam to ethene?

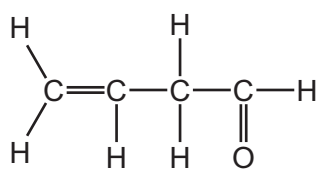
- A** No gases that cause global warming are produced.
- B** The ethanol that is produced is pure.
- C** Fermentation is a fast process.
- D** Fermentation uses renewable raw materials.

39 The diagram shows the structure of a polymer.

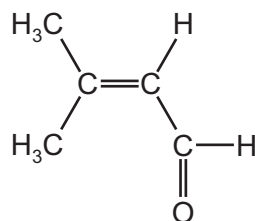


Which structure represents the monomer for this polymer?

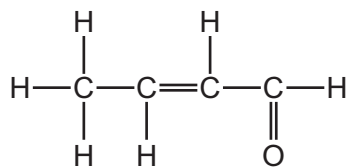
A



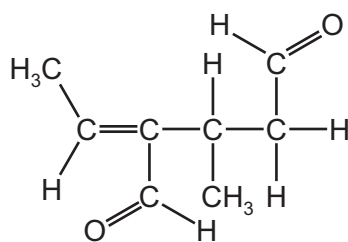
B



C



D



40 Which piece of apparatus is used to measure 24.5 cm^3 of gas produced during a reaction?

- A** beaker
- B** conical flask
- C** measuring cylinder
- D** volumetric pipette

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

Cambridge IGCSE™ (9–1)

CHEMISTRY (9–1)**0971/22**

Paper 2 Multiple Choice (Extended)

May/June 2024**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	B	1
3	C	1
4	C	1
5	D	1
6	D	1
7	B	1
8	A	1
9	B	1
10	B	1
11	D	1
12	C	1
13	D	1
14	C	1
15	B	1
16	C	1
17	C	1
18	D	1
19	C	1
20	C	1
21	B	1
22	C	1
23	A	1
24	A	1
25	D	1
26	D	1
27	A	1
28	B	1

Question	Answer	Marks
29	A	1
30	B	1
31	A	1
32	A	1
33	B	1
34	A	1
35	A	1
36	C	1
37	B	1
38	D	1
39	C	1
40	C	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

October/November 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Oxygen melts at -219°C and boils at -183°C .

At which temperature is oxygen a liquid?

- A** -225°C **B** -189°C **C** -175°C **D** 25°C

- 2 The pressure of a sample of gas is decreased. The temperature is kept constant.

Which row describes the effects on the particles?

	movement of particles	collisions between particles
A	slower	occur less often
B	slower	occur with more force
C	no change in speed	occur less often
D	no change in speed	occur with more force

- 3 Rubidium has two isotopes, $^{85}_{37}\text{Rb}$ and $^{87}_{37}\text{Rb}$.

Which statement explains why both isotopes have the same chemical properties?

- A** They have the same number of protons.
B They have the same electronic configuration.
C They have different numbers of neutrons.
D They have different mass numbers.

- 4 Which pair of elements react to form a compound with a strong attraction between oppositely charged ions?

- A** carbon and bromine
B carbon and nitrogen
C sodium and oxygen
D sodium and potassium

5 Four substances, P, Q, R and S, are described.

- P is diatomic.
- Q is a good conductor of electricity when solid and when molten.
- R is a silver solid with a very high melting point.
- S reacts with oxygen to form a brown gas.

Which substances are metals?

- A** P and Q **B** P and S **C** Q and R **D** R and S

6 Which diagram shows the covalent bonding in a molecule of carbon dioxide?

- A** O—C—O **B** O=C—O **C** O=C=O **D** O≡C≡O

7 The bonding, structure and melting point of sodium chloride and sulfur dichloride are shown.

compound	bonding	structure	melting point / °C
sodium chloride	ionic	giant lattice	801
sulfur dichloride	covalent	simple molecular	–121

Why does sulfur dichloride have a lower melting point than sodium chloride?

- A** The covalent bonds in sulfur dichloride are weaker than the attractive forces between molecules in sodium chloride.
- B** The covalent bonds in sulfur dichloride are weaker than the ionic bonds in sodium chloride.
- C** The attractive forces between molecules in sulfur dichloride are weaker than the attractive forces between molecules in sodium chloride.
- D** The attractive forces between molecules in sulfur dichloride are weaker than the ionic bonds in sodium chloride.

8 Diamond and graphite have giant covalent structures of carbon atoms.

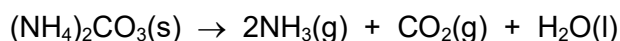
Which statement describes graphite?

- A** It has a strong, rigid three-dimensional structure.
- B** It has four strong covalent bonds between each carbon atom.
- C** It has layers, which can slide over each other.
- D** It has no delocalised electrons so does **not** conduct electricity.

- 9 Which row explains the malleability and electrical conductivity of a solid metal?

	malleability	electrical conductivity
A	Delocalised electrons can move freely through the structure.	Delocalised electrons can move freely through the structure.
B	Delocalised electrons can move freely through the structure.	Positive ions can move freely through the structure.
C	Rows of positive ions can slide over each other.	Delocalised electrons can move freely through the structure.
D	Rows of positive ions can slide over each other.	Positive ions can move freely through the structure.

- 10 The equation for the decomposition of ammonium carbonate, $(\text{NH}_4)_2\text{CO}_3$, is shown.



[M_r : $(\text{NH}_4)_2\text{CO}_3$, 96]

The **total** volume of gas produced is 360 cm^3 at r.t.p.

Which mass of ammonium carbonate, $(\text{NH}_4)_2\text{CO}_3$, is decomposed?

- A** 0.24 g **B** 0.48 g **C** 0.96 g **D** 1.44 g
- 11 What is the empirical formula of a compound that contains 3.66 g of hydrogen, 37.8 g of phosphorus and 58.5 g of oxygen?
- A** $\text{H}_6\text{P}_2\text{O}_6$ **B** H_4PO_4 **C** H_3PO_3 **D** HPO
- 12 Aqueous copper(II) sulfate is electrolysed using graphite electrodes.

Which row identifies the product and observations at each electrode during the electrolysis?

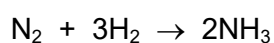
	anode		cathode	
	product	observation	product	observation
A	oxygen	bubbles of gas	copper	electrode turns pink
B	copper	electrode turns pink	oxygen	bubbles of gas
C	none	electrode dissolves	copper	electrode turns pink
D	oxygen	bubbles of gas and electrode dissolves	hydrogen	bubbles of gas

13 Molten sodium chloride is electrolysed using inert electrodes.

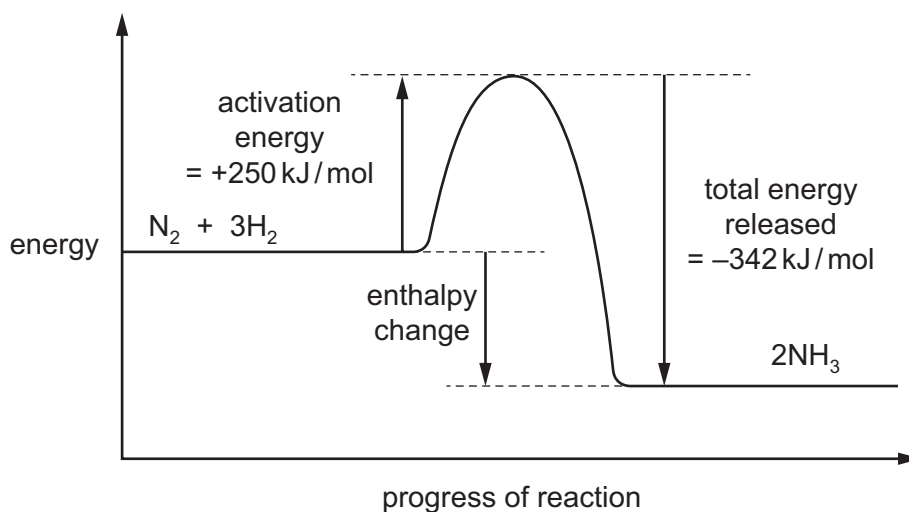
Which row shows the products formed at the cathode and anode?

	cathode	anode
A	chlorine	hydrogen
B	chlorine	sodium
C	hydrogen	chlorine
D	sodium	chlorine

14 The equation for the formation of ammonia is shown.



The reaction pathway diagram for the reaction is shown.



What is the enthalpy change for the reaction?

- A** -592 kJ/mol
- B** -92 kJ/mol
- C** +92 kJ/mol
- D** +592 kJ/mol

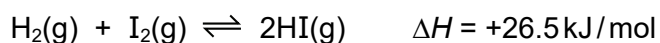
15 Sulfur dioxide is converted to sulfur trioxide in the Contact process.

The conditions used are 450 °C and 200 kPa with a vanadium(V) oxide catalyst.

Which row describes and explains the effect of changing conditions on the rate of reaction?

	change in conditions	effect on rate	explanation
A	no catalyst	lower	the activation energy is higher
B	higher pressure	higher	the particles have more kinetic energy
C	lower temperature	lower	the particles collide more frequently
D	lower pressure	higher	there are more particles per unit volume

16 Hydrogen gas reacts with iodine gas to form hydrogen iodide gas in an equilibrium reaction.

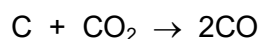


Which changes increase the yield of HI at equilibrium?

- 1 adding a catalyst
- 2 adding more hydrogen gas
- 3 increasing the pressure
- 4 increasing the temperature

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

17 The equation for the reaction of carbon with carbon dioxide is shown.



Which row identifies the carbon atom that is reduced and its change in oxidation number?

	atom that is reduced	change in oxidation number
A	carbon in CO ₂	+2 → +4
B	carbon in CO ₂	+4 → +2
C	elemental carbon, C	0 → +2
D	elemental carbon, C	+2 → 0

- 18** Aqueous iron(II) sulfate is added to acidified potassium manganate(VII). The purple colour of the potassium manganate(VII) disappears.

Aqueous potassium iodide is added to acidified potassium dichromate(VI). A dark brown solution forms.

Which row identifies the role of the iron(II) sulfate and the potassium dichromate(VI) in these reactions?

	iron(II) sulfate	potassium dichromate(VI)
A	oxidising agent	oxidising agent
B	oxidising agent	reducing agent
C	reducing agent	reducing agent
D	reducing agent	oxidising agent

- 19** Which row shows the difference between a weak acid and a strong acid?

	weak acid	strong acid
A	fully dissociated	partially dissociated
B	concentrated	dilute
C	dilute	concentrated
D	partially dissociated	fully dissociated

- 20** Which substance turns methyl orange red?

- A** aqueous ammonia
- B** dilute hydrochloric acid
- C** aqueous sodium hydroxide
- D** distilled water

- 21** Which row describes zinc oxide and calcium oxide?

	zinc oxide	calcium oxide
A	basic	acidic
B	acidic	basic
C	amphoteric	acidic
D	amphoteric	basic

22 Which row shows the properties of a transition element?

	catalyst	colour of oxide	electrical conductivity
A	yes	red	good
B	yes	green	poor
C	no	yellow	good
D	no	white	poor

23 Fluorine is the element at the top of Group VII of the Periodic Table.

Which statement describes fluorine?

- A** It is inert.
- B** It is monatomic.
- C** It is non-metallic.
- D** It is a solid at room temperature.

24 When aluminium is placed in dilute hydrochloric acid, there is no reaction.

When zinc is placed in dilute hydrochloric acid, bubbles of gas are immediately given off.

Which statement correctly explains these observations?

- A** Aluminium is coated with a layer of aluminium oxide.
- B** Aluminium is more reactive than hydrogen.
- C** Aluminium is less reactive than zinc.
- D** Zinc is less reactive than hydrogen.

25 Which statements about the use of sacrificial protection to prevent iron from rusting are correct?

- 1 A more reactive metal than iron is used as a sacrificial protector because it undergoes reduction before iron.
- 2 Zinc is used as a sacrificial protector because it gains electrons more readily than iron.
- 3 Copper is **not** used as a sacrificial protector because it is less reactive than iron.
- 4 Magnesium is used as a sacrificial protector because it loses electrons more readily than iron.

- A** 1 and 2
- B** 1 and 4
- C** 2 and 3
- D** 3 and 4

26 Aluminium is extracted from its ore by electrolysis.

What is the role of cryolite in this process?

- A to lower the operating temperature
- B to lower the boiling point of bauxite
- C to raise the melting point of bauxite
- D to act as a catalyst

27 Which row identifies two greenhouse gases and three processes by which they contribute to global warming?

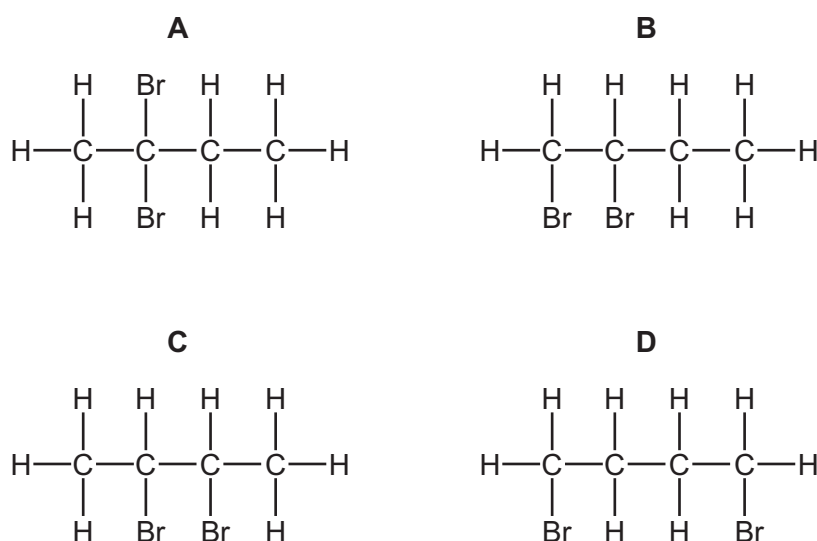
	two greenhouse gases	three processes
A	carbon dioxide and methane	absorption, creation and reflection of thermal energy
B	carbon dioxide and oxygen	absorption, creation and reflection of thermal energy
C	carbon dioxide and methane	absorption, emission and reflection of thermal energy
D	methane and oxygen	absorption, emission and reflection of thermal energy

28 Which mixture contains all of the elements in a typical NPK fertiliser?

- A ammonium nitrate and calcium phosphate
- B ammonium phosphate and potassium chloride
- C potassium nitrate and ammonium chloride
- D potassium carbonate and ammonium nitrate

29 Bromine reacts with but-2-ene.

What is the displayed formula of the product of this reaction?



30 Which statement is correct?

- A Bitumen is used as a fuel for ships.
- B Coal, natural gas and oxygen are all fuels.
- C Hydrogen is the main constituent of natural gas.
- D Petroleum is separated into useful substances by fractional distillation.

31 Which statement explains why ethanoic acid is saturated?

- A The molecule dissociates completely in water.
- B There is a carbon–oxygen double bond in the molecule.
- C The carbon–carbon bond in the molecule is a single bond.
- D All the carbon–hydrogen bonds in the molecule are single bonds.

32 Which statement about compounds in the same homologous series is correct?

- A They have the same chemical properties because they have the same number of carbon atoms.
- B They have the same physical properties because they have the same number of carbon atoms.
- C They have different chemical properties because they have different numbers of carbon atoms.
- D They have different physical properties because they have different numbers of carbon atoms.

33 Which row shows the properties of methane?

	soluble in water	state at room temperature	gives a positive test with aqueous bromine
A	no	gas	no
B	no	gas	yes
C	yes	liquid	no
D	yes	liquid	yes

34 The table shows two methods used to make ethanol.

method	type of process	conditions			source of raw material
		temperature / °C	pressure / atm	catalyst	
fermentation	batch	35	1	yeast	sugar cane
adding steam to ethene	continuous	300	60	acid	petroleum

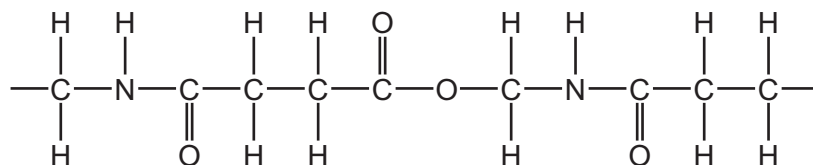
Which statement gives an advantage of preparing ethanol by fermentation rather than by adding steam to ethene?

- A Fermentation takes several days to complete.
- B Little energy is used in the fermentation process.
- C The fermentation of glucose from sugar cane produces pure ethanol.
- D Fermentation uses a non-renewable raw material.

35 Which equation represents an addition reaction?

- A $\text{CH}_3\text{CHO} + \text{HCN} \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CN}$
- B $\text{C}_6\text{H}_6 + \text{Br}_2 \rightarrow \text{C}_6\text{H}_5\text{Br} + \text{HBr}$
- C $\text{NH}_4\text{Br} \rightarrow \text{NH}_3 + \text{HBr}$
- D $\text{C}_{14}\text{H}_{30} \rightarrow \text{C}_2\text{H}_4 + \text{C}_8\text{H}_{18} + \text{C}_4\text{H}_8$

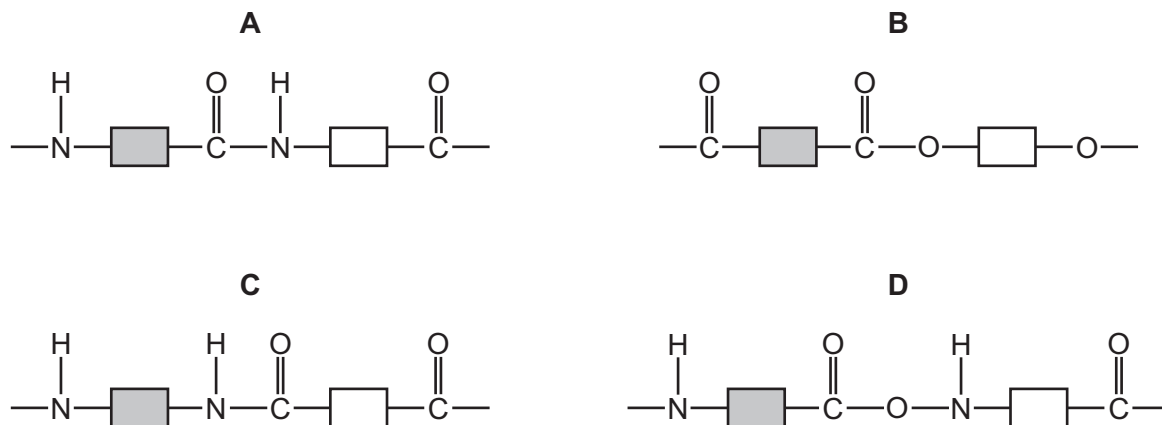
36 The structure of part of a polymer is shown.



How many amide and ester linkages are included in the structure shown?

	amide linkages	ester linkages
A	1	0
B	1	1
C	2	1
D	2	2

37 Which structure represents part of a protein?



38 Which piece of apparatus can only measure a single fixed volume?

- A** a 250 cm³ beaker
- B** a 50 cm³ burette
- C** a 100 cm³ measuring cylinder
- D** a 25 cm³ volumetric pipette

39 Pure solid copper(II) nitrate can be obtained from a mixture of copper(II) nitrate and copper powder.

Three stages in the method are listed.

- X add water and stir
- Y crystallise
- Z filter

After the three stages, the copper(II) nitrate is washed and dried.

What is the correct order of stages X, Y and Z to obtain pure solid copper(II) nitrate from the mixture?

- A** X → Y → Z
- B** X → Z → Y
- C** Y → X → Z
- D** Z → X → Y

40 Which row describes a test and the observation for aqueous sulfate ions?

	test	observation
A	add dilute nitric acid	a gas is produced which turns limewater cloudy
B	add dilute nitric acid and aqueous barium nitrate	white precipitate forms
C	add dilute nitric acid and aqueous potassium manganate(VII)	solution decolourises
D	add dilute nitric acid and aqueous silver nitrate	white precipitate forms

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

The Periodic Table of Elements

Group																		
I	II											III	IV	V	VI	VII	VIII	
		1Hhydrogen1																
		Key atomic number atomic symbol name relative atomic mass																
3Li lithium 7	4Be beryllium 9																	
11Na sodium 23	12Mg magnesium 24																	
19K potassium 39	20Ca calcium 40	21Sc scandium 45	22Ti titanium 48	23V vanadium 51	24Cr chromium 52	25Mn manganese 55	26Fe iron 56	27Co cobalt 59	28Ni nickel 59	29Cu copper 64	30Zn zinc 65	31Ga gallium 70	32Ge germanium 73	33As arsenic 75	34Se selenium 79	35Br bromine 80	36Kr krypton 84	
37Rb rubidium 85	38Sr strontium 88	39Y yttrium 89	40Zr zirconium 91	41Nb niobium 93	42Mo molybdenum 96	43Tc technetium —	44Ru ruthenium 101	45Rh rhodium 103	46Pd palladium 106	47Ag silver 108	48Cd cadmium 112	49In indium 115	50Sn tin 119	51Sb antimony 122	52Te tellurium 128	53I iodine 127	54Xe xenon 131	
55Cs caesium 133	56Ba barium 137	57–71lanthanoids	72Hf hafnium 178	73Ta tantalum 181	74W tungsten 184	75Re rhenium 186	76Os osmium 190	77Ir iridium 192	78Pt platinum 195	79Au gold 197	80Hg mercury 201	81Tl thallium 204	82Pb lead 207	83Bi bismuth 209	84Po polonium —	85At astatine —	86Rn radon —	
87Fr francium —	88Ra radium —	89–103actinoids	104Rf rutherfordium —	105Db dubnium —	106Sg seaborgium —	107Bh bohrium —	108Hs hassium —	109Mt meitnerium —	110Ds darmstadtium —	111Rg roentgenium —	112Cn copernicium —	113Nh nihonium —	114Fl flerovium —	115Mc moscovium —	116Lv livermorium —	117Ts tennessine —	118Og oganesson —	
lanthanoids		57La lanthanum 139	58Ce cerium 140	59Pr praseodymium 141	60Nd neodymium 144	61Pm promethium —	62Sm samarium 150	63Eu europium 152	64Gd gadolinium 157	65Tb terbium 159	66Dy dysprosium 163	67Ho holmium 165	68Er erbium 167	69Tm thulium 169	70Yb ytterbium 173	71Lu lutetium 175		
actinoids		89Ac actinium —	90Th thorium 232	91Pa protactinium 231	92U uranium 238	93Np neptunium —	94Pu plutonium —	95Am americium —	96Cm curium —	97Bk berkelium —	98Cf californium —	99Es einsteinium —	100Fm fermium —	101Md mendelevium —	102No nobelium —	103Lr lawrencium —		

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

Cambridge IGCSE™ (9–1)

CHEMISTRY**0971/22**

Paper 2 Multiple Choice (Extended)

October/November 2024**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	C	1
3	B	1
4	C	1
5	C	1
6	C	1
7	D	1
8	C	1
9	C	1
10	B	1
11	C	1
12	A	1
13	D	1
14	B	1
15	A	1
16	D	1
17	B	1
18	D	1
19	D	1
20	B	1
21	D	1
22	A	1
23	C	1
24	A	1
25	D	1
26	A	1
27	C	1
28	B	1

Question	Answer	Marks
29	C	1
30	D	1
31	C	1
32	D	1
33	A	1
34	B	1
35	A	1
36	C	1
37	A	1
38	D	1
39	B	1
40	B	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/21

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages.



- 1 A gas is placed in a sealed container with a fixed volume.

The gas is heated.

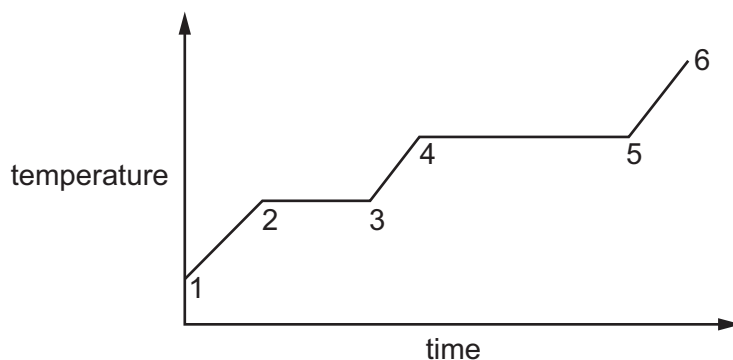
Which statements are correct?

- 1 The gas molecules move around faster.
- 2 The gas particles collide with the walls of the container more frequently.
- 3 The pressure in the container decreases.

A 1 and 2 **B** 1 only **C** 2 and 3 **D** 2 only

- 2 A sample of ethanol is heated at a constant rate.

The heating curve for this sample is shown.



Which statement about the heating curve is correct?

- A** Ethanol changes its physical state between 3 and 4.
B Ethanol condenses between 2 and 3.
C Energy is absorbed between 3 and 4 but **not** absorbed between 4 and 5.
D More energy is absorbed between 4 and 5 than between 2 and 3.

- 3 Which atom has an equal number of protons, neutrons and electrons?

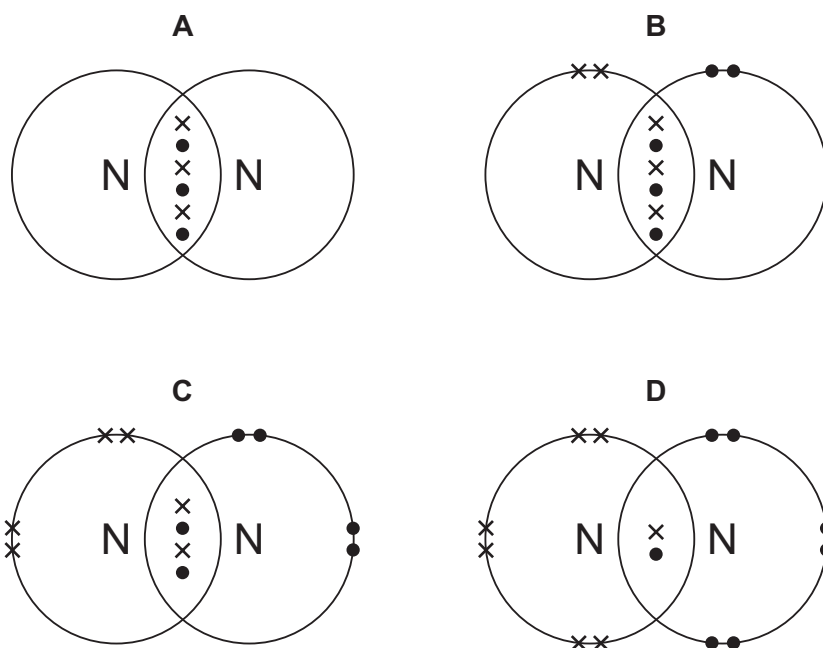
A ^{40}Ar **B** ^1H **C** ^{23}Na **D** ^{14}N

- 4 The only naturally occurring isotopes in a sample of europium are ^{151}Eu and ^{153}Eu . This sample of europium has a relative atomic mass, A_r , of 152.

What is the percentage of ^{151}Eu present in this sample to three significant figures?

A 0.987% **B** 50.0% **C** 76.0% **D** 98.7%

- 5 Which statement about potassium chloride is correct?
- A** It has a high melting point because it is a giant structure with strong covalent bonds.
- B** It conducts electricity when solid because of its oppositely charged ions.
- C** It conducts electricity when molten because its ions can move.
- D** It is insoluble in water because of its strong bonding.
- 6 Which dot-and-cross diagram shows the outer shell electron arrangement in a molecule of nitrogen?



- 7 Which statement about silicon(IV) oxide is correct?
- A** It contains silicon ions and oxide ions arranged in a giant lattice.
- B** It is used in electrical circuits because it can conduct electricity.
- C** Its bonds each contain one shared pair of electrons.
- D** Its atoms are arranged in the same way as the atoms in graphite.

8 Three organic molecules are described.

- The molecular formula of propanoic acid is $\text{CH}_3\text{CH}_2\text{COOH}$.
- The molecular formula of lactic acid is $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$.
- A glucose molecule consists of 6 carbon atoms, 6 oxygen atoms and 12 hydrogen atoms only.

Which molecules have the same empirical formula?

- A** propanoic acid, lactic acid and glucose
B propanoic acid and lactic acid only
C propanoic acid and glucose only
D lactic acid and glucose only

9 Calcium carbonate reacts with dilute hydrochloric acid.

The equation for the reaction is shown.

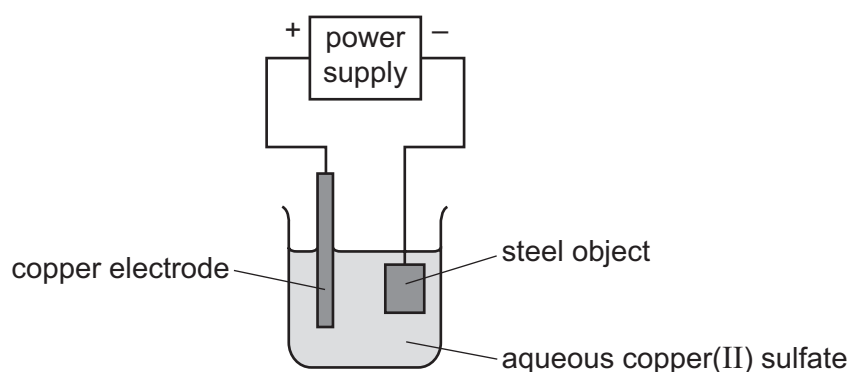


1.00 g of calcium carbonate is added to 50.0 cm^3 of 0.0500 mol/dm^3 hydrochloric acid.

Which volume of carbon dioxide is made in this reaction?

- A** 30 cm^3 **B** 60 cm^3 **C** 120 cm^3 **D** 240 cm^3

10 The diagram shows the electroplating of a steel object.



A student makes three statements.

- 1 The steel object turns a reddish-brown colour.
- 2 The aqueous copper(II) sulfate changes to a paler blue colour.
- 3 The copper electrode becomes thinner.

Which statements are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 11 Concentrated aqueous sodium chloride is electrolysed using inert electrodes.

Which equation shows the correct ionic half-equation for the reaction at the anode?

- A $\text{Cl}^- + \text{e}^- \rightarrow \text{Cl}$
- B $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
- C $\text{H}^+ \rightarrow \text{H} + \text{e}^-$
- D $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$

- 12 Which equation represents the overall reaction in a hydrogen–oxygen fuel cell?

- A $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- B $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$
- C $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}_2$
- D $\text{H}_2\text{O}_2 \rightarrow \text{H}_2 + \text{O}_2$

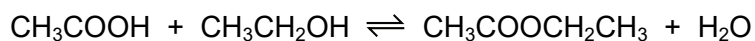
13 Water is added to anhydrous copper(II) sulfate in a test-tube.

The mixture becomes hot.

Which type of reaction and reaction pathway diagram apply to this reaction?

	type of reaction	reaction pathway diagram
A	endothermic	
B	endothermic	
C	exothermic	
D	exothermic	

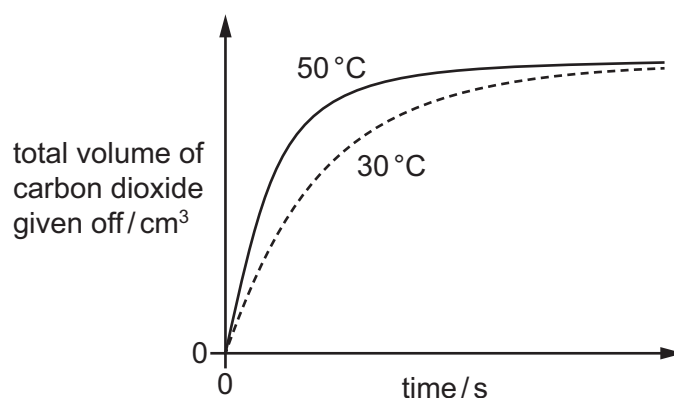
14 The equation represents the reaction between ethanoic acid and ethanol.



Which type of reaction does this show?

- A** addition
- B** neutralisation
- C** reduction
- D** reversible

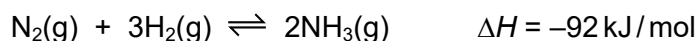
- 15 The graph shows the results of adding excess calcium carbonate to hydrochloric acid at 50 °C and at 30 °C. Only the temperature is changed.



Which row describes the reacting particles at 30 °C compared to those at 50 °C?

	frequency of collisions between particles	average kinetic energy of particles
A	higher	higher
B	higher	lower
C	lower	higher
D	lower	lower

- 16 Industrially, ammonia is made by the Haber process. The equation is shown.



The conditions used are 450 °C and 200 atm.

Which statements about the conditions are correct?

- 1 Temperatures below 450 °C give a lower rate of reaction.
- 2 Temperatures below 450 °C give a lower yield of ammonia.
- 3 Pressures above 200 atm give a lower yield of ammonia.
- 4 Pressures above 200 atm are too expensive to maintain.

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

17 Ethanol is warmed with acidified aqueous potassium manganate(VII).

Which row describes the type of reaction of the ethanol and the colour change observed in the reaction mixture?

	type of reaction	colour change
A	oxidation	colourless to purple
B	oxidation	purple to colourless
C	reduction	colourless to purple
D	reduction	purple to colourless

18 Two experiments involving acids R or S are done.

- 0.10 mol of acid R completely dissolves in 500 cm³ of water. 0.0048 mol of acid R is dissociated in the solution.
- 0.10 mol of acid S completely dissolves in 500 cm³ of water. 0.0179 mol of acid S is dissociated in the solution.

Which statement about acids R and S is correct?

- A** Acid R is a strong acid, and acid S is a weak acid.
- B** Acid R is a weak acid, and acid S is a strong acid.
- C** Acid R and acid S are both strong acids.
- D** Acid R and acid S are both weak acids.

19 Lead(II) sulfate is an insoluble salt.

Which process is **not** used to prepare a pure sample of this salt?

- A** crystallisation
- B** drying
- C** filtration
- D** precipitation

20 The formulae of two hydrated salts are shown.

- $\text{CuSO}_4 \cdot x\text{H}_2\text{O}$
- $\text{CoCl}_2 \cdot y\text{H}_2\text{O}$

What are the values of x and y ?

	x	y
A	5	5
B	5	6
C	6	5
D	6	6

21 The properties of the elements change from left to right across the Periodic Table.

Which statement describes how the properties change across Period 3?

- A** Elements change from gases to solids.
- B** Elements change from non-conductors to electrical conductors.
- C** Elements change from metal to non-metal.
- D** Elements have a decreasing number of electron shells.

22 Which property decreases down Group VII of the Periodic Table, from chlorine to iodine?

- A** atomic number
- B** density
- C** reactivity
- D** melting point

23 Some properties of elements are listed.

- 1 They have good electrical conductivity.
- 2 They have high melting points.
- 3 They are malleable.
- 4 They have ions with variable oxidation numbers.

Which properties are correct for **both** iron and lithium?

- A** 1 and 3
- B** 1 and 4
- C** 2 and 3
- D** 2 and 4

24 Two statements about noble gases are listed.

- 1 Noble gases are reactive, monatomic gases.
- 2 Noble gases all have a full outer shell of electrons.

Which statements about noble gases are correct?

- A Both statements are correct, and statement 2 explains statement 1.
- B Both statements are correct but statement 2 does **not** explain statement 1.
- C Statement 1 is correct but statement 2 is **not** correct.
- D Statement 2 is correct but statement 1 is **not** correct.

25 Which statement about the properties and uses of copper is correct?

- A It is used in electrical wiring, and it is ductile.
- B It is mixed with tin to make the alloy brass.
- C It is used in overhead electrical cables because it has a low density.
- D It is used in takeaway food containers because of its resistance to corrosion.

26 Three statements about solid Z are listed.

- It reacts with dilute hydrochloric acid to produce hydrogen.
- It reacts very slowly with cold water and quickly with steam.
- It is a good thermal conductor.

What is solid Z?

- A gold
- B magnesium
- C potassium
- D silver

27 Four metals J, K, L and M are added separately to aqueous solutions of their ions.

The results are shown.

	aqueous ions				
metal	J ²⁺	K ²⁺	L ²⁺	M ²⁺	
J	x	✓	✓	✓	key ✓ = reaction x = no reaction
K	x	x	x	✓	
L	x	✓	x	✓	
M	x	x	x	x	

Which statement about the metals is correct?

- A J has a greater tendency to form positive ions than K.
- B L has a greater tendency to form positive ions than J.
- C K has a greater tendency to form positive ions than L.
- D M has a greater tendency to form positive ions than K.

28 Which process is used in the extraction of iron from hematite?

- A electrolysis
- B precipitation
- C reduction
- D fractional distillation

29 Galvanising prevents iron from rusting.

Which statements about galvanising are correct?

- 1 Galvanising involves coating the iron with silver.
- 2 Galvanising prevents oxygen and water from coming into contact with the iron.
- 3 Galvanising works because the metal coating loses electrons more readily than iron.

- A 1, 2 and 3 B 1 and 3 only C 2 and 3 only D 2 only

30 Nitrogen, phosphorus and potassium are essential elements for improved plant growth.

Which mixture provides all **three** essential elements?

	mixture	formulae
A	ammonium phosphate + potassium chloride	$(\text{NH}_4)_3\text{PO}_4$ + KCl
B	ammonium phosphate + ammonium nitrate	$(\text{NH}_4)_3\text{PO}_4$ + NH_4NO_3
C	ammonium phosphate + ammonium chloride	$(\text{NH}_4)_3\text{PO}_4$ + NH_4Cl
D	ammonium nitrate + potassium chloride	NH_4NO_3 + KCl

31 Which statements help explain why an increasing amount of carbon dioxide or methane in the atmosphere causes global warming?

- 1 Carbon dioxide absorbs thermal energy from the Earth.
- 2 Methane does **not** absorb thermal energy from the Sun.
- 3 Carbon dioxide does **not** reflect thermal energy from the Sun to the Earth.
- 4 Methane reflects thermal energy from the Earth back to the Earth.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

32 The exhaust gases from cars contain oxides of nitrogen.

How are these oxides of nitrogen formed?

- A** Nitrogen and oxygen from the air react together at the high temperatures in the engine.
- B** Nitrogen and oxygen from the petrol react together in the car exhaust.
- C** Nitrogen from the petrol reacts with oxygen at the high temperatures in the engine.
- D** Nitrogen reacts with oxygen from the air in the catalytic converter.

33 The structural formula of methyl ethanoate is $\text{CH}_3\text{COOCH}_3$.

Which compounds are structural isomers of methyl ethanoate?

- 1 $\text{HCOOCH}_2\text{CH}_3$
- 2 $\text{CH}_3\text{CH}_2\text{COOH}$
- 3 $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH}$

A 1, 2 and 3 **B** 1 and 2 only **C** 2 and 3 only **D** 3 only

34 What is the **total** number of covalent bonds in one molecule of propane?

A 8 **B** 9 **C** 10 **D** 11

35 Industrially, ethanol is produced by fermentation or by the reaction of ethene with steam.

Which row is correct?

	fermentation	ethene + steam
A	uses a temperature of 100°C	uses a temperature of 350°C
B	needs the presence of yeast	does not need a catalyst
C	slow reaction	fast reaction
D	high yield of ethanol	low yield of ethanol

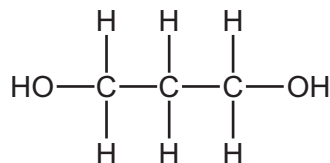
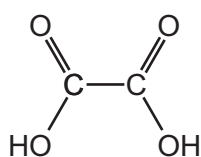
36 Some properties of an organic compound, J, are listed.

- It is a liquid at room temperature.
- It is soluble in water.
- A solution of J reacts with calcium carbonate to form carbon dioxide.
- A solution of J has a pH of 3.

In which homologous series does J belong?

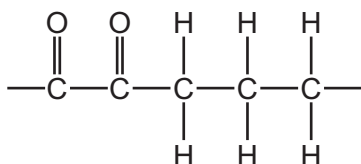
- A** alkane
- B** alkene
- C** alcohol
- D** carboxylic acid

37 The structures of two monomers that are used to make a condensation polymer are shown.

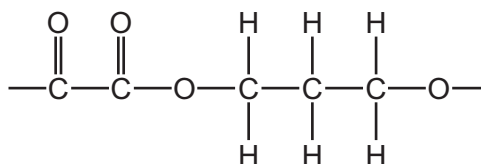


Which structure represents one repeat unit of the condensation polymer formed?

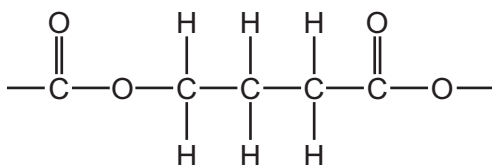
A



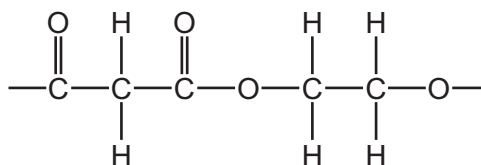
B



C



D

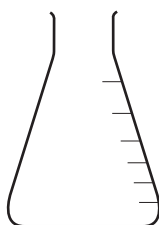


38 Which item of apparatus is used to measure 25.0 cm³ of aqueous sodium hydroxide?

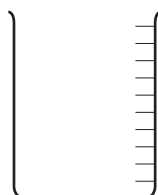
A



B



C



D

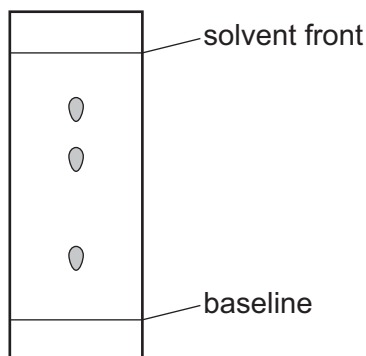


39 Which method is used to obtain copper(II) sulfate crystals from an aqueous solution of copper(II) sulfate?

- A** chromatography
- B** condensation
- C** evaporation
- D** filtration

40 A mixture of four different colourless amino acids is analysed by paper chromatography.

The final chromatogram is shown.



Why does the chromatogram only show three spots?

- A A locating agent is **not** used.
- B One of the amino acids is insoluble in the solvent.
- C The solvent front is too near the top of the paper.
- D Two of the amino acids have the same R_f value.

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										1 H hydrogen 1	5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20	2 He helium 4
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40		
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —		

lanthanoids		57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
		89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

Cambridge IGCSE™ (9–1)

CHEMISTRY (9–1)**0971/21**

Paper 2 Multiple Choice (Extended)

May/June 2025

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	D	1
3	D	1
4	B	1
5	C	1
6	B	1
7	C	1
8	D	1
9	A	1
10	C	1
11	B	1
12	A	1
13	C	1
14	D	1
15	D	1
16	B	1
17	B	1
18	D	1
19	A	1
20	B	1
21	C	1
22	C	1
23	A	1
24	D	1
25	A	1
26	B	1
27	A	1
28	C	1

Question	Answer	Marks
29	C	1
30	A	1
31	B	1
32	A	1
33	B	1
34	C	1
35	C	1
36	D	1
37	B	1
38	D	1
39	C	1
40	D	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0971/22

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

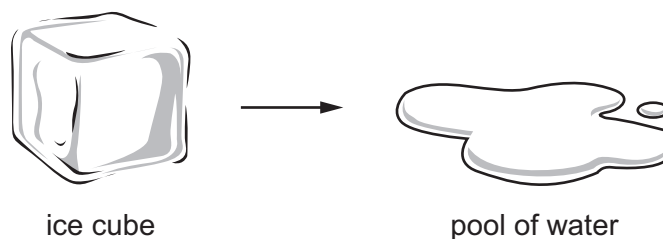
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 An ice cube melts.

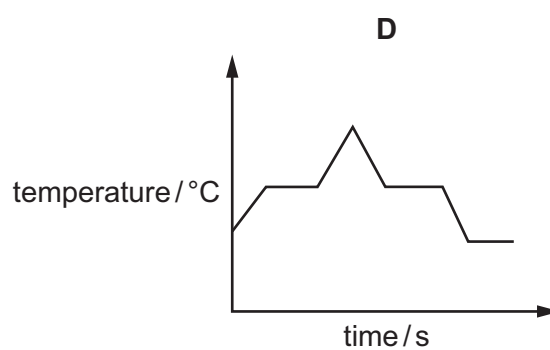
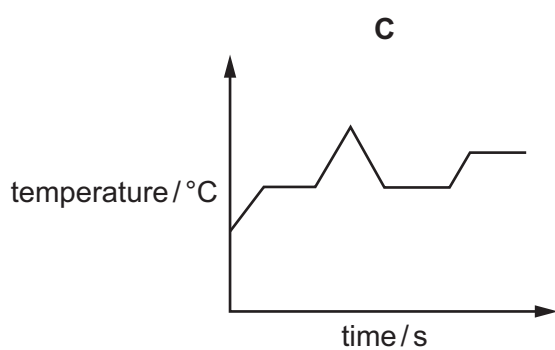
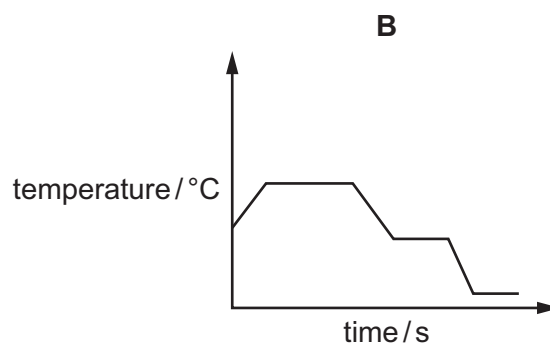
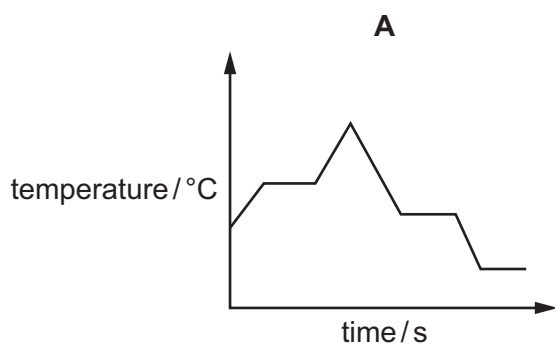


What happens to the molecules of water in the ice cube?

- A They evaporate.
 - B They dissolve.
 - C They gain energy.
 - D They lose energy.
- 2 A sample of a liquid, X, is heated to a temperature above its boiling point.

X is then cooled, so that it condenses and then freezes.

Which graph describes the change in temperature of X over time?



- 3 Which row gives the number of protons, electrons and neutrons in an atom of zinc?

	protons	electrons	neutrons
A	30	30	35
B	30	35	35
C	35	30	30
D	35	35	30

- 4 Which particles have the electronic configuration 2,8,8?

- 1 an argon atom, Ar
- 2 an aluminium ion, Al^{3+}
- 3 a sodium ion, Na^+
- 4 a chloride ion, Cl^-

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

- 5 A sample of element Q has two isotopes.

Their relative masses and abundances are shown.

relative mass of isotope	abundance /%
238	66
244	34

What is the relative atomic mass of this sample of Q to three significant figures?

A 240 **B** 241 **C** 242 **D** 243

- 6 Which statement about silicon(IV) oxide, SiO_2 , is correct?

- A** It conducts electricity because it contains delocalised electrons.
- B** It has a giant covalent structure with each silicon atom bonded to four oxygen atoms.
- C** It is a simple covalent molecule.
- D** Its structure is similar to that of graphite.

- 7 When 65 g of zinc reacts with 32 g of sulfur, 97 g of zinc sulfide is produced.

When 65 g of zinc reacts with 40 g of sulfur, the mass of zinc sulfide produced is still 97 g.

Which statement explains this observation?

- A Some of the zinc sulfide evaporates.
- B The reaction rate is slow.
- C The reaction stops before it is complete.
- D Zinc is the limiting reactant.

- 8 Aluminium reacts with iron(III) oxide to form aluminium oxide and iron.

Which chemical equation for the reaction between aluminium and iron(III) oxide is correct?

- A $Al + FeO \rightarrow AlO + Fe$
- B $2Al + Fe_2O \rightarrow Al_2O + 2Fe$
- C $Al + Fe_2O_3 \rightarrow Al_2O_3 + Fe$
- D $2Al + Fe_2O_3 \rightarrow Al_2O_3 + 2Fe$

- 9 The concentration of a solution of aqueous sodium hydroxide is 0.50 mol/dm^3 .

Which mass of sodium hydroxide is used to make 500 cm^3 of this solution?

- A 10 g B 20 g C 40 g D 160 g

- 10 Molten zinc oxide is electrolysed using inert electrodes.

Which row identifies the product at each electrode?

	anode	cathode
A	zinc	oxygen
B	hydrogen	oxygen
C	oxygen	zinc
D	oxygen	hydrogen

11 Dilute aqueous sodium chloride is electrolysed using platinum electrodes.

What is the half-equation for the reaction at the cathode?

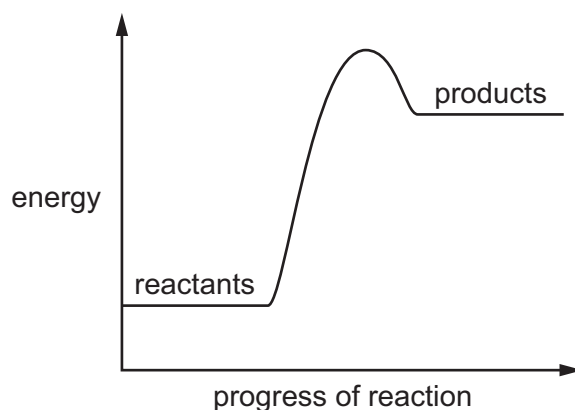
- A** $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
- B** $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$
- C** $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
- D** $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$

12 Hydrogen–oxygen fuel cells can be used to power vehicles.

Which statement about hydrogen–oxygen fuel cells is correct?

- A** The equation for the overall reaction is $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}_2$.
- B** The only chemical products are water and carbon dioxide.
- C** Chemical energy in the fuel is converted into electrical energy.
- D** The hydrogen used in hydrogen–oxygen fuel cells is a fossil fuel.

13 A reaction pathway diagram is shown.



Which row identifies the type of reaction and how the temperature of the surroundings changes during the reaction?

	type of reaction	temperature of the surroundings
A	endothermic	decreases
B	endothermic	increases
C	exothermic	decreases
D	exothermic	increases

- 14 The average bond energy for the C–H bond is 413 kJ/mol.

What is the enthalpy change when 1.0 mol of methane molecules is formed from carbon and hydrogen atoms?

- A –1652 kJ B –413 kJ C +413 kJ D +1652 kJ

- 15 Which statement describes the effect of adding a catalyst to a chemical reaction?

- A The activation energy, E_a , of the reaction is increased.
 B The enthalpy change, ΔH , of the reaction stays unchanged.
 C The frequency of collisions between the particles is decreased.
 D The kinetic energy of the particles is increased.

- 16 Which row describes the conditions used in the manufacture of sulfuric acid by the Contact process?

	catalyst	pressure	temperature
A	iron	high	high
B	iron	low	low
C	vanadium(V) oxide	high	low
D	vanadium(V) oxide	low	high

- 17 Which equations show the underlined species acting as a reducing agent?

- 1 Fe₂O₃ + 3CO → 2Fe + 3CO₂
 2 Fe + CuSO₄ → FeSO₄ + Cu
 3 5Fe²⁺ + MnO₄[–] + 8H⁺ → 5Fe³⁺ + Mn²⁺ + 4H₂O
 4 Fe(OH)₂ + H₂SO₄ → FeSO₄ + 2H₂O

- A 1 and 2 B 1 and 4 C 2 and 3 D 3 and 4

- 18 A farmer incorrectly adds two substances to the soil at the same time.

They react together to form a gas which turns damp red litmus paper blue.

What are the two substances?

- A a basic oxide and a potassium salt
 B a basic oxide and an ammonium salt
 C an acidic oxide and a potassium salt
 D an acidic oxide and an ammonium salt

- 19 Which row shows the equations for the dissociation of hydrochloric acid and of ethanoic acid in aqueous solution?

	hydrochloric acid	ethanoic acid
A	$\text{HCl}(\text{aq}) \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$	$\text{CH}_3\text{COOH}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$
B	$\text{HCl}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$	$\text{CH}_3\text{COOH}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$
C	$\text{HCl}(\text{aq}) \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$	$\text{CH}_3\text{COOH}(\text{aq}) \rightarrow \text{H}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$
D	$\text{HCl}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$	$\text{CH}_3\text{COOH}(\text{aq}) \rightarrow \text{H}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$

- 20 Which three oxides are **all** acidic?

- A** CaO, NO₂, SO₂
B CaO, CO₂, Na₂O
C CO₂, NO₂, SO₂
D CO₂, Na₂O, SO₂

- 21 The table shows some properties of noble gases.

noble gas	density in g/dm ³	boiling point in °C
helium	0.2	−269
neon	0.9	−246
argon	1.8	−186
krypton		
xenon	5.9	−108

Which row predicts the density and the boiling point of krypton?

	density in g/dm ³	boiling point in °C
A	1.4	−75
B	2.4	−195
C	3.8	−153
D	8.5	−213

22 Sodium is a Group I metal.

What is a physical property of sodium?

- A non-conductor of electricity
- B high melting point
- C high density
- D malleable

23 Tennessine (atomic number 117) is a manufactured element that is below astatine in Group VII of the Periodic Table.

What is the expected state of tennessine at room temperature and pressure?

- A a diatomic gas
- B a liquid
- C a monatomic gas
- D a solid

24 When chlorine is bubbled into aqueous sodium bromide, a displacement reaction occurs.

Which description of this reaction is correct?

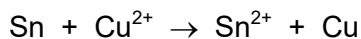
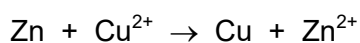
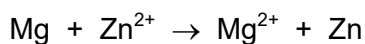
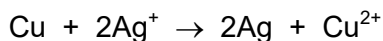
- A Chloride displaces bromide.
- B Chlorine displaces bromine.
- C Chloride displaces bromine.
- D Chlorine displaces bromide.

25 Brass is an alloy of copper.

Which statement about brass or copper is correct?

- A Brass does **not** conduct electricity because it is a compound.
- B Copper conducts electricity because its atoms are free to move.
- C Brass is harder than copper because its layers of atoms **cannot** easily slide over each other.
- D Copper is stronger than brass because it is a pure metal.

26 The ionic equations for some reactions between metals and aqueous metal ions are shown.



Which further information can be used to place copper, magnesium, silver, tin and zinc in order of their reactivity?

- A Zinc reacts with Sn^{2+} ions.
- B Copper does **not** react with Sn^{2+} ions.
- C Magnesium reacts with Cu^{2+} ions.
- D Magnesium reacts with Ag^+ ions.

27 Zinc is used to galvanise iron to prevent it from rusting.

Which statements are correct?

- 1 Galvanising is an example of a barrier method.
- 2 If the zinc is scratched, the iron will rust very quickly.
- 3 Galvanising is an example of sacrificial protection.
- 4 Zinc is more reactive than iron and so accepts electrons more readily.

- A 1, 2 and 4 B 1 and 3 C 2, 3 and 4 D 2 and 3 only

28 Which equation represents a reaction that occurs in a blast furnace during the extraction of iron from hematite?

- A $\text{Ca} + \text{CO}_2 \rightarrow \text{CaO} + \text{CO}$
- B $\text{Fe}_2\text{O}_3 + 3\text{Ca} \rightarrow 2\text{Fe} + 3\text{CaO}$
- C $\text{Ca}(\text{OH})_2 \rightarrow \text{CaO} + \text{H}_2\text{O}$
- D $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$

29 Water is added separately to anhydrous copper(II) sulfate and to anhydrous cobalt(II) chloride.

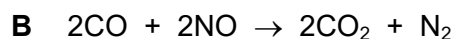
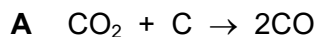
Which row shows the colour changes?

	anhydrous copper(II) sulfate	anhydrous cobalt(II) chloride
A	blue to white	blue to pink
B	blue to white	pink to blue
C	white to blue	blue to pink
D	white to blue	pink to blue

30 Gases which cause acid rain form in car engines.

These gases can be removed using a catalytic converter.

Which equation represents the removal of one of these gases using a catalytic converter?



31 Which statements about carbon dioxide are correct?

- 1 Carbon dioxide absorbs thermal energy from the Earth.
- 2 Carbon dioxide increases the thermal energy loss to space.
- 3 The level of carbon dioxide in the atmosphere has increased due to the burning of fossil fuels.
- 4 Carbon dioxide is the only gas that causes global warming.

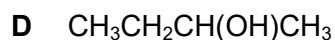
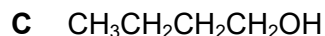
A 1 and 2

B 1 and 3

C 2 and 3

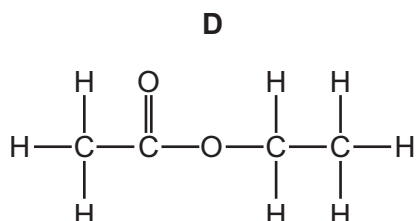
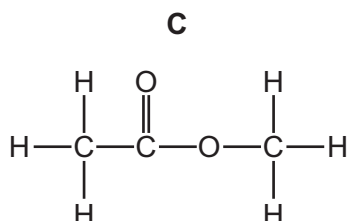
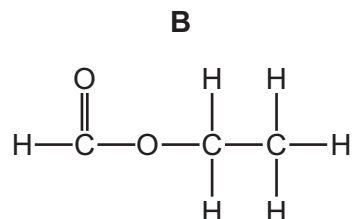
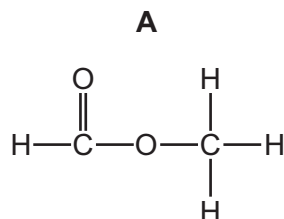
D 2 and 4

32 What is the structural formula of butan-2-ol?



- 33** Methanol reacts with ethanoic acid to form an ester.

What is the displayed formula for this ester?



- 34** Petroleum can be separated into useful fractions by fractional distillation.

Which row shows a correct use of the named fraction?

	fraction	use
A	bitumen	lubricant
B	naphtha	fuel for home heating
C	kerosene	jet fuel
D	refinery gas	making chemicals

- 35** Ethane reacts with chlorine in the presence of ultraviolet light.

Which products are formed?

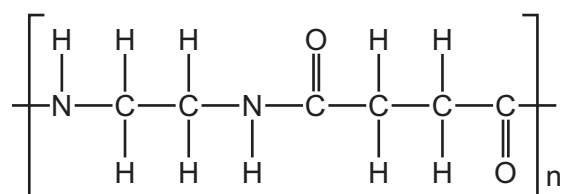
- A** $\text{ClCH}_2\text{CH}_2\text{Cl}$ and H_2
B $2\text{CH}_3\text{Cl}$ and H_2
C $\text{C}_2\text{H}_5\text{Cl}$ and HCl
D $\text{CH}_2=\text{CH}_2$ and 2HCl

- 36** Ethanol can be made by the fermentation of aqueous glucose and by the catalytic addition of steam to ethene.

What are two advantages of making ethanol by the catalytic addition of steam to ethene rather than by the fermentation of aqueous glucose?

- A** faster reaction and renewable raw materials
- B** purer product and faster reaction
- C** renewable raw materials and continuous process
- D** uses more energy and forms a purer product

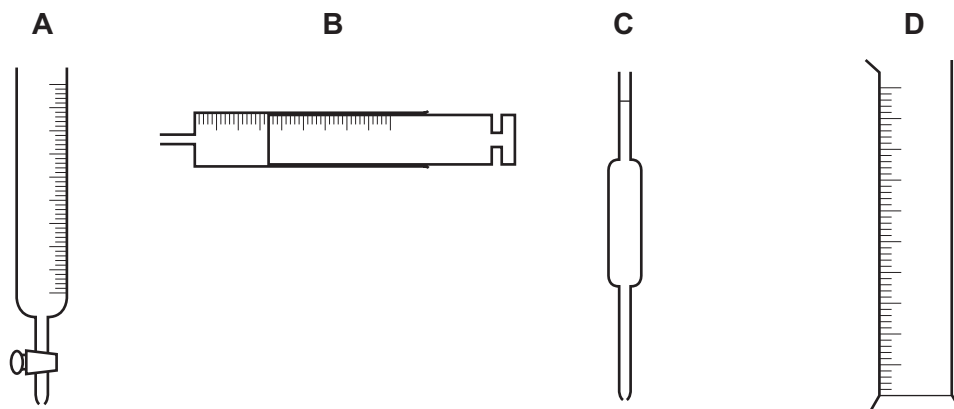
- 37** The structure of a polymer repeat unit is shown.



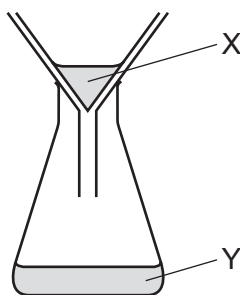
Which pair of monomers is used to make this polymer?

	monomer 1	monomer 2
A	$\begin{array}{cccc} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{N}- & \text{C}- & \text{C}- & \text{N} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{H} \end{array}$	$\begin{array}{ccccc} & \text{H}-\text{O} & & \text{O}-\text{H} & \\ & & & & \\ & \text{C} & - & \text{C} & \\ & & & & \\ & \text{O} & & \text{O} & \end{array}$
B	$\begin{array}{cccc} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{N}- & \text{C}- & \text{C}- & \text{N} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{H} \end{array}$	$\begin{array}{ccccccc} \text{H}-\text{O} & & \text{H} & & \text{H} & & \text{O}-\text{H} \\ & & & & & & \\ \text{C} & - & \text{C} & - & \text{C} & - & \text{C} \\ & & & & & & \\ \text{O} & & \text{H} & & \text{H} & & \text{O} \end{array}$
C	$\begin{array}{ccccccc} \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & & \text{H} \\ & & & & & & \\ \text{N}- & \text{C}- & \text{C}- & \text{C}- & \text{C}- & & \text{N} \\ & & & & & & \\ \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & & \text{H} \end{array}$	$\begin{array}{ccccc} \text{H}-\text{O} & & & & \text{O}-\text{H} \\ & & & & \\ \text{C} & - & \text{C} & & \\ & & & & \\ \text{O} & & \text{O} & & \end{array}$
D	$\begin{array}{ccccccc} \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & & \text{H} \\ & & & & & & \\ \text{N}- & \text{C}- & \text{C}- & \text{C}- & \text{C}- & & \text{N} \\ & & & & & & \\ \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & & \text{H} \end{array}$	$\begin{array}{ccccccc} \text{H}-\text{O} & & \text{H} & & \text{H} & & \text{O}-\text{H} \\ & & & & & & \\ \text{C} & - & \text{C} & - & \text{C} & - & \text{C} \\ & & & & & & \\ \text{O} & & \text{H} & & \text{H} & & \text{O} \end{array}$

38 Which item of apparatus is used to measure exactly 26.3 cm^3 of a liquid?



39 A mixture containing an aqueous salt and an insoluble salt is filtered.



Which row describes X and Y?

	X	Y
A	solute	pure water
B	solute	filtrate
C	residue	pure water
D	residue	filtrate

40 Pure ethanol has a melting point of -114°C and a boiling point of 78°C .

What are the melting and boiling points of a sample of ethanol with glucose dissolved in it?

	melting point/ $^\circ\text{C}$	boiling point/ $^\circ\text{C}$
A	-116	77
B	-116	79
C	-112	77
D	-112	79

BLANK PAGE

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

Group

57	La	lanthanum	139	58	Ce	cerium	140	59	Pr	praseodymium	141	60	Nd	neodymium	144	61	Pm	promethium	—	62	Sm	samarium	150	63	Eu	euroium	152	64	Gd	gadolinium	157	65	Tb	terbium	159	66	Dy	dysprosium	163	67	Ho	holmium	165	68	Er	erbium	167	69	Tm	thulium	169	70	Yb	ytterbium	173	71	Lu	lutetium	175
89	Ac	actinium	—	90	Th	thorium	232	91	Pa	protactinium	231	92	U	uranium	238	93	Np	neptunium	—	94	Pu	plutonium	—	95	Am	americium	—	96	Cm	curium	—	97	Bk	berkelium	—	98	Cf	californium	—	99	Es	einsteinium	—	100	Fm	fermium	—	101	Md	merdelevium	—	102	No	nobelium	—	103	Lr	lawrencium	—

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

Cambridge IGCSE™ (9–1)

CHEMISTRY (9–1)**0971/22**

Paper 2 Multiple Choice (Extended)

May/June 2025**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	D	1
3	A	1
4	B	1
5	A	1
6	B	1
7	D	1
8	D	1
9	A	1
10	C	1
11	A	1
12	C	1
13	A	1
14	A	1
15	B	1
16	D	1
17	C	1
18	B	1
19	A	1
20	C	1
21	C	1
22	D	1
23	D	1
24	B	1
25	C	1
26	A	1
27	B	1
28	D	1

Question	Answer	Marks
29	C	1
30	B	1
31	B	1
32	D	1
33	C	1
34	C	1
35	C	1
36	B	1
37	B	1
38	A	1
39	D	1
40	B	1



Cambridge IGCSE™ (9–1)

CHEMISTRY

0971/22

Paper 2 Multiple Choice (Extended)

October/November 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

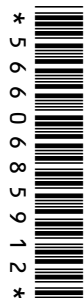
INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

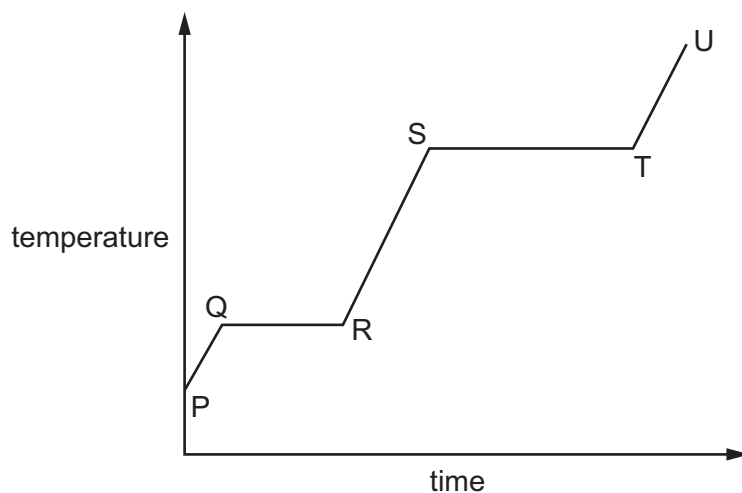
This document has **16** pages.



- 1 What happens to the volume of a fixed mass of gas when the temperature is increased or the pressure is decreased?

	increased temperature	decreased pressure
A	volume decreased	volume decreased
B	volume decreased	volume increased
C	volume increased	volume decreased
D	volume increased	volume increased

- 2 The diagram shows the heating curve for a substance.



Between which two points do the particles of the substance change from being in fixed positions to being mobile?

- A** P and Q **B** Q and R **C** S and T **D** T and U

- 3 A student investigates the diffusion of ammonia gas, NH_3 , and hydrogen chloride gas, HCl .

Two sets of apparatus are set up as shown at room temperature and pressure.



The damp red litmus paper in apparatus 1 changes colour after 30 seconds.

How long does it take for the damp blue litmus paper in apparatus 2 to change colour?

- A 21 seconds
- B 30 seconds
- C 64 seconds
- D The damp blue litmus paper does **not** change colour.

- 4 Which row describes elements, compounds and mixtures?

	elements	compounds	mixtures
A	all the atoms are the same type	contain different atoms chemically joined together	can be separated using physical processes
B	all the atoms are the same type	can be separated into new substances using chemical processes	contain different atoms chemically joined together
C	can be separated into new substances using chemical processes	all the atoms are the same type	contain different atoms chemically joined together
D	contain different atoms chemically joined together	all the atoms are the same type	can be separated using physical processes

- 5 Which statement about atoms or ions is correct?

- A The mass number of an atom is the total number of nucleons and protons.
- B The electronic configuration of an aluminium ion, Al^{3+} , is 2,8,3.
- C Hydrogen is the only atom that has an incomplete first electron shell.
- D Most of the volume of an atom is taken up by its nucleus.

- 6 Which two values are used to calculate the relative atomic mass of a sample of boron?

	value 1	value 2
A	the total number of boron isotopes	the total number of boron atoms
B	the relative abundance of each boron isotope in the sample	the relative mass of each boron isotope
C	the relative abundance of each boron isotope in the sample	the total number of boron atoms
D	the total number of boron isotopes	the relative mass of each boron isotope

- 7 Which statement describes the structure of sodium chloride?

- A** a lattice of alternating positive and negative ions
B a lattice of positive ions surrounded by a 'sea' of delocalised electrons
C a mixture of sodium and chlorine atoms
D a regular lattice of sodium and chlorine atoms

- 8 Graphite and diamond are both forms of the element carbon.

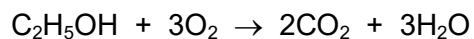
Which row shows the number of other carbon atoms that each carbon atom is covalently bonded to in graphite and in diamond?

	graphite	diamond
A	3	3
B	3	4
C	4	3
D	4	4

- 9 Which row describes the structure and bonding in cobalt?

	structure	bonding
A	giant	ionic
B	simple	ionic
C	giant	metallic
D	simple	metallic

- 10 The equation for the combustion of ethanol is shown.



The relative molecular mass, M_r , of ethanol is 46.

Which statement about the reaction is correct?

- A** 4.6 g of ethanol produces 4.4 g of carbon dioxide.
B 92 g of ethanol produces 88 g of carbon dioxide and 108 g of water.
C 138 g of ethanol reacts with 32 g of oxygen.
D 184 g of ethanol produces 216 g of water.
- 11 A compound contains 40.0% carbon, 6.7% hydrogen and 53.3% oxygen only.

The relative molecular mass, M_r , of the compound is 60.

Which row shows the empirical formula and the molecular formula of the compound?

	empirical formula	molecular formula
A	CH_2O	CH_2O
B	CH_2O	$\text{C}_2\text{H}_4\text{O}_2$
C	$\text{C}_3\text{H}_7\text{O}_3$	$\text{C}_3\text{H}_7\text{O}_3$
D	$\text{C}_3\text{H}_8\text{O}$	$\text{C}_3\text{H}_8\text{O}$

- 12 Hydrogen–oxygen fuel cells are used in electric cars.

Three statements about these fuel cells are listed.

- 1 The process in the cell is called electrolysis.
- 2 Water is broken down in the cell to produce hydrogen, oxygen and electricity.
- 3 Water is the only chemical product formed in the reaction.

Which statements are correct?

- A** 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

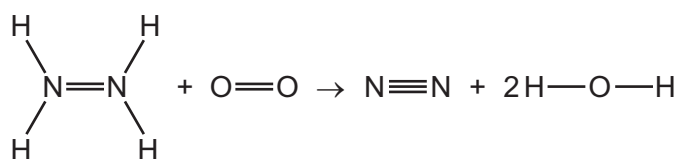
13 A solution of copper(II) sulfate can be electrolysed using copper electrodes or carbon electrodes.

Which statements are correct?

- 1 Using copper electrodes, oxygen gas forms at the anode.
- 2 Using copper electrodes, copper atoms lose electrons at the anode.
- 3 Using carbon electrodes, copper metal forms at the cathode.
- 4 Using carbon electrodes, copper ions gain electrons at the cathode.

A 1 and 2 **B** 1 and 3 **C** 2, 3 and 4 **D** 4 only

14 The equation for the combustion of hydrazine, N_2H_4 , is shown.



bond	bond energy in kJ/mol
N–H	391
N=N	409
N≡N	944
O–H	463
O=O	496

What is the overall enthalpy change for the combustion of one mole of hydrazine?

- A** –327 kJ/mol
B –111 kJ/mol
C +111 kJ/mol
D +327 kJ/mol

- 15** Dissolving ammonium chloride in water is an endothermic change.

Which row shows the energy change and temperature change of the mixture during the dissolving of ammonium chloride?

	energy change	temperature change
A	energy is absorbed	decrease
B	energy is absorbed	increase
C	energy is released	decrease
D	energy is released	increase

- 16** Aqueous sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$, reacts with dilute hydrochloric acid to form a yellow precipitate of sulfur.



The precipitate forms more quickly when the reactants are heated.

Which statements explain this observation?

- 1 The reacting particles collide more frequently.
- 2 The collisions between reacting particles have more energy.
- 3 The activation energy of the reaction is lower.

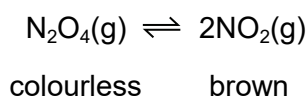
- A** 1 and 2 **B** 1 and 3 **C** 1 only **D** 2 and 3

- 17** Which statement about the conditions used in the Haber process and in the Contact process is correct?

- A** The typical pressure used in the Haber process is similar to that used in the Contact process.
- B** The typical temperature used in the Haber process is similar to that used in the Contact process.
- C** The Haber process uses a vanadium(V) oxide catalyst, and the Contact process uses an iron catalyst.
- D** Both processes produce an acidic gas.

- 18 The equation shows the equilibrium between dinitrogen tetroxide, N_2O_4 , and nitrogen dioxide, NO_2 .

The colours of the reactant and product are shown.



The forward reaction is endothermic.

Which statement is **not** correct?

- A** At equilibrium, the concentrations of the reactant and the product stay constant.
- B** At equilibrium, the rate of the forward reaction is equal to the rate of the reverse reaction.
- C** When the pressure is increased, a darker brown colour is seen.
- D** When the temperature is increased, a darker brown colour is seen.
- 19 What is the oxidation number of chromium in the $\text{Cr}_2\text{O}_7^{2-}$ ion?
- A** -2 **B** -1 **C** +6 **D** +12
- 20 Which method is used to prepare a pure, dry sample of lead(II) chloride?
- A** mix aqueous lead(II) nitrate and dilute hydrochloric acid → filter → crystallise and dry the filtrate
- B** mix aqueous lead(II) nitrate and dilute hydrochloric acid → filter → wash and dry the residue
- C** mix solid lead(II) hydroxide and aqueous sodium chloride → filter → crystallise and dry the filtrate
- D** mix solid lead(II) hydroxide and aqueous sodium chloride → filter → wash and dry the residue
- 21 Phosphorus is an element in Group V of the Periodic Table.

It burns in air to form an oxide that dissolves in water to form a solution with a pH of 1.

Which row describes this oxide of phosphorus?

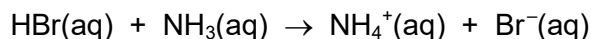
	metal oxide	non-metal oxide	acidic oxide	basic oxide
A	✓	x	✓	x
B	✓	x	x	✓
C	x	✓	✓	x
D	x	✓	x	✓

key

✓ = yes

x = no

- 22** The equation shows the reaction between aqueous hydrogen bromide and aqueous ammonia.



Which statement describes the role of aqueous hydrogen bromide?

- A** It is a catalyst.
- B** It is a reducing agent.
- C** It is a proton acceptor.
- D** It is a proton donor.

- 23** Which statement about the elements in Group I is correct?

- A** Hydrogen is produced when Group I elements react with water.
- B** Ions of Group I elements have a 1– charge.
- C** Sodium is more reactive than potassium.
- D** Solid sodium is a poor electrical conductor.

- 24** Which row describes the colour and state of iodine at room temperature and pressure?

	colour	state
A	green	liquid
B	grey-black	solid
C	purple	solid
D	red-brown	liquid

- 25** Part of the Periodic Table is shown.

Which element is an inert gas?

A simplified periodic table with 18 columns and 4 rows. The columns are grouped into four sets of two, three, six, and two columns respectively. Element A is in the first column of the second row. Element B is in the third column of the second row. Element C is in the sixth column of the second row. Element D is in the sixth column of the third row.

26 Four different metals are separately added to water.

The oxide of each metal is separately heated with carbon.

The results are shown.

metal	metal with water	metal oxide with carbon
Q	no reaction	brown solid forms
R	rapid fizzing	no reaction
S	a few bubbles	no reaction
T	a few bubbles	grey solid forms

What is the order of reactivity of the four metals, least reactive first?

A Q → T → S → R

B Q → S → T → R

C R → S → T → Q

D R → T → S → Q

27 An iron nail is fully covered with zinc to prevent it from rusting.

Which row describes this method of protection and explains why zinc is used to protect the iron nail?

	method of protection	reason zinc is used
A	barrier method and sacrificial protection	zinc is more reactive than iron
B	barrier method and sacrificial protection	iron is more reactive than zinc
C	sacrificial protection only	zinc is more reactive than iron
D	barrier method only	iron is more reactive than zinc

28 Iron is extracted from hematite in the blast furnace.

Which reaction increases the temperature in the blast furnace to over 1500 °C?

A calcium carbonate → calcium oxide + carbon dioxide

B calcium oxide + silicon(IV) oxide → calcium silicate

C carbon + oxygen → carbon dioxide

D carbon dioxide + carbon → carbon monoxide

29 Aluminium is extracted by the electrolysis of purified bauxite.

Which row shows the half-equations at the electrodes and explains why one of the electrodes has to be replaced regularly?

	half-equation at anode	half-equation at cathode	electrode replacement
A	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	cathode replaced as it gets coated with aluminium
B	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	anode replaced as it burns away in oxygen
C	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	cathode replaced as it burns away in oxygen
D	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	anode replaced as it gets coated with aluminium

30 Which flow chart describes the correct treatment sequence of the domestic water supply?

- A** filtration → chlorination → sedimentation
- B** sedimentation → chlorination → filtration
- C** chlorination → filtration → sedimentation
- D** sedimentation → filtration → chlorination

31 Which statement about oxides of nitrogen is correct?

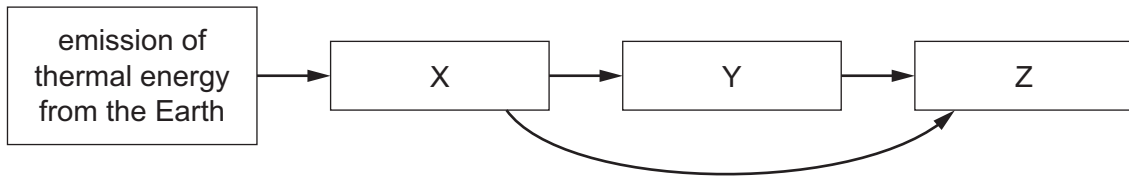
- A** They are made in car engines by the incomplete combustion of gasoline at a high temperature.
- B** They are made in car engines when nitrogen in gasoline reacts with oxygen in air at a high temperature.
- C** They are removed from car exhausts by catalytic converters and form nitrogen and sulfur dioxide.
- D** They are removed from car exhausts when they react with carbon monoxide in catalytic converters.

32 The alkenes are a homologous series.

Which statement about the members of the homologous series of the alkenes is correct?

- A** They have the same functional group.
- B** They have the same relative molecular mass.
- C** They have different chemical properties.
- D** They have different general formulae.

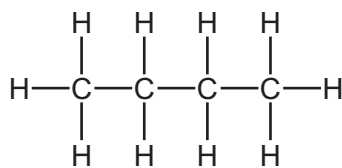
- 33 The diagram represents two processes, X and Y, and their effect, Z, in increasing global warming.



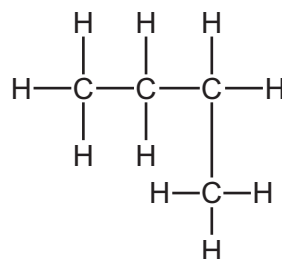
Which row identifies X, Y and Z?

	X	Y	Z
A	absorption of thermal energy by greenhouse gases	emission of thermal energy to the Earth	reduced thermal energy loss to space
B	emission of thermal energy by greenhouse gases	emission of thermal energy to space	reduced thermal energy loss to space
C	absorption of thermal energy by greenhouse gases	emission of thermal energy to the Earth	increased thermal energy loss to space
D	emission of thermal energy by greenhouse gases	emission of thermal energy to space	increased thermal energy loss to space

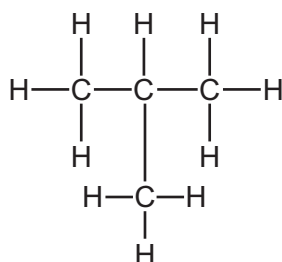
34 Which structures are structural isomers of each other?



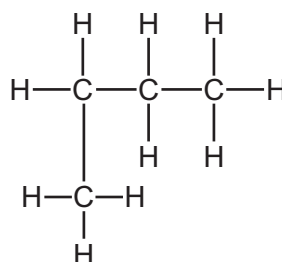
1



2



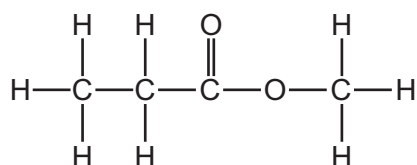
3



4

- A** 1, 2, 3 and 4
B 1, 2 and 4 only
C 1 and 3 only
D 2 and 4 only

35 The structure of an ester is shown.



What is the name of this ester?

- A** ethyl methanoate
B methyl ethanoate
C propyl methanoate
D methyl propanoate

36 Which row shows the products obtained by the cracking of a large alkane?

	alkene	hydrogen	water
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key

✓ = yes

✗ = no

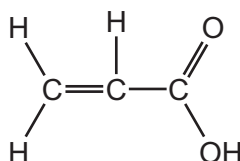
37 Ethanol is produced by the fermentation of aqueous glucose or by the catalytic addition of steam to ethene.

Which row shows some advantages and disadvantages of these processes?

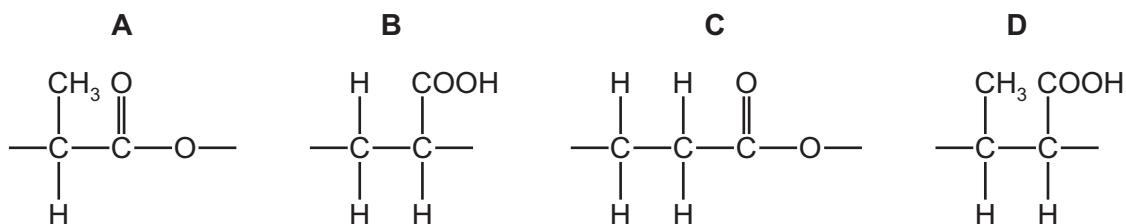
	fermentation of aqueous glucose		catalytic addition of steam to ethene	
	advantage	disadvantage	advantage	disadvantage
A	fast reaction	ethanol produced continuously	almost pure ethanol formed	renewable raw material
B	ethanol produced in batches	slow reaction	fast reaction	ethanol produced continuously
C	renewable raw material	ethanol produced in batches	almost pure ethanol formed	slow reaction
D	renewable raw material	dilute aqueous ethanol formed	fast reaction	non-renewable raw material

38 Propenoic acid is an organic compound that is both an alkene and a carboxylic acid.

The structure of propenoic acid is shown.



Which structure represents one repeat unit of the addition polymer formed from propenoic acid?



39 Which item of apparatus is used to measure 13.7 cm^3 of dilute hydrochloric acid?

- A a balance
- B a burette
- C a conical flask
- D a volumetric pipette

40 Chromatography is used to separate and analyse mixtures of soluble substances.

The first steps of the process are listed.

- apply sample to baseline
- develop chromatogram in solvent

What is the next step in the process to detect soluble **colourless** substances?

- A calculate the R_f values
- B compare the results with known substances
- C count the number of spots
- D use a suitable locating agent

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

The Periodic Table of Elements

Group																		
I	II											III	IV	V	VI	VII	VIII	
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										1 H hydrogen 1	5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —	
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —	

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
actinoids	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

Cambridge IGCSE™ (9–1)

CHEMISTRY**0971/22**

Paper 2 Multiple Choice (Extended)

October/November 2025**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	B	1
3	C	1
4	A	1
5	C	1
6	B	1
7	A	1
8	B	1
9	C	1
10	D	1
11	B	1
12	D	1
13	C	1
14	A	1
15	A	1
16	A	1
17	B	1
18	C	1
19	C	1
20	B	1
21	C	1
22	D	1
23	A	1
24	B	1
25	C	1
26	A	1
27	A	1
28	C	1

Question	Answer	Marks
29	B	1
30	D	1
31	D	1
32	A	1
33	A	1
34	C	1
35	D	1
36	B	1
37	D	1
38	B	1
39	B	1
40	D	1